Confidence-Aware Episodic Memory with Self-Improvement for Progressive Hallucination Reduction in Large Language Models

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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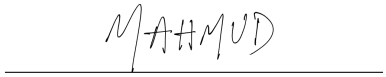
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Abstract

Large language models produce confident but factually incorrect outputs across deployment domains: asked for supporting literature, chatbots return references that do not exist [43], for between 28.6 and 91.4 percent of the references they generate in systematic-review searches [65], a failure mode that increased parameter count alone does not close [39]. This thesis proposes *Confidence-Aware Episodic Memory with Self-Improvement (CAEM)*, an architectural intervention that treats hallucination reduction as a system-level problem rather than a post-hoc detection problem. A single query passes through a verified episodic memory store, a confidence-aware three-tier router with an independent safety override, and a nine-signal verifier ensemble spanning internal calibration, sample-set agreement, external grounding, and question-answer relevance. The ensemble feeds a per-benchmark calibrated-probability composite (per-signal isotonic regression, James-Stein shrinkage at $\alpha_{\text{shrink}} = 0.6$, regularised logistic boost over the per-signal log-odds [82]) which gates storage at two fixed cut-points, $\tau_{\text{store}} = 0.60$ and $\tau_{\text{defer}} = 0.45$. Thresholds remain constant across cycles; the cycle-boundary refit absorbs distribution drift into the signal-to-probability mapping, not the threshold. The design replaced an earlier split-conformal alternative [77] that was rejected on the registered calibration-to-evaluation distribution-shift evidence. Self-improvement is delivered through low-rank adaptation [36] at rank thirty-two and scaling factor sixty-four over the attention and feed-forward projections, with the backbone frozen; after every fine-tune the loop refits temperature, per-signal isotonic curves, and per-benchmark boost weights on the re-scored calibration fold (the same held-out fold re-scored under the updated model, not a new fold) and runs retroactive re-verification before the next cycle.

Empirical validation uses Qwen-2.5-3B-Instruct on a five-benchmark panel split into a training panel of three (FEVER, TriviaQA, CommonsenseQA) and a held-out transfer panel of two (TruthfulQA, StrategyQA). Cycle zero locks a configuration with storage precision 0.796 over 289 stored entries on the cycle-zero validation fold, drawn from a one-hundred-variant pre-specified sweep. Against the matched zero-shot baseline on the same backbone, the within-trajectory best cycle reaches a pooled exact-match gain of +20.3 percent and a pooled composite hallucination metric reduction of −29.4 percent on the training panel; the full-panel reading at the same cycle is +7.4 percent exact match and −11.0 percent composite hallucination, with the margin compressed by a TruthfulQA regression in which dense retrieval returns passages adjacent to the question's misconception rather than refutations. On the transfer panel, StrategyQA's composite hallucination falls by 17.5 percent at trajectory end, a reduction that does not reach statistical significance. Against the locked external eight-baseline panel, CAEM leads every baseline on the pooled composite hallucination axis at both the best and final cycles and on pooled exact match at the best cycle, and at the trajectory-end cycle leads the fine-tuned baseline on pooled exact match (0.517 against 0.490) while trailing the active-retrieval baseline by 0.008; the per-benchmark exact-match significance panel reports fourteen wins and three defeats at the best-cycle anchor and nine wins and three defeats at the trajectory-end anchor under Holm-corrected paired McNemar tests with percentile bootstrap intervals, with every defeat confined to the misconception-bait TruthfulQA surface where the architecture trades exact match for calibrated abstention.

The architectural contribution is paired with a formal analysis of the storage class. Two results are proven: pool purity is bounded below by calibration-anchored empirical precision under upward-only retroactive update, and storage makes the pool purer than the generator's raw output exactly when the verifier's balanced accuracy exceeds one half. Design heuristics, stated with their idealising assumptions, describe the expected trajectory across cycles: storage rate rises with verifier discrimination; the precision gap on the committed-cycle subsequence stays within a bounded stationary band, with the sharper geometric rate registered as a longer-horizon receipt; and the memory-direct tier's asymptotic hallucination rate is bounded by the complement of the stored pool's calibration-fold precision, independent of deployment scale. The architecture's value over plain exact-match fine-tuning at matched labelling budget rests on six differentiators: a per-query precision contract fine-tuning cannot produce, a memory-direct tier that compounds cost savings as the pool grows, a zero-shot tier that distils verified retrieval-augmented reasoning into parametric capacity, asymmetric rollback that preserves memory while reverting the adapter, the verifier as a sample-efficiency multiplier (validating a precision contract rather than carrying gradient), and inference-time error catching that survives parametric saturation. Every served answer carries the calibrated probability of correctness alongside a four-class decision tag. The realised six-cycle trajectory, the cycle-zero validation gate verdict, the external baseline panel under matched protocol, and the retrieval-utility classifier ablation are reported in the empirical chapter and appendices.

Keywords: hallucination reduction; episodic memory; calibrated probability; fixed-threshold gating; self-improvement; retrieval-augmented generation; large language models; verifier ensemble; isotonic regression; low-rank adaptation; continual learning; abstention.

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List of Acronyms

Architecture and methods

CAEM	Confidence-Aware Episodic Memory with Self-Improvement
RUC	Retrieval-Utility Classifier (Stage 3b)
RAG	Retrieval-Augmented Generation
CoT	Chain-of-Thought prompting
LoRA	Low-Rank Adaptation
KLE	Kernel Language Entropy
NTM	Neural Turing Machine
A-MEM	Agentic Memory
FLARE	Forward-Looking Active Retrieval
SI / MAS	Synaptic Intelligence; Memory-Aware Synapses
SIL	Self-Improvement Loop
NLI	Natural Language Inference
DPR	Dense Passage Retrieval
EWC	Elastic Weight Consolidation
MC	Monte Carlo (MC-Dropout)
STaR	Self-Taught Reasoner (**V-STaR**: verifier)
DNC	Differentiable Neural Computer
FActScore	Atomic-fact factuality score
Self-RAG	Self-reflective RAG
CP	Conformal Prediction (**split-CP**; registered then replaced by fixed-threshold storage rule)

Models and judges

LLM	Large Language Model
MiniCheck	Flan-T5-Large NLI judge for claim-support
SBERT	Sentence-BERT (cosine encoder)
MNLI / SNLI	Multi-genre / Stanford NLI corpora

Benchmarks

FEVER	Fact Extraction and VERification
TQA	TriviaQA
CSQA	CommonsenseQA
HotpotQA	Multi-hop QA (dropped at cycle zero)
HaluEval-QA	Hallucination-Evaluation QA (RUC training only)
MMLU	Massive Multitask Language Understanding (retention probe)
BIG-bench	Beyond the Imitation Game
SQuAD	Stanford Question Answering Dataset
NQ	Natural Questions (dropped at cycle zero; RUC training only)
TruthfulQA	Truthful QA (Haiku-judged EM)
StrategyQA	Multi-hop strategy QA
OpenBookQA	Open-Book QA (RUC training only)
HaluEval	Hallucination Evaluation benchmark

Metrics, statistics, infrastructure

EM	Exact Match
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
CES	Composite Evaluation Score (geometric mean of ACC, EPI, RET, CAL, VER; **EPI** $= 1 -$ CHM)
AUROC	Area Under ROC curve
F_1	Token-level F_1
CHM	Composite Hallucination Metric
ECE	Expected Calibration Error
TPR / FPR / TNR	True / False / True-Negative Rate

TPR	True-Positive Rate	**CI**	Confidence Interval
ID / OOD	In- / Out-of-Distribution	**EMA**	Exponential Moving Average
FPR	False-Positive Rate	**FAISS**	Facebook AI Similarity Search (**IVF**: inverted-file)
TTL	Time To Live (deferred-buffer expiry)	**QA**	Question Answering

List of Symbols

Confidence, verifier signals, thresholds

u_{pre}	Pre-routing confidence	u_{stored}	Calibrated post-generation composite
u_{token}	Mean token log-probability	u_{dropout}	MC-Dropout variance, K=5
u_{internal}	$0.5u_{\text{tok}}+0.5(1-u_{\text{drop}})$	s_{avg}	Mean pairwise cosine, M=3
p_{entail}	Chain-to-answer NLI entailment	h_{norm}	Normalised semantic entropy, K=10
$p_{\text{ground,max}}$	Top-1 passage entailment	$p_{\text{ground,mean}}$	Mean entailment over top-3
$p_{\text{ground,atomic}}$	Length-gated atomic-fact entailment	$q_{\text{a,rel}}$	Question-answer relevance

h_{norm} (normalised semantic entropy, K=10) was registered then retired post Cycle-0 audit (boost weight $\approx -5 \times 10^{-4}$); deployed verifier consumes nine signals.

$\tau_{\text{store}}, \tau_{\text{defer}}$	Fixed store / defer cut-points on u_{stored} (0.60 / 0.45)	τ_{train}	Strict training-pool threshold (0.70)
τ_{retro}	Retroactive prune threshold (0.50)	τ_{ttl}	Deferred-buffer TTL (cycles)
τ_{cons}	Memory consolidation cosine	ϕ_{pg}	Abstention floor on top-passage entailment (0.20)
ϕ_{safe}	Pre-routing safety floor on u_{pre} (0.38; per-benchmark refit)		
η_u, η_g	Early-exit thresholds	τ_{RUC}	Retrieval-utility classifier threshold (0.375)

Calibrated probability composite

$\alpha_{\text{store}}, \alpha_{\text{defer}}$ (split-CP miscoverage targets) and α_{ema} (EMA refit factor) were registered in the conformal storage gate then retired when the gate was replaced by the fixed-threshold rule above; the calibration-versus-evaluation distribution shift broke the marginal-coverage premise of split-CP on the five-benchmark panel. The composite refit absorbs distribution drift through the per-signal isotonic curves and the per-benchmark boost coefficients on a fresh per-cycle fold.

$n_{\min}$	Minimum top-k for fit (20)	$n_{\text{cal}}, n_{\text{eval}}$	Calibration / evaluation fold sizes
$\pi_{\text{store}}, \pi_{\text{defer}}$	Calibration-fold empirical precision	π_{retro}	Retroactive-survivor precision
π_t, π_∞	Pool purity at cycle t, asymptote	P_c, P_c^{theory}	Empirical / Bayes pool purity
P_j, iso_j	Per-signal isotonic probability	$L_{i,j}$	Calibration log-odds matrix
$\mathbf{w}, b, w_j$	Boost weights, intercept	C	Boost L_2 strength (0.01)
T	Logit-space temperature	$\sigma(\cdot), \ell$	Logistic; composite log-odds

Self-improvement, router, memory

$\boldsymbol{\theta}, \boldsymbol{\theta}^{(t)}$	Generator parameters at cycle t	$\boldsymbol{\theta}^\star$	Pre-retention-guard argmin
λ_{anc}	L_2 anchor strength	$\mathcal{L}_{\text{CE}}, \mathcal{L}$	CE loss; total per-cycle loss
$\rho, \rho_{\min}$	Retention ratio; floor (0.93)	$\rho_{\text{pre/post}}$	Pre / post fine-tune probe accuracy
$\mathcal{T}$	Training pool above τ_{train}	$\mathcal{M}, \mathcal{M}_t$	Episodic memory; pool at cycle t
$\mathcal{M}_t^{\text{kept}}, \mathcal{N}_{t+1}$	Survivors; new admissions	$\mathcal{B}, K_{\text{def}}$	Deferred buffer; capacity

λ	Routing similarity weight (0.70)	$s, \mathcal{S}$	Cosine; dispatch score $\lambda s+(1-\lambda)u_{\text{stored}}$
$\tau_1(t), \tau_1^*(D)$	Tier-1 hit rate; asymptote	$\varepsilon_{\text{arch}}$	Architectural false-negative rate
$H_{\text{mem}}(t), H_t, H_\infty$	Memory / total / asymptotic hallucination	M, K	Chain count (3); answer count (10)
p_{RAG}	RUC retrieval-helpfulness score	p_{IK}	Self-knowledge probe (Kadavath-style)

Theorem trajectory and statistics

d, d_t	Cohen's d at cycle t	$d_{\text{id}}, d_{\text{transfer}}$	ID and transfer Cohen's d
μ_+, μ_-, ς	Within-class means; shared std	p_+, p_-	Correct / incorrect base rates
$\Phi(\cdot), \phi(\cdot)$	Std-normal CDF; density	σ_t	Empirical storage rate at cycle t
r	Geometric contraction rate	G_t	Cycle-t gap $\lvert\pi_t-\pi_\infty\rvert$
$A, k, c^*(\varepsilon)$	Decay fit; time-to-equilibrium	p_c, α_c	Base correctness; verifier balanced acc.
$\text{TPR}_c, \text{FPR}_c, \text{TNR}_c$	Storage-gate rates at cycle c	$f(p)$	Recurrence $p\alpha/(p\alpha+(1-p)(1-\alpha))$
$x(\lvert\theta\rvert, C), K(C)$	Asymptotic correctness; supportable set	$\chi^2_{\text{McN}}, b, c$	McNemar; discordant pair counts
B	Bootstrap replicates (1 000)	α	Significance level (overloaded)
EM, F_1	Per-task correctness	$\text{CHM}, \text{CES}, \text{ECE}$	Composite hallucination / eval / cal. error
N, T_{cycles}	Cycle index; registered cap (10); realised closure at 6	$\lvert\theta\rvert$	Backbone parameters (3.086×10^9)

Chapter 1

Introduction

This chapter introduces the problem the thesis addresses, motivates the choice to tackle it through architectural rather than incremental means, formalises the problem statement, declares the objectives, summarises the proposed system at a high level, and registers the scope within which the work is bounded.

1.1 Background

1.1.1 Capability and the persistent failure mode

Large language models trained on broad text corpora and instruction-tuned for assistant-style interaction now perform competitively on a wide range of natural language tasks, including factoid question answering, multi-step reasoning, summarisation, and program synthesis. Frontier systems such as GPT-4, Claude, and PaLM reach or exceed reported human accuracy on standardised academic benchmarks, and their accessibility through chat interfaces has made them part of everyday research, education, and professional workflows [44], [45], [58]. The same systems, however, generate confident but factually incorrect statements at deployment-relevant rates. The accepted name for this failure mode is hallucination, although the term covers heterogeneous causes ranging from retrieval limitations to confabulation under genuine uncertainty.

A practical consequence is that the same fluency that makes large language models useful also makes their hallucinations dangerous. A false answer phrased with the same confidence as a true one is harder to detect than a false answer that sounds uncertain. Users without independent verification, particularly in high-stakes domains, take the model's confident phrasing as a signal of reliability and then act on incorrect information. This thesis takes the position that hallucination is therefore not merely a benchmark-accuracy problem but a deployment-safety problem, and that addressing it requires architectural change rather than incremental decoding adjustments.

1.1.2 The hallucination problem at scale

The empirical record on hallucination is consistent across benchmarks and domains. The TruthfulQA benchmark of Lin and colleagues isolates questions on which humans themselves are prone to hold confident misconceptions, and reports that the largest frontier model evaluated achieved only 58 percent on the truthfulness axis and 21 percent on the combined truthful-and-informative axis, against a human baseline of 94 and 87 percent respectively [39]. The benchmark also documented an inverse-scaling relationship in which larger models were more, not less, prone to echo human falsehoods, indicating that scale alone does not solve the problem.

Long-form generation shows the same pattern at finer granularity. The atomic-fact decomposition protocol of Min and colleagues, evaluating chat-style models on biographical generation, found that only 58 percent of the atomic facts in ChatGPT's biographies were supported by the reference knowledge source [57]. Domain-specific evaluations are still more concerning. Asked to support a clinical explanation with references, ChatGPT returned five citations of which none existed [43], and a study of systematic-review searches found that 39.6, 28.6 and 91.4 percent of the references produced by GPT-3.5, GPT-4 and Bard respectively were hallucinated [65], far above any threshold a responsible deployment would tolerate.

These results indicate that hallucination is neither an artefact of small models nor a residual that scale alone will eliminate; the survey literature describes it as an artefact of neural generation that persists in large, instruction-tuned and preference-tuned models [49], [88]. This motivates changing the structure of the system around the model rather than relying on further training alone.

1.1.3 The stateless gap

Three structural shortcomings of the current inference paradigm shape every other limitation a deployed model exhibits. First, standard inference is stateless: every query is processed independently, no representation of past successful answers is retained across sessions, and identical questions can produce inconsistent answers from one interaction to the next. This wastes computation on queries the system has effectively already solved and prevents the accumulation of domain-specific knowledge through use.

Second, modern neural networks are systematically overconfident. Calibration studies report that modern neural classifiers assign confidence that exceeds their actual accuracy, with expected calibration errors above 10 percentage points on standard image benchmarks [17], and that language models fine-tuned with reinforcement learning from human feedback appear badly miscalibrated until their outputs are rescaled [37]. The confidence a model states in words discriminates correct from incorrect answers only weakly: for GPT-4 it averages an area under the receiver operating curve of about 0.63, close to the 0.5 of random guessing, and no single elicitation method is consistently best [83]. A user therefore cannot reliably tell which answers are likely to be wrong from the model's stated confidence alone.

Third, deployed models are static. Once trained they neither learn from their mistakes nor adapt to the patterns of an institution that uses them, because naive online fine-tuning incurs catastrophic forgetting on the broad capabilities the base training

instilled [19]. The implication is that verification effort spent today produces no compounding benefit tomorrow: a thousand answers verified by humans this morning will not change a single answer the model gives this afternoon.

These three shortcomings reinforce one another. Without memory the model cannot reuse verified knowledge. Without calibrated confidence the system cannot recognise when memory is missing. Without self-improvement the verifications that humans or external systems do produce decay into one-shot interventions. The remainder of this thesis develops a system that breaks the cycle along all three axes simultaneously. The next section argues why a combined intervention at the architectural level is the appropriate response.

1.2 Rationale of the Study or Motivation

1.2.1 The deployment gap

The capability gap that motivates this thesis is the distance between what large language models do well and what reliable deployment requires of them. Fluency, breadth of topic coverage, and on-demand reasoning are already present in current frontier models. What is not present is the discipline of distinguishing facts that the model has correctly internalised from confabulations that emerge under genuine uncertainty [70], and the discipline of accumulating verified knowledge across interactions instead of regenerating answers from scratch every time. The gap is not aesthetic. In educational deployments, students who learn from confidently incorrect tutoring risk acquiring misconceptions. In healthcare information services, a hallucinated drug interaction or treatment dosage would produce decisions whose costs are measured in patient outcomes rather than in benchmark points. In enterprise knowledge workflows, fabricated citations and missing provenance erode the trust that the entire deployment rests on.

A second, less visible cost is computational. The stateless protocol of a typical large-model deployment regenerates every answer from first principles, including answers whose substance has already been verified in a previous interaction. As deployment scale grows, the cost of unnecessary regeneration accumulates against any computational budget. A system that can recognise a familiar query and serve a previously verified answer at near-zero cost converts the same compute envelope into more delivered value, and that conversion is itself part of the rationale.

1.2.2 Why prior pipelines do not close the gap

Three families of prior systems have attempted to reduce hallucination, and each has a structural limitation that prevents it from closing the gap on its own. Retrieval-augmented generation grounds the model in retrieved evidence [32], which raises factual recall when the retrieval succeeds but is silent when retrieval misses or returns evidence the model summarises incorrectly. The output remains a single-pass generation whose quality is bounded by retrieval quality, and the system has no representation of which past answers were correct. Prompting interventions, including chain-of-thought reasoning and self-consistency over sampled chains [40], [61], expose internal reasoning and exploit agreement across resampled answers, but

they provide no persistence: each session begins with no memory of prior verified answers, and the same question asked tomorrow triggers the same reasoning effort as today. Post-generation verification [55], [70] detects hallucinations after they occur and supports refusal or revision, but it does not feed back into model weights or memory, so the same hallucination patterns recur in the next session.

The implication is that hallucination reduction at deployment-relevant rates is unlikely to come from incremental improvement of any one family. Stronger retrievers, longer chains of thought, and sharper post-generation verifiers each remain bounded by the structural limitation of their family when used in isolation. What is missing is a system in which verification feeds memory, memory feeds routing, routing feeds generation, and improvement compounds across cycles instead of evaporating at session boundaries. That system is an architectural intervention, and arguing for it is the work this thesis attempts.

1.2.3 The architectural opportunity

Three capabilities, taken together, would close the gap that the existing three families leave open. The first is a verified episodic memory whose admission rule is finer than a single binary threshold. An answer that is plausible but has not earned the precision required for retrieval should not be discarded, because the same query may be answered more confidently after the model has been updated; it should instead enter a deferred buffer and be reconsidered at the next cycle boundary. A four-outcome decision (store, defer, abstain, discard) gives the system somewhere to put borderline outputs without contaminating the high-precision pool. The second capability is confidence-aware routing that reads pre-generation uncertainty as an independent safety signal alongside the combined memory similarity and stored quality. It adds a per-query retrieval-utility decision on the queries that would otherwise fall to the retrieval-augmented tier, so that retrieval is consulted only where it is expected to help rather than unconditionally. When the pre-generation uncertainty is high, the routing layer should force grounded retrieval before generation has started, rather than rely on post-generation review to catch an answer that should never have been produced. The third capability is a constrained self-improvement loop that fine-tunes a small low-rank adapter on a frozen backbone using a verified high-quality pool, so that the bounded adapter plays the role that the parameter anchor plays in elastic weight consolidation [19], and that rolls the adapter back if a held-out retention probe indicates general capability has eroded. Together the three capabilities convert verification from an event that fires once per query into a discipline that compounds across cycles.

A practical consequence of the joint architecture, named the confidence-aware episodic memory with self-improvement (CAEM), is that the three serving tiers are intended to act not as static cost categories but as a compute ladder that reorganises itself under the cycle loop, and that two distinct statelessness problems of the standard inference paradigm are converted into two stateful mechanisms by two different architectural elements. The first statelessness problem is that retrieval-augmented generation pays the full retrieval-and-generation cost in every session because the model never absorbs what retrieval just taught it; CAEM closes this by feeding verified retrieval-augmented outputs into the self-improvement loop's training pool,

so each cycle's low-rank-adapter update folds the previously retrieval-dependent reasoning into the generator's parametric capacity. The design expectation is that the next cycle answers part of the previously retrieval-dependent queries from the zero-shot tier alone, without paying the retrieval cost. The second statelessness problem is that even when the model already knows the answer, the standard protocol re-generates it from scratch on every recurrence and offers no guarantee that two sessions produce the same answer; CAEM closes this through the verified episodic memory, where exact and near-duplicate queries are served directly from the stored answer at embedding-lookup cost, and where the retroactive re-verification pass at every cycle boundary refreshes the stored score of every cached entry against the updated generator, so that the cached answer's quality tracks the model's current capability rather than freezing at the moment of admission. In this design the zero-shot tier absorbs queries that the self-improvement loop has taught the generator to answer, the memory-direct tier absorbs queries that recur and whose cached answer survives the retroactive pass, and the retrieval-augmented tier retains the novel queries that neither mechanism yet covers. On the realised six-cycle trajectory this redistribution is muted on the evaluation fold: the retrieval-augmented tier served 51.3 percent of evaluation queries at cycle zero and between 44.7 and 55.1 percent at later cycles, and the memory-direct tier served at most 0.1 percent, because the evaluation questions are content-disjoint from the stored episodes by construction (Table 5.10). The memory-direct tier is built for repeated and paraphrased questions: on a separate probe of 100 questions built from episodes stored by cycle five, it fired on 42 of 50 exact repeats and 19 of 50 paraphrases, and its answers were correct in 41 of the 42 and all 19 cases.

The user-facing surface mirrors the same logic. Every served answer carries the calibrated probability of correctness as a continuous percentage alongside the four-class decision tag, so a user sees both the categorical reliability label (verified, provisional, conflicting, or insufficient) and the underlying probability that the answer is right. When the calibrated probability and the grounding evidence both fall short of the registered floors, the system abstains explicitly rather than serving a guessed answer with an inflated confidence label. The architecture therefore targets three deployment-time properties at once: hallucination that falls across cycles through the verifier-gated self-improvement loop, per-query cost that falls as queries migrate from the retrieval-augmented tier into the cheaper zero-shot and memory-direct tiers, and answers that always carry the system's own calibrated estimate of their reliability.

There is also a theoretical case for why this combined intervention can succeed even when verification is itself imperfect. By Bayes' rule, the stored pool is purer than the generator's raw output whenever the storage gate is more likely to admit a correct answer than an incorrect one, however imperfect the gate is otherwise. The formal statement is Proposition 4.2; how the property is expected to carry across cycles is stated in Chapter 4 as a design heuristic rather than a proven result. For the rationale here, the relevant point is that the architecture does not require a perfect verifier; it requires a verifier whose balanced accuracy exceeds one half on the distribution of claims actually scored, and Chapter 5 reports the evidence on whether that condition holds on the five-benchmark panel.

The scope of the intervention is bounded so that it can be argued empirically within

an undergraduate research budget: a single small open generator (Qwen-2.5-3B-Instruct), five public factual question-answering benchmarks, and a single consumer-grade graphics processor, so that the result is reproducible without specialised infrastructure; Section 1.6 sets out the full scope. The next section formalises the problem the intervention is designed to solve.

1.3 Problem Statement

1.3.1 Formal problem

Let a deployed question-answering system be specified by a generator distribution and a retrieval substrate, both held fixed at deployment time. A query arrives and produces an answer through the system's per-query pipeline, after which the system must decide whether to commit the produced answer into a persistent state that will influence future answers. The thesis addresses the following problem. Given a stream of queries drawn from a knowledge-intensive distribution, design a deployable system whose committed answer pool is dominated by correct answers, whose user-facing answers carry a calibrated probability of being correct that is refitted on labelled data at every cycle, so that its precision can be checked against held-out data, whose computational cost per query falls as the system processes more of the stream, and whose general capability does not erode as it accumulates domain-specific knowledge. The system is required to satisfy these properties without per-domain calibration at deployment, without external service calls, and on a single small instruction-tuned generator.

The objects under control are the storage decision applied to each generated answer, the calibration of the confidence used to make that decision, the routing of incoming queries among a memory tier and one or more generation tiers, and the cycle-boundary update of the model parameters. The objects not under control are the generator's prior over factual content, the retrieval corpus, and the eventual stream distribution. Within the controlled set the design must produce a falsifiable empirical claim against a panel of external baselines under matched protocol.

1.3.2 Shortcomings addressed and exclusions

The system must address the three shortcomings set out in Section 1.1.3 together: stateless operation that discards verified answers, stated confidence that does not separate correct from incorrect answers, and a static model that cannot learn from verified use. Because the three reinforce one another, addressing any one in isolation yields a system that is better on one axis and unchanged on the other two: a memory without calibrated confidence cannot tell when it is worth consulting, and calibration without self-improvement leaves the underlying error rate unchanged. The thesis therefore commits to a joint treatment in which memory feeds routing, routing feeds generation, and verification feeds both memory and the cycle-boundary update.

Two classes of approach are explicitly excluded from the present scope. The first is changes to the generator's pre-training corpus or pre-training objective. The thesis treats the generator as fixed and instruments the system around it, so that the architectural claim is separable from any pre-training advantage. The second is

reliance on external commercial verification services. Every verification signal the system uses must be reproducible from the open weights of a small specialised model and a publicly available retrieval corpus, so that the result is reproducible without paid infrastructure. The boundary of the problem the thesis solves is therefore an architecture that operates on a fixed open generator, with open verification machinery, on standard hardware, and that is designed to satisfy, and evaluated against, the four properties registered in Section 1.3.1 on the five-benchmark factual question-answering panel of Section 5.1.1. The next section converts this problem statement into the specific objectives that organise the rest of the work.

1.4 Objective

1.4.1 Primary objective

The primary objective of this thesis is to develop a deployable architecture that reduces confident factual hallucination in knowledge-intensive question answering on a single small instruction-tuned generator, while preserving the generator's general capability across self-improvement cycles. The architecture, CAEM, addresses the three reinforcing shortcomings registered in Section 1.3 through a verified episodic memory paired with a calibrated probability composite, a four-outcome storage decision, a confidence-aware three-tier router with an independent safety override, retroactive cycle-boundary review of stored content, and a constrained self-improvement loop guarded by a multi-modal retention probe, a panel of held-out probes that spans three task formats (general-knowledge multiple choice, open-text factoid, and commonsense multiple choice). The success of the architecture is measured against a panel of external baselines under matched protocol on a five-benchmark factual question-answering panel partitioned into three training benchmarks and two transfer benchmarks, using a composite hallucination metric that aggregates eight measurable subtypes of confident-error failure mode, together with per-tier latency and the per-cycle retention ratio as auxiliary diagnostic axes.

1.4.2 Specific objectives

The primary objective decomposes into seven specific objectives. Each is a separate engineering or scientific deliverable that contributes to the joint claim and that admits an independent empirical check in the later chapters.

1. **Verified episodic memory with a four-outcome decision:** Design an episodic memory whose admission rule is finer than a single binary threshold. A generated answer enters one of four mutually exclusive outcomes determined by a calibrated probability composite, a confabulation early-exit rule that fires when high internal confidence coincides with low external grounding, and a deferral path that holds borderline answers for cycle-boundary reconsideration. Storage uses a similarity-indexed encoder over query and reasoning content with a novelty filter and a value-based pruning policy that combines recency, retrieval frequency, and stored-answer quality.

2. **Pre-generation confidence, three-tier routing with safety override, and a retrieval-utility classifier:** Compute a pre-generation confidence

estimate from a short greedy decode over the query that fuses the token-probability aggregate of the first thirty-two decoded tokens with a layer-convergence ratio read from the hidden states over the query, calibrate it via per-benchmark temperature scaling on a disjoint fold, and use it independently from the memory similarity score and the stored-answer quality. The router combines query-to-memory similarity with stored-answer quality into a single dispatch score that selects between memory-direct retrieval, zero-shot generation, and retrieval-augmented generation. An independent safety override forces the conservative tier whenever pre-generation confidence falls below a fixed floor, so that grounded retrieval is available before generation begins and is not entangled with the quality of memory the query happens to land near. A retrieval-utility classifier (Stage 3b) then intercepts the safety-override dispatch and the score-formula fall-through to the retrieval-augmented tier and refines the default into either the zero-shot tier or the retrieval-augmented tier on a per-query basis (Section 4.2.5); its contribution is measured by the on/off ablation in Chapter 5.

3. **Nine-signal post-generation verifier with a calibrated probability composite:** Combine nine complementary signals into a per-answer record across four families: internal calibration of the generator, sample-set agreement across resampled chains, external grounding against retrieved evidence under three aggregation rules, and question-answer relevance. The signals enter a calibrated probability composite that fits per-signal isotonic regression both on the pooled training-panel fold and on each per-benchmark slice, blends each per-benchmark child curve with the pooled fit through a James-Stein style shrinkage prior, and aggregates the per-signal calibrated probabilities through a regularised logistic regression layer fitted on the same calibration fold. The learned aggregation follows the multi-scoring approach of [82], which combines the logits of several hallucination scores through logistic regression, in place of a fixed weighted sum.

4. **Fixed-threshold storage gate with two registered cut-points:** Convert the calibrated probability composite into the storage and deferral decisions through two cut-points and one abstention floor, all fixed before the main run and held constant across cycles. The storage cut-point is chosen on the cycle-zero calibration fold and sets how sure the system must be before an answer enters memory; the deferral cut-point, set lower, sends a wider band of borderline answers to the deferred buffer for reconsideration at the next cycle boundary.

5. **Retroactive review and deferred reconsideration at cycle boundaries:** Because the verifier's internal-calibration signals read the generator's hidden state directly, stored composites are model-relative snapshots that drift after every fine-tune. A retroactive re-verification pass at every cycle boundary rescores stored episodes under the updated model, overwrites each stored composite with its new value, raised or lowered, and prunes any entry whose composite has fallen below a prune threshold set beneath the storage threshold. A companion pass over the deferred buffer promotes entries that have crossed the storage threshold and discards entries whose age has exceeded a fixed time-to-live.

6. **Constrained self-improvement with a parameter-efficient adapter and a multi-modal retention guard:** Curate the fine-tuning pool from stored episodes that exceed a strict quality threshold set above the storage cut-point and pass the curated pool through a benchmark-balancing reweighting layer that prevents single-benchmark dominance. Fine-tune the generator through a low-rank adaptation layer on the attention and feed-forward projections rather than through full-parameter fine-tuning [36], with the trainable footprint at roughly two percent of the backbone parameter count and the underlying backbone frozen across the cycle. The bounded adapter parameter budget, layered on a frozen backbone, acts as the implicit cycle-to-cycle drift bound in the spirit of elastic weight consolidation [19], without the hand-tuned quadratic anchor coefficient that a full-parameter fine-tune would require. Probe a registered set of held-out distributions before and after each fine-tune and abort the cycle with an adapter rollback when the worst probe ratio falls below a fixed floor; the memory is preserved across rollbacks, so accumulated knowledge survives abort events even when the adapter does not.

7. **Empirical validation against external baselines and architectural ablations:** Evaluate the architecture on the five-benchmark panel and the three-probe retention panel described in Section 1.6.1. Report the composite hallucination metric, the per-tier behaviour, the retention ratio, and the calibration trajectory cycle by cycle. Compare against eight external baselines under matched protocol using paired statistical tests with multiple-comparison correction within each benchmark family, and run the pre-specified retrieval-utility-classifier on/off ablation, with its acceptance gates declared in advance and its bundled-confound scope disclosed.

1.4.3 Pre-specified hypotheses

The thesis registers five falsifiable hypotheses ahead of the empirical chapters. Each hypothesis names a specific quantity, a deciding direction, and a deciding statistic, so that the empirical chapters either confirm or refute the hypothesis without ambiguity. The first four hypotheses pair with the formal statements of Chapter 4, which comprise two proven results, design heuristics, and remarks on idealised models; the fifth tests the operational claim about deployed cost. Below each hypothesis, a short note says what it tests and how it is decided.

1. **Verifier balanced accuracy exceeds one half on the calibration-matched distribution at every cycle:** Storing answers makes the pool purer than the generator's raw output only if the gate admits correct answers more often than incorrect ones, which is the same as a balanced accuracy above one half (Proposition 4.2). The deciding statistic is the gate's balanced accuracy at the storage threshold, measured on each cycle's calibration fold and reported per benchmark. The hypothesis is falsified by any cycle in which the condition fails on the pooled training panel.

2. **Pool purity is non-decreasing across cycles whenever the precondition is satisfied:** If the condition of H1 holds and the verifier does not change, an idealised model of repeated filtering (Remark 4.9) predicts that each cycle's pool is at least as pure as the last. The deciding statistic is the pool purity

at each cycle. The hypothesis is falsified by any cycle in which the condition holds and pool purity falls. Chapter 4 explains why the realised system, which refits its composite at every cycle, may fluctuate around this prediction rather than follow it exactly.

3. **Late-cycle pool-purity increments enter a bounded stationary band on the committed-cycle subsequence:** The convergence heuristic (Heuristic 4.4) expects pool purity to settle. A convergence rate cannot be estimated from six cycles, so the test is whether purity stays inside a narrow band over the committed cycles, that is, the cycles whose fine-tune was kept rather than rolled back. The deciding statistic is the purity on those cycles and the width of the band. The sharper test of a geometric convergence rate needs at least fifteen committed cycles and is left to future work.

4. **The composite hallucination metric at the realised cycle horizon approaches a parameter-bounded asymptote:** The full-pipeline hallucination decomposition (Remark 4.10) splits the user-facing hallucination rate into a part served from memory and a part generated afresh, and predicts a floor set by the memory-tier hit rate and by the correctness ceiling of the generator. The deciding statistic is the gap between the composite hallucination metric at the final cycle and that predicted floor.

5. **Per-cycle Tier 1 hit rate grows monotonically as memory accumulates, producing a measurable per-query cost reduction:** Serving a query from memory costs only an embedding lookup, so a growing memory share should lower the average cost per query. The deciding statistic is the share of queries served by the memory tier at each cycle and the corresponding pooled latency.

The empirical chapters report each hypothesis with its deciding statistic and tag each as supported, partially supported, or unsupported at the trajectory's end. Together the seven specific objectives and the five hypotheses constitute the complete falsifiable content of the architectural claim, each anchored to a measurement reported in Chapter 5. Table 1.1 summarises the registration in compact form.

1.5 Methodology in Brief

1.5.1 Architecture overview

The architecture organises a single query into an eight-stage pipeline that is traversed once per query and wrapped in a per-cycle update loop. The first stage produces a pre-routing confidence from a short greedy decode over the query, fusing the decode's token-probability aggregate with a layer-convergence ratio. The second stage encodes the query with a sentence-embedding encoder and searches a similarity-indexed episodic memory for the nearest stored entry, returning a similarity score and the neighbour's stored-quality score. The third stage combines the similarity score and the stored-quality score into a single dispatch score and routes the query to the memory-direct tier when the score is high, the zero-shot tier in a middle band, and the retrieval-augmented tier when it is low, with an independent safety override that forces the retrieval-augmented default whenever the pre-routing confidence falls below a fixed floor. A retrieval-utility classifier then intercepts both the safety-override

ID	Hypothesis	Deciding statistic	Formal claim
H1	Verifier balanced accuracy exceeds one half on the calibration-matched distribution at every cycle.	Per-cycle verifier balanced accuracy at the fitted storage threshold, pooled training panel.	Proposition 4.2 precondition
H2	Pool purity is non-decreasing across cycles whenever the precondition is satisfied.	Per-cycle pool-purity trajectory under the precondition-satisfied regime.	Remark 4.9
H3	Late-cycle pool-purity increments enter a bounded stationary band on the committed-cycle subsequence.	Per-cycle pool-purity reading on the committed-cycle subsequence and realised band width.	Heuristic 4.4
H4	The composite hallucination metric at the realised cycle horizon approaches a parameter-bounded asymptote.	Gap between final-cycle composite hallucination metric and the predicted asymptote.	Remark 4.10
H5	Per-cycle Tier 1 hit rate grows monotonically as memory accumulates, producing a measurable per-query cost reduction.	Per-cycle Tier 1 share and per-cycle pooled latency.	Operational claim (Section 3.8.3)

Table 1.1: The five pre-specified hypotheses with deciding statistic and the theoretical or operational claim each tests. Verdicts are reported in Table 5.38.

dispatch and the score-formula fall-through to the retrieval-augmented tier and refines that default into either the zero-shot tier or the retrieval-augmented tier on a per-query basis, so that retrieval is consulted only on queries it is expected to help; the classifier is specified in Section 4.2.5. The fourth stage executes the chosen tier: the memory-direct tier serves the neighbour's stored answer; the zero-shot tier runs a generation on the fine-tuned generator; the retrieval-augmented tier retrieves passages from a public corpus and runs a grounded generation. The fifth stage passes every generated answer through a nine-signal post-generation verifier whose output is a calibrated probability that the answer is correct. The sixth stage applies a four-outcome decision (store, defer, abstain, discard) to that probability, with a confabulation early-exit rule that takes precedence over the thresholds. Both stages are summarised below. The seventh stage executes the implied memory and user-facing actions. The eighth stage runs at cycle boundaries and is detailed below. Figure 1.1 gives the high-level view; the full per-stage pipeline appears in Chapter 4.

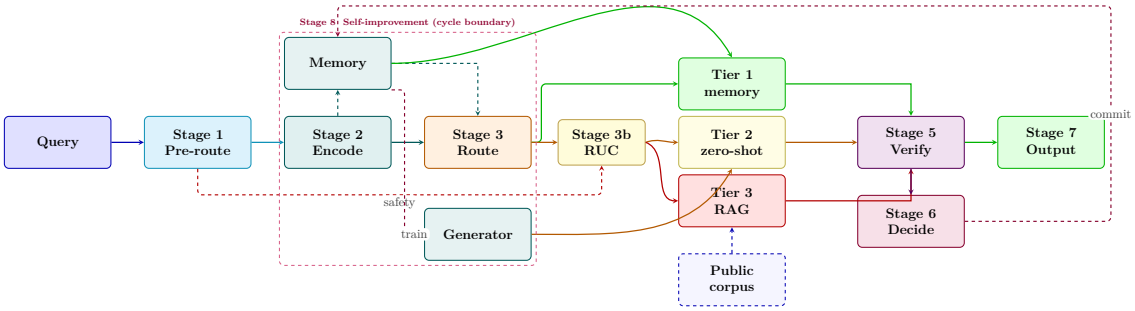


Figure 1.1: Eight-stage pipeline: pre-route, encode, route, Stage 3b retrieval-utility classifier on the score-formula and safety-override fall-through to the retrieval-augmented tier, tier execution, verify, decide, output, and a cycle-boundary self-improvement block; full specification in Chapter 4.

1.5.2 Verification and the storage decision

The verifier evaluates nine complementary signals across four families. Internal calibration signals read the generator's own state and include token-level uncertainty, dropout-based disagreement, and an internal-state composite. Sample-set signals measure agreement across resampled chains, including average pairwise similarity and chain-to-answer entailment. External grounding signals measure entailment between retrieved passages and the answer under three aggregation rules: top-one entailment, pooled mean entailment, and a weakest-link atomic-fact decomposition that fires only when the answer is long enough to admit decomposition. A question-answer relevance signal closes the off-topic-but-grounded failure mode by checking whether the answer addresses the question. The signals enter a calibrated probability composite that maps each per-signal score to a calibrated probability via isotonic regression and aggregates them through a regularised logistic regression layer fitted on a disjoint calibration fold; the calibrated output is the storage score. The confabulation early-exit rule discards an answer outright when high internal confidence coincides with low external grounding. Otherwise the four-outcome decision maps the storage score to one of four mutually exclusive actions: commit to memory, defer to a bounded buffer

with a time-to-live, abstain explicitly, or discard. The user-facing action that follows depends on the outcome, so that the user receives a stored answer, a generated answer, or a calibrated refusal rather than a silently uncertain response.

1.5.3 Calibration discipline

The storage score is converted into the storage and deferral decisions by two fixed cut-points, both chosen before the main run, the storage cut-point on the cycle-zero calibration fold, and held constant across cycles (Section 4.2.8 explains why fixed cut-points replaced an earlier conformal formulation). What changes from cycle to cycle is the calibrated probability composite: at every cycle boundary a labelled calibration fold refits it, per benchmark under a shrinkage prior toward the pooled fit, while the verifier-curated answers produced by the pipeline supply the training signal for the fine-tune. The research trajectory reuses one cold-start calibration fold of 1 500 labelled samples, re-scored under each cycle's fine-tuned model; the production design in Chapter 4 instead draws a fresh stratified labelled fold from each cycle's admissions. Because the cut-points are fixed and only the composite is refitted, the meaning of the storage threshold stays the same as the score distribution shifts.

1.5.4 Cycle-boundary self-improvement loop

The cycle loop iterates the per-query pipeline over a fixed workload, accumulates verified episodes into memory, and runs three cycle-boundary passes before fine-tuning. The retroactive re-verification pass rescores every stored episode under the updated generator, updates each stored composite to its new value, and prunes entries whose composite has fallen below the prune threshold. The deferred reconsideration pass examines every entry in the bounded deferral buffer and either promotes, expires, or requeues the entry. The memory consolidation pass collapses near-duplicate stored episodes that exceed a similarity threshold, picking the highest-quality representative within each near-duplicate cluster. Fine-tuning then draws a training pool from stored episodes whose composite score exceeds a strict quality threshold above the storage threshold, ensuring the training pool is held to a higher bar than the user-facing retrieval tier. The fine-tune is regularised implicitly through the bounded low-rank-adapter parameter budget layered on a frozen backbone, in the spirit of elastic weight consolidation [19] but without an explicit quadratic anchor coefficient. Before and after the fine-tune, the cycle probes the three held-out retention surfaces and computes the worst per-probe retention ratio. When that ratio falls below a fixed floor, the cycle aborts: weights are rolled back to the previous checkpoint and the retroactive pass for the next cycle is skipped. The memory is never rolled back, so accumulated knowledge survives abort events even when weights do not. This asymmetry decouples weight-level drift from memory-level accumulation and is what makes the self-improvement loop safe to iterate.

Figure 1.2 illustrates the trajectory across early, mid, and late phases. Each phase runs the same per-query pipeline over a fresh workload of queries, then applies the cycle-boundary update passes under the retention guard. The three boxes are representative phases rather than literal per-cycle depictions; the actual trajectory is reported cycle by cycle in Chapter 5.

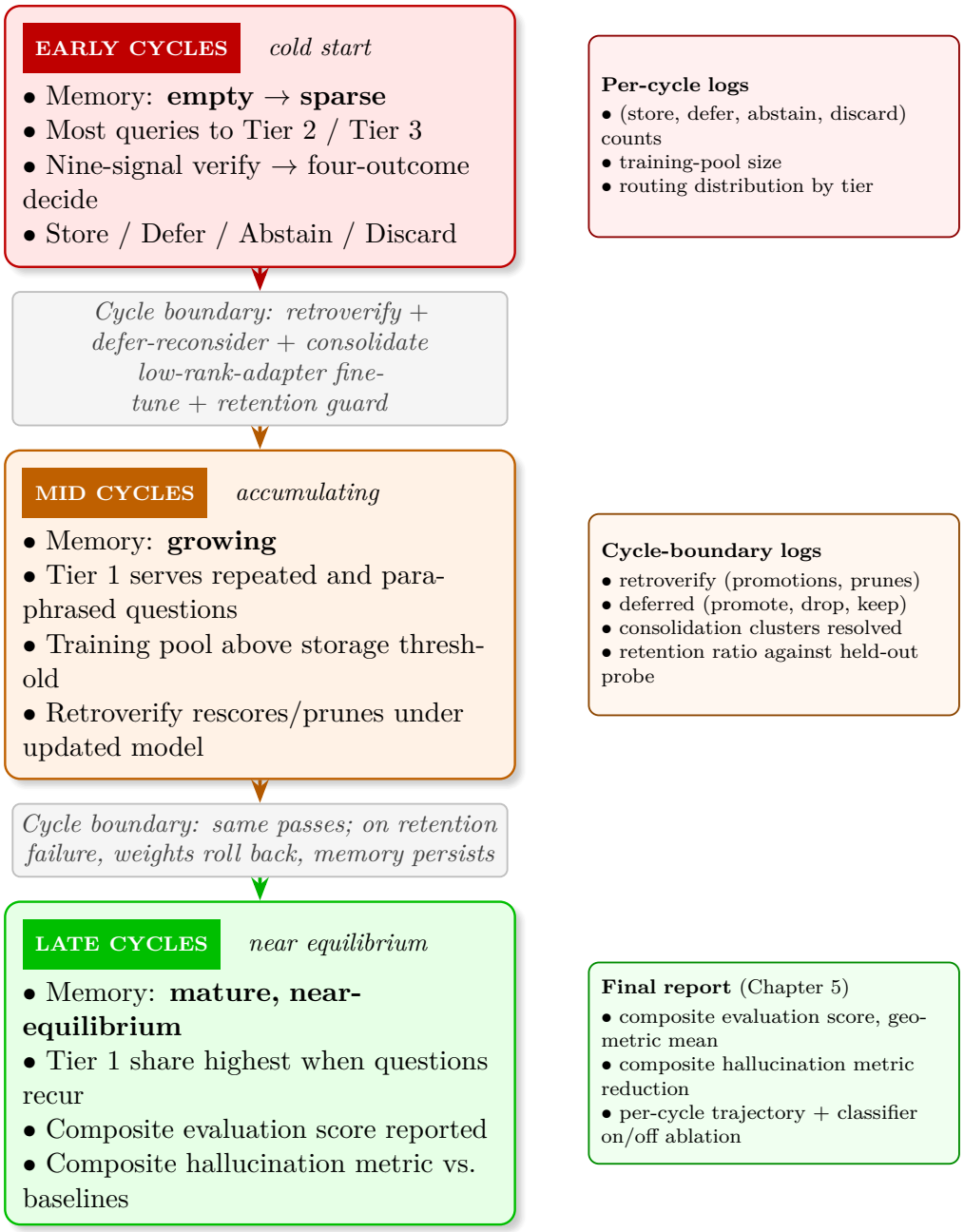


Figure 1.2: Intended self-improvement trajectory across early, mid, and late phases (design view). Tier 1 answers only repeated or paraphrased questions, so on the evaluation fold, whose questions are distinct from the stored episodes, it rarely fires; the realised per-cycle readings are in Chapter 5. Each phase runs the per-query pipeline and, at cycle boundaries, the retroactive re-verification, deferred reconsideration, consolidation, and adapter fine-tune under the multi-modal retention guard. Full specification appears in Chapter 4; headline results in Chapter 5.

1.6 Scopes and Challenges

1.6.1 Research scope

The architecture targets factual question answering, defined as the task of producing answers that admit a definite correctness verdict against a reference. The five eval-

uation benchmarks adopted in this work span factual claim verification (FEVER), open-domain trivia (TriviaQA), multiple-choice commonsense reasoning (CommonsenseQA), truthfulness on common-misconception probes (TruthfulQA), and multi-step commonsense reasoning (StrategyQA), partitioned into a three-benchmark training panel (FEVER, TriviaQA, CommonsenseQA) and a two-benchmark transfer panel (TruthfulQA, StrategyQA). Two further benchmarks, HotpotQA [22] and Natural Questions, were evaluated at cycle zero and dropped before the main run: the base generator answered only 9 and 17 percent of their questions correctly, respectively, and the storage gate kept just 26 and 12 of 500 answers, of which half or fewer were correct, too few and too noisy to train on. Their cycle-zero results are preserved in the project archive, and the Natural Questions training split is reused only to train the retrieval-utility classifier. Each of the five panel benchmarks admits an objective scoring rule, namely exact match or label accuracy, depending on the task family; TruthfulQA's free-text answers are judged against the benchmark's reference answers by a language-model judge, following the judge-based evaluation introduced with the benchmark [39], and the same judge scores every system. A multi-modal retention probe panel is reserved as held-out retention control rather than as a target benchmark: the fine-tuning loop is not optimised on any probe, and the retention guard aborts any cycle in which the worst per-probe accuracy ratio falls below a fixed fraction of the cycle-zero baseline. The probe panel comprises a general-knowledge multiple-choice surface (MMLU), an open-text factoid surface drawn from the TriviaQA test fold, and a commonsense multiple-choice surface drawn from the CommonsenseQA test fold; the worst-probe rule is the registered abort criterion.

The scope explicitly excludes tasks whose evaluation is intrinsically subjective, including creative generation, open-ended dialogue, and stylistic rewriting. Tasks that require knowledge beyond the pre-training cutoff, such as breaking news, market data, and recently issued documents, are excluded because the system does not include a continuous-update mechanism for ingesting external knowledge sources. Code generation and formal mathematical reasoning are excluded so that the evaluation surface remains within natural-language reasoning and the verifier signals retain a uniform interpretation across benchmarks. The system operates on English text only; multilingual transfer and code-switched evaluation are deferred to future work.

Resource scope is set by the engineering envelope of an undergraduate thesis. The generator is a three-billion-parameter open-weight instruction-tuned decoder-only model, chosen for two reasons. First, it admits parameter-efficient fine-tuning on a single consumer-grade graphics processor with thirty-two gigabytes of memory, under sixteen-bit mixed-precision arithmetic, through a low-rank adaptation layer on the attention and feed-forward projections, with the trainable footprint at roughly two percent of the backbone parameter count and the underlying backbone frozen across the cycle; full-parameter fine-tuning on the same backbone size does not fit the engineering envelope and is registered as the rejected alternative on the joint retention-and-feasibility axis. Second, as an instruction-tuned model it has competitive baseline accuracy on factual question answering for its size, which the self-improvement argument requires (see the baseline-performance challenge below). The ten-cycle horizon is sized to capture the early-cycle calibration phase, the mid-cycle accumulation phase, and the late-cycle equilibrium phase expected by the

convergence heuristic in Chapter 4, while remaining feasible within a twelve-month compute envelope; an equilibrium-saturation gate at every cycle boundary can stop the run earlier, and the realised trajectory closed at cycle five under that gate, consistent with reports that iterative self-improvement saturates within a few rounds [92]. The episodic memory is sized to a target capacity in the millions of entries with a tiered index that promotes from a flat structure to a quantised inverted-file structure above a threshold; the workloads reported in this thesis grow the memory to 12 175 entries at cycle five (Table 5.17), below the size at which the index switches to the quantised structure and far below the capacity at which value-score pruning triggers, so the realised run uses exact similarity search and never invokes pruning.

1.6.2 Technical challenges

The implementation faces six primary technical challenges that the architecture must address.

- **Verifier reliability and stored-quality drift:** The nine-signal post-generation verifier produces a calibrated storage score that doubles as a retrieval-time quality proxy and as a quality filter on the training pool. Three of the nine signals belong to the internal-calibration family and read the generator's hidden state directly, so their distributions shift after every fine-tune. The stored composite of every episode is therefore a model-relative snapshot, not a permanent attribute of the entry. The retroactive re-verification pass at each cycle boundary refreshes every stored composite under the updated generator, raising or lowering it to the new value, and prunes entries whose composite has fallen below the prune threshold. The cost of this discipline is a full re-scoring of every stored episode at each cycle boundary, which grows with the size of the memory.
- **Catastrophic forgetting under iterative fine-tuning:** Fine-tuning on domain-specific verified examples risks degrading the generator's general-purpose capabilities. The architecture mitigates this risk through three layers of defence: a strict training-pool quality filter that holds the training pool to a higher bar than the user-facing storage tier, a bounded low-rank-adapter parameter envelope layered on a frozen backbone that acts as the implicit cycle-to-cycle drift bound (in the spirit of elastic weight consolidation [19] but without an explicit quadratic anchor coefficient on the shipped path), and a multi-modal retention probe panel whose worst per-probe ratio gates the commit. The retention probe panel spans a general-knowledge multiple-choice surface, an open-text factoid surface drawn from a training-benchmark test fold, and a commonsense multiple-choice surface drawn from a second training-benchmark test fold; the cycle aborts when any per-probe ratio falls below the fixed floor, weights are rolled back to the previous checkpoint, and the memory is preserved so that a fine-tune that fails in any given cycle does not lose the verified episodes accumulated during it. The three layers are not ablated individually within the realised horizon; the retention guard's effect is visible in the cycle-four rollback reported in Chapter 5, and the only architectural ablation reported there is the retrieval-utility-classifier on/off contrast.
- **Pre-routing confidence calibration:** The pre-routing confidence is used

both as a one-dimensional input to the dispatch score and as the sole input to the safety override that forces grounded retrieval, so its calibration influences the system's behaviour twice. The architecture fuses two cheap signals, namely the token-probability aggregate of a short greedy decode that scores the first thirty-two tokens of a draft answer and a layer-convergence ratio computed from the hidden states over the query, and temperature-scales the fused scalar per benchmark on a disjoint fold by minimising binary negative-log-likelihood. The non-trivial design choice is that the pre-routing signal does not require the full multi-chain sampling that the post-generation verifier uses, so its cost is paid per query regardless of which tier eventually serves the answer; the cost of the more expensive sample-based signals is deferred to the post-generation verifier on the zero-shot and retrieval-augmented paths only.

- **Memory retrieval quality and novelty filtering:** At deployment scale the episodic memory uses approximate nearest-neighbour search, which raises the possibility of missing a semantically similar entry whose embedding happens to fall near a partition boundary; at the scale of the realised run the search is exact. Near-duplicate writes are filtered at storage time by a novelty threshold on cosine similarity, and the consolidation pass at each cycle boundary collapses near-duplicate clusters by retaining only the highest-quality representative within each cluster. The novelty threshold is a tuning knob with a clear failure mode in either direction: a threshold set too high admits paraphrase-level duplicates that bloat the index without adding coverage, while a threshold set too low rejects legitimate paraphrases that would have improved retrieval. The chosen value is fixed before the main run and held constant across cycles.
- **Baseline performance dependency for self-improvement:** The self-improvement argument depends on the base generator performing meaningfully above random on the target benchmarks, because training on verified-but-imperfect episodes can only compound accuracy when the verifier's expected false-positive rate is low enough that the training pool is cleaner than the generator's untrimmed outputs. The risk is mitigated by selecting an instruction-tuned backbone with competitive baseline accuracy on factual question answering, by selecting benchmarks on which the base generator clears that threshold, and by composing the storage decision from the calibrated probability composite, the confabulation early-exit gate, and the in-distribution training-pool gate that excludes transfer-benchmark episodes from fine-tuning. Each component contracts the storage class below the rate that an unrestricted score would admit, and the conjunction is what makes the precision floor on the storage tier achievable.
- **Evaluation metric selection:** Measuring hallucination reduction requires a metric design that is more discriminative than aggregate accuracy. The composite hallucination metric used in this thesis is the equal-weighted mean over an eight-axis failure-mode taxonomy that includes confident confabulation, factual fabrication, logical fabrication, off-topic answers, defensive evasion, template leakage, false refusal, and length-padded over-generation. A ninth axis is retained in the taxonomy but excluded from the default denominator because the corresponding subtype is structurally unreachable under the verifier backend used here, and its rate is reported separately for completeness. A

uniform reduction across eight structurally distinct failure modes is a stronger architectural claim than a larger reduction on any single axis. The composite evaluation score is a geometric mean across accuracy, epistemic quality, retention, calibration, and verifier reliability; the geometric form penalises any axis that collapses, which matches the design constraint that no single axis may be sacrificed for the others. Comparison against the baseline panel is conducted under matched protocol, with the same evaluation samples, backbone, answer format, and scoring rules, so that effects are not confounded by re-implementation choices.

Each mitigation above is built into the architecture rather than added as a downstream patch; Chapter 5 reports the evidence available for each on the realised trajectory.

1.7 Chapter signpost

This chapter has registered the deployment-safety problem, the architectural opportunity that closes the stateless-overconfident-static triad, the seven specific objectives that decompose the primary objective, and the five pre-specified hypotheses that the empirical chapters adjudicate. Chapter 2 establishes the technical foundations on which the architectural commitments rest (the transformer backbone, the hallucination taxonomy, natural language inference, sentence embeddings, self-consistency decoding, and catastrophic forgetting) and surveys the empirical literature across hallucination measurement, verification, confidence estimation, memory-augmented architectures, continual learning, and evaluation benchmarks. The survey locates the architectural contribution of this thesis against the prior work that the seven specific objectives draw from, so the methodology can be read as a single integrated response to a literature whose individual contributions do not close the joint gap. Chapter 3 states the functional requirements, the societal and environmental impact, and the project plan. Chapter 4 specifies the architecture stage by stage together with its formal analysis. Chapter 5 reports the realised six-cycle trajectory, the baseline comparisons, the retrieval-utility-classifier ablation, and the verdict on each hypothesis. Chapter 6 summarises the findings, the limitations, and the future work, and the appendices collect the implementation reference, the per-cycle evidence, the significance-test details, and the reproducibility recipe.

Chapter 2

Literature Review

2.1 Preliminaries

Chapter 1 laid out the three limitations of current LLMs that CAEM addresses (stateless operation, poor confidence calibration, absence of self-improvement) and the three architectural commitments (verified episodic memory, confidence-aware routing, constrained self-improvement) that answer them. This chapter supplies the two kinds of background a reader needs to evaluate those commitments in detail. The Preliminaries section that follows establishes foundational concepts and terminology: the transformer backbone, the hallucination taxonomy, natural language inference, sentence embeddings, self-consistency decoding, and catastrophic forgetting. The Review of Existing Research section that follows it then surveys the empirical literature across six areas (hallucination measurement, verification, confidence estimation, memory-augmented architectures, continual learning, and evaluation benchmarks), locating CAEM's contribution relative to prior work. Together they prepare the ground for the formal methodology in Chapter 4.

2.1.1 Large Language Models and Transformer Architecture

Neural networks that have been trained on quite big text collections to predict and produce natural language are known as large language models. The Transformer architecture [20], which processes sequences using a self-attention mechanism allowing all the tokens in the input to take account of every other token, is adopted by new LLMs. The encoder and decoder layers in this architecture are arranged in layers and each one has feed forward networks and multi-head self-attention.

In this study, we utilize Qwen-2.5-3B-Instruct [80], a three-billion-parameter decoder-only transformer that has already undergone instruction-tuning on a broad mixture of task-formatted demonstrations before reaching the CAEM pipeline. Decoder-only models factorise the joint probability of the output tokens left-to-right and produce the answer autoregressively from the question prompt, with the question and the generated rationale co-existing in a single causal attention stream. Qwen-2.5-3B-Instruct is chosen over larger open checkpoints because it is the largest decoder-only backbone that fits the CAEM self-improvement loop on a single consumer-grade GPU when held in sixteen-bit brain-float precision with the MC-Dropout sampling and the

nine-signal verifier active on the same device, and it is chosen over smaller decoder-only or encoder-decoder alternatives because its instruction-tuned initialisation makes the cycle-zero verifier outputs sharp enough for the four-outcome storage decision to separate high-purity from borderline items from the first cycle onwards rather than requiring an additional warm-up stage.

Fine-tuning adapts pre-trained models to specific tasks or domains by continuing training on task-specific data with a smaller learning rate. The model parameters $\boldsymbol{\theta}$ are adjusted to reduce loss on the new data while preserving the general abilities learned during pre-training. Standard fine-tuning updates every parameter; parameter-efficient approaches keep the underlying backbone frozen and adapt only a small fraction of the parameter count through trainable side-modules. CAEM ships the parameter-efficient path through the low-rank adaptation layer of [36] on the attention and feed-forward projections, with the trainable footprint at roughly two percent of the backbone parameter count and the underlying backbone frozen across every cycle of the self-improvement loop. The empirical evidence motivating that choice over full-parameter fine-tuning is reviewed in Section 2.2.6.

2.1.2 Hallucination Taxonomy

Hallucinations in language model outputs can be understood along two main dimensions. The first dimension separates *intrinsic hallucinations*, which directly contradict information provided in the source input, from *extrinsic hallucinations*, which introduce information that can be neither supported nor contradicted by the source [49]. For instance, if a biography states that "Einstein was born in Germany" and the model generates "Einstein was born in France," this is an intrinsic hallucination because it contradicts the stated source. But if the model generates "Einstein played violin exceptionally well" without any evidence in the source, this is an extrinsic hallucination even if the claim happens to be true: the defining characteristic is that it cannot be verified from the input, not that it contradicts anything in it.

The second dimension distinguishes *factuality hallucinations*, which conflict with verifiable world knowledge, from *faithfulness hallucinations*, which contradict user input or task context [88]. For example, a model that states "photosynthesis produces oxygen from carbon dioxide and water" is factually accurate, whereas one that states "photosynthesis occurs in mitochondria" produces a factuality hallucination. Faithfulness focuses on whether the output of the model remains consistent with the supplied input or context, especially when summarizing documents or following instructions.

This research focuses on factuality hallucinations, which occur in knowledge-based reasoning tasks. Here, correct answers can be verified against trusted sources or logical rules. Faithfulness to retrieved evidence also matters for retrieval-augmented generation, where outputs must be grounded in retrieved documents.

Within factuality hallucinations, four subclasses are operationally distinguished because CAEM's nine-signal verifier is organised in four families that map onto them. *Confabulations*, in the sense of [70], are outputs the model could equally have produced differently on a resample even when individual token probabilities look confident, and are detected by the sample-set agreement family of two signals (s_{avg},

the mean unique-pair cosine similarity over M=3 resampled chains in the sentence-embedding space, and p_{entail}, the mean entailment of each resampled chain against the served answer under the natural-language-inference judge).[1] *Ungrounded factual claims* are answers without support in retrieved evidence, detected by the external-grounding family of three signals ($p_{\text{ground_max}}$, $p_{\text{ground_mean}}$, and the length-gated weakest-link atomic-fact entailment $p_{\text{ground_atomic}}$ that runs only on hypotheses whose token length exceeds the registered atomic-decomposition gate). *Off-topic answers* that drift from the question are detected by the question-answer relevance signal $q_{\text{a,rel}}$ that scores the served answer against the question through the same cross-encoder used for passage rerank. *Confidently-wrong* factual outputs (internally confident but externally unsupported) are caught by the early-exit confabulation gate that combines a high internal-calibration signal with a low max-match grounding score. The deployed configuration ships MiniCheck [81] as the entailment backend across all hypothesis lengths because the long-hypothesis Qwen judge fails the registered Pearson floor on the registered 500-sample calibration overlap fold and is ablated in the locked configuration; the contradiction probability p_{contra} is structurally zero under MiniCheck (the unary supported-against-context output is the only label class the judge produces), the corresponding hard-veto branch was retired on 2026-04-22, and the contradiction-scoring call was further short-circuited at the verifier compute path so the signal does not enter the gradient or the composite log-odds at all; contradicted factual claims are caught indirectly through the calibrated grounding signals' contribution to the composite. Outside this scope, CAEM does *not* target instruction-following failures, intermediate-reasoning-step errors inside chain-of-thought traces (only the final answer is verified), intrinsic faithfulness hallucinations in summarisation or translation (where the grounding source is the input document rather than world knowledge), or input-contradiction hallucinations that conflict with user-supplied premises. These exclusions set the failure-mode boundary within which the results in Chapter 5 should be interpreted.

2.1.3 Natural Language Inference

Natural Language Inference (NLI) is the task of determining whether a hypothesis sentence logically follows from a premise sentence [8]. The PASCAL Recognising Textual Entailment challenges framed this as a binary decision (entailed or not entailed); later NLI corpora, including the Multi-Genre NLI corpus on which RoBERTa-Large-MNLI is fine-tuned, use three classes. Given premise P and hypothesis H, a three-class NLI model classifies the relationship as:

- **ENTAILMENT**: P implies H is true
- **CONTRADICTION**: P implies H is false
- **NEUTRAL**: P neither confirms nor denies H

[1] The nine signals are defined formally in the Unified Verifier subsection of Chapter 4 (table of signal definitions). This chapter refers to them by their canonical symbol only where needed for motivation. The normalised semantic entropy signal of [70] was registered earlier in the project alongside the same family but retired before the main run because its cycle-zero per-benchmark Pearson correlation with exact-match labels was indistinguishable from zero on every per-benchmark slice of the calibration fold; the kernel-language-entropy upgrade of [78] was registered as a candidate replacement and is also retired in the shipped architecture.

For example, if we assume "The Beatles were formed in Liverpool in 1960" and hypothesis is "The Beatles were an English band," then this relationship is ENTAILMENT. Given hypothesis "The Beatles were formed in London," the relationship is CONTRADICTION. But if the given hypothesis is "The Beatles recorded many albums," then the relationship is NEUTRAL.

As the verifier judge in CAEM, we employ MiniCheck-Flan-T5-Large [81], a Flan-T5-Large checkpoint fine-tuned on synthetically decomposed language-model-generated claim-support data. MiniCheck is binary supported-versus-unsupported by construction; this is intentional in the deployed configuration, because the Cycle 0 calibration retired the contradiction-veto branch (the hard p_{contra} rule) once the head-to-head fold confirmed the contradiction signal collapses to zero under the binary judge. A frozen Qwen-3B long-hypothesis judge was registered alongside MiniCheck to handle hypothesis lengths above the MiniCheck encoder's effective range, but it failed the pre-specified Pearson floor on the registered 500-sample calibration overlap fold and is ablated; every hypothesis is therefore dispatched through MiniCheck, with documented 512-token truncation on the rare hypotheses that exceed the encoder range. The legacy generic-NLI baseline RoBERTa-Large-MNLI [24], fine-tuned on Multi-Genre Natural Language Inference and reaching approximately 90.2% on the MNLI matched development set, is retained as a Chapter 5 ablation target (the generic-NLI-backend ablation variant registered in Section 5.4.2) so the backend choice is defended by trajectory evidence rather than by citation alone. The NLI-style judge, whichever backend is active, serves two purposes in CAEM. Firstly, it verifies factual correctness by checking whether generated claims are supported by retrieved evidence. Secondly, it helps group answers that mean the same thing by testing for bidirectional entailment: if P entails H and H entails P, then P and H are considered semantically equivalent [52].

2.1.4 Sentence Embeddings and Semantic Similarity

Sentence embeddings map the length of variable text sequences to fixed dimensional dense vectors in a semantic space where similar meanings have high cosine similarity. Sentence BERT (SBERT) [26] produces sentence embeddings by fine-tuning BERT in a siamese network structure, first on natural language inference data (SNLI and MultiNLI) and optionally on semantic textual similarity data. The model we employ, all-mpnet-base-v2 from the same Sentence-Transformers family (built on an MPNet backbone rather than the original BERT), generates 768-dimensional embeddings.

Cosine similarity between vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is computed as:

$$\text{sim}(\mathbf{v}_1, \mathbf{v}_2) = \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\|\|\mathbf{v}_2\|} = \cos(\theta) \tag{2.1}$$

where θ is the angle between vectors. Similarity ranges from -1 (opposite) to 1 (identical), with 0 indicating orthogonality.

Semantic similarity is different from string similarity. For instance, the questions "Is 17 prime?" and "Is the number 17 a prime number?" have low string overlap but high semantic similarity (approximately 0.98). CAEM uses semantic similarity for key operations like memory retrieval (finding similar questions), novelty detection (avoiding redundant storage), and self-consistency verification (measuring reasoning

chain agreement). The system uses FAISS (Facebook AI Similarity Search) [35] for fast, efficient approximate nearest-neighbour search, allowing it to retrieve sub-millisecond results over thousands of stored episodes.

2.1.5 Self-Consistency Decoding

Self consistency [61] improves reasoning quality by first generating multiple independent reasoning chains for the same question, then it selects the most common answer. Unlike standard greedy decoding which produces a single output, self-consistency samples M chains using temperature $T > 0$ to introduce diversity.

For CAEM, we adapt self consistency technique for verification rather than choosing an answer. Given question q and generated answer a, we generate M additional chains and measure agreement via average pairwise similarity:

$$s_{\text{avg}} = \frac{2}{M(M-1)} \sum_{1 \le i < j \le M} \text{sim}(\mathbf{v}_i, \mathbf{v}_j) \tag{2.2}$$

where $\mathbf{v}_i$ is the embedding of chain i, and only unique off-diagonal pairs are used (self-similarity diagonal terms are excluded). High consistency ($s_{\text{avg}} > 0.85$) indicates robust reasoning where multiple paths converge, while low consistency suggests uncertainty or errors. The s_{avg} symbol is the canonical name for this signal throughout Chapters 3–5; some older literature writes it as u_{SC}. CAEM pairs s_{avg} with the chain-to-answer entailment signal p_{entail} (computed on the same M=3 resampled pool to amortise the decode cost) as the two-signal sample-set agreement family that feeds the calibrated probability composite. The normalised semantic entropy signal of [70] was registered earlier in the project as a third sample-set member computed over K=10 chains at temperature T=1.0, and the kernel-language-entropy generalisation of [78] was registered as a candidate replacement that uses graded semantic-similarity kernels rather than hard cluster assignments; both were retired before the main run on the cycle-zero calibration evidence that the entropy signal contributed no discriminative information beyond what s_{avg} and p_{entail} on the same resampled pool already provided, so the deployed verifier consumes nine signals across the four families, of which the sample-set agreement family contributes two.

2.1.6 Catastrophic Forgetting and Continual Learning

Catastrophic forgetting [5] occurs when neural networks learn new tasks and suddenly lose their ability to perform previously learned tasks. Standard gradient descent focuses only on improving performance for the current task and does not protect earlier knowledge, so parameter updates that help the new task tend to weaken the old ones. In large language models, fine-tuning on a new domain likewise erodes performance outside that domain, and the erosion grows with training duration [64].

Elastic Weight Consolidation (EWC) [19] addresses this by identifying essential parameters for previous tasks using the Fisher Information Matrix, then adding quadratic penalties preventing large updates to important parameters:

$$\mathcal{L} = \mathcal{L}_{\text{new}} + \frac{\lambda}{2} \sum_i F_i(\theta_i - \theta_i^*)^2 \tag{2.3}$$

where F_i measures parameter importance, θ^* are previous parameters, and λ controls regularization strength. A cheaper alternative replaces the full Fisher matrix with an isotropic L_2 anchor, $\frac{\lambda}{2}\|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{prev}}\|^2$, applied to the adapter parameters of the previous cycle rather than to the full backbone; this is the uniform-Fisher special case, which Kirkpatrick et al. show to be weaker than EWC because it protects all weights equally and leaves little capacity for the new task [19]. This penalty is registered in CAEM as the parameter envelope for the inactive full-fine-tune fallback (when it engages, $\lambda > 0$); on the shipped low-rank-adapter path it collapses by design ($\lambda = 0$) because the bounded adapter parameter envelope layered on a frozen backbone is itself the cycle-to-cycle drift bound, and an explicit weight-anchor would otherwise penalise deviation from a zero-anchored full-backbone reference whose weights the loop is not updating. CAEM therefore combines the bounded-adapter envelope with two further mechanisms whose joint effect bounds the cycle-to-cycle drift in the underlying generator. The first is the parameter-efficient adaptation primitive itself: low-rank adaptation [36] on the attention and feed-forward projections, with the trainable footprint at roughly two percent of the backbone parameter count and the underlying backbone frozen across the cycle, forgets less of the base model's general capability than full-parameter fine-tuning on the empirical evidence reviewed in Section 2.2.6. The second is a multi-modal retention guard whose probe set spans a general-knowledge multiple-choice surface, an open-text factoid surface from a held-out training-benchmark test fold, and a commonsense multiple-choice surface from a second held-out training-benchmark test fold; the cycle aborts and rolls the adapter back to the previous snapshot whenever the worst probe ratio against the pristine baseline falls below a fixed floor. The accumulated episodic memory is preserved across rollbacks, so verified knowledge survives an adapter-rollback event even when the weight update does not. This three-mechanism combination addresses the challenge of learning from imperfectly verified data without degrading through error reinforcement.

The conceptual basis for understanding CAEM's nine-signal post-generation verifier, pre-routing confidence, episodic-memory architecture, and constrained self-improvement loop is established by the groundwork above. The following sections describe how prior research develops and validates these individual components, preparing for their integration in Chapter 4.

2.2 Review of Existing Research

Where the Preliminaries section above defined the concepts and notation (hallucination taxonomy, natural language inference, sentence embeddings, self-consistency, catastrophic forgetting), this section surveys the prior empirical and methodological literature across six areas essential to understanding CAEM's design: hallucination characterization and measurement, verification methods for reasoning validation, confidence estimation techniques, memory-augmented architectures, continual learning approaches, and evaluation benchmarks. Each subsection identifies key contributions and limitations that motivate CAEM's architectural choices, so that a reader who has already absorbed the Preliminaries can move directly to the comparative positioning without re-reading definitional material.

2.2.1 Hallucination Problem and Measurement

Understanding the nature and characteristics of hallucination in LLMs requires methodologies like taxonomic frameworks and quantitative measurement. Early studies of hallucination focused on summarization and data-to-text generation. Ji et al. [49] published the first major survey in ACM Computing Surveys. It distinguishes intrinsic hallucinations, which contradict the source material, from extrinsic hallucinations, which can be neither supported nor contradicted by it. Their work shows that hallucinations arise at several stages, including data collection that embeds factual errors, training procedures that reinforce misleading correlations, and inference methods that fail to account for uncertainty.

Huang et al. [88] extended this study specifically for large language models in ACM Transactions on Information Systems, distinguishing factuality hallucinations from faithfulness hallucinations. Their analysis traces the causes of hallucination across the model lifecycle to data, training and inference, and ties each family of mitigation methods to the cause it addresses. This suggests that effective mitigation should intervene at several stages instead of relying solely on detection applied after the model has already produced an answer, which supports CAEM's multi-component strategy combining memory, verification, and iterative learning.

Quantitative measurement studies of hallucination prevalence reveal concerning statistics across domains and model scales. Lin et al. [39] introduced TruthfulQA at ACL2022, a benchmark of 817 questions across 38 categories which has been specifically designed to elicit imitative falsehoods where models reproduce common human misconceptions found in training data. Their result demonstrates an inverse scaling effect: the best GPT-3 175B configuration achieved only 21% on the truthful-and-informative metric (and 58% on truthfulness alone), against 87% and 94% respectively for the human baseline. Larger models often performed worse than smaller ones. This suggests that scaling alone is not sufficient for truthfulness, which motivates architectural innovation.

The inverse scaling result has major result for hallucination mitigation strategies. If adding parameters reduces truthfulness on some tasks then mitigation strategies must operate beyond parameter growth. CAEM addresses this by incorporating external verification and memory components that operate independently of model scale, providing corrections and verified knowledge regardless of base model size.

Min et al. [57] introduced FActScore at EMNLP 2023, which evaluates long-form generations by decomposing the answer into atomic claims and reporting the share supported against retrieved sources rather than scoring the answer holistically. The method achieves less than two-percent error against human evaluation, and applied to biography generation reports a fifty-eight percent FActScore for ChatGPT, indicating that even highly capable models produce substantial proportions of unsupported atomic facts. CAEM adopts the FActScore convention at the storage gate: the weakest-link atomic-fact entailment $p_{\text{ground_atomic}}$ is computed by decomposing the served answer into atomic claims, scoring each against the reranked passages, and propagating the minimum claim-level entailment as the answer's grounding floor. The decomposition is registered with a length gate so that short answers (below the gate's token threshold) bypass the atomic step, on the principle that short answers contain a single claim whose mean-match score is already a faithful summary of grounding;

this length-conditioning is what keeps the atomic decomposition's compute cost bounded on the five-benchmark panel.

Domain-specific studies show even higher hallucination rates in specific contexts. Alkaissi and McFarlane [43] asked ChatGPT to explain a clinical finding and to supply supporting references; none of the five paper titles it returned existed. Chelli et al. [65] checked the references that chatbots generated for systematic reviews and reported hallucination rates of 39.6% for GPT-3.5, 28.6% for GPT-4, and 91.4% for Google Bard.

Recent benchmark provided comprehensive evaluation across Several dimensions. One of them being legal sector. Factual accuracy has been measured in legal sector. According to Google's FACTS benchmark (2025), it shows that even top performing systems maintain 6.4% hallucination rates on legal queries. Furthermore, the Vectara Hallucination Leaderboard tracks document summarization hallucination rates ranging from 3% for GPT-4 to 27.2% for Google PaLM 2 Chat. It thus demonstrates wide variation among models and confirm that hallucination remains an issue even in state of the art systems.

The information above reveals three critical takeaways that shape CAEM's design. Firstly, hallucination occurs across models, tasks, and domains, with rates ranging from 3% to over 90% depending on context. Secondly, larger models do not automatically produce fewer hallucinations, which implies that architectural intervention is necessary. Lastly, verifying information at the level of atomic facts is far more informative than evaluating full responses, motivating CAEM's commitment to fact-level NLI-based verification.

2.2.2 Verification Methods

Verification methods are used to check whether a model's reasoning is correct before the information is stored or shown to users. This subsection explains four approaches (chain-of-thought prompting, self-consistency decoding, semantic entropy, and natural-language inference) that together supply the signals aggregated inside CAEM's composite verifier.

Chain of thought prompting, which was introduced by Wei et al. [40] at NeurIPS 2022, improves reasoning quality by prompting models with worked examples whose intermediate steps precede the final answer. Showing these intermediate steps leads to stronger reasoning and provides better starting points for verification. When tested on the PaLM 540B model, chain of thought greatly improved performance on math reasoning tasks, increasing accuracy on GSM8K from about eighteen percent to about fifty-seven percent. The authors found that the technique is an emergent ability of model scale: below roughly 100 billion parameters chain-of-thought prompting did not help and could lower performance, a pattern that held for the instruction-tuned InstructGPT models in their GPT-3 series as well. Kojima et al. [38] later showed that the single trigger "Let's think step by step" elicits chain-of-thought reasoning without worked examples. Instruction tuning does not remove the small-scale effect: Chung et al. [68] report that their instruction-tuned Flan-T5-Large (780 million parameters) still scores lower with chain-of-thought prompting than with direct answering on MMLU (40.5 versus 45.1 percent) and on BIG-Bench Hard (31.5 versus

37.5 percent). Whether chain-of-thought prompting helps CAEM's three-billion-parameter instruction-tuned backbone, Qwen-2.5-3B-Instruct [80], is therefore an empirical question, which the chain-of-thought baseline in Chapter 5 answers.

For CAEM, chain of thought serves two purposes. Firstly, it improves base model reasoning quality. It also provides better starting points for verification. Secondly, it gives explicit reasoning chains that enables step by step verification through NLI which also checks whether each reasoning step is supported by evidence. The explainability of chain of thought reasoning proves essential for diagnosis when verification fails, enabling error analysis that identifies which reasoning steps caused failures.

Wang et al. [61] expanded this idea in 2023 by introducing self consistency. Instead of producing a single reasoning path, the model generates several different explanations and then selects the most common answer. The key idea is that difficult and complex problems usually have multiple valid reasoning paths that lead to the same correct answer. On the other hand, incorrect answers rarely achieve consistency across independent derivations. Reported gains over chain-of-thought prompting reach +17.9 percentage points on GSM8K, +11.0 on SVAMP, +12.2 on AQuA, and +6.4 on StrategyQA; these are the largest gains across the four models tested, and for PaLM 540B the corresponding gains are +17.9, +7.6, +12.5 and +6.3.

CAEM uses self consistency in different ways. It chooses reasoning chain agreement through semantic similarity rather than selecting the majority answer. If the chains mostly agree, the reasoning is then considered reliable and suitable for storage. But if they differin opinion then the system treats the result as uncertain and may apply further checks or reject it. As a result, this method is efficient since it only requires a small number of extra generations which makes it practical for regular verification. And it also provides useful confidence signals.

Another important method is semantic entropy introduced bt Farquhar et al. [70] in 2024. It represents the most rigorous treatment of uncertainty based hallucination detection. Their key innovation addresses a primary limitation of token level entropy: Answers that express the same idea in different wording are grouped together using inference checks, and uncertainty is measured across these meaning based groups.

The methodology achieves AUROC of 0.790 for fabrication detection on question answering tasks. It substantially outperforms token entropy (0.691) and verbalized probability methods (0.698). Even though the performance remains robust across model families including LLaMA 2 (7B-70B), Falcon (7B-40B), and Mistral 7B, the Nature publication validates semantic entropy as a cutting edge technique for uncertainty quantification, motivating its inclusion in CAEM despite computational overhead.

Semantic entropy serves as the third verification layer for CAEM. It catches reports where both NLI and self consistency fail. Let us consider a model which is asked "What is the capital of Australia?" that consistently always generates "Sydney" across multiple chains with high individual probabilities. NLI verification may pass if the model generates factually correct supporting statements about Sydney. Here, self consistency shows perfect agreement. But still the semantic entropy will generate 10 diverse samples and will measure cluster entropy. This may reveal

that the model also produces "Canberra" and "Melbourne" at high temperature thus indicating underlying uncertainty. This detection capability for systematic confabulation justifies the computational cost.

Natural language inference provides direct verification by checking whether a claim is supported by evidence. Thorne et al. [21] introduced the FEVER dataset at NAACL 2018, a fact-verification task closely related to NLI, with 185,445 claims generated from Wikipedia; unlike NLI, the evidence for each claim must first be retrieved from Wikipedia rather than being given. Systems must classify claim evidence relationships as SUPPORTS, REFUTES, or NOT ENOUGH INFO. This three way classification would map directly to CAEM's storage decisions. The decision would be to support claims merit storage, refute claims are rejected, and insufficient evidence trigger uncertainty handling.

Pre trained NLI models such as RoBERTa-Large-MNLI has achieved approximately 90% accuracy on the MNLI benchmark. It thus provides reliable verification for factual claims. But the limitation is that even though NLI verifies individual factual statements, they may miss logical reasoning errors that do not contradict facts. For example, a model could state "Sydney is Australia's largest city" which is true and conclude "therefore Sydney is the capital" which is an invalid reference. NLI would verify the factual premise while missing the logical error thus motivating the need for complementary verification layers.

A more recent strand of work brings conformal prediction to the language-model claim level. Mohri and Hashimoto [77] introduced *Language Models with Conformal Factuality Guarantees* at ICML 2024, decomposing each generated answer into atomic sub-claims, scoring each, and removing claims below a calibrated threshold τ such that the surviving sub-claims are jointly correct with probability at least $1 - \alpha$ on a small annotated calibration set. Their framework is the closest published method that explicitly crosses an eighty-percent factuality target without filter-time gold labels. Yadkori et al. [84] extended this with conformal-risk-control to bound the hallucination rate at a user-specified target through learned abstention rather than claim removal, providing the direct precedent for CAEM's abstention class in the four-outcome decision tree. Quach et al. [79] present *Conformal Language Modeling* at ICLR 2024, calibrating a stopping rule, a rejection rule, and a confidence rule simultaneously through Learn-then-Test multiple-hypothesis testing. Cherian et al. [67] sharpen the Mohri–Hashimoto framework with conditional guarantees and a boosted score, learning a linear combination of several per-claim scores by differentiating through the conformal procedure; the closest precedent for CAEM's boost layer over the per-signal log-odds in Section 4.2.7 is the logistic-regression multi-scoring of Valentin et al. [82]. CAEM considered the split-conformal formulation as the storage-gate primitive and rejected it on the registered cycle-zero evidence reviewed in Section 4.2.8: the calibration-versus-evaluation distribution shift on the five-benchmark panel breaks the marginal-coverage premise, with the realised eval-fold precision falling fifteen to fifty percentage points below the conformal target on training benchmarks. The deployed gate is therefore a fixed-threshold rule on the calibrated probability, in the lineage of post-hoc calibration analyses by [7] and the selective-classification framing of [16].

The integration of these methods provides complementary strengths inside CAEM's

nine-signal verifier. Chain of thought improves reasoning interpretability and gives the verifier explicit intermediate claims to score. Self-consistency and chain-level entailment contribute the sample-set agreement family of two signals: the average pairwise similarity s_{avg} over $M{=}3$ resampled chains, and the chain-to-answer entailment probability p_{entail} averaged over the same resampled pool. NLI against retrieved evidence contributes the external-grounding family of three signals: a max-match entailment probability $p_{\text{ground_max}}$, a mean-match entailment probability $p_{\text{ground_mean}}$, and a length-gated weakest-link atomic-fact entailment probability $p_{\text{ground_atomic}}$ that decomposes the answer into atomic claims under the FActScore convention [57] and runs only when the answer's token length exceeds the registered atomic-decomposition gate; below the gate the atomic signal collapses to the mean-match value. The contradiction-probability branch was retired in the locked configuration: MiniCheck's binary supported-versus-unsupported judge yields $p_{\text{contra}} = 0$ structurally, the registered head-to-head calibration confirmed the signal carries no calibrated information once the long-hypothesis Qwen judge is ablated, the corresponding hard-veto rule was removed on 2026-04-22, and the contradiction-scoring call itself was short-circuited at the verifier compute path so the signal does not enter the gradient or the composite. The three external-grounding signals join three internal-calibration signals (token probability, MC-Dropout variance, and a derived internal-state composite), the two sample-set signals, and Branch C's question-answer relevance signal $q_{\text{a,rel}}$ to give the nine-signal record, whose *calibrated-probability* composite u_{stored} feeds the four-outcome storage decision through a fixed-threshold rule on the calibrated probability scale. The composite is not a fixed weighted sum: each signal is mapped to a calibrated $\Pr(\text{em} = 1 \mid \text{signal})$ through per-signal isotonic regression with auto-detected direction [7], the per-signal curves are blended with a per-benchmark child fit through a James-Stein style shrinkage prior, and the per-signal log-odds are aggregated through an L_2-regularised logistic boost layer in the lineage of Cherian-Gibbs-Candès conditional boosting [67]. The fixed-threshold rule then reads the calibrated composite at two registered cut-points (the storage cut-point and the deferral cut-point) and emits the four outcomes Store, Defer, Abstain, Discard; the realised cut-point values and the architectural cost of the calibration discipline are reported in Chapter 5. The point for this chapter is that the literature supplies four well-validated families of signals whose failure modes are substantially disjoint, which is what justifies a calibrated multi-signal composite over any single-signal detector.

A complementary strand of architectural literature concerns the prompt-design conditions under which a calibrated verifier can score the served answer at all. Two failure modes register as upstream of any signal-level diagnostic. The first, empty-answer non-compliance, occurs when the prompt structure causes the generator to emit empty or whitespace-only completions on a non-trivial share of queries; the verifier then has no hypothesis to score and the calibrated composite degenerates to its prior. The second, open-QA over-abstention, occurs when a refusal-aware prompt template instructs the generator too aggressively to abstain on uncertainty, with the consequence that the storage class is starved of the in-distribution training signal it requires to fit a calibrated composite. CAEM's locked configuration adopts two registered prompt-design fixes against these failure modes: a constrained-emit grammar that requires a non-empty completion in the answer span before the verifier is dispatched, and an open-QA template that elicits a best-effort answer alongside

the abstention option rather than instead of it. Both fixes are pre-specified, evaluated against the pre-fix configuration on the same evaluation fold, and the empirical compliance and abstention deltas are reported in Chapter 5.

2.2.3 Confidence Estimation

Reliable confidence estimation helps systems tell the difference between answers that can be trusted and ones that need extra checking from outside sources. This section looks at basic methods for measuring uncertainty and explains how these ideas have been adapted for use with large language models.

Gal and Ghahramani [14] laid the theoretical foundations for practical uncertainty estimation in their ICML 2016 paper "Dropout as a Bayesian Approximation." Their key contribution explains that dropout usually during training can be understood as approximate Bayesian inference over model parameters. By keeping dropout active during testing and running the model multiple times, it becomes possible to measure how much the outputs vary.This variation can be used to estimate epistemic uncertainty, which refers to uncertainty caused by limited knowledge about the model itself. And this is different from aleatoric uncertainty, which comes from noise or randomness in the data. But it cannot be reduced.

The theoretical framework shows that a network with dropout before each weight layer approximates a deep Gaussian process. This predicted average and variance can be approximated through Monte Carlo sampling. It describes as performing K forward passes with different dropout masks and computing output variance. For better understanding, this provides uncertainty estimates without requiring group training or modifying the loss function. Because this approach works with any network that already uses dropout, it is especially useful for adapting existing large language models.

For CAEM, MC Dropout is one of the three internal-calibration signals inside the post-generation verifier (u_{token}, u_{dropout}, and the derived $u_{\text{internal}} = 0.5\,u_{\text{token}} + 0.5\,(1 - u_{\text{dropout}})$ that combines them). It quantifies model uncertainty through output variance: high variance indicates the model is unstable with respect to its own weights and suggests that the composite score should be pulled down, while low variance indicates confident generation. This signal is insufficient on its own, since models can be confidently wrong, but the computational cost of the registered five forward passes is acceptable inside the verifier, which only runs on queries that have already been dispatched to Tier 2 or Tier 3 generation. MC Dropout is deliberately *not* used in the pre-routing confidence u_{pre}, which must be cheap enough to run on every query before tier selection.

A major problem with neural network confidence is that models are often too confident in their answers. Guo et al. [17] demonstrated this at ICML 2017. They showed that modern deep networks predict confidence scores that substantially exceed their actual accuracy; a ResNet-110 on CIFAR-100, for example, has an expected calibration error of 16.5 percentage points before calibration. They popularised the Expected Calibration Error (ECE) of Naeini et al. [11] as the standard metric, dividing predictions into bins by confidence and measuring the gap between confidence and accuracy within each bin.

Their proposed solution, temperature scaling, achieves remarkable effectiveness with minimal complexity. The method divides logits by a learned scalar temperature $T > 0$ optimized on a validation set: $\hat{y} = \sigma(\mathbf{z}/T)$ where $\mathbf{z}$ are logits and σ is softmax. Temperature scaling preserves accuracy (argmax unchanged) of classification while calibrating probabilities. Even though only a single parameter is used, it outperforms more complex methods including histogram binning, isotonic regression, and Bayesian approaches. The simplicity and effectiveness make temperature scaling the standard calibration method applied after training is complete.

CAEM uses temperature scaling as the final step in each stage of confidence estimation: once on the pre-routing u_{pre} and once on the post-generation composite u_{stored}. This step adjusts the raw score so that it matches the real accuracy observed on a held-out calibration split, so that when either stage reports eighty percent confidence, about eighty percent of those predictions are actually correct. This makes it possible to set reliable thresholds for tier selection, for the four-outcome storage decision, and for the training-pool filter τ_{train}.

Kadavath et al. [37] from Anthropic explored self-evaluation in large language models in a 2022 preprint. They introduced two methods for extracting confidence from LLMs themselves. First is P(True) asks models to evaluate the probability their own previous answers are correct. Second, P(IK), the probability that "I know" the answer, is predicted by a model trained for the purpose before any answer is proposed. Their findings reveal nuanced calibration properties. Larger models tend to show better calibration on multiple choice questions which means their confidence levels line up more closely with how often they are actually correct. However, policies fine-tuned with reinforcement learning from human feedback appear badly miscalibrated, because reinforcement learning collapses predictions toward the behaviours that receive the most reward, although a single temperature adjustment largely restores calibration.

The idea that reinforcement learning from human feedback can hurt calibration is important. Models are trained to be helpful. But they may express high confidence to appear authoritative even when uncertain which is exactly the opposite of desired behavior for high stakes applications. Because of this conflict between sounding helpful and being truthful, CAEM relies on several outside confidence signals instead of trusting the model's own confidence alone.

Xiong et al. [83] conducted a detailed study of confidence estimation methods for large language models. They provided comprehensive approaches into white box which use internal information like token logits and sequence probabilities, and black box methods, which rely on external signals such as verbalized confidence or consistency checks. Their systematic evaluation across multiple benchmarks reveals striking findings. Verbalised confidence is overconfident and discriminates poorly: even for GPT-4 its average AUROC for telling correct answers from incorrect ones is about 0.63, close to the 0.5 of random guessing. Sampling-consistency signals do much better on some tasks (up to 0.93 AUROC on GSM8K), but no single method consistently outperforms the others, and all of them struggle on tasks that require professional knowledge.

CAEM's multi-signal approach is influenced by this finding that no individual signal is reliable across tasks. The system splits confidence into two stages. The first stage,

pre-routing confidence u_{pre}, fuses two cheap signals (a token-probability estimate from a short draft decode, and an encoder-side convergence ratio C_{conv} over the final attention layers) through a temperature-scaled sigmoid, and is computed on every query. It exists to drive tier selection and the independent safety override. The second stage is the nine-signal post-generation verifier, which runs only after Tier-2 or Tier-3 generation and computes three internal-calibration signals (the mean token log-probability u_{token}, the MC-Dropout variance u_{dropout} [14], and the derived internal-state composite $u_{\text{internal}} = 0.5\,u_{\text{token}} + 0.5\,(1 - u_{\text{dropout}})$), two sample-set agreement signals (s_{avg}, p_{entail} on the same M=3 resampled pool), three external-grounding signals ($p_{\text{ground_max}}$, $p_{\text{ground_mean}}$, $p_{\text{ground_atomic}}$), and Branch C's question-answer relevance signal $q_{\text{a,rel}}$ that guards against off-topic generations. These nine signals feed the calibrated-probability composite u_{stored} through per-signal isotonic regression with a per-benchmark shrinkage prior and the L_2-regularised boost layer; the contradiction probability p_{contra} is excluded from the composite signal set in the locked configuration because the MiniCheck-only deployment makes it structurally zero. The realised per-signal isotonic curves, the boost-layer weights, and the fixed-threshold cut-points at the locked configuration are reported in Chapter 5.

Recent work by Tian et al. [59] asks how calibrated confidence can best be elicited from models fine-tuned with human feedback. Their study found that the raw token probabilities of such models are consistently overconfident, and that confidence verbalised as output tokens is typically better calibrated, often reducing the expected calibration error by a relative 50%. Because the best confidence source differs between model families and tasks, this reinforces CAEM's strategy of combining implicit signals like token probability, dropout, and self-consistency with explicit ones, and of recalibrating every signal rather than trusting any raw score.

A complementary 2024 survey by Geng et al. [71] catalogues white-box and black-box confidence-estimation methods for language models across the same axes that drive CAEM's verifier ensemble, and is the positioning document for CAEM's multi-signal design choice. Two recent single-signal methods deserve specific mention as alternatives we considered. Chen et al. [66] introduced INSIDE / EigenScore at ICLR 2024, computing the log-determinant of the covariance matrix over middle-layer hidden states across K resampled responses; the method reports AUROC of 0.80 to 0.83 on TriviaQA and clear gains over other unsupervised signals on CoQA, SQuAD and Natural Questions (on TriviaQA, perplexity scores slightly higher), and is registered in CAEM as a candidate replacement for the dropout signal that was not adopted in the locked configuration. Nikitin et al. [78] introduced Kernel Language Entropy at NeurIPS 2024, generalising Farquhar et al.'s hard-cluster semantic entropy to a soft-kernel von Neumann entropy over pairwise semantic similarities; KLE is registered as a future-work candidate should CAEM re-introduce the semantic-entropy family in a later iteration (the corresponding h_{norm} signal of [70] was retired before the main run on cycle-zero evidence that it contributed no discriminative information beyond s_{avg} and p_{entail} on the same resampled pool), but it is not adopted in the locked configuration because the empirical AUROC gain is bounded above the cost of the additional kernel computation. The selective-prediction precision-versus-coverage tradeoff that CAEM reports in Chapter 5 follows the form pioneered by Kamath, Jia, and Liang [30] for selective question answering under domain shift, where a calibrator over feature signals predicts $P(\text{correct})$ and the selective rule answers only

when the calibrator's confidence exceeds the registered precision target.

The confidence estimation literature establishes several key findings for CAEM's design. Firstly, relying on a single confidence signal does not provide enough accuracy for safe routing, which makes combining multiple signals necessary. Secondly, systematic overconfidence requires calibration applied after training through per-signal isotonic regression and through a calibrated-probability composite. Thirdly, white-box methods that access internal model information offer complementary insights compared to black-box approaches based on output consistency. Lastly, RLHF training may degrade calibration, so pre-RLHF models or instruction-tuned checkpoints such as Qwen-2.5-3B-Instruct are expected to give more reliable confidence signals than RLHF-aligned chat models of comparable scale. Together, these points shape CAEM's two-stage confidence architecture. The first stage is a temperature-scaled pre-routing score u_{pre} that fuses token probability with encoder convergence and runs on every query. The second stage is a nine-signal post-generation verifier producing the calibrated-probability composite u_{stored} through per-signal isotonic regression (with a per-benchmark shrinkage prior toward the pooled fit) and an L_2-regularised boost layer, whose output is then read by a fixed-threshold rule at two registered cut-points to emit the four-outcome storage decision; both stages are fitted on a held-out calibration split with no live label information at deployment time.

2.2.4 Memory-Augmented Architectures

External memory architectures give neural networks a place to store information outside their usual parameters. It helps them learn and fetch facts. This subsection explains how these ideas developed. Starting with early systems that used differentiable memory and ending with the modern approach known as retrieval augmented generation.

Graves et al. [10] introduced Neural Turing Machines (NTMs) in 2014. NTMs ventured into the idea of giving a neural network external memory that it could read from and write to in a smooth, trainable way. The architecture combines a controller network with an addressable memory matrix, supporting both content-based addressing (via cosine similarity) and location-based addressing (via shift operations). The key innovation is a read/write mechanism that enables end-to-end training via backpropagation. As a result, NTMs successfully learned algorithmic tasks including copying, sorting, and associative recall.

The importance of CAEM is that it shows how external memory can give a model abilities that normal models do not have. NTMs were mainly used for learning algorithms. But, it can store facts as well. CAEM uses this idea in its episodic memory. It stores verified question and answer pairs along with the reasoning behind them. This lets the model call them up when needed without any delay.

Weston et al. [13] introduced Memory Networks at ICLR 2015. Their model scaled to millions of memory slots. It also showed that a system could perform several rounds of reasoning to answer a question. The architecture consists of four learned components. The components are: turning the input into features, updating the memory, turning the memory output into features, and finally producing a response. Furthermore, Memory Networks performed strongly on bAbI question-answering

tasks that required reasoning over multiple facts. Sukhbaatar et al. [12] extended this model with end-to-end Memory Networks at NeurIPS 2015. They removed the need for supervision on which memories to retrieve by using a continuous attention mechanism over the memory. This line of work motivates CAEM's use of stored knowledge as an explicit, inspectable memory.

Graves et al. [15] presented the Differentiable Neural Computer (DNC) in Nature 2016. It built on the idea of Neural Turing Machines but added smarter memory tools like dynamic allocation. It also keeps track of the order in which things were written. DNCs achieved a 3.8% mean error on the bAbI tasks, compared with 7.5% for the best previous jointly trained model, which suggests that learned memory management improves performance.

Lewis et al. [32] introduced Retrieval-Augmented Generation (RAG) at NeurIPS 2020. They showed that combining what a model knows in its parameters with information stored outside the model can make its answers much more accurate. RAG adds a large searchable index of Wikipedia to a BART model with four hundred million parameters. It uses Dense Passage Retrieval to look up relevant documents. Then it works out the final answer by considering the generation probability across all the documents it retrieved. The architecture achieves substantial improvements over parametric-only models: 44.5% exact match on Natural Questions (versus 41.5% prior SOTA), 68.0% on TriviaQA (versus 60.5%), and 45.5% on WebQuestions (versus 44.7%).

Critically, RAG demonstrates index hot-swapping: updating the document index without retraining achieves 70% accuracy on questions about updated world knowledge, proving that non-parametric memory can be updated without expensive retraining. This property directly motivates CAEM's episodic memory, which can be updated continuously through verification without model fine-tuning, though CAEM additionally employs iterative fine-tuning to consolidate learned patterns.

The limitation of standard RAG for CAEM's purposes is that retrieved knowledge is not verified before use and does not accumulate verified reasoning chains. CAEM extends RAG by adding verification layers and storing verified outputs for future retrieval, creating a growing knowledge base rather than relying solely on static external corpora.

Johnson et al. [35] developed FAISS to handle similarity search at the scale of billions of items in 2019, enabling efficient implementation of memory-augmented systems. The library achieves an 8.5× speedup over prior GPU methods through optimised k-selection algorithms. FAISS also supports several index types, such as IVF with product quantisation, which allows the system to search in sub-linear time. For CAEM, FAISS is the tool that makes real-time memory search possible across the low tens of thousands of stored episodes accumulated across the trajectory; keeping search time below one tenth of a second is important for the system to remain usable in practice.

Reimers and Gurevych [26] introduced Sentence-BERT (SBERT) at EMNLP 2019. SBERT produces semantically meaningful sentence embeddings through siamese-network fine-tuning. The speed improvement over standard BERT is substantial: finding the most similar sentence pair in a group of ten thousand sentences takes

about sixty-five hours with standard BERT, but only about five seconds with SBERT, because one can compare the precomputed embeddings directly instead of running a heavy cross-encoder over every pair. This efficiency gain is what makes CAEM's memory search possible: each query is turned into a 768-dimensional embedding, and the system can look through the entire memory index in real time.

More recent work focuses on writeable episodic stores and on insertion-time memory augmentation, both of which are direct architectural neighbours of CAEM's memory module. Das et al. [69] introduced Larimar at ICML 2024, an associative key-value episodic memory with one-shot edits, selective forgetting, and an order-of-magnitude faster knowledge-editing path than direct fine-tuning; Larimar establishes that a writeable episodic store is a feasible alternative to parametric storage but does not associate a quality scalar with each entry. Xu et al. [94] introduced A-MEM, a Zettelkasten-inspired agentic memory that writes structured textual notes at insertion and evolves links between memories over time; A-MEM is the closest architectural neighbour to CAEM in the sense that both write structured metadata at insertion, but A-MEM's metadata is textual (tags, links) while CAEM's is a single calibrated scalar storage score that drives a tiered router rather than evolving link structure. Jeong et al. [74] introduced Adaptive-RAG at NAACL 2024, a three-tier router that classifies queries by complexity and dispatches to no-retrieval, single-step retrieval, or multi-step retrieval accordingly; Adaptive-RAG shares CAEM's surface tier count and a backbone-family neighbourhood but routes on query-side complexity at inference time, not on insertion-time stored confidence on a shared memory, and that distinction is what CAEM's three-tier router differentiates against in Section 4.2.4.

These memory architecture developments establish the feasibility and benefits of external knowledge storage. NTMs prove differentiable memory enables structured reasoning. Memory Networks scale to millions of facts with multi-hop reasoning. RAG demonstrates that non-parametric knowledge improves factual accuracy and supports dynamic updates. FAISS and Sentence-BERT provide the efficient infrastructure for practical deployment. CAEM synthesizes these advances and extends them in three specific ways. First, retrieval is gated by a combined similarity-and-confidence score $\mathcal{S} = \lambda\, s + (1 - \lambda)\, u_{\text{stored}}$ rather than similarity alone, so high-similarity matches that were stored with low verifier confidence still route to Tier 3 grounding. Second, routing is overridden by an independent safety gate on the pre-routing confidence u_{pre} that forces Tier 3 whenever u_{pre} falls below the registered safety floor, regardless of how similar the best memory match is. Third, the memory itself is populated only through a four-outcome storage decision (Store, Defer, Abstain, Discard) governed by a fixed-threshold rule on the calibrated probability composite, with a bounded deferred-reconsider buffer (capacity K_{def}, time-to-live τ_{ttl}) for borderline items, so the knowledge base grows only from items that clear the registered storage cut-point rather than from all retrievals. These extensions distinguish CAEM from static-corpus RAG and motivate the rest of the methodology.

2.2.5 Selective and Adaptive Retrieval

The retrieval-augmented architectures of the previous subsection retrieve unconditionally on every query: every incoming question pays the retrieval cost, and the

retrieved passages enter the generator's context whether the query benefits from external evidence or not. A complementary line of work argues that retrieval is not uniformly helpful and proposes architectures that decide whether to retrieve on a per-query basis. The selective-retrieval literature anchors the architectural design of Section 4.2.5, which adds a binary retrieval-utility classifier between CAEM's three-tier router and its retrieval-augmented fallback so that queries on which retrieval would amplify the question's misconception framing route to the zero-shot tier instead.

Mallen et al. [54] established the empirical anchor for the selective-retrieval position at ACL 2023. They introduced PopQA, a fourteen-thousand-question benchmark of entity-centric factoid queries, and showed that large language models handle popular entities through their parametric memory more reliably than through retrieval, while the same models fail systematically on long-tail entities that retrieval can rescue. The architectural lesson is that retrieval utility is a function of the subject entity's popularity rather than a global hyperparameter, and the corresponding routing rule is a per-relation popularity threshold derived from Wikipedia pageview ranks. With GPT-3 davinci-003, the adaptive retrieval rule reaches forty-six and a half percent accuracy on PopQA, 5.3 percentage points above the best non-adaptive method. The architectural lesson for CAEM is the empirical receipt that retrieval-utility prediction is feasible from cheap question-side features alone (an entity popularity lookup) and does not require a heavy language-model forward pass at routing time, the design constraint that Section 4.2.5 adopts.

Wang et al. [62] extended this line to a learned classifier at EMNLP Findings 2023. Their self-knowledge routing system collects training labels by running each candidate question through both a no-retrieval pipeline and a retrieval-augmented pipeline, recording the per-pipeline exact-match score, and labelling the question as *retrieval-helpful* when the retrieval-augmented exact match exceeds the no-retrieval exact match and as *retrieval-hurt* otherwise. A classifier is then built on the labelled set (the paper compares direct prompting, in-context learning, a BERT-base classifier and a k-nearest-neighbour rule, of which the last performs best), and at inference the classifier dispatches each query to retrieval or to the zero-shot generator according to its prediction. The training-label rule (compare exact match across the two routings) is the direct precedent for the empirical-delta labelling rule that CAEM's retrieval-utility classifier inherits in Section 4.2.5; the architectural difference is that CAEM's classifier sits downstream of the existing tier-1 and tier-2 dispatch outcomes and gates only the retrieval-augmented fallback, while Wang's classifier replaces the dispatch entirely.

Jeong et al. [74] introduced Adaptive-RAG at NAACL 2024 as a three-class extension of the binary routing decision. Their classifier (a T5-large language model fine-tuned on automatically-collected complexity labels) dispatches each query to one of three classes (no retrieval, single-step retrieval, multi-step retrieval), trading routing accuracy for the cost of running a large model at routing time. The reported cost-accuracy frontier on a six-dataset open-domain question-answering suite (SQuAD, Natural Questions, TriviaQA, HotpotQA, 2WikiMultiHopQA, MuSiQue) beats single-strategy baselines and the unconditional retrieval-augmented baseline; for CAEM's purposes the T5-large routing cost is the design's main weakness, while the authors themselves single out the quality of the automatically assigned complexity labels. The

architectural lesson for CAEM is that a learned classifier on automatically-collected outcome labels can decide the routing question, but that putting a large model at routing time costs more than the architecture gains; the follow-up of Moskvoretskii et al. [89] extends Adaptive-RAG to the regime where the classifier carries no language-model forward pass and matches the routing accuracy of the language-model-based variant on the same suite. CAEM's retrieval-utility classifier sits at the lightweight-classifier-only end of this design space, with the only language-model dependency being an offline self-knowledge probe that is fit once at cycle zero and consumed as a fixed feature thereafter.

Asai et al. [63] introduced Self-RAG at ICLR 2024 as the end-to-end fine-tuned alternative to the external-classifier approach. Self-RAG trains a single language model to emit four kinds of reflection tokens during generation (a `Retrieve` token that triggers retrieval, an `IsREL` token that judges passage relevance, an `IsSUP` token that judges support, and an `IsUSE` token that judges usefulness), so the routing decision is made by the same model that generates the answer. The deployed signal for retrieval routing is the probability the model assigns to the `Retrieve` token at decoding time; removing retrieval at inference time drops PopQA accuracy by roughly forty-five percent relative in the authors' ablation, the empirical receipt for the joint design's load-bearing role. The architectural cost is that the reflection tokens are tied to the LLaMA-family backbone Self-RAG fine-tunes on, and retargeting them to a different backbone changes enough of the system to invalidate the architectural comparison; this is the same constraint that prevents Self-RAG from entering the external baseline panel of Chapter 5 as a numerical baseline rather than a citation-level reference.

Baek et al. [85] introduced Probing-RAG at NAACL Findings 2025 as a midpoint between the heavy classifier and the in-generator reflection tokens. They train a linear probe on the intermediate hidden states of the generator (rather than on its output tokens), with the probe trained under a balanced positive-negative protocol that explicitly enforces a fifty-fifty class ratio. The probe is trained to predict, from the hidden states, whether the model's current answer is correct and hence whether additional retrieval is needed; the reported accuracy across five open-domain question-answering datasets beats single-step retrieval, FLARE, DRAGIN, and Adaptive-RAG on the same suite while retrieving roughly half as often. The architectural lesson for CAEM is the most directly relevant of the five: the closest cousin of CAEM's retrieval-utility classifier on the feature axis is the hidden-state-as-routing-signal pattern that Probing-RAG establishes, with the methodological difference that the latter probes intermediate layers while CAEM consumes the last-token hidden state under the disable-adapter context as a single self-knowledge feature alongside thirty-five other cheap question-side and passage-side features.

The empirical ceiling that this literature collectively establishes for retrieval-utility prediction is a held-out area-under-curve of approximately 0.80 to 0.86 in distribution and 0.65 to 0.75 out of distribution, with the highest reported in-distribution readings on Probing-RAG and on the SeaKR uncertainty-decoder family. No published system convincingly exceeds an in-distribution area-under-curve of 0.90 for generalised retrieval-utility prediction on a multi-benchmark held-out fold. CAEM's retrieval-utility classifier targets this ceiling under the lightweight-classifier-only operating point, with a deployed nested-cross-validation area-under-curve of 0.85 on a three-thousand-two-hundred-and-eight-row held-out fold reported in Section 4.2.5.

2.2.6 Self-Improvement and Continual Learning

Self-improvement mechanisms enable models to learn from interaction and feedback, while continual learning addresses the challenge of acquiring new knowledge without forgetting previous capabilities. This subsection reviews methods for iterative refinement and catastrophic forgetting mitigation that inform CAEM's self-improvement loop.

Kirkpatrick et al. [19] introduced Elastic Weight Consolidation (EWC) at PNAS 2017, providing a principled solution to catastrophic forgetting in neural networks. When learning sequential tasks, standard gradient descent optimizes exclusively for current task performance, causing parameter updates that improve new task accuracy while dramatically degrading previous task performance. EWC addresses this by estimating parameter importance for previous tasks using the Fisher Information Matrix, then adding quadratic penalties preventing large updates to important parameters.

The EWC loss function augments standard task loss with a regularization term:

$$\mathcal{L}_{\text{EWC}} = \mathcal{L}_{\text{task}} + \frac{\lambda}{2} \sum_i F_i(\theta_i - \theta_i^*)^2 \tag{2.4}$$

where F_i approximates the Fisher Information measuring how much loss increases when parameter θ_i changes, θ^* are previous task parameters, and λ controls regularization strength. Parameters important for previous tasks (high F_i) receive strong penalties against change, while less important parameters adapt freely to new tasks.

Applied to Atari game learning, EWC maintained approximately 90% of single-task performance when sequentially learning 10 games, compared to severe degradation with standard SGD. For CAEM, EWC supplies the conceptual motivation for bounding cycle-to-cycle drift in adapter space; the isotropic L_2 anchor $\frac{\lambda}{2}\|\theta - \theta_{\text{prev}}\|^2$ that approximates EWC under a uniform-Fisher assumption is registered for the inactive full-fine-tune fallback ($\lambda > 0$), but on the shipped low-rank-adapter path it collapses by design ($\lambda = 0$). The reason is that the bounded adapter parameter envelope layered on a frozen backbone is itself the cycle-to-cycle drift bound: an explicit weight-anchor would otherwise penalise deviation from a zero-anchored full-backbone reference whose weights the loop is not updating, which is meaningless arithmetic. Together with the parameter-efficient adapter envelope and the multimodal retention guard described below, this bounded-envelope mechanism is sufficient to meet the retention target on the empirical evidence reported in Chapter 5.

The choice to fine-tune through a low-rank adapter rather than the full backbone is a separate and equally load-bearing decision in the self-improvement loop. Hu and colleagues introduced low-rank adaptation [36] as a parameter-efficient alternative to full fine-tuning that injects trainable low-rank matrices into the weight matrices of a frozen backbone; the original paper adapts only the attention projections and reports quality on par with full fine-tuning at less than one percent of the trainable parameter count on GPT-3 175B, and later work extends the adapters to the feed-forward projections. The choice between low-rank adaptation and full fine-tuning has empirical consequences for continual learning that the recent literature has measured directly. Biderman and colleagues compare low-rank adaptation with full fine-tuning for Llama-2-7B on code and mathematics, sweeping ranks from sixteen to

two hundred and fifty-six [64], and report two findings relevant to CAEM's setting: low-rank adaptation learns less of the target domain than full fine-tuning, but it forgets less of the base model's general capability, measured as average HellaSwag, ARC-Challenge, and WinoGrande accuracy. In code instruction tuning on Magicoder data, for example, full fine-tuning falls to 0.414 against 0.509 for low-rank adaptation at rank sixty-four. Forgetting grows with training duration in both cases, and in mathematics low-rank adaptation at rank 256 forgets nearly as much as full fine-tuning, so higher rank trades retention for fit. Dettmers and colleagues' QLoRA work [46] establishes the second decisive practical point: applying low-rank adaptation to all linear transformer-block layers, both attention and feed-forward, rather than to the attention projections alone is the most critical configuration choice for matching full fine-tuning, whereas the rank itself has little effect once all layers are adapted. CAEM's adapter target set spans every linear projection inside the transformer block (the four attention projections together with the three feed-forward projections of the gated multi-layer perceptron), which is the all-linear-layer target set that this literature converges on, fitted at rank thirty-two with $\alpha = 2r$; with this choice the effective scale α/r stays at two, so the rank-driven gradient collapse that motivates the rank-stabilised alternative for fixed-α adapters does not arise, and rank thirty-two is far below the large ranks at which the theory of [51] predicts instability for a constant scale. The low-rank-adaptation primary path is what permits CAEM to fit ten cycles of self-improvement on a single consumer-grade graphics processor in the project's compute envelope; full fine-tuning at the same backbone size would not fit, and the agent-research audit accompanying this thesis registers full fine-tuning as the rejected alternative on the joint retention-and-feasibility axis. The within-adapter analogue of catastrophic forgetting under sequential fine-tuning, documented by Wang and colleagues [60] and orthogonal-decomposition fixes for it, is registered as future work; the present thesis combines the bounded low-rank-adapter parameter envelope on a frozen backbone, the multi-modal retention guard with adapter rollback, and the strict training threshold above the storage cut-point as the three-mechanism defence against iterative-adapter drift across the realised cycle trajectory.

The practical implementation in CAEM sets the adapter L_2 anchor coefficient to $\lambda = 0$ on the shipped low-rank-adapter path (the $\lambda > 0$ branch is registered for the inactive full-fine-tune fallback only) and draws training pairs entirely from verified episodes that exceed a strict training threshold, with a benchmark-balancing reweighting layer that prevents single-benchmark dominance through a temperature-mixed softmax over per-benchmark counts, a registered minimum floor on every benchmark's contribution, a bounded upsampling cap on each benchmark's effective count, and a hard ceiling on any single benchmark's share of the pool. The earlier 90/10 mix that paired episode-specific examples with general-capability examples was retired in favour of the reweighting layer because the adapter's small trainable footprint already bounds drift away from the pretraining mass; the redesign trades a fixed mixing ratio for a benchmark-balanced training pool whose composition adapts to the cycle's actual episode counts. A registered set of held-out probes supplies the retention guard: a general-knowledge probe on the multitask language understanding distribution, a held-out test slice of the open-text factoid-question-answering training benchmark, and a held-out test slice of the multiple-choice commonsense-reasoning training benchmark. Before the first cycle a pristine baseline is recorded for every probe; after each cycle's fine-tune the post-cycle accuracy is recorded and the per-

probe ratio against the pristine baseline is computed. The cycle commits the adapter only when the worst probe ratio clears the floor $\rho_{\min} = 0.93$; otherwise the adapter is rolled back to the previous cycle's snapshot and retroactive re-verification is skipped for that cycle. The worst-probe rule replaces the earlier single-probe rule: a single general-knowledge probe missed the failure mode where the multiple-choice accuracy stayed high while open-text generation degraded silently, and the retention guard's purpose is to catch precisely that asymmetric drift before it propagates into the next cycle's gradient. Crucially, the episodic memory is never rolled back even when the adapter update is rejected, so verified knowledge accumulated this cycle remains available for future routing; this asymmetry is deliberate and distinguishes CAEM's retention mechanism from standard continual-learning rollback schemes.

Natarajan et al. [9] provided theoretical foundations for learning with noisy labels at NeurIPS 2013. Their unbiased estimator method constructs proxy loss functions recovering optimal classifiers despite label corruption. The key insight is that if the noise process is known or can be estimated (e.g., what fraction of positive labels are actually negative), modified loss functions achieve consistent parameter estimation. On a synthetic non-separable dataset their methods achieve more than 88% accuracy even when 40% of the labels are corrupted, and they are competitive with other methods for learning under label noise.

For CAEM, this framework directly applies to verified episodic memory: no practical verifier is perfect, so a known fraction of stored episodes will carry incorrect labels (false positives that nonetheless pass the calibrated composite). The stricter training threshold τ_{train}, which is registered strictly above the storage cut-point τ_{store} on the calibrated probability scale, is designed exactly to lift the purity of the training pool above the purity of the memory as a whole, and the noise-tolerant-learning literature is what justifies training on the resulting pool at all: it proves that training on pools whose purity is well below 100% can still improve the model rather than degrading it, provided the corruption rate is below a method-dependent tolerance. The cycle-zero calibration-fold empirical precision at the storage cut-point bounds the pool corruption rate from above in the calibration-consistent sense projected forward by exchangeability between the calibration fold and the cycle's candidate pool, which is the architectural reason CAEM commits to a fixed registered cut-point on the calibrated probability rather than a hand-tuned threshold on raw signals. The realised cut-point, the empirical purity of CAEM's training pool, and the resulting effect on fine-tune outcomes are reported in Chapter 5.

When Zelikman et al. [42] introduced STaR (Self Taught Reasoner) at NeurIPS 2022, they showed successful bootstrapping using iterative self improvement. STaR loop generates rationales for problems, filters for correct answers, fine tunes on correct rationale-answer pairs and then repeats. A key innovation is rationalization. For problems the model fails to answer correctly, it is given the correct answer as a hint and generates a rationale that leads to it. This data augmentation adds training signal from problems the model could not yet solve, which the basic loop discards.

In CommonsenseQA test, STaR achieved +35.9% improvement over few-shot baseline through iterative refinement. The method shows that bootstrapping from the model's own output works when coupled with appropriate filtering. The direct parallel to CAEM is evident: generate predictions, verify correctness, fine-tune on verified

examples, and repeat. CAEM extends STaR in two specific ways. First, simple correctness filtering is replaced by the nine-signal verifier whose calibrated-probability composite u_{stored} drives the four-outcome storage decision, so the filter is itself a learned verifier rather than a binary match. Second, verified items persist in episodic memory for future retrieval rather than being consumed purely as training data, which turns each cycle into a cumulative knowledge gain rather than a one-off weight update.

The recent work of examining intrinsic self-correction capabilities reveals important limitations. Huang et al. [73] found at ICLR 2024 that prompting LLMs to review and correct their own reasoning does not improve accuracy without external feedback, and at times degrades it. Kamoi et al. [75] analyzed when LLMs can correct mistakes at TACL 2024. They found that intrinsic self-correction often fails or degrades performance. The bottleneck is reliable feedback generation. So, models that produce incorrect initial answers usually cannot generate reliable correction signals without external validation.

This limitation motivates CAEM's architectural approach. Instead of depending on intrinsic self-correction using prompting alone, CAEM employs external verification through an NLI cross-encoder, multi-chain self-consistency, and semantic-entropy analysis over stochastic re-samples. These external signals provide the reliable feedback the self-correction literature identifies as a precondition for effective self-improvement, and their failure modes are sufficiently disjoint that aggregating them into a single composite score is expected to be substantially more reliable than any one of them alone, and substantially more reliable than the model's own self-evaluation. The end-to-end verifier accuracy is reported in Chapter 5. Together with the four-outcome storage decision and the training threshold $\tau_{\text{train}} > \tau_{\text{store}}$, this enables the virtuous improvement cycle in which verified generation produces higher-purity training data, which in turn improves generation.

The self-improvement literature also points out successful cases. When external feedback is available (for example code execution results, formal verification, retrieval of sources), models can effectively adapt the corrections. CAEM's Tier 3 routing with RAG provides exactly the same external feedback by retrieving evidence that grounds generation and enables meaningful verification. The system learns not just from its own outputs but from the interaction between generation and evidence based verification.

Domain adaptation research provides additional relevant findings. Gururangan et al. [29] demonstrated at ACL 2020 that continued pre-training on domain-specific unlabeled data (domain-adaptive pretraining) followed by task-specific fine-tuning outperforms direct fine-tuning on limited labeled data. While CAEM does not perform domain-adaptive pretraining due to computational constraints, the finding supports iterative refinement: progressive training on accumulated verified examples enables gradual domain adaptation.

The distinction between CAEM's approach and standard continual learning deserves emphasis. Traditional continual learning addresses learning sequential tasks A, B, C without forgetting previous tasks. CAEM's scenario differs: the task remains constant (question answering), but the model progressively improves at that task through accumulated verified examples. This setting is closer to curriculum learning

and iterative refinement than task-sequential continual learning, though catastrophic forgetting concerns still apply since fine-tuning on domain-specific verified examples risks degrading general capabilities.

Several 2024–2025 contributions tighten the theoretical and architectural framing of self-improvement in ways directly relevant to CAEM's theoretical analysis and to its retention guard. Hosseini et al. [72] introduced V-STaR at COLM 2024, training a DPO verifier over correct and incorrect self-generated solutions and using it to rank test-time candidates; CAEM generalises V-STaR by using the verifier for training-data admission rather than just test-time ranking, and by adding the L_2 anchor and the episodic memory. Song et al. [92] formalise the *generation–verification gap*, the improvement obtained by reweighting a model's generations with its own verification, and show empirically that self-improvement requires a positive gap, that small models fail to achieve one, and that iterative self-improvement saturates within two or three rounds; this is the conceptual motivation for CAEM's $\alpha > 1/2$ verifier-purity precondition in Section 4.3. Yi et al. [95] prove theoretically, in linear regression, that a verifier filter over synthetic data prevents model collapse in self-consuming training loops and yields near-term improvement, while in the long run the model converges to the verifier's knowledge centre, so that gains plateau or reverse unless the verifier is unbiased; this is the strongest published safety anchor for CAEM's storage-gate-filtered SIL, and also its main caution. Das et al. [86] prove at ICML 2025 that retraining a model on its own predicted hard labels increases population accuracy in a binary linearly-separable classification setting with randomly flipped labels, and show empirically that retraining only on samples where the prediction agrees with the given label helps further; this is the closest published analogue of CAEM's pool-purity inequality (Theorem 4.1), and CAEM extends the idea from binary classification with self-predicted labels to open-ended generation with an external multi-signal verifier. Huang, Block, Foster et al. [87] formalise self-improvement as a sharpening mechanism at ICLR 2025 and prove convergence-style amortisation-via-self-training bounds that are the closest published analogue of CAEM's bounded-band stationarity heuristic on the committed-cycle subsequence (Heuristic 4.4). Lang, Sontag, and Vijayaraghavan [76] provide a complementary expansion-based bound at NeurIPS 2024 for when a strong student trained on weak-teacher pseudolabels outperforms the teacher; their bound requires expansion assumptions rather than a verifier-accuracy threshold and is therefore complementary to, rather than redundant with, CAEM's data-purity formulation.

Synthesis of these methods informs CAEM's self-improvement loop design. EWC supplies the conceptual motivation for bounding cycle-to-cycle drift in adapter space; the explicit L_2 anchor, the uniform-Fisher special case that the EWC paper shows to be weaker than EWC itself, is registered for the inactive full-fine-tune fallback only, and on the shipped low-rank-adapter path the bounded adapter parameter envelope on a frozen backbone acts as the implicit drift bound without the hand-tuned anchor coefficient that the full-parameter fine-tune would require. The low-rank-adaptation primitive bounds the trainable footprint at roughly two percent of the backbone parameter count and frozen-backbone semantics, which the parameter-efficient-fine-tuning literature shows forgets less of the base model's general capability than full fine-tuning. Noisy-label learning shows that learning from imperfectly labelled data remains possible when the label noise is bounded, and motivates the stricter

training threshold $\tau_{\text{train}} > \tau_{\text{store}}$ that boosts training-pool purity relative to memory purity. STaR demonstrates successful bootstrapping patterns and motivates the cycle structure. The self-correction research validates the need for external verification rather than intrinsic review and motivates the nine-signal verifier over any single-signal confidence proxy. Finally, the multi-modal retention guard with worst-probe floor $\rho_{\min} = 0.93$ and adapter rollback (but not memory rollback) closes the loop by converting the catastrophic-forgetting risk into a detectable, rejectable event rather than a silent degradation, with a probe set spanning a general-knowledge multiple-choice surface, an open-text factoid surface, and a commonsense multiple-choice surface so that asymmetric drift across answer formats is caught at the cycle boundary. The resulting improvement trajectory across the realised cycle horizon, and the retention margin actually achieved across the multi-probe panel, are reported in Chapter 5.

Table 2.1 positions CAEM against the closest published systems on the architectural axes that distinguish them: whether the system stores a per-entry confidence at insertion, what signal the router uses, what filter the training-data admission applies, what regulariser preserves general capability under self-improvement, and whether a formal precision or convergence guarantee is supplied. No single prior system shares CAEM on every axis simultaneously; the table makes the differentiation explicit.

System	**Stored conf.**	**Routing signal**	**Training filter**	**CL guard**	**Formal guarantee**
CAEM (this work)	yes (calibrated u_{stored})	combined sim + u_{stored}, u_{pre} override	9-signal cal-prob composite + fixed-threshold gate (per-bench shrinkage; LoRA SIL)	bounded LoRA adapter envelope + multi-modal retention rollback	calibration-consistent precision floor; purification condition; trajectory heuristics
A-MEM [94]	no (textual tags)	similarity + link evolution	—	—	—
Larimar [69]	no (key-value addr.)	key match	one-shot edit	—	—
Adaptive-RAG [74]	no	query complexity classifier	—	—	—
V-STaR [72]	no	—	DPO verifier (test-time rank)	—	—
Mohri-Hashimoto [77]	no	—	split-conformal sub-claim filter	—	conformal factuality
MiniCheck [81]	no	—	NLI judge (fact verification)	—	—

Table 2.1: Positioning of CAEM against the closest prior systems. Insertion-time stored confidence, combined-score routing on a shared memory, the calibrated-probability composite at a fixed-threshold gate, the LoRA self-improvement primitive with adapter-rollback continual-learning guard against a multi-modal retention probe panel, and the joint calibration-consistent precision floor and convergence guarantees together constitute CAEM's defensible novelty axis.

2.2.7 External Baselines and Competitive Positioning

The related work reviewed above clusters into three families relevant to CAEM's evaluation: prompt-level reasoning methods, retrieval-augmented generation methods, and training-time self-improvement methods. Chapter 5 instantiates one representative from each family as an external baseline so that every mechanism CAEM adds on top of a plain language model is compared against at least one prior system that already uses that mechanism in isolation. This subsection establishes what each baseline is, what it isolates, and what a fair comparison would look like; the numerical results appear in Chapter 5.

The prompt-level family contributes three baselines. The first, **zero-shot Qwen-2.5-3B-Instruct**, uses the same decoder-only backbone CAEM fine-tunes, with no CoT prefix, no retrieval, and no verification. This is the empirical floor: any gain CAEM reports must exceed the zero-shot score on the same backbone, otherwise the gain is attributable to model capacity rather than to the CAEM mechanism. The second, **Chain-of-Thought** prompting [40], adds the standard zero-shot "Let's think step by step" trigger [38] without any further modification. CoT isolates the contribution of intermediate reasoning on the same backbone and controls for the possibility that CAEM's gains reflect better rationale generation rather than the verifier-memory architecture. The third, **Five-shot Chain-of-Thought**, prepends five in-context $(\text{question}, \text{reasoning}, \text{answer})$ demonstrations from the TriviaQA train split before the CoT trigger and is the strongest prompting-only comparison against CAEM's retrieval-free Tier-2.

The retrieval family contributes three baselines. **DPR-RAG** [31], [32] retrieves the top $k = 5$ Wikipedia passages from a dense DPR index, concatenates them into a 384-token context, and generates from the same Qwen-2.5-3B-Instruct backbone through the Tier-3 prompt builder; this isolates the contribution of external grounding without any verification or memory. **CoT+DPR-RAG** combines the CoT trigger with DPR retrieval and isolates the contribution of reasoning over retrieved evidence. **FLARE** [50] extends DPR-RAG with active retrieval: rather than retrieving once at the start of generation, FLARE generates one sentence at a time and triggers a retrieval whenever the token-level confidence on the upcoming sentence falls below a registered threshold, with retrieved passages prepended to the generation context for the next sentence; this isolates the contribution of selective and step-conditioned retrieval over single-pass retrieval. A fourth retrieval system, **Self-RAG** [63], is acknowledged as the most ambitious retrieve-and-critique pipeline in the literature but is included only as a citation-level reference rather than as a replicated numerical baseline, because Self-RAG's critic tokens are tied to a LLaMA-family backbone and retargeting them to Qwen-2.5-3B-Instruct would change enough of the system to invalidate the architectural comparison.

The training-time family contributes one baseline. **Vanilla fine-tuning** trains Qwen-2.5-3B-Instruct on the verified episode pool for the same realised cycle horizon as CAEM at matched per-benchmark chunk size, but without the multi-modal retention probe panel and rollback guard, without the bounded low-rank-adapter envelope, and without the episodic memory or the verifier at inference time, so every selected episode is treated as a clean label. This isolates the contribution of CAEM's retention-protection machinery: if vanilla fine-tuning degrades on the

retention probe panel or on the later benchmark cycles while CAEM does not, the retention guard and the bounded adapter envelope are responsible. The confidence-calibration family contributes one baseline. **Semantic entropy** [70] computes a model-internal uncertainty score by sampling multiple generations under temperature and clustering them into semantically-equivalent groups, then reports the entropy of the cluster distribution as the per-query uncertainty signal; the baseline runs the same backbone as the zero-shot system and reports the same answer, but exposes an uncertainty surface that is the most direct external comparator to CAEM's multi-signal calibrated probability composite. **STaR** [42] is discussed above as the closest prior method in spirit and is run only as an optional ceiling reference conditional on remaining compute budget: its rationalisation step and its binary correctness filter differ enough from CAEM's nine-signal verifier and four-outcome decision that a large gap to STaR would mix architectural contribution with rationalisation-quality artefacts, so STaR is reported where possible as an aspirational ceiling for what a pure iterative-refinement scheme can reach rather than as a primary competitor.

Taken together, these eight running external baselines (B1–B8 in Section 5.4.1) span the space of prior systems that share at least one mechanism with CAEM: three isolate the prompt-level component (B1 zero-shot, B2 CoT, B5 five-shot CoT), three isolate the retrieval component (B3 DPR-RAG, B4 CoT+DPR-RAG, B7 FLARE), one isolates the training-time component (B6 vanilla fine-tuning), and one isolates a sampling-based uncertainty calibration on the same backbone (B8 semantic entropy). Self-RAG is referenced at citation level only, not run as a numerical baseline, because retargeting its critic tokens to Qwen-2.5-3B-Instruct would change enough of the system to invalidate the architectural comparison; STaR is reported separately as an optional ceiling when compute permits, and neither system enters the significance panel's Holm correction. No single baseline combines all four families, which is precisely the gap CAEM occupies and which Chapter 5's retrieval-utility-classifier on/off ablation isolates against the remainder of the architecture.

2.2.8 Evaluation Benchmarks

Rigorous evaluation requires benchmarks testing diverse reasoning capabilities with objective correctness criteria. This subsection reviews the five primary factual-QA benchmarks that constitute CAEM's locked evaluation panel (FEVER, TriviaQA, CommonsenseQA training, and TruthfulQA, StrategyQA transfer) and the MMLU general-knowledge probe that sits inside the multi-modal retention guard, explaining how each tests a different aspect of hallucination reduction. Natural Questions is also described for completeness: it and HotpotQA were evaluated at cycle zero and dropped before the main run because the base generator answered too few of their questions correctly for the storage gate to build a usable training pool (Section 1.6.1).

Thorne et al. [21] created the FEVER dataset at NAACL 2018 for fact verification, providing 185,445 claims generated from Wikipedia requiring classification as SUPPORTS, REFUTES, or NOT ENOUGH INFO relative to Wikipedia evidence. Claims are designed to test factual knowledge; an illustrative pair (not taken from the dataset) is "Barack Obama was the 44th President of the United States" (SUPPORTS) and "The Beatles formed in Manchester" (REFUTES, since they formed in Liverpool). The NOT ENOUGH INFO category is particularly valuable,

testing whether systems recognise insufficient evidence rather than hallucinating conclusions. FEVER directly evaluates the external-grounding family of CAEM's verifier, because the task structure matches the premise-hypothesis entailment check that produces p_{ground} and p_{entail}. FEVER is also the benchmark on which the four-outcome storage decision is most directly exercised: SUPPORTS claims map to STORE, REFUTES claims to DISCARD via the low composite arising from contradicted retrieved evidence, and NOT ENOUGH INFO claims to ABSTAIN via the low-composite-and-low-grounding rule.

Joshi et al. [18] introduced TriviaQA at ACL 2017, providing about 95,000 trivia question-answer pairs authored by trivia enthusiasts, which with on average six supporting documents each from the web and Wikipedia give over 650,000 question-answer-evidence triples. Questions span an unusually wide topical range (history, science, geography, entertainment) and the provided evidence is noisy (only a subset of the retrieved documents actually answers the question), making TriviaQA a realistic test of retrieval-augmented generation under distractor documents. For CAEM, TriviaQA tests whether the Tier-3 retrieval-augmented path can ground answers against noisy evidence and whether the mean-match grounding signal $p_{\text{ground_mean}}$ remains calibrated when retrieval quality varies across queries.

Kwiatkowski et al. [23] introduced Natural Questions (NQ) at TACL 2019, providing real anonymised queries to the Google search engine paired with Wikipedia-page long answers and span-level short answers. Unlike synthetic benchmarks, NQ captures the distribution of questions users actually ask, with the accompanying variability in specificity, ambiguity, and presupposition. NQ is the dominant open-domain QA benchmark in recent retrieval-augmented-generation literature [31], [32]. CAEM evaluated NQ at cycle zero and dropped it before the main run: the base generator answered only 17 percent of its questions correctly, too few for the storage gate to build a usable training pool. Its training split is reused only to train the retrieval-utility classifier.

Lin et al. [39] designed TruthfulQA specifically to measure how models mimic human falsehoods, providing 817 questions across 38 categories in health, law, finance, politics, and conspiracies. Questions are adversarially constructed to elicit common misconceptions: "What happens if you crack your knuckles?" where the false but common answer is "you get arthritis" and the true answer is "nothing harmful happens; no proven link to arthritis." The inverse scaling finding (larger models often less truthful) makes TruthfulQA the most challenging benchmark for parametric scaling approaches. Human performance reaches 94% truthful and 87% truthful-and-informative, while the best GPT-3 175B configuration achieves 58% and 21% respectively. For CAEM, TruthfulQA was registered as the benchmark where the sample-set family of signals was expected to pay off, on the reasoning that a misconception would make the model alternate between the correct answer and the misconception across stochastic decodes, which the sample-set agreement signals s_{avg} and p_{entail} would detect. The literature predicts otherwise: semantic entropy targets arbitrary, sample-to-sample errors and explicitly does not cover models that are consistently wrong because of common misconceptions [70], since such a model gives the same wrong answer on every decode and agreement-based signals then read it as confident. The realised results agree with the literature rather than with the registered expectation: entries stored on TruthfulQA at cycle zero are correct only

52.2% of the time, against 75 to 82% on the training panel (Table 5.31), and the semantic-entropy baseline is among the highest-scoring baselines on TruthfulQA exact match rather than being penalised for heavy abstention (Chapter 5). TruthfulQA is therefore the benchmark that exposes the blind spot of agreement-based verification.

Geva et al. [33] introduced StrategyQA at TACL 2021, presenting 2,780 questions requiring implicit reasoning where decomposition steps are not mentioned in questions. For example, the dataset question "Could a llama birth twice during the War in Vietnam?" requires knowing Vietnam War duration (implicit: 1955–1975) and llama gestation period (implicit: approximately 11 months), then computing whether two births fit the timeframe. Human accuracy reaches 87% while best models achieve approximately 66%. The implicit reasoning requirement tests whether models genuinely understand underlying concepts or merely perform shallow pattern matching. For CAEM, StrategyQA evaluates the sample-set agreement signal s_{avg} because implicit reasoning should converge across multiple chains if the underlying world-knowledge is consistent; divergence across chains is a strong hallucination signal specifically on this benchmark.

Hendrycks et al. [34] introduced MMLU in 2021, a 57-subject multiple-choice benchmark covering humanities, STEM, social sciences, and professional domains. MMLU is used in CAEM as one of three components of the *multi-modal retention probe panel*, not as a target benchmark. The fine-tune is not optimised for MMLU; instead, MMLU is probed before the first cycle (the pristine baseline) and after every cycle's fine-tune, alongside two further probes drawn from held-out test slices of the training panel: an open-text factoid probe on the held-out TriviaQA test fold, and a commonsense multiple-choice probe on the held-out CommonsenseQA test fold. The cycle aborts and rolls the adapter back to the previous snapshot whenever the worst per-probe ratio against the pristine baseline falls below the fixed floor $\rho_{\min} = 0.93$; the worst-probe rule replaces an earlier single-probe rule because a general-knowledge multiple-choice probe alone missed the failure mode where MCQ accuracy stayed high while open-text generation degraded silently. The episodic memory is preserved across rollbacks. The choice of MMLU as one of the three probes is deliberate: catastrophic forgetting manifests most visibly as erosion of capability on tasks the model was not trained on this cycle, so a broad out-of-distribution probe is a more sensitive trigger than in-distribution QA, and the held-out training-bench test folds add the asymmetric-drift sensitivity that the broad probe alone would miss.

Benchmark complementarity ensures comprehensive evaluation across the registered five-benchmark panel. FEVER tests entailment-based grounding, including the mapping from NOT ENOUGH INFO to the ABSTAIN outcome. TriviaQA tests retrieval-augmented generation under noisy evidence. CommonsenseQA tests commonsense multi-step reasoning over short multiple-choice options. TruthfulQA exercises the semantic-entropy proxy and the q-a relevance signal on imitative falsehoods, as a transfer-only benchmark held out from the SIL training pool. StrategyQA exercises sample-set agreement on implicit multi-hop reasoning, also as a transfer-only benchmark. MMLU, the open-text TriviaQA test fold, and the multiple-choice CommonsenseQA test fold close the catastrophic-forgetting loop through the multi-modal retention probe panel. Together, they cover the spectrum of factual reasoning tasks where hallucination reduction matters most, and success across the panel demonstrates that CAEM's improvements generalise across reasoning types rather

than overfitting to specific task characteristics.

2.3 Summary of Key Findings

The literature review both motivates and supplies the technical premises for CAEM's architectural orientation. Hallucination-measurement studies report prevalence rates varying widely by domain, with up to 91.4% of chatbot-generated references hallucinated in systematic-review searches [65] and fabricated references in clinical writing [43]. TruthfulQA's inverse-scaling finding [39], in which the best GPT-3 175B configuration achieved only 21% on the combined truthful-and-informative metric, is the clearest demonstration that parametric scaling alone is insufficient for truthfulness, and FActScore [57] shows that even competent models support only about 58% of their atomic facts, which is why CAEM verifies atomic facts rather than holistic responses. Together these findings motivate a multi-component architectural intervention rather than a pure scaling approach.

The verification literature supplies four families of signals with substantially disjoint failure modes. Chain-of-thought prompting [40] produces the intermediate claims that downstream verification operates on. Self-consistency [61] shows that agreement across sampled reasoning chains is an uncertainty signal; CAEM's sample-set agreement signal s_{avg} adapts it to embedding space as the mean pairwise cosine similarity of the sampled chains. Chain-to-answer entailment under the natural-language-inference judge provides the second sample-set signal p_{entail}, computed on the same $M{=}3$ resampled pool to amortise the decode cost. The normalised semantic entropy of [52], [70] and the kernel-language-entropy generalisation of [78] were registered earlier in the project as candidate sample-set members but retired before the main run on cycle-zero evidence that they contributed no discriminative information beyond what s_{avg} and p_{entail} already provide on the same resampled pool. Claim-support verification against retrieved evidence provides the external-grounding family of three signals ($p_{\text{ground_max}}$, $p_{\text{ground_mean}}$, length-gated atomic-fact $p_{\text{ground_atomic}}$). The thesis uses MiniCheck-Flan-T5-Large [81] as the deployed entailment backend across all hypothesis lengths because the registered long-hypothesis Qwen judge fails the pre-specified Pearson floor on the registered 500-sample calibration overlap fold and is ablated in the locked configuration; MiniCheck is fine-tuned specifically on language-model-generated claim-support data and outperforms prior specialised fact-checkers on claims of the kind the verifier actually scores (its raw scores are bimodal rather than calibrated, which CAEM's per-signal isotonic step corrects), whereas generic NLI models trained on MultiNLI and SNLI (including RoBERTa-Large-MNLI [24]) exhibit systematic miscalibration on language-model hallucinations: [91] report a one-hundred-percent false-positive rate at ninety-five-percent recall for a near-state-of-the-art generic NLI model on HaluEval [53], corresponding to a verifier balanced accuracy at or below the chance threshold, which would falsify the empirical premise of the purification condition. The legacy backend is retained in Chapter 5's verifier-calibration diagnostic so the backend choice is defended by trajectory evidence rather than citation alone. CAEM organises nine signals into a post-generation verifier across four families (three internal-calibration signals, two sample-set agreement signals, three external-grounding signals, and Branch C's question-answer relevance signal $q_{\text{a,rel}}$), aggregates them through per-signal isotonic regression with a per-

benchmark shrinkage prior toward the pooled fit and an L_2-regularised boost layer into the calibrated-probability composite u_{stored}, and reads the composite through a fixed-threshold rule at two registered cut-points to emit the four-outcome storage decision (STORE, DEFER, ABSTAIN, DISCARD). A confabulation early-exit gate in which the sharply-asymmetric $(u_{\text{internal}}, p_{\text{ground,max}})$ conjunction is checked before the composite catches the confidently-wrong failure mode that would otherwise route through the gate. The contradiction probability p_{contra} is excluded from the composite signal set in the locked configuration because the MiniCheck-only deployment makes it structurally zero, the corresponding hard-veto rule was retired on 2026-04-22, and the contradiction-scoring call itself was short-circuited at the verifier compute path so the signal does not enter the gradient or the composite log-odds.

Composite verification approaches that fuse multiple individual signals are recent but published precedent exists. [82] publish the closest analogue: a multi-scoring hallucination detector that calibrates about ten scores with multicalibration and combines their logits through logistic regression, evaluated on TriviaQA, FEVER, HaluEval, and BIG-bench, with a Spearman cross-score correlation heat-map that reveals clusters of related scores. [93] benchmark around thirty uncertainty-estimation methods head-to-head and report that sample-diversity methods such as semantic entropy dominate open-generation tasks such as TriviaQA and CoQA, while verbalised-uncertainty and maximum-softmax-probability methods dominate constrained-output tasks such as MMLU. This heterogeneity across output regimes is the principled justification for composite verification over single-signal detection, and motivates the five-benchmark panel evaluated in Chapter 5 rather than a single-benchmark target. The nine-by-five Spearman matrix across CAEM's five benchmarks (scheduled in Chapter 5 as a Phase 1 Full deliverable) is a direct extension of the Valentin 2024 four-benchmark cross-score matrix to the retrieval and MC-Dropout signal families that their score set did not cover.

The confidence-estimation literature shows that verbalised confidence discriminates correct from incorrect answers only weakly (an AUROC of about 0.6) and that no single confidence signal is consistently best [83], that modern networks are systematically overconfident and require calibration applied after training [17], and that models tuned with reinforcement learning from human feedback appear badly miscalibrated until their outputs are rescaled [37], with raw token probabilities that are consistently overconfident [59]. CAEM's backbone, Qwen-2.5-3B-Instruct, is itself preference-tuned (its post-training uses DPO and GRPO [80]), which is why CAEM never consumes a raw confidence: the pre-routing score is temperature-scaled and every verifier signal is isotonically recalibrated on a held-out fold. These findings shape CAEM's two-stage confidence architecture: a cheap temperature-scaled pre-routing score u_{pre} that fuses token probability with encoder convergence and runs on every query, and the nine-signal post-generation verifier whose calibrated-probability composite u_{stored} runs only after generation. The pre-routing score drives tier selection through the combined score $\mathcal{S} = \lambda\, s + (1-\lambda)\, u_{\text{stored}}$ and an independent safety override that fires whenever u_{pre} falls below the registered safety floor ϕ_{safe}.

Memory-augmented architectures establish the feasibility and benefit of external knowledge storage. Neural Turing Machines [10], Memory Networks [13], and DNCs [15] demonstrate that differentiable memory enables structured reasoning; RAG [32] shows that non-parametric knowledge improves factual accuracy (44.5% vs 41.5%

on Natural Questions) and supports index updates without retraining; FAISS [35] and Sentence-BERT [26] make similarity search cheap enough to run in real time. CAEM extends this tradition in three specific ways: the memory is populated only through a four-outcome storage decision governed by a fixed-threshold rule on the calibrated probability composite, with a bounded deferred-reconsider buffer (capacity K_{def}, time-to-live τ_{ttl}) for borderline items rather than from every retrieval; routing combines similarity with stored-verifier confidence rather than using similarity alone; and an independent safety override forces Tier 3 grounding whenever pre-routing confidence falls below the registered safety floor, regardless of the best memory match.

Continual-learning and self-improvement research supplies the theoretical backbone for iterative refinement. EWC [19] maintains performance on earlier tasks while learning new ones and supplies the conceptual motivation for bounding cycle-to-cycle drift in adapter space; the explicit L_2 anchor, the uniform-Fisher special case that the EWC paper itself shows to be weaker than EWC, is registered for the inactive full-fine-tune fallback only (Fisher-matrix computation at every cycle boundary on a three-billion-parameter Qwen-2.5-3B-Instruct backbone is prohibitive on a single consumer-grade GPU), while on the shipped low-rank-adapter path the bounded adapter parameter envelope on a frozen backbone acts as the implicit drift bound without the hand-tuned anchor coefficient that the full-parameter fine-tune would require. The parameter-efficient-fine-tuning literature reviewed at Section 2.2.6 establishes that low-rank adaptation [36] forgets less of the base model's general capability than full-parameter fine-tuning [64], and that adapters on all linear layers are needed to match full fine-tuning quality [46], which is the empirical anchor for CAEM's choice of the LoRA primary path with the L_2 anchor placed on the adapter parameters rather than on the full backbone. Empirical studies at pre-trained transformer scale suggest that weight-regularisation methods add little once the backbone is pre-trained: [56] find that pre-trained weights themselves ease forgetting by leading to wider minima, and [41] find across five pre-trained language models that EWC performs worst among the continual-learning methods they compare, while experience replay is the most robust. Within-adapter forgetting under sequential fine-tuning is documented by [60], whose orthogonal-subspace decomposition is registered as a future-work upgrade if the multi-modal retention probe panel reports persistent drift on the trajectory's later cycles. Noisy-label learning [9] shows that a classifier can still be learned from substantially corrupted labels, provided the class-conditional noise rates sum to less than one and the loss is corrected for them; CAEM does not correct its loss, and instead lowers the noise rate of the training pool directly through the registered stricter training threshold $\tau_{\text{train}} > \tau_{\text{store}}$ that lifts training-pool purity above memory purity, where τ_{store} is the registered fixed-threshold cut-point on the calibrated probability scale. STaR [42] demonstrates the bootstrap loop of generate–filter–fine-tune that CAEM generalises by replacing simple correctness filtering with the nine-signal calibrated-probability composite u_{stored} (per-signal isotonic regression with a per-benchmark shrinkage prior plus an L_2-regularised boost layer; the contradiction probability p_{contra} is excluded from the composite signal set because the MiniCheck-only deployment makes it structurally zero), and by persisting verified items in episodic memory rather than only in the adapter. The self-correction critiques of [73] and [75] establish that intrinsic prompt-based self-correction rarely improves accuracy without external feedback, which is exactly

the feedback CAEM supplies through the external verifier. Together, these pieces motivate CAEM's cycle-boundary mechanics: retroactive re-verification of stored items against the refreshed adapter, deferred-buffer reconsideration with a three-way branch (promote, drop, keep), pool reweighting with a temperature softmax and a hard share cap, parameter-efficient fine-tuning on the high-purity training pool through the bounded LoRA adapter envelope, and a multi-modal retention guard with floor $\rho_{\min} = 0.93$ on the worst-probe ratio that aborts the cycle and rolls back the adapter (but not the memory) when general capability erodes.

The evaluation benchmark panel is chosen to exercise these mechanisms on substantially disjoint failure modes. FEVER [21] tests the external-grounding family and the three-way mapping from SUPPORTS, REFUTES, or NOT ENOUGH INFO onto STORE, DISCARD, or ABSTAIN respectively. TriviaQA [18] tests Tier-3 retrieval-augmented generation under noisy evidence on the open-text factoid surface and contributes a held-out test fold to the multi-modal retention probe panel. CommonsenseQA [28] tests commonsense multi-step reasoning over five-option multiple-choice items and contributes a second held-out test fold to the retention probe panel. TruthfulQA [39] and StrategyQA [33] are held out from the SIL training pool as transfer-only benchmarks, exercising the cross-encoder relevance signal on imitative falsehoods and sample-set agreement on implicit multi-hop reasoning respectively. MMLU [34] is the third probe in the multi-modal retention panel, supplying a broad general-knowledge multiple-choice surface that catches catastrophic forgetting on tasks the cycle was not trained on. The headline hallucination metric is the *Composite Hallucination Metric* (CHM), the equal-weighted mean across an eight-subtype failure-mode taxonomy (confident confabulation, factual fabrication, logical fabrication, off-topic, defensive evasion, template leak, false refusal, over-long), aggregated across benchmarks into a geometric Composite Evaluation Score whose five factors capture accuracy, epistemic quality, retention, calibration, and verifier reliability. Specific quantitative results on this panel are reported in Chapter 5.

Taken together, the literature establishes that hallucination remains unsolved despite substantial investment, that parametric scaling is demonstrably insufficient, and that individual methods achieve only limited discriminative accuracy. CAEM's response is to combine verified episodic memory with a four-outcome storage decision under a fixed-threshold rule on the calibrated probability composite, two-stage confidence (cheap pre-routing score plus nine-signal post-generation verifier), a combined-score router with independent safety override, and a constrained self-improvement loop whose hardest protection is a multi-modal retention-guard abort with adapter rollback but memory preservation. Each piece is directly motivated by a specific limitation surfaced in this chapter.

2.4 Chapter signpost

The preliminaries section and the literature review have established the technical foundations on which the architecture rests and have located CAEM's joint intervention against the verification, retrieval, memory-augmented, and continual-learning prior work whose individual contributions do not close the deployment-safety gap registered in Chapter 1. Chapter 3 translates the architectural opportunity surfaced by the literature into a registered specification: seven functional requirements paired with

named architectural components, five non-functional requirements with empirical anchors, six production-parity design commitments, and the societal, environmental, ethical, standards, project-management, risk, and economic readings that bound the engineering envelope. The specification is then discharged end-to-end by the methodology chapter that follows.

Chapter 3

Requirements, Impact, and Project Management

3.1 Final Specifications and Requirements

3.1.1 Functional requirements

The architecture is required to perform seven functions at deployment, each derived from the problem statement registered in Chapter 1. First, the system must produce a calibrated answer for every admissible query, where calibration means that the answer is accompanied by a storage score whose value reliably tracks the empirical likelihood of correctness. Second, the system must abstain explicitly when calibration falls short of the registered storage and deferral thresholds, returning a refusal rather than a hedged or speculative answer. Third, the system must commit verified episodes to a persistent episodic memory, storing the question, the answer, the supporting context, and the storage score as an indivisible record. Fourth, the system must recall the relevant stored episode on subsequent queries that fall within its similarity neighbourhood, and must defer to the zero-shot or retrieval-augmented tier when no neighbour clears the dispatch threshold, with a retrieval-utility classifier deciding on a per-query basis between zero-shot generation and the retrieval-augmented tier on the residual queries so that retrieval is invoked only when its expected contribution clears the registered classifier-side threshold. Fifth, the system must perform regularised parameter-efficient fine-tuning at each cycle boundary through a low-rank adaptation layer, drawing only from stored episodes whose storage score exceeds the strict training threshold and passing the curated pool through a benchmark-balancing reweighting layer. Sixth, the system must enforce a retention guard against a registered multi-modal probe panel and roll back the adapter when the worst per-probe ratio falls below the fixed floor, while preserving the memory across the rollback so that the verified episodes accumulated during a failed cycle are not lost. Seventh, the system must render the calibrated probability of correctness to the user as a continuous percentage alongside the four-class decision tag on every served answer, so that the user sees both the categorical reliability label and the underlying confidence number; the abstain decision suppresses the answer text and substitutes a registered insufficient-evidence message while still rendering the calibrated probability for inspection. Table 3.1 summarises the seven functional requirements together with

the architectural component that delivers each.

ID	Requirement	Architectural component (specified in Chapter 4)
FR1	Calibrated answer per query	Nine-signal verifier feeding the per-benchmark calibrated probability composite (with shrinkage prior)
FR2	Explicit abstention under uncertainty	Fixed-threshold storage gate + confabulation early-exit rule
FR3	Commit verified episodes to memory	Four-outcome decision tree (store branch)
FR4	Recall on subsequent queries	Three-tier router with combined-score dispatch and safety override, refined by the retrieval-utility classifier that intercepts the safety-override and score-formula fall-through dispatches at the registered threshold $\tau_{\text{RUC}} = 0.375$
FR5	Cycle-boundary regularised fine-tune	Self-improvement loop with low-rank adapter, pool-reweighting layer, and strict training threshold
FR6	Retention guard with adapter rollback	Multi-modal retention probe panel (worst-probe rule) and asymmetric memory-vs-adapter rollback
FR7	Calibrated probability rendered to the user with every answer	Production response envelope: four-class decision tag + continuous u_{stored} percentage; insufficient-evidence message substitutes the answer text under abstain

Table 3.1: Functional requirements and the architectural components that deliver them.

3.1.2 Non-functional requirements

Five non-functional requirements bound the operating envelope of the architecture. The latency envelope per query is tier-dependent: the memory-direct tier resolves in the time of an embedding lookup and a similarity search, the zero-shot tier adds the cost of a single decode, and the retrieval-augmented tier adds the cost of passage retrieval and a grounded decode; per-tier latency budgets are reported empirically in Chapter 5. The storage precision floor anchors the calibration-fold empirical precision of the storage class against the registered cycle-zero threshold sweep at the locked cut-point $\tau_{\text{store}} = 0.60$,

$$\Pr\Big(\text{EM} = 1 \,\Big|\, u_{\text{stored}} \geq \tau_{\text{store}}\Big)_{\text{cal}} \;=\; \pi_{\text{store}}, \tag{3.1}$$

and the empirical receipt of this anchor is the cycle-zero reading $\pi_{\text{store}} = 0.796$ over 289 stored entries on the held-out cycle-zero training-panel validation fold (500 samples per benchmark, 1500 in total) under the locked composite. The locked cut-point was selected against the cycle-zero threshold sweep that compared $\tau = 0.60$ against the alternative $\tau = 0.65$: the stricter cut admits roughly a factor of four fewer entries at the same per-store precision shelf, so the registered configuration

gains substantially more correct memories at no cost in per-store precision at cycle close. The retention floor requires the post-fine-tune accuracy on every probe in the registered multi-modal panel to remain at no less than 93% of the pristine baseline, with the cycle aborting on the worst per-probe ratio,

$$\min_{p \in \mathcal{P}} \rho_p(M_t) \;:=\; \min_{p \in \mathcal{P}} \frac{\mathrm{EM}(M_t,\, p)}{\mathrm{EM}(M_0,\, p)} \;\geq\; \rho_{\min} = 0.93, \tag{3.2}$$

where M_t is the post-fine-tune generator at cycle t and $\mathcal{P}$ is the multi-modal probe set comprising a general-knowledge multiple-choice probe, a held-out open-text factoid probe drawn from a training-benchmark test fold, and a held-out commonsense multiple-choice probe drawn from a second training-benchmark test fold; any fine-tune that violates this floor on any probe rolls the adapter back and resumes the per-query phase. The reproducibility envelope requires every artefact released alongside the thesis, including the locked storage cut-points, the boost intercept, the per-benchmark shrinkage curves, the seed-pool composition, and the per-cycle snapshots, to be regenerable from a single recorded random seed and a published code commit. The integrity envelope requires that no benchmark sample appearing in the evaluation fold may have appeared in the calibration fold, the seed pool, or the per-cycle stream chunks, so that calibration-fold and evaluation-fold precision are measured on disjoint distributions. Table 3.2 summarises the five non-functional requirements with their registered targets and the empirical receipt available from cycle zero.

ID	Requirement	Registered target	Cycle-zero receipt
NFR1	Per-query latency envelope	Tier-dependent: embedding lookup (Tier 1), single decode (Tier 2), retrieval + grounded decode (Tier 3)	Per-tier trajectory in Chapter 5
NFR2	Storage precision anchor at $\tau_{\text{store}} = 0.60$	Held-out π_{store} registered against the cycle-zero threshold sweep that selected 0.60 over 0.65 for substantially more correct memories at the same per-store precision shelf	$\pi_{\text{store}} = 0.796$ over 289 stored entries on the cycle-zero training-panel validation fold
NFR3	Retention floor (multi-modal probe panel)	≥ 0.93 of pristine baseline on every probe in the panel; worst-probe rule aborts the cycle	Per-cycle trajectory in Chapter 5
NFR4	Reproducibility envelope	Every artefact regenerable from a single seed and a published commit	Locked configuration JSONs + public-host snapshot
NFR5	Integrity envelope	Disjointness between evaluation, calibration, seed, and per-cycle stream-chunk folds	Cross-fold disjointness verified per benchmark

Table 3.2: Non-functional requirements with registered targets and cycle-zero receipts.

3.1.3 Constraints and design commitments

The functional and non-functional requirements together imply six production-parity commitments that the engineering programme adopts in advance. The single-backbone commitment fixes the underlying generator backbone across every cycle and every tier; only the low-rank adaptation layer is updated by the self-improvement loop, so that all reported gains are attributable to memory accumulation, calibration discipline, and adapter-level fine-tuning rather than to a backbone swap. The frozen-cut-point commitment fixes the storage and deferral cut-points $\tau_{\text{store}} = 0.60$ and $\tau_{\text{defer}} = 0.45$ on the calibrated probability scale once before the main run begins and holds them constant across every cycle, so that the operating point of the storage decision does not drift even as the underlying score distribution shifts; only the calibrated probability composite refits at each cycle boundary on the cycle's own labelled fold, with per-benchmark child curves blended toward the pooled fit through the registered shrinkage prior so that benchmarks with weaker per-bench evidence regress toward the pooled fit rather than overfitting on small per-bench folds. The local-execution commitment forbids any external model interface in the inference path; the verifier ensemble, the embedding encoder, the passage index, and the generator all run from open weights on a single rented graphics processor. The bounded-cycle commitment caps the self-improvement loop at ten cycles and admits an early stop only on an empirical equilibrium diagnostic; the choice is justified in Chapter 4 through the convergence analysis. The matched-protocol commitment requires that every external baseline be evaluated under the same samples, prompts, and scoring rules as the architecture itself, so that the comparison panel reported in Chapter 5 is not confounded by re-implementation choices. The single-seed reproducibility commitment caps the random-state surface at one recorded seed, with a published per-cycle artefact catalogue and a snapshot of the pre-step-seven state on a public dataset host, so that the trajectory can be regenerated end to end by a third party.

3.2 Societal Impact

3.2.1 Deployment benefits

The primary societal benefit of the architecture is a measurable reduction in confident misinformation produced by deployed language models on factual question answering. A model whose answers are routed through the calibrated probability composite and the fixed-threshold storage gate either returns an answer whose calibrated probability clears the registered cut-point, or abstains explicitly when the storage score falls short. This explicit refusal channel is itself a societal benefit, because it converts an otherwise opaque uncertainty into a user-facing trust signal that downstream consumers, including educators, journalists, and information specialists, can read and act on. In domains where fluency without grounding has produced documented harm, such as health-information retrieval [43], [65], and in other high-stakes settings such as legal triage and educational tutoring, an architecture that abstains rather than fabricates is a strict improvement on the deployed alternative. The episodic memory further amortises verification cost across users: an answer that is verified once on the first query that asks it can be reused under a high-confidence retrieval

action on every subsequent matching query, so that the verification cost incurred by an early user becomes a public good for the user community that follows.

3.2.2 Dual-use considerations

A verified-memory system that learns from its own interactions can in principle memorise harmful content if the verifier is misconfigured or if its training pool admits adversarially produced inputs. Three architectural mitigations bound this risk in the deployed configuration. First, the retroactive re-verification pass at each cycle boundary re-scores every stored episode under the updated generator and prunes entries whose calibrated storage score has fallen below the storage threshold, so that a misconfigured snapshot does not persist across cycles unchallenged. Second, the strict training threshold above the storage threshold keeps any episode whose storage score lies in the marginal band out of the training pool, even when the same episode would be admissible for retrieval. Third, the retention guard against the multi-modal probe panel (general-knowledge multiple-choice plus held-out test slices of two training benchmarks) flags fine-tunes that have measurably degraded any of the probed capabilities and aborts them with an adapter rollback under the worst-probe rule, so that a single poisoned cycle cannot silently propagate forward. Together these mitigations make the architecture's failure mode an honest abstention rather than confident harm, but the limits of this guarantee depend on the verifier's coverage, which the thesis reports empirically and discusses as a threat to validity in Chapter 5.

3.2.3 Marginalised-user impact

The harm caused by confident hallucination is not evenly distributed across the user population. Users who lack independent means to verify a confident-sounding answer, including students consulting language-model tutors, patients consulting language-model health agents, and applicants consulting language-model administrative agents, are disproportionately exposed because they cannot fall back on a domain expert to check the model. The architecture's calibrated abstention class is a partial response to this inequality: by routing low-confidence answers to an explicit refusal rather than a hedged speculation, the system removes the most dangerous failure mode for the very users who would otherwise have no defence against it. The benefit is structural rather than rhetorical, because the abstention is governed by a fixed-threshold rule on a per-benchmark calibrated probability composite registered before the main run rather than a soft heuristic, and the abstention rate is reported in the four-outcome decision distribution in Chapter 5. The architecture does not solve the broader problem of unequal access to verification, but it removes one specific channel through which fluency was previously laundered into false assurance.

3.3 Environmental Impact

3.3.1 Compute footprint of the experiment

The compute envelope of the empirical programme is dominated by four phases: the cycle-zero calibration phase that fits the per-benchmark calibrated probability

composite under the shrinkage prior on a fifteen-hundred-sample fold and registers the storage and deferral cut-points against the threshold sweep, the main run that iterates the per-query pipeline and the cycle-boundary update passes under the registered ten-cycle cap with realised closure at six cycles under the equilibrium-saturation gate, the external baseline panel that re-runs every contrast under matched protocol, and the architectural ablation that runs the retrieval-utility classifier on/off contrast at the cycle-three close. The end-to-end programme runs on a single rented graphics processor of the consumer-grade thirty-two-gigabyte class, with the cycle-zero phase complete in tens of hours, the main run sized to several days under the locked batched-calibration configuration, and the baseline and classifier-ablation panels each scaled in proportion to the workload they replicate. The carbon-equivalent footprint per phase is reported by combining the recorded wall-clock duration with the published wattage envelope of the rented hardware and a public grid-emissions conversion factor, following the methodology of Strubell et al. [27],

$$\mathrm{kgCO_2e}_{\phi} \;=\; T_{\phi} \cdot P_{\mathrm{TDP}} \cdot \varepsilon_{\mathrm{grid}}, \tag{3.3}$$

where T_{ϕ} is the wall-clock duration of phase ϕ in hours, P_{TDP} is the rated power of the graphics processor in kilowatts, and $\varepsilon_{\mathrm{grid}}$ is the local grid emissions factor in kilograms of carbon-dioxide equivalent per kilowatt-hour. The per-phase totals are tabulated in Appendix E together with the conversion factor used.

3.3.2 Cycle budget rationale

The ten-cycle registered cap is not an arbitrary stopping point. It is the point at which the convergence analysis in Chapter 4 predicts the trajectory has crossed into its near-equilibrium phase, and at which the equilibrium diagnostic that the implementation runs at every cycle boundary is expected to fire. Stopping earlier would produce a noisier reading on the late-cycle behaviour that the architecture's claims rest on; running longer would incur a measurable marginal carbon cost without a matching diagnostic justification. The cycle budget is therefore an environmental decision as much as it is an experimental one: each additional cycle adds its own fine-tuning energy and its own retroactive re-verification energy, and each must be earned by a diagnostic that says the trajectory has not yet converged. The honest disclosure here is that early-stopping inside the cap is permitted by the runbook when the equilibrium diagnostic fires, and the realised horizon under this discipline was six cycles closed on the equilibrium-saturation gate at the cycle-five boundary (with cycle four aborted under the retention guard and its weights rolled back); the realised trajectory is reported in detail in Chapter 5.

3.3.3 Retention versus retraining

The retention guard with weight rollback is, in addition to a quality control, a carbon-saving mechanism. A fine-tuning loop without a retention guard either continues to update under regressed weights, which compounds the energy cost of every subsequent cycle that has to recover the lost capability, or eventually requires a full restart from a clean checkpoint, which discards every cycle of fine-tuning energy that came after the regression. The architecture's rollback rule restores weights from the previous checkpoint when the retention ratio falls below the fixed floor, while

preserving the memory across the rollback so that the verified episodes accumulated during the failed cycle are not lost. The carbon implication is that a failed cycle costs only the fine-tuning energy of the failed cycle itself, not the energy of every successor cycle that would have been needed to recover, and the verified-episode accumulation that survives the rollback shortens the time to convergence on the next attempt. This argument is qualitative at the chapter-one level and is quantified in the deployment-cost projection in Section 3.8.

3.4 Ethical Issues

3.4.1 Alignment and refusal posture

The architecture treats explicit abstention as a first-class user-facing action rather than a fallback, and this design choice has an ethical justification independent of its accuracy effect. A model that hedges or speculates under uncertainty exposes the user to harm without informing them that uncertainty is present, while a model that abstains transfers the uncertainty to the user as a visible signal that they can act on by seeking a second opinion or escalating to a human expert; this connects to the recent literature on whether language models know what they know [37]. The architecture's storage and deferral cut-points are registered scalars on the calibrated probability scale fitted against a labelled calibration fold, anchored against the cycle-zero threshold sweep that selected the registered configuration on empirical-precision and stored-volume evidence rather than on a hand-tuned heuristic. The ethical commitment behind this posture is that, in the regime where a model cannot be confident, it should not occupy the user's attention as if it were.

3.4.2 Bias propagation through episodic memory

A learning system that commits its own outputs to memory and then trains on that memory can amplify the biases present in either the verifier or the underlying training distribution. Three mechanisms in the architecture bound this risk. The retroactive re-verification pass at every cycle boundary re-scores every stored episode under the updated adapter, so a bias that the verifier acquires across cycles does not persist in the stored pool unchallenged. The per-cycle refit of the calibrated probability composite, with per-benchmark child curves blended toward the pooled fit through the shrinkage prior, tracks score-distribution drift on the cycle's own labelled fold rather than freezing the cycle-zero composite shape across the trajectory, while the storage cut-point itself stays frozen so the operating point of the gate does not drift. The strict training-pool quality filter holds the training pool to a higher bar than the user-facing storage tier, so episodes whose storage score lies in the marginal band cannot enter the next cycle's fine-tune even if they pass the retrieval gate. None of these mechanisms eliminates bias inherited from the pre-training corpus, and the thesis discusses this limitation explicitly in Chapter 5; what they do is bound the rate at which new bias can be introduced through the self-improvement loop itself.

3.4.3 Data provenance and licensing

The empirical programme uses publicly licensed benchmarks and publicly licensed retrieval corpora throughout. The five evaluation benchmarks are FEVER [21], TriviaQA [18], CommonsenseQA [28], TruthfulQA [39], and StrategyQA [33], all of which are distributed under licences that permit research use and redistribution under attribution; the licence terms are catalogued in Appendix E together with the access date and the dataset version hash. Two further benchmarks, HotpotQA and Natural Questions, were evaluated at cycle zero and dropped before the main run (Section 1.6.1); their cycle-zero results are preserved in the project archive under the same licence terms. The retrieval substrate uses a Wikipedia-derived public passage corpus distributed under a Creative Commons licence. No private user data, no scraped social-media content, and no proprietary corpus enters either the calibration fold, the seed pool, the rehearsal buffer, or the cycle pool. The cold-start memory used to bootstrap the self-improvement loop is generated entirely from the licensed benchmark train splits under the same access terms as the evaluation folds, with disjointness from the evaluation fold enforced at the sample level. The thesis releases its locked configuration files, its per-cycle artefact catalogue, and the snapshot of the pre-self-improvement state on a public dataset host, so that the research community can audit the data path end to end without relying on privately held resources.

3.5 Standards and Codes of Practice

3.5.1 Software-engineering standards

The implementation follows the Python community's PEP 8 style convention for source layout, naming, and import order, with the deviations limited to the conventional eighty-eight character line length and the use of structural pattern matching where it materially improves readability. All third-party dependencies are pinned to exact versions in a single requirements file, which is committed alongside the source tree so that the dependency closure that produced the reported results is itself a versioned artefact rather than a moving target. Random-state surface is reduced to a single seed at process start, propagated explicitly through the generator, the sampler, the dropout layers, and the FAISS index, so that two consecutive runs under the same seed produce bit-identical artefacts modulo non-deterministic kernels. Version control discipline is enforced through a single linear branch-per-phase convention, with a tagged checkpoint at every cycle boundary and at every artefact-producing step, so that the empirical record is reconstructible from the commit graph alone.

3.5.2 Reproducibility standards

Reproducibility is treated as a first-class deliverable rather than a courtesy. The thesis releases four classes of artefact in support: a step-by-step reproducibility recipe in Appendix E that lists the exact command sequence required to regenerate every reported number from the recorded seed; a public-dataset-host snapshot of the entire pre-self-improvement state, including the cold-start memory, the cycle-zero calibrated probability composite, and the encoder weights at the operating point; a per-cycle artefact tree that pairs each cycle's run-complete summary with its

memory snapshot, its retention-probe readings, its empirical-precision audit, and its decision-distribution log; and the locked configuration file that records the storage and deferral cut-points, the per-signal shrinkage prior, the boost-layer regulariser, and the LoRA adapter parameters used in every downstream scoring and training site. Together these artefacts allow an independent reader to verify any specific empirical claim in Chapter 5 by re-executing the relevant stage rather than by trusting the reported number.

3.5.3 Evaluation-protocol standards

The empirical comparisons in this thesis follow four pre-specified protocol standards. The matched-protocol convention requires every external baseline to be evaluated under the same samples, the same prompt template family, and the same scoring rules as the architecture, so that the comparison panel reported in Chapter 5 is not confounded by re-implementation choices that vary across published baselines. The pre-specification convention requires that the calibration sweep grid, the storage and deferral threshold targets, and the architectural ablation expected-effect bands be recorded in the runbook before any sweep results are read; the locked configuration is then applied verbatim across every cycle of the main run, and any out-of-band outcome is reported as an honest disclosure rather than a tuning opportunity. Throughout the thesis, a quantity described as pre-specified or registered was fixed in the project's configuration and runbook before the main evaluation run; the term does not refer to an entry in an external pre-registration registry. The selective-prediction convention reports the precision-versus-coverage tradeoff in the form pioneered by [30], with the abstention class measured against a registered precision floor rather than a hand-tuned operating point. The multiple-comparison convention requires Holm-Bonferroni correction [4] across the baseline contrasts within each benchmark family, so that a significance verdict on a single benchmark cannot be inflated by the family-wise error rate; the bootstrap ninety-five-percent confidence interval on the exact-match delta is reported alongside the corrected p-value so that effect size is visible in addition to significance, with the paired McNemar test under continuity correction [1] as the per-benchmark significance test.

3.6 Project Management Plan

3.6.1 Phase breakdown

The project decomposes into five phases, sequenced so that each phase produces an artefact that the next phase consumes. The pre-thesis phase covers literature review, problem-statement registration, methodology design, and the construction of the per-stage pipeline that the empirical phases depend on. The calibration phase, governed by the cycle-zero discipline, fits the per-benchmark calibrated probability composite under the shrinkage prior on a disjoint calibration fold, registers the storage and deferral cut-points against the threshold sweep, and locks the architectural configuration that the main run will apply verbatim while permitting the composite to refit per cycle. The main run phase iterates the per-query pipeline across up to the registered ten-cycle cap under the locked configuration, with cycle-boundary update passes at every cycle boundary and an equilibrium-saturation gate that

terminates the trajectory early when the post-rollback subsequence stabilises, and produces the per-cycle artefact catalogue that the comparative phase consumes. The comparative phase runs the external baseline panel and the retrieval-utility-classifier on/off architectural ablation under the matched protocol, and runs the statistical analysis that licenses the architectural claims registered in Chapter 1. The write-up phase consolidates the artefact catalogue and the comparative results into the thesis report, the supervisor-facing presentation, and the journal manuscript that derives from this work.

3.6.2 Milestone schedule

Each phase is anchored by dated milestones recorded in Appendix E together with planned and actual completion dates. The pre-thesis phase milestone, the locked methodology design, was reached on schedule. The calibration phase milestone, the cycle-zero validation gate verdict, was reached after one full re-iteration of the calibration sweep: the first iteration failed the pre-specified correlation floor on the long-hypothesis judge calibration and the second iteration, re-run with the patched signal-dump path and the wider parameter grid, produced the locked configuration that the main run uses. The schedule slip incurred by this iteration is reported honestly here rather than absorbed into a smoothed timeline, because the iteration is itself evidence of method discipline; Chapter 5 reports the failed iteration's diagnostic numbers alongside the locked configuration's diagnostic numbers so that the reader can compare the two. The main run phase milestone, the realised six-cycle trajectory closed under the equilibrium-saturation gate, and the comparative phase milestone, the eight-baseline panel complete alongside the retrieval-utility classifier ablation, are reported under the realised dates that the empirical chapter records. The write-up phase milestone is the submission of this report and its companion artefacts on the registered submission date.

3.6.3 Deliverables

The project releases six deliverables on completion. The source-code repository is published under an open-source licence at https://github.com/aksaN000/caem-thesis and contains the full implementation of the architecture, the cycle-zero calibration scripts, the cycle-boundary update passes, the external baseline drivers, and the architectural ablation drivers. The locked artefact bundle is hosted on the Google Drive folder at https://drive.google.com/drive/folders/1xErhsRDYe_qO0ZgUBn0WXK81eHc5jZCa, with the realised six-cycle trajectory under a dated main-run subtree, and contains the cycle-zero calibrated probability composite, the registered storage and deferral cut-points, the cold-start memory snapshot, and the per-cycle artefact catalogue alongside the bulky Wikipedia passage FAISS index mirrored on the Hugging Face dataset host (Section A.1). The thesis report itself is the document that the reader is currently holding. The supervisor-facing slide deck distils the headline empirical claim, the methodology, and the threats to validity into a defence-ready presentation. The journal version of this work, sized for the conference and journal venue most appropriate to the architecture's contribution, is prepared from the consolidated comparative chapter and is submitted after the thesis defence. A live demonstration that exercises the per-query pipeline on a small representative workload is included

as a supplementary deliverable for the supervisor and the external examiner.

3.7 Risk Management

3.7.1 Technical risks

Four technical risks are registered against the empirical programme. Calibration drift across cycles is the risk that the calibrated probability composite, refitted at every cycle boundary on a fresh per-cycle fold, loses its empirical-precision anchor at the frozen storage cut-point under the distribution shift introduced by each cycle's adapter update. Verifier and judge disagreement is the risk that the post-generation verifier and the long-hypothesis judge produce inconsistent storage scores on the same answer, leaving the storage decision under-determined on long answers; the empirical realisation of this risk drove the long-hypothesis judge ablation reported in Chapter 5. Out-of-memory regression on long-hypothesis paths is the risk that a long answer triggers a verifier branch whose memory footprint exceeds the rented graphics processor's capacity, halting the cycle; this risk surfaced empirically during the calibration phase on the long-hypothesis judge fitting step and was contained by chunking the long-hypothesis batch. Equilibrium diagnostic mis-fire is the risk that the cycle-boundary diagnostic either stops the run before equilibrium is reached, producing a noisy late-cycle reading, or fails to stop a non-converging run, producing a runaway compute burn; the diagnostic's pre-specified tolerance band and its two-consecutive-cycles requirement bound this risk in both directions. Retrieval-utility classifier distribution shift is the risk that the classifier, fitted under nested cross-validation on a training distribution that intentionally includes benchmarks outside the architecture's evaluation panel to broaden retrieval-utility regime coverage, generalises imperfectly onto the evaluation panel and routes a fraction of retrieval-helpful queries to the zero-shot tier or a fraction of retrieval-hurt queries to the retrieval-augmented tier; the nested-cross-validation protocol, the registered threshold sweep, and the on/off ablation against the architecture without the classifier on the same trajectory bound the realised cost of this risk.

3.7.2 Budget and infrastructure risks

Three infrastructure risks are registered against the empirical programme. Rented graphics-processor credit exhaustion before the registered ten-cycle trajectory and the comparative panels complete is the most consequential, because the architecture depends on a single backbone fine-tuned across cycles and a partial run cannot be salvaged into a meaningful comparison against external baselines. Node reclamation by the cloud provider mid-run is the risk that the rented machine is preempted between cycle boundaries, costing the cycle in flight and any post-cycle update passes that had not yet committed; the runbook commits per-cycle artefacts at every cycle boundary precisely so that a preempted cycle costs the cycle in flight rather than the entire run. Storage exhaustion on the public dataset host is the risk that the per-cycle artefact catalogue plus the model snapshots plus the passage index exceed the storage quota of the host that publishes them; the artefact catalogue is sized in advance to fit the host's free-tier quota, with the larger memory snapshots stored under separate quotas if needed.

3.7.3 Mitigation register

Each registered risk is paired with a registered mitigation. Calibration drift is mitigated by the cycle-boundary composite refit that re-fits the per-signal isotonic curves and the per-benchmark boost coefficients on a fresh per-cycle fold (with the per-benchmark child curves blended toward the pooled fit through the registered shrinkage prior), and by the empirical-precision audit at the locked cut-points that runs once per cycle on the same recalibrated fold so that the calibration-fold precision π_{store} at the frozen $\tau_{\text{store}} = 0.60$ is itself a measurable quantity tracked across the trajectory. Verifier and judge disagreement is mitigated by the long-hypothesis judge ablation, which removes the disagreement source from the deployed configuration when the judge fails its pre-specified correlation floor, and the disagreement reading itself is reported in Chapter 5 as a future-work caveat. Out-of-memory regression on long-hypothesis paths is mitigated by chunking the long-hypothesis batch and by a hard out-of-memory abort that triggers a clean restart from the previous checkpoint rather than a corrupted continuation. Equilibrium diagnostic mis-fire is mitigated by the diagnostic's pre-specified tolerance band, its two-consecutive-cycles requirement, and the upper-cap of ten cycles that bounds the worst case in both directions. Credit exhaustion and node reclamation are mitigated by the archive-before-overwrite discipline, the pre-self-improvement snapshot on the public dataset host that allows the run to resume from the snapshot rather than from cycle zero, and the per-cycle artefact commit that bounds the loss from a preempted cycle to the cycle in flight. Storage exhaustion is mitigated by sizing the artefact catalogue against the host's free-tier quota in advance and by tiered storage of the larger snapshots under separate quotas where the catalogue alone would exceed the budget. Table 3.3 consolidates the risk register and its registered mitigations, with the realised column recording which risks materialised during the empirical programme and the form their realisation took.

3.8 Economic Analysis

3.8.1 Compute budget

The compute budget for the empirical programme is recorded in graphics-processor hours at the rented rate of the consumer-grade thirty-two-gigabyte class machine used for the entire run. The cycle-zero calibration phase consumed tens of hours, dominated by the verifier-signal sweep over the calibration fold; the realised main run closed at six cycles under the equilibrium-saturation gate inside the registered ten-cycle cap and consumed several days under the locked batched-calibration configuration, dominated by the per-cycle fine-tune and the cycle-boundary retroactive re-verification pass; the external baseline panel scales in proportion to the number of samples evaluated and the number of decode passes per sample, with the chain-of-thought variants and the retrieval-augmented variants contributing the bulk of the cost; the realised architectural ablation consists of a single head-to-head contrast at the cycle-three close that runs the retrieval-utility classifier on and off against the same matched-protocol panel. The full budget is recorded against the rented-credit envelope reported in Appendix E, and a downgraded-device envelope is reported alongside it that estimates the same programme on a sixteen-gigabyte device with parameter-efficient fine-tuning,

Risk	Mechanism	Registered mitigation	Realised
Calibration drift across cycles	Composite loses empirical-precision anchor at frozen τ_{store} under adapter-induced shift	Cycle-boundary composite refit (per-signal isotonic + per-bench shrinkage + boost); empirical-precision audit at locked cut-points per cycle	Pending main run
Verifier and judge disagreement	Post-generation verifier and long-hypothesis judge produce inconsistent scores	Long-hypothesis judge ablation when correlation floor fails; future-work caveat	**Yes** (judge ablated; Pearson $0.5843 < 0.70$)
Out-of-memory regression on long paths	Long answer triggers verifier branch exceeding GPU capacity	Chunking on long-hypothesis batch; hard-abort with clean-restart	**Yes** (long-hypothesis batch chunking implemented; degenerate sample skip on OOM)
Equilibrium diagnostic mis-fire	Premature stop or runaway compute burn	Pre-specified tolerance band, two-consecutive-cycles requirement, ten-cycle upper cap	Pending main run
GPU credit exhaustion	Run does not complete inside the rented credit envelope	Phase budgeting against the recorded credit; archive-before-overwrite discipline	Pending main run
Node reclamation by cloud provider	Cycle in flight lost on preemption	Per-cycle artefact commit; pre-self-improvement snapshot on the public dataset host	Pending main run
Storage exhaustion on dataset host	Artefact catalogue + snapshots exceed the host's free-tier quota	Free-tier-aware sizing; tiered storage under separate quotas for larger snapshots	Pending main run
Calibration iteration failure at validation gate	Cycle-zero composite or conformal fit fails the pre-specified gate	Re-iterate with patched signal-dump path and wider parameter grid	**Yes** (first calibration iteration failed on the signal-dump bug; second iteration locked)

Table 3.3: Risk register with registered mitigations and the realised status as of the cycle-zero validation gate. Pending entries are resolved against the empirical record in Chapter 5.

so that the practical cost on a downgraded device can be derived from the published numbers without re-running the experiment.

3.8.2 Cost-benefit on the storage gate

The locked operating point admits approximately one in five incoming queries to the storage class on the cycle-zero validation fold (the realised cycle-zero reading is 289 stored entries over the 1500-sample training-panel validation fold, 500 samples per benchmark, a storage rate near 19%), and the realised per-cycle stream-chunk storage rate sits at about 52% to 53% of the 4700-sample stream chunk over cycles one to three of the trajectory; this is a deliberately conservative regime relative to the unrestricted-admission baseline that trades volume for precision. The economic structure of this trade is direct: writing c_{ver} for the verifier cost paid on every incoming query and c_{gen} for the average decode cost of a zero-shot or retrieval-augmented generation, the marginal cost of admitting one additional stored episode is bounded above by zero new work because every signal it consumes was already computed, while the marginal benefit per stored episode is bounded below by the cumulative cache-hit savings it produces over its lifetime,

$$\Delta C_{\text{store}} \;\leq\; 0 \qquad \text{and} \qquad \Delta B_{\text{store}} \;\geq\; n_{\text{hit}} \cdot c_{\text{gen}}, \tag{3.4}$$

where n_{hit} is the number of future queries whose dispatch score selects this episode in the memory-direct tier and saves the full decode cost. The conservative storage rate is therefore not a missed-opportunity cost but a quality-controlled investment: episodes that pass the gate are precise enough to be useful at retrieval time, and the precision floor is what makes the cache-hit savings $n_{\text{hit}} \cdot c_{\text{gen}}$ real rather than nominal.

3.8.3 Deployment cost projection

The headline economic claim of the architecture is a deployed cost-per-query that decays with the memory share of incoming queries. At cold start the memory is small and most queries route to the zero-shot or retrieval-augmented tier, producing a per-query cost dominated by the decode and the passage retrieval. As the memory accumulates verified episodes across cycles, the share of queries served by the memory-direct tier rises, and each memory-direct query incurs only the cost of an embedding lookup and a similarity search rather than a decode. The total cost-per-query at cycle t is therefore a weighted average over the three tiers,

$$\bar{c}(t) \;=\; \pi_1(t)\, c_1 \;+\; \pi_2(t)\, c_2 \;+\; \pi_3(t)\, c_3, \qquad \pi_1(t) + \pi_2(t) + \pi_3(t) = 1, \tag{3.5}$$

where $\pi_i(t)$ is the share of incoming queries dispatched to tier i at cycle t and c_i is the average per-query cost on that tier. The architecture predicts $\pi_1(t)$ rises and $\pi_3(t)$ falls as cycles progress; the empirical realisation of $\bar{c}(t)$ is reported in Chapter 5 so that the projection can be read off the trajectory rather than asserted in advance. Where deployment workloads concentrate on a recurring query distribution, the memory-direct share grows fastest and the deployed cost converges toward the retrieval-only floor; this is the regime in which the architecture is most economically attractive, and it is the regime that the deployment-study future-work item in Chapter 6 would test under live user traffic.

3.9 Chapter signpost

The functional and non-functional requirements, the production-parity design commitments, the societal and environmental readings, the ethical framing, the standards and reproducibility envelope, and the risk and economic envelopes registered in this chapter together fix the operating constraints under which the architecture must be specified. Chapter 4 discharges this specification end-to-end: the eight-stage per-query pipeline with the Stage 3b retrieval-utility classifier, the nine-signal verifier ensemble and its calibrated probability composite, the fixed-threshold storage gate and the four-outcome decision tree, the cycle-boundary update passes, the constrained self-improvement loop under the multi-modal retention guard, and the pool-purity theorem, the purification condition, and the trajectory heuristics with their proofs or argument sketches and pre-specified empirical receipts.

Chapter 4

Methodology and Design

4.1 Design Process or Methodology Overview

4.1.1 The eight-stage pipeline narrative

The architecture organises a single query into an eight-stage pipeline that is traversed once per query and wrapped in a per-cycle update loop. Stage 1, pre-routing, computes a calibrated confidence on the query from a single short forward pass that fuses a short-greedy-decode token-probability aggregate with an encoder-layer convergence ratio; the design and rationale appear in Section 4.2.2. Stage 2, encode and lookup, encodes the query into a sentence-embedding vector and probes the episodic memory for the nearest stored entry, returning a similarity score and the neighbour's stored-quality score; the substrate is detailed in Section 4.2.1 and Section 4.2.3. Stage 3, route, combines the similarity and stored-quality scores into a single dispatch score and selects one of three tiers, with an independent safety override that forces dispatch off the score-formula path whenever the pre-routing confidence falls below a fixed floor; the router is specified in Section 4.2.4. Stage 3b, retrieval-utility classifier consultation, runs only when the dispatch lands on the retrieval-augmented tier and refines that default into either the zero-shot tier or the retrieval-augmented tier under a learned binary classifier on the query; the classifier is specified in Section 4.2.5. Stage 4 executes the tier chosen by Stage 3 or refined by Stage 3b: the memory-direct tier serves the neighbour's stored answer; the zero-shot tier runs a generation on the fine-tuned generator; the retrieval-augmented tier retrieves passages from the public corpus and runs a grounded generation. Stage 5, verify, passes every generated answer through the nine-signal post-generation verifier whose four signal families feed the calibrated probability composite; the verifier is specified in Section 4.2.6 and the composite in Section 4.2.7. Stage 6, decide, applies the four-outcome decision tree to the composite output, with a confabulation early-exit rule that takes precedence over the composite thresholds; the decision tree is specified in Section 4.2.9 and the fixed-threshold storage gate in Section 4.2.8. Stage 7, output, executes the implied user-facing action, returning the stored or generated answer or the calibrated refusal. Stage 8 runs at cycle boundaries and consists of the retroactive re-verification pass, the deferred-entry reconsideration pass, the memory consolidation pass, and the regularised fine-tune under the retention guard; the cycle-boundary passes are specified in Section 4.2.10.

4.1.2 The cycle loop walk-through

A single cycle traces the same trajectory in every iteration. The cycle ingests a fixed workload of fresh queries from the registered training pool. Each query is served by Stages 1 through 7 of the per-query pipeline, accumulating verified episodes in the episodic memory, deferred entries in the bounded deferral buffer, abstentions in the user-facing log, and discards in the discard log. After the workload is exhausted, the cycle-boundary block runs: the retroactive re-verification pass rescores every stored episode under the current generator and applies the upward-only update rule that promotes entries whose composite has crossed the storage threshold and prunes those that have fallen below it; the deferred-entry reconsideration pass examines every entry in the deferral buffer and either promotes, expires, or requeues the entry; the memory consolidation pass collapses near-duplicate stored episodes into a highest-quality representative within each near-duplicate cluster. The fine-tune then draws a training pool from stored episodes whose composite score exceeds the strict training threshold above the storage threshold, applies the parameter-efficient update through the bounded low-rank-adapter envelope on the frozen backbone, and probes the multi-modal retention probe panel to compute the worst per-probe retention ratio. When the retention ratio falls below the registered floor, the cycle aborts and rolls back the weights; the memory is preserved across the rollback so that the verified episodes accumulated during the failed cycle survive, and only the next per-query phase resumes. The cycle then advances, with the locked architectural configuration applied verbatim and only the calibrated probability composite refit on a fresh per-cycle fold. The rationale for this cycle structure is that it decouples the user-facing per-query path from the slow expensive cycle-boundary updates, while the upward-only retroactive rule keeps the stored pool's quality monotonic across cycles even when the verifier itself is improving.

4.1.3 A worked example

The trajectory of two real entries from the cycle-zero artefact bundle anchors the reader before the formal specification in Section 4.2. The first is a stored FEVER episode and the second is a discarded TriviaQA question; both are recorded with their full per-signal readings in the released cold-start memory catalogue and the released cycle-zero rescored-evaluation catalogue (the literal on-disk paths are listed in the reproducibility appendix).

Consider the FEVER claim *The starting point of the French Revolution was before 1792*, paired with the instruction to label the claim as supports, refutes, or not enough information. Stage 1 computes the pre-routing confidence from the encoder's short-prefix token-probability aggregate and its encoder-layer convergence ratio; the fused and temperature-scaled scalar comfortably clears the safety floor, so no override is triggered. Stage 2 encodes the claim into a sentence-embedding vector and probes the episodic memory. On the cycle-zero pass with an empty memory there is no neighbour to retrieve, so Stage 3 defaults the query to the retrieval-augmented tier; Stage 3b consults the retrieval-utility classifier, which refines the default to the zero-shot tier because the claim is short, self-contained, and rated by the classifier as not benefiting from passage retrieval; Stage 4 generates the label *supports*; Stage 5 evaluates the nine signals across the four families. The top-passage entailment of

the answer against the retrieved evidence is high (above 0.94); the pooled-mean and atomic entailment are lower because only a fraction of the supporting evidence is in the immediate context window; the semantic consistency across resampled chains is high (above 0.85); the question-answer relevance score is moderate (around 0.69). The calibrated probability composite aggregates the per-signal calibrated probabilities through the regularised logistic boost layer and produces a storage score of approximately 0.51. Stage 6 then applies the four-outcome decision tree: the score clears the deferral threshold but falls just below the storage threshold at the locked operating point, so the entry is committed to the deferred buffer for cycle-boundary reconsideration. After the cycle-boundary retroactive pass under the updated generator the same entry crosses the storage threshold and is promoted into the cold-start memory under entry identifier zero. Stage 7 returns the original answer *supports* to the user, which is the correct label.

Consider next the TriviaQA question *The song "If I Ruled The World" comes from which musical?*, which appears in the cycle-zero evaluation fold. Stage 1 produces a pre-routing confidence that does not clear the safety floor, because the encoder's convergence ratio on the question is low. The safety override fires on the score-formula path; Stage 3b then consults the retrieval-utility classifier on the same query and confirms the retrieval-augmented dispatch because the base-model self-knowledge probe reads low and the dense passage substrate carries supporting evidence. Stage 4 retrieves passages from the public corpus and runs a grounded generation; the generator returns an empty answer, signalling an explicit refusal. Stage 5 evaluates the verifier signals on the empty answer: the token-level uncertainty signal is very low (around 0.02), the top-passage entailment is low (around 0.11), the atomic entailment is low (around 0.09), and the question-answer relevance score sits at the chance level (around 0.50). The calibrated probability composite produces a storage score of approximately 0.07, well below both the deferral and storage cut-points at the locked operating point. Stage 6 then routes the entry: the storage cut-point and the deferral cut-point are both unmet, and the top-passage entailment is below the registered abstention floor, so the decision tree returns the abstention outcome rather than a silent discard. Stage 7 maps the abstention outcome to the user-facing refusal string registered for the deployment and the raw model prediction is suppressed from the user surface. The two trajectories together exercise the architecture's defining behaviour: the verifier-mediated path that promotes a tentatively answered claim to memory after retroactive review, and the verifier-mediated path that turns a low-confidence question into a calibrated refusal rather than a fabricated answer.

4.2 Preliminary Design or Design (Model) Specification

4.2.1 Generator and retrieval substrate

The architecture is built on four substrate components whose architectural identity is fixed across every cycle of the self-improvement loop. The generator is a three-billion-parameter open-weight instruction-tuned decoder-only model from the Qwen 2.5 family [80], used for all generation in the zero-shot tier and the retrieval-augmented tier and used as the source of internal-calibration signals in the verifier. The choice of

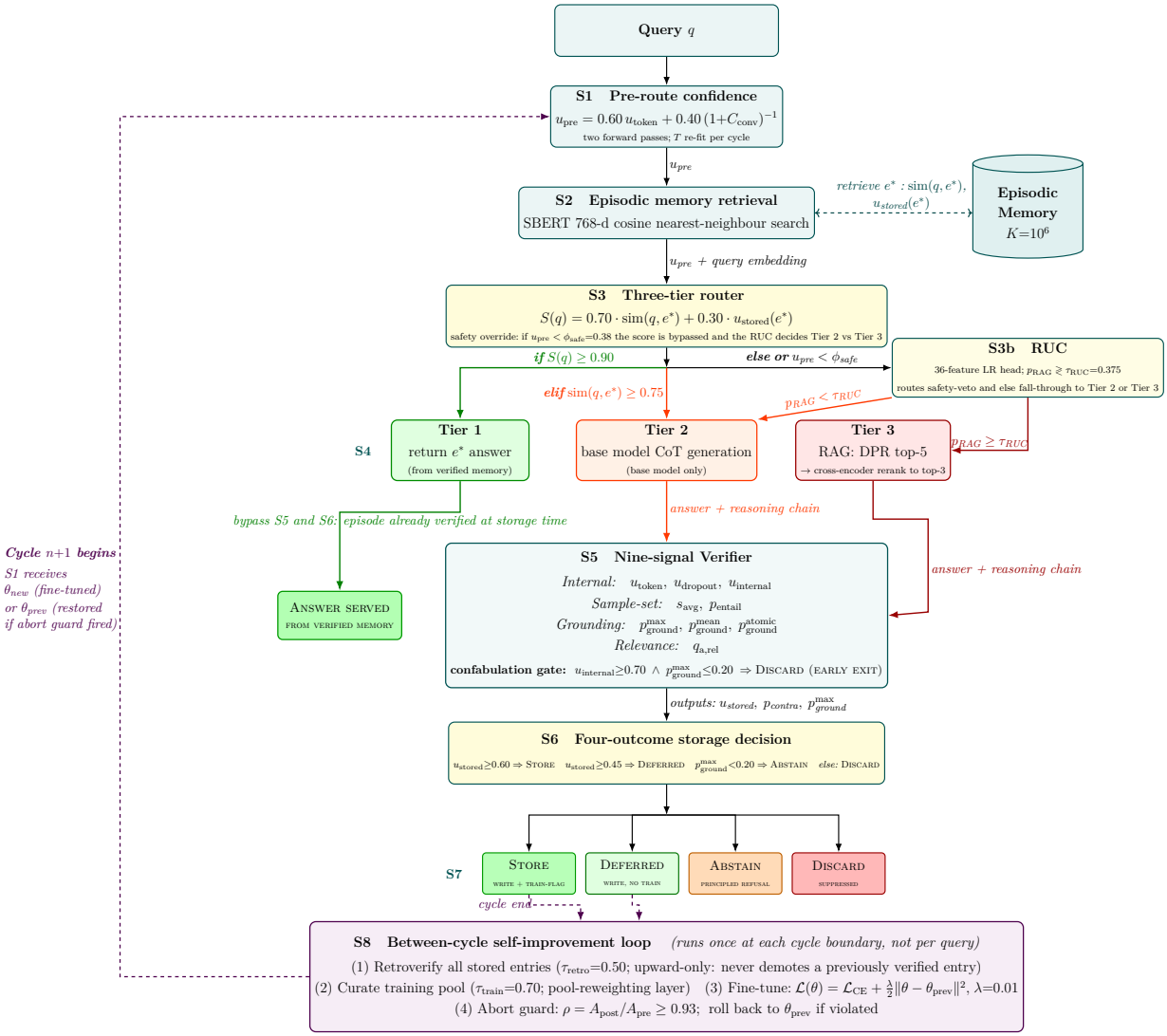


Figure 4.1: The eight-stage CAEM pipeline. Stages S1–S8 are described in Chapter 4; the retrieval-utility classifier (Stage S3b, RUC) of Section 4.2.5 sits between the three-tier router and the retrieval-augmented tier, intercepting the safety-veto and score-formula fall-through dispatches and redirecting retrieval-hurt queries to the zero-shot tier. The nine deployed signals at S5 are the registered set minus h_{norm} (registered then retired at the cycle-zero audit; see the list-of-symbols footnote) and p_{contra} (structurally zero under the deployed MiniCheck judge). The outer dashed arrow indicates the cycle-boundary handoff from the close of one cycle to the start of the next.

a small generator rather than a frontier-scale model is deliberate. A small generator admits parameter-efficient fine-tuning on a single consumer-grade graphics processor through low-rank adaptation on the attention and feed-forward projections, with the trainable footprint at roughly two percent of the full backbone parameter count, which is the necessary condition for the cycle-boundary fine-tune to run inside the project's compute envelope. Equally, a small generator's baseline accuracy on the five-benchmark panel registered for the main run is high enough that the verifier's expected false-positive band is non-degenerate, so the storage gate has a usable signal-to-noise ratio without depending on a frontier-scale capacity that the project cannot afford. The adapter parameters are the single locus of cycle-to-cycle drift; the underlying backbone is frozen and everything else in the substrate is held constant.

The dense passage index is a single sentence-embedding-encoded snapshot of a public Wikipedia-derived corpus, structured as a dense-retrieval index in the spirit of [31]. The encoder is a sentence-transformer model of moderate dimension [26] that is also used to encode the user's query at lookup time, so query and passage embeddings live in the same space. The index is a tiered structure: a flat in-memory index for capacity below the registered switching threshold, promoted to an inverted-file product-quantised index above that threshold so that lookup latency stays within the per-query envelope at the deployment-scale memory size projected for production. The retrieval substrate is shared by the retrieval-augmented tier and by the external grounding family of verifier signals, so a single passage retrieval per query feeds both the grounded generation and the entailment scoring.

The natural-language-inference judge that the verifier consumes for chain-to-answer entailment and for the external grounding signals is realised by the MiniCheck checkpoint [81], a unary supported-against-context scorer trained on language-model-generated claim-support data and therefore distributionally matched to the kind of generated answers the verifier is asked to score. The judge is a unary model that returns a single supported probability rather than a three-way label distribution, which is why the contradiction signal of the verifier ensemble has been retired at the architectural level: the contradiction class never receives non-zero mass under this judge.

The cross-encoder reranker is a separately trained encoder-only model from the bge-reranker family that scores text pairs by relevance. It is used in two places in the architecture, both as a single shared instance to amortise its load cost: as the optional reranker on the retrieval-augmented tier and on the external grounding family, where the verifier reranks the top-twenty retrieved passages down to the top-three before scoring entailment, and as the question-answer relevance signal in the verifier. The reranker and the natural-language-inference judge are distinct models with distinct training distributions; combining them as separate substrate components is what allows the verifier to keep the question-relevance signal architecturally orthogonal to the grounding family, so that an off-topic answer that is incidentally entailed by a retrieved passage is still caught by the relevance signal. The capacity and latency profile of the substrate as a whole is reported empirically in Chapter 5, and the rationale for the substrate choices over alternatives is registered against the constraints in Section 3.1.3: the single-backbone commitment, the local-execution commitment, and the bounded-cycle commitment together fix the substrate to the components specified here.

4.2.2 Pre-routing confidence

The pre-routing confidence is a single scalar computed from a single short forward pass on the query, used at two points in the per-query pipeline: as a one-dimensional input to the dispatch score that selects the tier, and as the sole input to the safety override that forces the retrieval-augmented tier when the confidence falls below a fixed floor. The signal is a fusion of two cheap quantities, both produced by the same forward pass over the query and a short greedy decode that follows it. The first is a short-greedy-decode token-probability aggregate, computed by running a thirty-two-token greedy decode under the chat template on the query and recording the geometric mean of the per-token log-probabilities of the chosen tokens; this captures how confidently the generator commits to its next-token choices under the question's framing. The second is an encoder-layer convergence ratio adapted from [90], computed from the hidden-state stack of the same forward pass as the variance ratio between the early-layer half and the late-layer half of the decoder; this captures how stably the representation has settled across the layers that drive the downstream confidence, with high variance in the early layers relative to the late layers reading as the model not having converged on a single interpretation of the question. The fusion uses fixed mixing weights of 0.60 on the token-probability aggregate and 0.40 on the inverse-one-plus convergence ratio, registered before the calibration phase rather than learned, so the fused scalar's interpretation does not drift with the fine-tune.

The fused scalar is calibrated to a probability through temperature scaling [17] fitted on a disjoint calibration fold by minimising binary negative-log-likelihood under the labels-correct-under-zero-shot binary signal. Temperature scaling is preferred over Platt scaling, isotonic regression, or histogram binning at this site for three reasons. Temperature scaling has a single scalar parameter, which prevents the calibration fold from over-fitting at this site and conserves the fold's degrees of freedom for the heavier post-generation calibration work. Temperature scaling preserves the relative ordering of inputs, which matters for the dispatch score that uses the scalar as a continuous input rather than a hard label. Temperature scaling has a closed-form interpretation as a uniform sharpening or softening of the score distribution, which makes the calibrated output diagnosable at each cycle boundary by comparing the temperature scalar across cycles.

The non-trivial design choice is that the pre-routing confidence does not require the multi-chain resampling that the post-generation verifier uses. The cost of the signal is therefore paid per query regardless of which tier eventually serves the answer (a single forward pass over the query plus a thirty-two-token greedy decode), in exchange for the dispatch score and the safety override having a usable confidence signal at every query. The cost of the more expensive sample-based signals, including the dropout-disagreement signal and the sample-set agreement family, is deferred to the post-generation verifier on the zero-shot and retrieval-augmented paths only, so memory-direct queries incur no decode-phase signal cost beyond the short pre-routing decode. The contribution to the hypotheses registered in Chapter 1 is direct: the safety override that this signal alone drives is what closes the high-uncertainty failure mode where neither memory nor zero-shot generation has earned the right to answer, and the dispatch score that uses this signal jointly with the memory similarity score is what permits the memory-direct tier to be cheap without sacrificing safety. Algorithm 1 states the fusion procedure as it is implemented in the per-query

pipeline.

Algorithm 1 Pre-routing confidence fusion (Stage 1).

Require: query q, generator G, max generation length $L = 32$, per-benchmark temperature scalar T_b, mixing weights $w_{\text{tok}} = 0.60$ and $w_{\text{conv}} = 0.40$

$\left(\{t_i\}_{i=1}^{n}, \{\log p_i\}_{i=1}^{n}\right) \leftarrow G.\text{generate}(q;\ L,\ \text{greedy},\ \text{scores})$ ▷ short greedy decode of up to L tokens; greedy disables sampling and *scores* returns token log-probabilities

$u_{\text{tok}} \leftarrow \exp\left(\frac{1}{n}\sum_{i=1}^{n}\log p_i\right)$ ▷ geometric mean of per-token log-probabilities of the chosen tokens

$\mathbf{H} \leftarrow G.\text{forward}(q,\ \texttt{output_hidden_states}=\text{True})$ ▷ full hidden-state stack from a single forward pass

$c_{\text{conv}} \leftarrow \text{Var}\left(\mathbf{H}^{(1:L/2)}\right) \,/\, \text{Var}\left(\mathbf{H}^{(L/2+1:L)}\right)$ ▷ variance ratio of early to late decoder layers, the Nandakishor convergence ratio

$z \leftarrow w_{\text{tok}} \cdot u_{\text{tok}} + w_{\text{conv}} \cdot \frac{1}{1+c_{\text{conv}}}$ ▷ fixed-weight fusion

$u_{\text{pre}} \leftarrow \sigma\left(\text{logit}(z)/T_b\right)$ ▷ per-benchmark temperature scaling on the logit

return u_{pre}

4.2.3 Episodic memory store

The episodic memory is the architecture's single source of long-horizon learning between cycle-boundary fine-tunes. The design is in the lineage of memory-augmented neural networks for sequence and question-answering tasks [10], [13], [15], of writeable episodic key-value stores for language models [69], and of insertion-time memory augmentation [94], but is specialised here as an external, query-keyed episodic store whose distinguishing property is the scalar storage score persisted with each entry. Each entry is an indivisible record that pairs the question, the served answer, the supporting reasoning chain, the sentence-embedding vector of the question, the verifier's per-signal readings at storage time, the calibrated storage score, and the entry's storage cycle. Storing the question and the answer together with the supporting context and the per-signal readings allows the cycle-boundary retroactive re-verification pass to recompute the storage score under the updated generator without re-running the original generation, which is the architectural property that makes upward-only updates affordable at scale.

Similarity search is performed by cosine similarity in the sentence-embedding space, against a tiered index that promotes from a flat in-memory structure to an inverted-file product-quantised structure above a registered switching threshold on the entry count. Cosine similarity is preferred over Euclidean distance because the sentence-transformer encoder is trained under a contrastive objective that places semantically equivalent questions on the same direction in embedding space, so angular separation tracks semantic distance more reliably than L_2 separation. The tiered index choice is justified by the deployment-scale projection: at the workloads reported in this thesis the memory grows to the low tens of thousands of entries and the flat index suffices, but the deployed architecture has a path to the millions-of-entries regime under the inverted-file quantised index without a code rewrite.

A novelty filter at write time bounds the rate at which the memory absorbs near-duplicate entries. When a candidate write's nearest neighbour in the existing memory

exceeds the novelty similarity threshold (set to a value above which two entries are considered paraphrases of the same query), the candidate is suppressed; this prevents the memory from bloating with paraphrase-level duplicates that consume capacity without adding coverage. The threshold is registered before the calibration phase and held constant across cycles. A complementary consolidation pass runs at every cycle boundary at a more conservative similarity threshold, which collapses near-duplicate clusters by retaining the highest-quality representative within each cluster and merging the source-benchmark provenance into the surviving entry.

A capacity policy bounds the memory at a registered upper limit on the number of stored entries. When the limit is reached, a value-score pruning rule selects entries for removal: the value score combines retrieval frequency, success rate at retrieval time, recency, and storage-time confidence into a single rank, and the lowest-ranked entries are pruned until the memory is back below the limit. The pruning rule is invoked only when the limit is exceeded, so the small memories reported in this thesis never trigger it; the policy nonetheless governs the architecture's deployment-scale behaviour and is verified empirically through the consolidation diagnostics in Chapter 5.

The contribution of the memory design to the registered hypotheses is twofold. First, by storing the per-signal readings together with the answer, the memory supports the upward-only retroactive update rule that gives the pool-purity lower bound of Theorem 4.1 its monotonic structure across cycles. Second, by enforcing the novelty and consolidation policies, the memory bounds the rate at which storage capacity accumulates without coverage, which is what makes the deployment-cost projection in Section 3.8 non-trivial: the per-query cost saving depends on memory entries being distinct from one another at retrieval time.

4.2.4 Three-tier router with the safety override

The router is the dispatch logic that sits between the encode-and-lookup stage and the answer-generating tiers. It takes three inputs from the upstream stages: the cosine similarity of the query to the nearest stored entry, the stored-quality score of that neighbour, and the calibrated pre-routing confidence; it produces one output, the tier index, together with a flag that records whether the safety override fired.

The combined dispatch score blends the similarity score and the neighbour's stored-quality score under a fixed weight registered before the calibration phase. The combined score uses a similarity weight of $\lambda = 0.70$ on the cosine similarity and the complementary weight $1 - \lambda = 0.30$ on the neighbour's stored-quality score. The choice to give similarity the larger weight reflects the empirical fact that retrieval errors at the lookup step degrade Tier 1 quality more than uncertainty in the stored-quality score does, because a wrong neighbour is a categorically different query while an under-confident correct neighbour is still a correct answer; the rationale for the specific weight is registered in the calibration runbook and the sensitivity analysis is reported in Chapter 5.

The dispatch rules execute in a fixed order. The safety override is checked first: when the pre-routing confidence falls below the registered floor of 0.38, the router dispatches to the retrieval-augmented tier and records the override flag, regardless of the combined score. The floor was lowered from the originally-registered value of 0.60

during the cycle-1 routing audit, which observed that the higher floor was forcing well-supported queries onto the retrieval-augmented tier under the post-temperature-cap pre-routing distribution and depressing the architecture's effective tier utilisation; the lowered floor preserves the architectural intent of the safety override while admitting a wider band of queries to the cheaper tiers. The ordering matters because the override expresses an architectural constraint that the system must not generate without grounded retrieval whenever its pre-generation readiness is low. If the override does not fire, the combined score is compared against the Tier 1 threshold of 0.90; queries above the threshold are dispatched to the memory-direct tier, which serves the neighbour's stored answer without invoking the generator. Queries that do not clear the Tier 1 threshold are checked against the Tier 2 similarity threshold of 0.75 on the similarity score alone; queries above this threshold are dispatched to the zero-shot tier, which runs a generation on the fine-tuned generator without grounded retrieval. Queries that do not clear either threshold fall through to the retrieval-augmented tier, which retrieves passages from the public corpus and runs a grounded generation. The retrieval-augmented tier is the architectural floor for the routing rule of this subsection: it is the path that the system falls back to whenever the cheaper paths cannot earn the right to answer, subject to the per-query retrieval-utility refinement of Section 4.2.5 that may redirect the safety-veto fall-through and the score-formula fall-through to the zero-shot tier whenever retrieval is judged unlikely to help.

The three-mechanism separation between the combined-score dispatch, the safety override, and the per-query retrieval-utility decision of Section 4.2.5 is what distinguishes the router from a single-threshold dispatcher and from prior three-tier routers such as Adaptive-RAG [74], whose three tiers are gated by a query-complexity classifier rather than by per-entry stored quality on a shared memory. A single-threshold dispatcher would conflate the question of whether the memory has a useful neighbour with the question of whether the model is ready to generate, and would either over-retrieve under high memory similarity with low pre-routing confidence or under-retrieve in the opposite regime. The three-mechanism design keeps the three questions separate: the combined score captures the memory's coverage of the incoming query, the safety floor captures the model's readiness to generate, the retrieval-utility decision captures whether retrieval is expected to help on the particular query, and a query is dispatched to the cheap memory-direct tier only when the first two mechanisms agree that the cheap path is safe while the third mechanism governs the choice between zero-shot and retrieval-augmented generation on the residual queries.

The expected per-tier hit rate envelope predicted by the architecture is read off the dispatch rules together with the self-improvement loop's effect on the generator's zero-shot capability. Early in a cycle, when the memory is small and the adapter has not yet absorbed cycle-zero verified reasoning, the Tier 1 share is bounded above by the fraction of queries with a stored neighbour whose combined score clears 0.90; the Tier 2 share is bounded above by the fraction of queries with a similar neighbour clearing 0.75 but no Tier 1-quality match, plus the share of residual queries that the retrieval-utility classifier of Section 4.2.5 redirects from the retrieval-augmented fallback to the zero-shot tier; the Tier 3 share absorbs the remaining residual on which the classifier judges retrieval helpful. As cycles progress, two coupled redistributions reduce the Tier 3 share. The Tier 1 share grows because the verified episodic store

accumulates cached answers for recurring queries that earlier cycles served through Tier 3 retrieval-augmented generation, and the retroactive re-verification pass keeps the cached composites fresh under the updated generator. The Tier 2 share grows because the self-improvement loop folds verified retrieval-augmented reasoning into the generator's parametric capacity at every cycle boundary, so queries that would have required Tier 3 retrieval at cycle zero are answered zero-shot from the fine-tuned generator at later cycles without paying the retrieval cost. The empirical realisation of this envelope is reported in Chapter 5 as the per-tier hit rate trajectory. Table 4.1 summarises the dispatch rules in the order of evaluation, and Algorithm 2 states the routing procedure as implemented.

Order	Tier	Condition	Output path
1	Tier 3	$u_{\text{pre}} < 0.38$ (safety override)	Retrieval-augmented generation
2	Tier 1	$\mathcal{S} = 0.7\,s + 0.3\,u_{\text{stored}} \geq 0.90$	Serve neighbour's stored answer
3	Tier 2	$s \geq 0.75$	Zero-shot generation
4	Tier 3	otherwise	Retrieval-augmented generation (fallback)

Table 4.1: Three-tier dispatch rules in order of evaluation. The safety override is checked first; the combined-score Tier 1 rule is checked second; the similarity-only Tier 2 rule is checked third; the retrieval-augmented tier absorbs the remainder, subject to the per-query retrieval-utility refinement of Table 4.2.

Algorithm 2 Three-tier router with safety override (Stage 3).

Require: similarity s, neighbour stored-quality u_{stored}, pre-routing confidence u_{pre}
Require: constants $\lambda = 0.70$, $\theta_1 = 0.90$, $\theta_{\text{sim}} = 0.75$, $\phi_{\text{safe}} = 0.38$
$\mathcal{S} \leftarrow \lambda\,s + (1-\lambda)\,u_{\text{stored}}$ ▷ combined dispatch score
if $u_{\text{pre}} < \phi_{\text{safe}}$ **then**
    return (Tier 3, `safety_override` = True) ▷ refined downstream by Algorithm 3
else if $\mathcal{S} \geq \theta_1$ **then**
    return (Tier 1, `safety_override` = False)
else if $s \geq \theta_{\text{sim}}$ **then**
    return (Tier 2, `safety_override` = False)
else
    return (Tier 3, `safety_override` = False) ▷ refined downstream by Algorithm 3
end if

4.2.5 Retrieval-utility classifier

The router of Section 4.2.4 dispatches every query that fails the Tier 1 combined-score gate and the Tier 2 similarity gate to the retrieval-augmented tier as the architectural fallback. The dispatch is the correct default whenever the dense passage substrate carries grounding evidence that helps the generator answer the query; it is the

wrong default whenever the retrieved evidence amplifies the question's misconception framing or distracts the generator from a stronger parametric answer. The empirical anchor for the wrong-default mode is the regression on misconception-bait benchmarks observed in the pre-rerun trajectory of the architecture without a routing refinement: TruthfulQA exact match under the Haiku-judged correctness rule declined cycle over cycle as the self-improvement loop folded retrieval-amplified answers into the verified pool. The same architectural diagnosis was reached independently by the selective-retrieval line of work reviewed in Section 2.2.5: Mallen et al. [54] on PopQA, Wang et al. [62] on self-knowledge routing, Jeong et al. [74] on Adaptive-RAG, and Baek et al. [85] on Probing-RAG all converge on the architectural conclusion that an unconditional retrieval dispatch leaves measurable accuracy on the table relative to a per-query routing decision. The retrieval-utility classifier is CAEM's architectural correction in that lineage. It sits between the two-mechanism dispatcher of Section 4.2.4 and the retrieval-augmented fallback, reads a per-query feature vector built from the query, the top-ranked passage, and a base-model self-knowledge probe, and decides whether the query is one on which retrieval helps the generator or one on which retrieval hurts. On the former it forwards the query to the retrieval-augmented tier as the router intended; on the latter it routes the query to the zero-shot tier, so that the generator's parametric knowledge serves the answer without paying the retrieval cost and without admitting passage-amplified errors into the self-improvement loop.

The classifier is a binary head with the two output classes RAG (route to the retrieval-augmented tier) and DIRECT (route to the zero-shot tier). It produces a calibration score $p_{\mathrm{RAG}} \in [0, 1]$ from a thirty-six-dimensional feature vector through standard-scaler normalisation followed by a logistic-regression scoring head with L2 regularisation. The deployed routing rule compares p_{RAG} against a registered threshold $\tau_{\mathrm{RUC}} = 0.375$ and dispatches to the retrieval-augmented tier when $p_{\mathrm{RAG}} \geq \tau_{\mathrm{RUC}}$ and to the zero-shot tier otherwise. The threshold is well below the unbiased operating point of 0.5 because the cost of a missed retrieval-helpful query (a query that needed retrieval but was sent to zero-shot generation) is asymmetric with the cost of a missed retrieval-hurt query (a query that should have stayed on zero-shot but was sent to retrieval-augmented generation): the first failure mode under-utilises retrieval evidence that was available, while the second admits passage amplification into a hard distribution where it had been measured to hurt. The threshold sweep registered at training time selects the operating point that jointly maximises the count of correctly-routed retrieval-helpful queries and the count of correctly-routed retrieval-hurt queries on the held-out folds; the realised maximum sits at the asymmetric value above.

The thirty-six features partition into three groups. The first group is computed inline from the question text and is always available at routing time: the question token length, a one-hot encoding over the seven canonical interrogative templates (*what*, *when*, *where*, *who*, *why*, *how*, *yes-or-no*), the presence of negation and of temporal or numerical cues, an entity count, the logarithm of the largest Wikipedia page-view count among the detected entities, a myth-regex hit indicator for the canonical misconception phrasings, and three linguistic markers (*adversarial phrasing*, *opinion-seeking*, *open-ended*) drawn from the question's surface form. The second group is computed from the top-ranked passage that the dense retriever returned for the query:

the top-one similarity score, the top-one passage–question entity overlap, six answer-type features between the question's expected answer type and the passage's surface (whether the question expects a year, whether the passage contains one, and whether the two match, with the same three indicators for numbers), and a Wikidata-entity-presence indicator that records whether the question's primary entity appears in the structured knowledge base. The third group is a battery of retrieval-quality and self-knowledge features that the routing-time pipeline assembles from the same retrieval substrate the retrieval-augmented tier would use: the top-five FAISS similarity statistics (maximum, mean, standard deviation), the cross-encoder rerank statistics at the same top-five (top-one rerank score and the mean and standard deviation over the top-five), three pairwise natural-language-inference statistics computed on the ten pairs of the top-five passages (mean, minimum, and standard deviation of the entailment probability), and a Kadavath-style self-knowledge probe p_{IK} that reads the base generator's last-token hidden state on the question alone, without adapter, and predicts whether the model knows the answer parametrically. The third group is the load-bearing axis of the classifier: the probe p_{IK} [37] carries the largest absolute coefficient in the trained scoring head (-1.77), so high parametric self-knowledge pushes the classifier strongly toward DIRECT, while passage-quality features with positive coefficients push toward RAG only when the parametric self-knowledge is low.

The training set is constructed from a content-hash-disjoint fresh pool that does not intersect the calibration fold, the seed pool, the per-cycle stream chunks, or the evaluation and test folds the rest of the architecture consumes. Each candidate row in the fresh pool is scored under the two routing destinations the classifier decides between: the question is answered both by a Tier 2 baseline (a zero-shot generation against the same backbone, without retrieval) and by a Tier 3 baseline (a retrieval-augmented generation against the same backbone, with the same dense substrate the deployment uses), and the two exact-match scores are recorded. The row's label is then assigned by a deterministic rule on the two exact-match scores: the row is a positive (*retrieval-helpful*, label one) when the Tier 3 exact match exceeds the Tier 2 exact match, a negative (*retrieval-hurt*, label zero) when Tier 2 exact match exceeds Tier 3, and is dropped when the two scores tie. The drop rule discards both kinds of ties: rows where both routings produced a correct answer (the classifier's decision is irrelevant) and rows where both routings produced a wrong answer (the classifier's decision is also irrelevant because the architecture's downstream gate will reject both at the storage step). The architectural intent of the drop rule is to give the classifier a training signal that strictly disambiguates the cases where the routing decision matters, and to let the architecture's downstream gate absorb the cases where it does not. The realised dataset spans eight benchmarks (FEVER, TriviaQA, Natural Questions, HaluEval-QA, CommonsenseQA, OpenBookQA, TruthfulQA, and StrategyQA), with three thousand two hundred and eight rows after the drop rule, of which one thousand eight hundred and forty-six are retrieval-helpful and one thousand three hundred and sixty-two are retrieval-hurt, a ratio of approximately 1.36 to one. TruthfulQA exact match in the label assembly uses the Haiku-judged correctness rule rather than the default ROUGE proxy, consistent with the architecture's downstream TruthfulQA scoring contract.

Three benchmarks in the training set (Natural Questions, HaluEval-QA, Open-

BookQA) are not part of the five-benchmark evaluation panel of Chapter 5: they are present in the classifier's training distribution only to broaden the coverage of retrieval-utility regimes the head must learn. This is a deliberate asymmetry between the classifier's training distribution and the architecture's evaluation distribution: the classifier learns from a broader pool than the architecture is scored on, so that the classifier's per-benchmark generalisation onto the evaluation panel is the load-bearing receipt the empirical chapter audits rather than an in-distribution recall measurement. The three training-only benchmarks do not feed any other component of the architecture, in particular the self-improvement training pool, so the asymmetry is contained within the classifier's training signal and does not leak into any downstream subsystem.

The classifier is fit under a nested five-fold stratified cross-validation. The outer folds partition the three thousand two hundred and eight rows by label into five equally-sized strata, holding out one stratum per outer split for an out-of-fold prediction; the inner folds inside each training stratum select the L2 regularisation strength from the grid $\{0.005, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0\}$ by the same stratified protocol on the inner training data, so that the regularisation choice is made without ever seeing the outer held-out fold. The final-fit regularisation strength is read off the mode of the per-outer-fold inner selections and lands at $C = 0.5$, paired with a class-balancing weight that compensates for the asymmetric label rate. The held-out operating-point threshold τ_{RUC} is then selected by a sweep over the candidate grid $\{0.20, 0.225, \ldots, 0.75\}$ on the concatenated out-of-fold predictions, choosing the value that maximises the joint correctly-routed count across the two label classes. The nested protocol decouples the regularisation choice from the threshold choice and prevents either choice from leaking the held-out fold into the trained classifier.

Three held-out scores anchor the classifier's calibration receipt and license the classifier as a deployment-ready architectural component. The pooled out-of-fold AUROC across the five outer folds is 0.850, with a per-fold standard deviation of 0.008 across the five outer folds and an in-sample-versus-out-of-fold drift of 0.011 that closes well inside the registered overfit ceiling. At the deployed threshold the joint correctly-routed accuracy is 77.2% of the three thousand two hundred and eight held-out rows, with a confusion matrix that reads one thousand five hundred and fifty-four correctly-routed retrieval-helpful queries against two hundred and ninety-two miss-routed ones (recall 0.842 on the retrieval-helpful class) and nine hundred and twenty-three correctly-routed retrieval-hurt queries against four hundred and thirty-nine miss-routed ones (specificity 0.678 on the retrieval-hurt class). Per-benchmark AUROC on the same out-of-fold predictions ranges from 0.686 on OpenBookQA at the low end to 0.936 on TriviaQA at the high end; the realised band on the five-benchmark evaluation panel reads FEVER 0.718, TriviaQA 0.936, CommonsenseQA 0.728, TruthfulQA 0.732, and StrategyQA 0.852. The TruthfulQA reading is the load-bearing transfer score because the regression on TruthfulQA is the empirical anchor that motivated the classifier in the first place; the realised 0.732 sits below the stretch target of 0.75 that the pre-specified classifier acceptance gate registered, and is reported in Chapter 5 alongside the realised TruthfulQA recovery that the classifier delivers under deployment. The self-knowledge probe p_{IK} used as a feature carries its own held-out calibration receipt: the probe is fit on eleven thousand nine hundred and eighty questions under the same nested five-fold protocol,

on the binary task of predicting whether the base generator answers the question correctly without retrieval, with a mean out-of-fold AUROC of 0.771.

The classifier consumes a frozen scoring head at deployment. It is fit once at the cycle-zero pre-launch stage on the fresh pool above and is not refit at any cycle boundary across the realised six-cycle trajectory. The frozen choice is deliberate: the classifier's input features are computed from quantities the architecture does not retrain (the question text, the dense passage substrate, the base generator's hidden state under the disable-adapter context, and a pre-trained cross-encoder reranker), so cycle-boundary drift in the classifier's input distribution is bounded by the architecture's substrate-level stability rather than by the adapter-level drift the self-improvement loop induces. The architectural advantage of the frozen design is that the classifier's per-cycle behaviour can be audited as a pure routing-distribution drift on a fixed scoring head rather than as a coupled drift in both the input distribution and the scoring head's coefficients. The corresponding architectural cost is that any benchmark-level distribution shift between the fresh pool and a downstream evaluation pool is absorbed entirely by the classifier's per-benchmark generalisation rather than by a cycle-boundary refit, and the receipt for this absorption is the per-benchmark AUROC band on the evaluation panel reported above and the routing distribution trajectory reported in Chapter 5. The lightweight-classifier-only operating point CAEM adopts is consistent with the empirical conclusion of Moskvoretskii et al. [89], who show that an external classifier with no language-model forward pass at routing time matches the routing accuracy of language-model-based selective retrievers on the standard suite; CAEM differs in that the architecturally heavy retrieval-quality features (cross-encoder rerank, pairwise NLI on top-five passages) are reused from the verifier ensemble's already-loaded substrate rather than computed afresh.

The router consumes the classifier at exactly two of the four dispatch outcomes of Table 4.1: the safety-override outcome and the score-formula fall-through outcome. Both outcomes share the property that the pre-classifier dispatch would unconditionally route to the retrieval-augmented tier; the classifier intercepts each one and gives it a chance to be redirected to the zero-shot tier instead. The Tier 1 outcome (the combined-score memory-direct dispatch) and the Tier 2 outcome (the similarity-only zero-shot dispatch) are never gated by the classifier, because both already serve a tier other than the retrieval-augmented fallback and the classifier's binary choice is between exactly those two tiers. The post-classifier dispatch order is summarised in Table 4.2 and the classifier's scoring and dispatch logic is stated in Algorithm 3.

The contribution of the retrieval-utility classifier to the registered hypotheses is twofold. First, by routing retrieval-hurt queries to the zero-shot tier, the classifier closes a known regression mode on misconception-bait benchmarks that the architecture without the classifier exhibits and that the deferred-buffer and retroactive re-verification mechanisms cannot close because both consume the architecture's output rather than its tier choice. Second, by separating the retrieval-helpful subpopulation from the retrieval-hurt subpopulation at routing time, the classifier sharpens the self-improvement training pool's signal-to-noise ratio: the pool admits Tier 3 answers on queries where retrieval helps and Tier 2 answers on queries where retrieval would have hurt, so the fine-tune cycle absorbs grounded reasoning on the former without absorbing passage-amplified errors on the latter. The empirical receipt for both contributions, including the realised TruthfulQA recovery, the per-cycle

Order	Tier	Condition	Output path
1	Tier 3 / 2	$u_{\text{pre}} < 0.38$ (safety override): consult classifier; tier = 3 if $p_{\text{RAG}} \geq \tau_{\text{RUC}}$ else tier = 2	Retrieval-augmented or zero-shot generation, gated by the classifier
2	Tier 1	$\mathcal{S} = 0.7\,s + 0.3\,u_{\text{stored}} \geq 0.90$	Serve neighbour's stored answer
3	Tier 2	$s \geq 0.75$	Zero-shot generation
4	Tier 3 / 2	otherwise (score-formula fall-through): consult classifier; tier = 3 if $p_{\text{RAG}} \geq \tau_{\text{RUC}}$ else tier = 2	Retrieval-augmented or zero-shot generation, gated by the classifier

Table 4.2: Post-classifier dispatch rules in order of evaluation. The retrieval-utility classifier intercepts the two dispatch outcomes (safety override and score-formula fall-through) that would otherwise route unconditionally to the retrieval-augmented tier. The Tier 1 and Tier 2 outcomes are unchanged from Table 4.1: a query that earns the memory-direct or zero-shot dispatch on its own combined score does not consult the classifier.

Algorithm 3 Retrieval-utility classifier scoring and dispatch refinement.

Require: question q, top-ranked passage $\mathbf{p}_1$, dense top-five passages $\mathbf{P}_5$, pre-routing decision $d_{\text{pre}} \in \{\text{Tier 1}, \text{Tier 2}, \text{Tier 3}\}$

Require: frozen scaler Σ, frozen logistic head w, threshold $\tau_{\text{RUC}} = 0.375$

if $d_{\text{pre}} \in \{\text{Tier 1}, \text{Tier 2}\}$ **then**
  return d_{pre} ▷ classifier does not gate Tier 1 or Tier 2 outcomes
end if
$\mathbf{x}_{\text{q}} \leftarrow$ question-surface features (q) ▷ interrogative one-hot, negation, temporal, numerical, myth-regex, entity count
$\mathbf{x}_{\text{p}} \leftarrow$ top-one-passage features $(q, \mathbf{p}_1)$ ▷ similarity, entity overlap, answer-type matches
$\mathbf{x}_{\text{r}} \leftarrow$ retrieval-battery features $(\mathbf{P}_5)$ ▷ FAISS top-five statistics, cross-encoder rerank statistics, pairwise NLI statistics, Wikidata
$p_{\text{IK}} \leftarrow$ base-generator self-knowledge probe (q) ▷ Kadavath-style probe under the disable-adapter context
$\mathbf{x} \leftarrow [\mathbf{x}_{\text{q}}, \mathbf{x}_{\text{p}}, \mathbf{x}_{\text{r}}, p_{\text{IK}}]$
$p_{\text{RAG}} \leftarrow \sigma\big(w^{\top}\,\Sigma(\mathbf{x})\big)$ ▷ logistic on standard-scaled features
if $p_{\text{RAG}} \geq \tau_{\text{RUC}}$ **then**
  return Tier 3 ▷ retrieval-augmented generation
else
  return Tier 2 ▷ zero-shot generation
end if

routing-distribution trajectory, and the head-to-head ablation against the architecture without the classifier on the same six-cycle trajectory, is reported in Section 5.4.2.

4.2.6 Verifier ensemble

The verifier ensemble is the post-generation block that converts a generated answer into a vector of nine complementary signals across four families. The four-family decomposition follows the white-box-versus-black-box and instance-versus-aggregate axes catalogued in the recent confidence-estimation survey for language models [71], but extends them with the explicit grounding-versus-relevance separation that the off-topic-but-grounded failure mode requires. The ensemble is the single source of every quantity that the calibrated probability composite consumes, and its design is governed by two principles: each signal must measure a different failure mode from a different evidence source, and no signal may be redundant with another in the same family. The nine signals together feed the per-signal isotonic regression and the regularised logistic boost layer specified in Section 4.2.7; they also feed the four-outcome decision tree's confabulation early-exit rule specified in Section 4.2.9. Table 4.3 summarises every signal, its family, its evidence source, the resampling configuration that generates it, and the failure mode it closes. Algorithm 4 states the ensemble's execution order as implemented in the unified verifier.

Signal	**Family**	**Evidence source**	**Resampling**	**Failure mode closed**
u_{token}	Internal calibration	Generator log-probs	Single decode	Token-level low-confidence generation
u_{dropout}	Internal calibration	MC-dropout in train mode	K=5 passes	Hidden-state instability
u_{internal}	Internal calibration	Composite of u_{token}, u_{dropout}	Derived	Internal-confidence summary
s_{avg}	Sample-set agreement	Generator resampled chains	M=3 at T=0.7	Disagreement across chains
p_{entail}	Sample-set agreement	NLI judge on chain $\rightarrow$ answer	Averaged over M	Chain-to-answer support
$p_{\text{ground,max}}$	External grounding	NLI judge on passage $\rightarrow$ answer	Top-1 reranked	No supporting passage
$p_{\text{ground,mean}}$	External grounding	NLI judge on passage $\rightarrow$ answer	Mean over top-3	Weak average grounding
$p_{\text{ground,atomic}}$	External grounding	NLI judge on atomic facts	Weakest link, length $\geq$ 12 tokens	Partial-support fabrication on long answers
$q_{\text{a-rel}}$	QA relevance	Cross-encoder on (question, answer)	Single scoring	Off-topic answer with matching passage

Table 4.3: The nine verifier signals that enter the calibrated probability composite, organised across four families. Resampling counts and length gate match the shipped verifier implementation; no fold is shared with the calibration fold.

Internal calibration signals

The internal-calibration family reads the generator's own state on the answer it produced. The token-level uncertainty signal is the mean log-probability of the answer's tokens under the generation, with higher values reading as higher confidence; this signal captures the most local form of generator self-doubt. The dropout-disagreement

Algorithm 4 Verifier ensemble (Stage 5).

Require: query q, answer a, generator G, NLI judge J, cross-encoder X, retrieved passages $\mathcal{P}$

Internal calibration family

$u_{\text{token}} \leftarrow$ mean token log-prob of a under G at the generation temperature

$u_{\text{dropout}} \leftarrow$ variance of token log-prob of a across $K{=}5$ MC-dropout passes through G

$u_{\text{internal}} \leftarrow 0.5\, u_{\text{token}} + 0.5\,(1 - u_{\text{dropout}})$

Sample-set agreement family

$\{a^{(m)}\}_{m=1}^{M=3} \leftarrow$ resample chains from G at temperature 0.7

$s_{\text{avg}} \leftarrow$ mean unique-pair cosine similarity across $\{a^{(m)}\}$ in the sentence-embedding space

$p_{\text{entail}} \leftarrow \frac{1}{M} \sum_{m=1}^{M} J(a^{(m)} \to a)$

External grounding family

$\{e_p\}_{p \in \text{top}-3(\mathcal{P})} \leftarrow J(p \to a)$ for each reranked passage p

$p_{\text{ground,max}} \leftarrow \max_p e_p$

$p_{\text{ground,mean}} \leftarrow \text{mean}_p\, e_p$

if $|a| \geq 12$ tokens **then**

    decompose a into atomic facts $\{f_i\}$

    $p_{\text{ground,atomic}} \leftarrow \min_i \max_p J(p \to f_i)$

else

    $p_{\text{ground,atomic}} \leftarrow p_{\text{ground,mean}}$ ▷ length-gate fallback

end if

Question-answer relevance

$q_{\text{a-rel}} \leftarrow X(q, a)$ ▷ cross-encoder scoring

return $(u_{\text{token}}, u_{\text{dropout}}, u_{\text{internal}}, s_{\text{avg}}, p_{\text{entail}}, p_{\text{ground,max}}, p_{\text{ground,mean}}, p_{\text{ground,atomic}}, q_{\text{a-rel}})$

signal is the variance of the answer's log-probability across the registered $K = 5$ Monte-Carlo dropout passes through the generator with dropout layers active, with lower variance reading as higher confidence; this is the Bayesian-approximation interpretation of dropout introduced by [14], and it captures the stability of the generator's hidden state under stochastic ablation. The internal-state composite signal aggregates the two preceding signals into a single scalar with the fixed combination $0.5\,u_{\text{token}} + 0.5\,(1 - u_{\text{dropout}})$, registered before the calibration phase; this signal acts as the input to the confabulation early-exit rule because it isolates the generator's internal-confidence channel from the external-grounding channel. The internal-calibration family is the only family that reads the generator's hidden state directly and is therefore the only family whose distributions shift after every cycle's fine-tune; this drift is what motivates the retroactive re-verification pass in Section 4.2.10. The hidden-state-eigenvalue alternative [66], which scores the log-determinant of the per-query response covariance, was considered as a candidate replacement for the dropout signal but was not adopted in the locked configuration for cost-efficiency reasons.

Sample-set agreement signals

The sample-set family probes the generator's stability by resampling answers under temperature and measuring agreement across the resampled chains. The semantic-consistency signal is the mean unique-pair cosine similarity across $M = 3$ chains at temperature $T = 0.7$ in the sentence-embedding space, with higher values reading as higher agreement; it is an embedding-space adaptation of the self-consistency agreement signal [61]. The chain-to-answer entailment signal is the mean entailment of each of the M resampled chains against the served answer under the natural-language-inference judge, with higher entailment reading as higher support; this signal closes the failure mode where the served answer is well-formed but is not the conclusion that the model's own resampling supports. The two signals share the same $M = 3$ resampling pool, so semantic-consistency and chain-to-answer entailment are amortised against a single decode-phase cost. The family adds non-trivial decode-phase work to the per-query cost, which is why pre-routing routes the query to the memory-direct tier when it can, deferring the sample-set work to queries that genuinely need it.

Empirical retirements at the sample-set family. Two semantic-dispersion signals were registered earlier in the project and retired before the main run on the registered cycle-zero calibration evidence. The normalised semantic entropy signal of [52], [70], computed over $K = 10$ chains at temperature $T = 1.0$, registered a Pearson correlation indistinguishable from zero with exact-match labels on every per-benchmark slice of the calibration fold and a boost-layer weight within five times ten to the negative four of zero on every fit. The retained signal contributed no discriminative information beyond what the semantic-consistency and chain-to-answer entailment signals on the same $M = 3$ chains already provided, and the $K = 10$ stochastic generations needed to compute it dominated the verifier's wall-clock cost. The kernel-language-entropy generalisation that uses graded semantic-similarity kernels rather than hard cluster assignments [78] was registered as a candidate upgrade for the signal but was not adopted in the shipped configuration. Both the entropy signal and the kernel-entropy upgrade are retired in the registered architecture; the

deployed verifier consumes nine signals across the four families and the calibrated probability composite ingests the same nine.

External grounding signals

The external-grounding family scores the answer against the retrieved passages under three aggregation rules. The top-passage entailment signal is the entailment score of the highest-scoring passage against the answer under the natural-language-inference judge, with higher entailment reading as stronger support; this signal captures the case where a single retrieved passage strongly supports the answer. The pooled-mean entailment signal is the mean entailment over the top-three reranked passages, with higher mean reading as broader support; this signal captures the case where multiple passages each contribute partial support and the answer is the integration over them. The atomic-fact entailment signal decomposes the answer into atomic facts and computes the weakest-link maximum entailment across the top reranked passages for each atomic fact, with the final score being the minimum across the atomic facts; this signal captures the case where an answer is mostly supported but contains a single fabricated detail that the pooled-mean signal would average away. The atomic-fact signal fires only on answers that exceed a length gate of twelve tokens, registered in the implementation, because the atomic decomposition of a short label answer reduces to the answer itself and adds noise rather than signal. Below the gate the atomic signal degrades to the pooled-mean signal, so that a shorter answer is not penalised by an incoherent decomposition.

Question-answer relevance

The question-answer relevance signal scores the served answer against the original question under a cross-encoder, with higher score reading as higher topical relevance. This signal is treated as a separate signal rather than folded into the grounding family because the failure mode it closes is structurally different: an off-topic answer can be perfectly grounded in a passage that is itself off-topic for the question, and the grounding family alone would not detect this case. The cross-encoder is shared with the reranker on the retrieval-augmented tier, so the per-query cost of this signal is small relative to the work the cross-encoder is already doing on the same query. The contribution to the registered hypotheses is direct: the question-answer relevance signal is what closes the off-topic-but-grounded failure mode that the composite hallucination metric's off-topic subtype tracks, and it is one of two signals (alongside the chain-to-answer entailment) whose calibrated weight in the boost layer is expected to be non-trivial under the empirical fits reported in Chapter 5.

4.2.7 Calibrated probability composite

The calibrated probability composite is the function that converts the nine verifier signals into a single calibrated storage score that the fixed-threshold gate consumes. The composite is a three-stage pipeline. The first stage is per-signal isotonic regression that maps each raw signal value to a calibrated probability of correctness, fitted independently per signal on a disjoint calibration fold. The second stage is a per-benchmark refinement that fits a child composite on each training-benchmark slice of the calibration fold and blends each child's monotone curve with the pooled

fit through a James-Stein style shrinkage prior, so that benchmarks with weaker per-bench evidence regress toward the pooled fit rather than overfitting on small per-bench folds. The third stage is a regularised logistic boost layer that aggregates the per-signal calibrated probabilities through a weighted log-odds sum, with the weights and the intercept fitted on the same calibration fold against exact-match labels. Algorithm 5 states the fit-time and predict-time procedures as implemented; the locked configuration parameters used in the main run are pulled from the cycle-zero composite calibration artefact.

Algorithm 5 Calibrated probability composite (fit and predict).

Fit stage. Given a calibration fold partitioned by training benchmark $\{(\mathcal{D}_b)\}_{b\in\mathcal{B}}$ with each $\mathcal{D}_b$ a set of pairs (x_i, y_i) where $x_i \in \mathbb{R}^9$ is the per-signal vector and $y_i \in \{0, 1\}$ is the exact-match label; let $\mathcal{D}_{\text{pool}} = \bigcup_b \mathcal{D}_b$ be the pooled fold.

Pooled fit.

for each signal $j \in \{1, \ldots, 9\}$ **do**
    direction $\leftarrow \text{sgn}(\text{Pearson}(x_{:,j}, y))$ on $\mathcal{D}_{\text{pool}}$ ▷ auto-detect monotone direction
    $\text{iso}_j^{\text{pool}} \leftarrow \text{IsotonicRegression}(\text{increasing} = \text{direction}, y_{\min}{=}0.02, y_{\max}{=}0.98)$
    $\text{iso}_j^{\text{pool}}.\text{fit}(x_{:,j}, y)$ on $\mathcal{D}_{\text{pool}}$
end for

Per-benchmark fit with shrinkage prior.

for each benchmark $b \in \mathcal{B}$ and each signal j **do**
    fit a child isotonic curve iso_j^b on $\mathcal{D}_b$ alone
    at every knot x of iso_j^b, blend the calibrated probability with the pooled curve, $\text{iso}_j^b(x) \leftarrow \alpha_{\text{shrink}} \cdot \text{iso}_j^b(x) + (1 - \alpha_{\text{shrink}}) \cdot \text{iso}_j^{\text{pool}}(x)$, with the locked shrinkage strength $\alpha_{\text{shrink}} = 0.6$
end for

Boost layer.

Build feature matrix $L \in \mathbb{R}^{N\times 9}$ with $L_{i,j} = \log\left(\frac{\text{iso}_j^{\text{pool}}(x_{i,j})}{1-\text{iso}_j^{\text{pool}}(x_{i,j})}\right)$ on the pooled fold
$(\mathbf{w}, b) \leftarrow \text{LogisticRegression}(C{=}0.01,\ \text{penalty} = \ell_2,\ \text{max_iter}{=}2000).\text{fit}(L, y)$
Repeat the boost fit on each per-benchmark fold using the corresponding shrunk isotonic curves to produce per-benchmark weights $(\mathbf{w}_b, b_b)$

Predict stage. Given a query's per-signal vector $x \in \mathbb{R}^9$ and benchmark tag $b^\star$:

if $b^\star \in \mathcal{B}$ select per-benchmark $(\text{iso}_j^{b^\star}, \mathbf{w}_{b^\star}, b_{b^\star})$, else fall back to the pooled $(\text{iso}_j^{\text{pool}}, \mathbf{w}, b)$
$\ell \leftarrow b + \sum_{j=1}^{9} w_j \cdot \log\left(\frac{\text{iso}_j(x_j)}{1-\text{iso}_j(x_j)}\right)$
$u_{\text{stored}} \leftarrow \sigma(\ell) = 1/(1 + \exp(-\ell))$
return u_{stored}

Per-signal isotonic regression

Per-signal isotonic regression is preferred over the alternatives at this site for three reasons. Isotonic regression admits a non-linear monotone mapping rather than the single-scalar log-odds shift of Platt scaling, which matters because the verifier signals do not all admit a logistic relationship to correctness; the top-passage entailment signal in particular saturates at the upper end of its range, and Platt scaling would absorb that saturation into a globally too-soft probability assignment. Isotonic

regression is data-efficient relative to histogram binning at the calibration-fold size used in this thesis, because it shares mass across adjacent bins through its monotone structure rather than partitioning the fold into independent bins that each need their own minimum sample count. Isotonic regression fits a monotonically increasing map from score to probability [7]; CAEM's implementation auto-detects the monotone direction from the Pearson correlation between the raw signal and the binary label, so a signal that reads high-confidence-as-low-value (such as the dropout-disagreement signal, where higher variance reads as lower confidence) is calibrated correctly without a registry of per-signal directions. The implementation clips the calibrated probability into the range [0.02, 0.98] to keep the log-odds finite and to bound the influence of any single signal on the composite. Each signal must contribute at least fifty calibration-fold samples for its isotonic fit to be admitted; below this floor the signal is dropped and its log-odds contribution defaults to zero in the composite, so that the composite degrades gracefully when a fold is too small.

Per-benchmark fit with shrinkage prior

The composite is fitted at two granularities on the same calibration fold: a pooled fit over the union of training-benchmark slices, and a per-benchmark fit over each slice separately. The pooled fit serves as both the fallback for any query whose benchmark tag is not in the registered training panel and as the prior toward which each per-benchmark child curve is regularised. The shrinkage rule is the James-Stein style convex blend, applied at every knot of the per-benchmark curve, that replaces the child knot value with a weighted combination of the child's own value and the pooled curve evaluated at the same input. The locked shrinkage strength is $\alpha = 0.6$, which keeps a moderate per-benchmark adaptation while pulling each child curve away from the per-bench overfit that pure local fitting produces on weak-signal panels. The shrinkage strength was selected by the cycle-zero diagnostic that fitted the pure per-bench composite ($\alpha = 1$) and observed that the per-benchmark area-under-the-receiver-operating-characteristic readings on the weak-signal training panels regressed below the pooled reading by ten to fifteen points, signalling that the per-bench fold size of approximately five hundred samples per benchmark was too small to support the more flexible per-bench isotonic envelope unaided. The blend at $\alpha = 0.6$ moves the regression-AUROC readings back into parity with the pooled fit on those panels while preserving a measurable per-bench improvement on the strong-signal benchmarks. The per-benchmark child also retains its own boost coefficients fitted on the bench-local fold, so the per-bench curve and the per-bench weights jointly capture the benchmark's own reliability profile rather than borrowing the pooled boost weights uniformly.

Boost layer with L2 regularisation

The boost layer aggregates the per-signal calibrated probabilities by fitting a logistic regression on the per-signal log-odds as features against the exact-match labels of the calibration fold, following the multi-scoring approach of [82]. The learned coefficients become the per-signal weights and the intercept becomes the composite's bias term; at predict time the weighted log-odds sum is sigmoid-back-projected to a probability in the unit interval. A learned linear combination over the per-signal log-odds is preferred over an identity-weighted sum at this site because the verifier signals carry

overlapping information across families: the chain-to-answer entailment signal and the semantic-consistency signal both read sample-set agreement, the top-passage and pooled-mean entailment signals both read external grounding, and the boost layer is what allows the composite to absorb this redundancy by down-weighting the redundant component of each signal's contribution. The L2 penalty is preferred over L1 at this site because every signal in the registered set carries non-zero discriminative signal under the registered hypotheses; the L2 penalty shrinks weights smoothly without setting any to zero, which preserves the architectural commitment to the nine-signal coverage and keeps the boost layer a reweighting rather than a feature-selection step. The locked penalty strength is $C = 0.01$, which corresponds to a strong L2 regulariser; this value was registered in the cycle-zero calibration sweep against the alternatives $C \in \{0.010, 0.025, 0.050, 0.100, 1.000\}$, and the empirical evidence for the strong-regulariser choice is reported under the calibration sweep panel in Chapter 5. The boost layer is fitted in parallel for the pooled and per-benchmark composites so that the per-benchmark child carries its own per-signal weights, capturing the case where the same raw signal reads in a different direction on a different benchmark; the pooled boost coefficients are used as the fallback whenever the verifier sees a query whose benchmark tag is not in the registered panel.

4.2.8 Fixed-threshold storage gate

The storage gate is the procedure that converts the calibrated storage score from Section 4.2.7 into a storage decision at a pre-specified cut-point on the calibrated probability scale. The gate is a fixed-threshold rule: an entry is admitted to the storage class whenever its calibrated composite satisfies the storage condition, deferred whenever it satisfies the deferral condition, and otherwise routed to abstention or discard by the rule specified in Section 4.2.9. The fixed-threshold formulation replaces the split-conformal formulation considered earlier in the project, which fitted the storage and deferral thresholds at pre-specified precision targets and re-fit them at every cycle boundary under exchangeability between the calibration fold and the live distribution. The empirical evidence accumulated through the cycle-zero calibration sweep showed that the calibration-versus-evaluation distribution shift on the five-benchmark panel breaks the marginal-coverage premise of the split-conformal procedure, with the realised eval-fold precision falling fifteen to fifty percentage points below the conformal target on training benchmarks and the per-benchmark fits collapsing to the rejection limit on the weak-signal panels. The fixed-threshold replacement is preferred over the split-conformal alternative on two grounds: it makes no marginal-coverage claim that the cal-versus-eval shift can silently violate, because its precision is audited directly on every evaluation fold, in the lineage of thresholding a calibrated probability [7] and of selective classification [16]; both assume that calibration and test data share a distribution, so under shift the realised precision at the threshold can still fall [25], [30], and the transfer-panel TruthfulQA reading in Chapter 5 is such a case; and it commits the gate's operating point to a single registered scalar that any reader can locate in the configuration without reconstructing the conformal fit.

The gate's two cut-points are registered in the configuration before the main run begins. The storage threshold $\tau_{\text{store}} = 0.60$ is the cut-point at which the storage class is admitted; the deferral threshold $\tau_{\text{defer}} = 0.45$ is the cut-point below which

an entry is held in the deferral buffer pending cycle-boundary re-evaluation; the abstention floor $\phi_{\text{pg}} = 0.20$ on the top-passage entailment signal separates the abstention outcome from the discard outcome below the deferral threshold. All three constants are read at every decision call from the configuration object, so a future re-tuning propagates without re-instantiating the verifier. The empirical calibration-fold precision at each cut-point is recorded by an empirical-precision audit that runs once per cycle on the cycle's labelled calibration fold and writes the audit table to the cycle's output directory; the audit is the empirical receipt for the calibration-consistent floor invoked by Theorem 4.1.

Storage threshold

The storage threshold is the operating point at which the storage class is admitted; an entry is admitted whenever its calibrated composite satisfies $u_{\text{stored}} \geq \tau_{\text{store}} = 0.60$. The choice of cut-point is registered against the cycle-zero threshold sweep, which compared two candidates on the labelled cycle-zero calibration fold of the calibration sweep run before the main trajectory: a stricter cut at 0.65 admitted seventy correct entries out of one hundred and five total admissions for an empirical precision of approximately sixty-seven percent, while the registered cut at 0.60 admitted three hundred and six correct entries out of four hundred and seventy-eight total admissions for an empirical precision of approximately sixty-four percent (the main run's own cycle-zero calibration fold admits 575 entries at 0.746, Table 5.16). The two operating points sit on the same precision shelf at the cycle-zero distribution, but the lower cut admits roughly 4.4 times as many correct memories at the same per-store precision and is the registered configuration for the main run. The storage class is what populates the episodic memory and what the training pool draws from at each cycle boundary, so the gate is paired with the strict training threshold above the storage threshold and with the retroactive prune threshold below it; the storage cut alone is not the only barrier between a candidate answer and the next cycle's gradient update.

Deferral threshold

The deferral threshold defines the catchment for the deferral buffer; an entry is buffered whenever its calibrated composite falls in the band $\tau_{\text{defer}} \leq u_{\text{stored}} < \tau_{\text{store}}$. The registered cut-point is $\tau_{\text{defer}} = 0.45$. The looser cut-point is chosen because the deferral class does not propagate to the training pool at the next cycle boundary; deferred entries are revisited under the updated generator at the cycle-boundary deferred-entry reconsideration pass specified in Section 4.2.10 and either promoted into the storage class or expired by the time-to-live rule. Setting the deferral threshold loosely admits a wider band of marginal entries into the buffer, where they can be re-evaluated under stronger evidence at the next cycle boundary, while keeping the storage class itself disciplined by the stricter storage cut. The decision tree that consumes both cuts enforces the ordering invariant $\tau_{\text{defer}} \leq \tau_{\text{store}}$ structurally; the configuration object refuses to load a state in which the two values would be swapped.

Cycle-boundary recalibration

The locked configuration draws a sharp line between architectural parameters and calibration parameters. Architectural parameters stay frozen across every cycle of the main run: the storage and deferral cut-points $\tau_{\text{store}} = 0.60$ and $\tau_{\text{defer}} = 0.45$, the abstention floor $\phi_{\text{pg}} = 0.20$, the retroactive prune threshold $\tau_{\text{retro}} = 0.50$, and the strict training threshold above the storage cut. Calibration parameters re-fit at every cycle boundary under a single composite-recalibration pass that consumes the cycle's freshly-scored calibration fold and writes the per-signal isotonic curves, the per-benchmark child curves, the shrinkage-blended knot tables, and the boost layer's weights and intercept to the cycle's output directory. The recalibrated composite is then promoted to the canonical path that the verifier reads on the next pipeline initialisation, so the cycle-N verifier scores entries through cycle-N calibration rather than the previous cycle's stale curves.

The composite refit is sequenced inside the cycle boundary so that the post-fine-tune calibration governs the retroactive re-verification pass rather than the previous cycle's stale calibration. After the self-improvement loop commits its weight update, the calibration fold is rescored under the now-improved generator, the recalibration script re-fits the per-signal isotonic curves and the boost coefficients against this freshly-scored fold, and the verifier is rebound to the cycle-N calibration block before retroactive re-verification reads through it. Without this ordering the retroactive pass would score memory entries through the previous cycle's isotonic envelope whose fit no longer covers the post-fine-tune raw signal distribution, and exact-match-correct stored entries with high post-fine-tune token confidence would map to artefactually-low calibrated probabilities and trip the prune threshold; the corrected sequence holds the retroactive pass under the cycle's own freshly-fitted curves and avoids that artefact. The empirical-precision audit at the locked cut-points runs once per cycle against the same recalibrated fold and writes the audit table that the calibration-consistent floor invoked by Theorem 4.1 reads.

The split between frozen architectural cut-points and adaptive composite calibration preserves the architectural anchor that Hypothesis 1 of Section 1.4.3 tests: the nine-signal composite design is the object under empirical test across the trajectory, and re-tuning the cut-points cycle-by-cycle would conflate verifier-architecture quality with operating-point drift correction. The split also surfaces the research-versus-production parity: production deployment runs the same cycle-boundary composite refit as the research trajectory, with the cycle horizon set by accumulation rather than by a fixed query-budget schedule. A production cycle ends when a registered number of storage-class admissions has accumulated, or on a calendar trigger, whichever is sooner; at the cycle boundary the same labelled calibration fold is constructed (a stratified sample of the cycle's accumulated admissions, sent for offline labelling), the same per-signal isotonic curves and per-benchmark children and shrinkage blend and boost coefficients are refitted on that fold, and the same retroactive re-verification pass runs against the freshly refitted verifier. The production pipeline therefore inherits the cycle-boundary calibration discipline from research; the only operational difference is the cycle-trigger policy and the labelling cadence, both of which are specified in the production runbook accompanying this thesis.

4.2.9 Four-outcome storage decision

The four-outcome decision tree converts the calibrated storage score and a small set of guard signals into one of four mutually exclusive actions: store, defer, abstain, or discard. The four outcomes are preferred over a binary store-or-discard rule for two reasons. First, a binary rule conflates the genuinely-uncertain case with the confidently-wrong case: a query whose composite falls just below the storage threshold is not the same as a query whose composite is near zero, and treating both as discard discards information that the next cycle's updated generator could resolve. The deferral outcome captures the genuinely-uncertain band into a bounded buffer that is revisited at the cycle-boundary deferred-entry reconsideration pass. Second, a binary rule does not separate the failure mode where the system has nothing to say from the failure mode where the system would say something but cannot ground it; the abstention outcome captures the explicit-refusal case as a first-class user-facing action rather than a silent discard, in the spirit of the conformal-abstention framework that bounds hallucination rate at a user-specified target [84].

The decision tree is governed by three thresholds and one early-exit rule. The two cut-points $\tau_{\text{store}} = 0.60$ and $\tau_{\text{defer}} = 0.45$ are inherited from Section 4.2.8 and govern the store and defer outcomes. A third threshold $\phi_{\text{pg}} = 0.20$ on the top-passage entailment signal separates the abstention outcome (no supporting passage) from the discard outcome (supporting passage but uncertain composite). The confabulation early-exit rule fires when the internal-confidence composite is high but the top-passage entailment is low,

$$\left(u_{\text{internal}} \geq 0.70\right) \wedge \left(p_{\text{ground,max}} \leq 0.20\right) \Rightarrow \texttt{DISCARD with early_exit_triggered=True}, \tag{4.1}$$

and is checked before the storage and deferral cut-points. The early-exit rule is the architectural defence against the most damaging failure mode in the verifier ensemble: a generator that is internally confident but externally ungrounded. Without this rule the calibrated composite alone might assign such an answer a marginal storage score and admit it under the deferral threshold, where it would survive into the next cycle's reconsideration pass; the early-exit rule routes it directly to discard with a flag that the cycle-level diagnostics can read. Algorithm 6 states the decision tree as implemented in the unified verifier.

The retroactive-review motivation that justifies the deferred path is structural rather than rhetorical. A deferred entry is rescored at the next cycle boundary under the updated generator (which is fitted on the pool that the current cycle's storage outcomes contributed to), so a deferred entry whose composite falls just below the storage threshold this cycle has a non-trivial chance of crossing the threshold next cycle as the generator's competence on its question class improves. Without the deferred path, every such entry would be permanently discarded after one cycle of borderline competence, and the fine-tune pool's growth rate would be bounded above by the storage rate at the strictest threshold the system was ever evaluated against. With the deferred path, the fine-tune pool grows monotonically as deferred entries are promoted, and the pool's quality is preserved because every promoted entry has cleared the storage threshold under the updated model rather than the older one. The deferred buffer is bounded in size by a registered capacity and entries that exceed a registered time-to-live without promotion are expired; the bound prevents

Algorithm 6 Four-outcome storage decision (Stage 6).

Require: storage score u_{stored}, internal-confidence composite u_{internal}, top-passage entailment $p_{\text{ground,max}}$
Require: fixed cut-points $\tau_{\text{store}} = 0.60$, $\tau_{\text{defer}} = 0.45$ from Section 4.2.8
Require: constants $\eta_u = 0.70$, $\eta_g = 0.20$, $\phi_{\text{pg}} = 0.20$
if $u_{\text{internal}} \geq \eta_u$ **and** $p_{\text{ground,max}} \leq \eta_g$ **then**
    return (`DISCARD`, `early_exit_triggered` = True) ▷ confabulation early-exit
end if
if $u_{\text{stored}} \geq \tau_{\text{store}}$ **then**
    return (`STORE`, `early_exit_triggered` = False)
end if
if $u_{\text{stored}} \geq \tau_{\text{defer}}$ **then**
    return (`DEFERRED`, `early_exit_triggered` = False)
end if
if $p_{\text{ground,max}} < \phi_{\text{pg}}$ **then**
    return (`ABSTAIN`, `early_exit_triggered` = False)
end if
return (`DISCARD`, `early_exit_triggered` = False)

the buffer from growing unboundedly while the cycle-boundary reconsideration pass clears most entries within a few cycles of admission.

4.2.10 Cycle-boundary maintenance

The three cycle-boundary passes are also the architectural mechanisms that close the two statelessness problems registered in Section 1.2.3. The first statelessness problem is that standard retrieval-augmented generation pays the full retrieval-and-generation cost in every session because the model never absorbs what retrieval just taught it; the self-improvement loop closes this by drawing its training pool from stored episodes whose answers were originally produced by the retrieval-augmented tier and were certified by the verifier, so the fine-tune folds verified retrieval-augmented reasoning into the generator's parametric capacity and the next cycle's zero-shot tier serves a measurable share of the previously retrieval-dependent queries without retrieval. The second statelessness problem is that even when the model already knows the answer, the standard protocol re-generates it from scratch on every recurrence and offers no consistency guarantee across sessions; the verified episodic store closes this through similarity-indexed cached answers served at embedding-lookup cost, with the retroactive re-verification pass at every cycle boundary refreshing the stored composite under the locked verifier ensemble and the cycle-boundary-refit composite against the updated generator's outputs, so that the cached answer's quality tracks the model's current capability rather than freezing at the moment of admission. The deferred-entry reconsideration pass is the bridge between the two mechanisms: it revisits entries that were close to the storage cut-point at admission time but did not clear it, so that the same query can be re-admitted under the updated generator without rerunning retrieval. The three passes share a logical role: each runs once at the end of every cycle and each consumes the updated generator that

the cycle's fine-tune produces. The retroactive re-verification pass re-scores every stored episode under the locked verifier ensemble against the updated generator's outputs and the cycle-boundary-refit composite, then prunes or refreshes accordingly. The deferred-entry reconsideration pass revisits every entry in the deferral buffer and either promotes, drops, or keeps the entry. The self-improvement loop draws a training pool from the storage class, runs the parameter-efficient fine-tune through the bounded low-rank-adapter envelope on the frozen backbone (the registered L_2 anchor coefficient is zero on this path; the full-fine-tune fallback for which the anchor is non-zero is registered but inactive in the shipped configuration), and either commits or rolls back the resulting weights against the multi-modal retention guard. All three passes are sequenced so that the fine-tune happens after the deferred-reconsider pass and before the retroactive re-verification pass for the next cycle, so that the retroactive pass always sees the freshly fine-tuned model rather than the model that produced the entries it is re-scoring.

Retroactive re-verification

The retroactive re-verification pass re-scores every stored episode under the current cycle's generator and applies four operations: prune entries whose reasoning chain trips the repetitive-loop filter, prune entries whose new storage score falls below the registered prune threshold, refresh the per-signal block of entries whose new score crosses the threshold, and update the stored composite to the new value. The first two operations remove entries that have lost their right to belong in memory; the third and fourth operations propagate the verifier's improved view of the entry forward to all downstream consumers, including the next cycle's training pool selection. The prune threshold is registered at $\tau_{\text{retro}} = 0.50$, below the deferral threshold so that an entry that is borderline-deferred-quality but no longer storage-quality is still kept for the deferred-reconsider pass; entries below the prune threshold are categorically removed. The full per-signal block (all nine verifier signals) is overwritten on every re-verification, not just the composite, so that the next cycle's calibration runs against the freshest verifier view of every stored episode. Algorithm 7 states the pass.

Deferred-entry reconsideration

The deferred-entry reconsideration pass revisits every entry in the deferral buffer at every cycle boundary. Each entry is re-scored under the updated generator and the four-outcome decision tree is applied with the locked thresholds: an entry whose new score crosses the storage threshold is promoted into memory, an entry whose new score is below the deferral threshold is dropped, and an entry that remains in the deferral band is kept for the next cycle's reconsideration. A registered time-to-live of $\tau_{\text{ttl}} = 2$ cycles bounds the longest possible residency: an entry that survives two reconsideration passes without crossing the storage threshold is dropped categorically, so the buffer cannot accumulate entries that the model will never resolve. The buffer's hard capacity is registered at $K_{\text{def}} = 10\,000$ entries; when the cap is reached, the oldest entry is evicted to make room for new admissions. Algorithm 8 states the pass.

The memory consolidation pass runs alongside the deferred-reconsider pass at every cycle boundary. It collapses near-duplicate stored episodes by clustering entries whose

Algorithm 7 Retroactive re-verification pass (Stage 8a).

Require: memory $\mathcal{M}$ of stored entries, current verifier function verify$(\cdot)$ under updated generator
Require: prune threshold $\tau_{\text{retro}} = 0.50$
$n_{\text{up}} \leftarrow 0$, $n_{\text{rm}} \leftarrow 0$
for each entry $e \in \mathcal{M}$ **do**
  if e.reasoning_chain trips the repetitive-loop filter **then**
    remove e from $\mathcal{M}$ **; continue**
  end if
  $\mathbf{x}_{\text{new}} \leftarrow \text{verify}(e)$ ▷ full nine-signal re-scoring under updated model
  if $\mathbf{x}_{\text{new}}.u_{\text{stored}} < \tau_{\text{retro}}$ **then**
    remove e from $\mathcal{M}$; $n_{\text{rm}} \leftarrow n_{\text{rm}} + 1$
  else
    overwrite e's per-signal block and stored composite with $\mathbf{x}_{\text{new}}$
    $n_{\text{up}} \leftarrow n_{\text{up}} + 1$
  end if
end for
return $(n_{\text{up}}, n_{\text{rm}})$

Algorithm 8 Deferred-entry reconsideration pass (Stage 8b).

Require: deferral buffer $\mathcal{B}$, current verifier verify$(\cdot)$
Require: storage threshold τ_{store}, deferral threshold τ_{defer}, time-to-live $\tau_{\text{ttl}} = 2$
$n_{\text{pr}} \leftarrow 0$, $n_{\text{dr}} \leftarrow 0$, $n_{\text{kp}} \leftarrow 0$
for each entry $e \in \mathcal{B}$ **do**
  $\mathbf{x} \leftarrow \text{verify}(e)$
  if $\mathbf{x}.u_{\text{stored}} \geq \tau_{\text{store}}$ **then**
    promote e into the memory store; $n_{\text{pr}} \leftarrow n_{\text{pr}} + 1$
  else if $\mathbf{x}.u_{\text{stored}} < \tau_{\text{defer}}$ **or** $e.\text{age} + 1 \geq \tau_{\text{ttl}}$ **then**
    drop e from $\mathcal{B}$; $n_{\text{dr}} \leftarrow n_{\text{dr}} + 1$
  else
    $e.\text{age} \leftarrow e.\text{age} + 1$; keep e in $\mathcal{B}$; $n_{\text{kp}} \leftarrow n_{\text{kp}} + 1$
  end if
end for
return $(n_{\text{pr}}, n_{\text{dr}}, n_{\text{kp}})$

pairwise cosine similarity in the sentence-embedding space exceeds the consolidation threshold $\tau_{\text{cons}} = 0.92$, retaining the highest-quality representative within each cluster and merging the source-benchmark provenance of the dropped entries into the survivor. The threshold is set higher than the novelty threshold at write time, so that entries that survived the looser write-time filter but turn out to be near-paraphrases at consolidation time are still resolved into a single representative.

Self-improvement loop

The self-improvement loop is the cycle-boundary block that consumes the cycle's verified pool and produces the updated generator. It is an instance of the broader family of model-self-training procedures that fine-tune a generator on its own filtered outputs [42], [47], [72], and related to distilling a teacher's filtered reasoning into a student [48], specialised here through the fixed-threshold storage gate that supplies the filter, the parameter-efficient adapter layer that absorbs the gradient update, the loss-reweighted training pool that prevents single-benchmark dominance, and the multi-modal retention guard that gates the commit. Its architectural role at the system level is the cost-amortisation mechanism that converts verified retrieval-augmented reasoning into zero-shot parametric capability. At cycle zero the storage class is dominated by entries that the retrieval-augmented tier produced, because the generator alone cannot yet answer most queries above the registered storage threshold. The training pool that the loop draws from these entries therefore consists of question-and-reasoning pairs whose correctness was established through retrieval. The fine-tune folds those pairs into the adapter, and the resulting generator at the next cycle is able to answer a measurable share of the same question distribution from the zero-shot tier without invoking retrieval. The redistribution of queries across the three serving tiers across cycles, reported empirically in Section 5.1.4, is therefore the operational reading of the loop's contribution: the increase in the zero-shot tier's share at the expense of the retrieval-augmented tier's share is the per-cycle measure of how much of the retrieval-mediated knowledge the generator has now internalised, and the cost saving on per-query latency is the deployment-side consequence. Recent work on whether such loops collapse or improve their generator finds empirically that self-improvement requires a positive generation–verification gap and saturates within a few rounds [92], and proves in linear regression that filtering synthetic data through an external verifier prevents collapse but draws the model toward the verifier's own knowledge, so that gains plateau unless the verifier is unbiased [95]; CAEM's external multi-signal verifier and the registered cut-points of Section 4.2.8 are designed to keep the admitted storage class above the purification precondition of Proposition 4.2.

The loop has five operations, in order. First, sample selection from memory: the loop draws training pairs only from stored episodes whose composite score exceeds a strict training threshold registered above the storage cut-point, so the training pool is held to a higher bar than the user-facing retrieval tier. The selection is filtered through an in-distribution gate that excludes episodes whose source benchmark is disjoint from the registered training panel, so transfer-only benchmarks never feed back into the gradient update. Second, pool reweighting: the selected pairs are passed through a benchmark-balancing layer that prevents any single benchmark from dominating the gradient. The layer composes a temperature-mixed softmax over per-benchmark

counts, a registered floor on the smallest per-benchmark contribution, a bounded upsampling cap on each benchmark's effective count, a hard ceiling on any single benchmark's share of the pool, and a cold-start fallback that injects gold-labelled pairs from the seed pool when a benchmark contributes too few verified episodes to clear the floor. The composition is iterative and produces a final pool whose share-by-benchmark distribution lies within the registered tolerance band of the target distribution. Third, the optimiser and the parameter envelope: the loop wraps the open-weight generator with a low-rank adaptation layer fitted on the attention and feed-forward projections, with a registered rank, scaling factor, and target-module list, and runs a thirty-two-bit decoupled-weight-decay AdamW under gradient checkpointing only on the adapter parameters at a learning rate ten times the backbone-fine-tune rate registered for the underlying generator. The full backbone weights are frozen across the cycle, which keeps the trainable footprint at roughly two percent of the backbone parameter count and bounds the cycle-to-cycle drift in the underlying generator's representation. The eight-bit optimiser path that would otherwise apply to a frozen-backbone-plus-full-fine-tune fallback is registered but inactive in the shipped configuration, because on the small adapter the per-parameter quantisation noise from eight-bit moment storage is relatively larger than the modest optimiser-state-memory saving against the thirty-two-gigabyte envelope. Fourth, the regulariser: the loss on the active low-rank-adapter path is cross-entropy on the verified training pool only,

$$\mathcal{L}(\boldsymbol{\theta}) \;=\; \mathcal{L}_{\mathrm{CE}}(\boldsymbol{\theta}) \;+\; \tfrac{\lambda_{\mathrm{anc}}}{2}\left\|\boldsymbol{\theta}-\boldsymbol{\theta}^{(t-1)}\right\|_2^2 \quad \text{with } \lambda_{\mathrm{anc}}=0 \text{ on the shipped adapter path,} \tag{4.2}$$

where $\boldsymbol{\theta}$ is the adapter parameter vector, $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss on the verified training pool, and $\boldsymbol{\theta}^{(t-1)}$ is the previous cycle's adapter. The quadratic anchor term is registered as the parameter envelope for the inactive full-fine-tune fallback (when it engages, $\lambda_{\mathrm{anc}} > 0$ and the bracketed term becomes the uniform-Fisher approximation to elastic weight consolidation [19]). On the shipped low-rank-adapter path the bracketed term vanishes by design: the bounded parameter envelope of the adapter, layered on a frozen backbone, is itself the cycle-to-cycle drift bound, and an explicit weight-anchor would otherwise penalise deviation from a zero-anchored full-backbone reference whose weights the loop is not updating, which is meaningless arithmetic. Fifth, the retention guard: the loop probes a registered set of held-out distributions before the first cycle (the pristine anchor) and after every cycle, and computes the retention ratio (defined in Equation (3.2)) on each probe. The cycle commits the new adapter only when the worst probe ratio clears the floor $\rho_{\min} = 0.93$; otherwise the adapter is rolled back to the previous cycle's snapshot and the memory store is preserved unchanged. The probe set is multi-modal and intentionally diverse: a general-knowledge probe captures distributional drift away from the pretraining mass, while held-out test slices of the training-panel benchmarks capture in-task forgetting under the SIL gradient. The worst-probe rule is conservative on purpose: a loop that improves the average probe at the cost of regressing one specialised probe is the failure mode that an earlier single-probe rule on a general-knowledge probe alone failed to catch. Algorithm 9 states the loop.

The asymmetry between adapter and memory is the design property that makes the loop safe to iterate. A failed cycle costs the cycle's fine-tune energy but does not lose the verified episodes that the cycle accumulated; the next cycle inherits the previous

Algorithm 9 Self-improvement loop with multi-modal retention guard (Stage 8c).

Require: stored memory $\mathcal{M}$, previous adapter parameters $\boldsymbol{\theta}^{(t-1)}$, registered probe set $\mathcal{P}$

Require: training threshold $\tau_{\text{train}} > \tau_{\text{store}}$, anchor strength λ_{anc}, retention floor $\rho_{\min} = 0.93$

$\mathcal{T}_0 \leftarrow \{e \in \mathcal{M} : e.u_{\text{stored}} \geq \tau_{\text{train}}\}$ filtered by the in-distribution gate

$\mathcal{T} \leftarrow$ pool-reweight$(\mathcal{T}_0)$ ▷ temperature softmax + floor + upsample + share cap + cold-start

for each probe $p \in \mathcal{P}$: $\rho^p_{\text{pre}} \leftarrow$ accuracy of $\boldsymbol{\theta}^{(t-1)}$ on p (use cached pristine anchor when available)

$\boldsymbol{\theta}^\star \leftarrow \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$ from Equation (4.2), initialised at $\boldsymbol{\theta}^{(t-1)}$

for each probe $p \in \mathcal{P}$: $\rho^p_{\text{post}} \leftarrow$ accuracy of $\boldsymbol{\theta}^\star$ on p

if $\min_{p\in\mathcal{P}} \rho^p_{\text{post}}/\rho^p_{\text{pre}} \geq \rho_{\min}$ **then**

    $\boldsymbol{\theta}^{(t)} \leftarrow \boldsymbol{\theta}^\star$ ▷ commit the updated adapter

else

    $\boldsymbol{\theta}^{(t)} \leftarrow \boldsymbol{\theta}^{(t-1)}$ ▷ rollback; memory preserved unchanged

end if

return $\boldsymbol{\theta}^{(t)}$

adapter together with the larger memory, and the retroactive re-verification pass at the next cycle boundary will refresh the stored composites under the rolled-back adapter. The contribution to the registered hypotheses is direct: this asymmetry is what licenses the memory-growth invariant of Heuristic 4.7, because memory accumulates strictly across cycles even when the adapter does not.

4.3 Theoretical analysis

The analysis is organised in two parts of different strength. The first part contains the two results this thesis claims as proven. Theorem 4.1 gives a calibration-consistent lower bound on the precision of the stored pool; it follows from the storage rule and from the exchangeability of the calibration fold with the candidate pool, in the spirit of the selective-classification risk bound of [16], which likewise holds when calibration and test data are drawn from the same distribution, with the per-signal post-hoc calibration on a held-out fold following [7]. Proposition 4.2 gives the condition under which filtering through an imperfect verifier makes the stored pool purer than the generator's raw output at all; it follows from Bayes' rule and connects to the analysis of learning from noisy labels [9] and to recent analyses of retraining on self-predicted labels [86] and on weak-teacher pseudolabels [76]. The second part (Section 4.3.3) records design heuristics for how the stored pool is expected to evolve across cycles: storage-rate sensitivity, a convergence envelope, a ceiling on the memory-direct tier's hallucination rate, self-correction under repeated re-verification, and the equilibrium under retention rollback, with the sharpening analysis of [87] as the closest formal treatment of self-improvement dynamics. These heuristics rest on idealising assumptions (Gaussian score classes, stationarity across cycles, and independence of verifier errors across passes) that the realised system does not fully satisfy and that a six-cycle horizon cannot test; they are therefore stated as heuristics with argument sketches rather than as proven results, and their empirical readings

in Chapter 5 are directional. To keep the receipts compact: a *calibration-consistent* qualifier on a receipt indicates a finite-sample lower bound that holds at every cycle within the realised horizon under exchangeability between the cycle's calibration fold and its candidate pool; an *asymptotic-projection* qualifier indicates an expected-value prediction whose trajectory shape is testable but whose limit is not directly observable within a short horizon.

Empirical limit of the realised trajectory: The receipts below are evaluated against the six-cycle trajectory that closed at cycle five, when the equilibrium-saturation gate of Section 3.3.2 terminated the run after the retention guard fired at cycle four and the post-rollback subsequence stabilised. This six-cycle window is the *empirical limit* of the present thesis, not an asymptote: statements whose conclusions are infinite-horizon are evaluated against the cycle-five reading rather than against a fitted asymptote, and any residual gap between the realised limit and the theoretical asymptote is attributed to three implementation regularisers registered earlier in the chapter rather than to a falsification of the underlying mathematics. These regularisers are the L_2 adapter anchor of Equation (4.2), which pins the post-fine-tune weights near the previous-cycle checkpoint and prevents unbounded parametric drift; the multi-modal retention guard with rollback registered in Section 5.1.5, which converts any cycle whose worst-probe ratio falls below the floor into a no-op on the parametric backbone; and the parametric capacity ceiling of the base generator and the dense passage index, which bound the pool's effective precision strictly below one even under an idealised verifier. The organisation below preserves this distinction explicitly: the two proven results have cycle-local content and are evaluated against the realised trajectory with empirical receipts, the design heuristics whose content is genuinely asymptotic are read against the empirical limit with the longer-horizon test deferred to Chapter 6, and mathematical scaffolding identities that do not admit a thesis-side empirical contribution are gathered into a separate subsection at the end of the section so the reader can locate the formal machinery without confusing it with a falsifiable receipt.

4.3.1 Pool purity

Pool purity is the share of stored episodes whose stored answer is correct against the gold. Theorem 4.1 gives the calibration-consistent lower bound under the exchangeability precondition; the Bayes identity of Proposition 4.2 in the next subsection gives the condition under which the stored pool is purer than the generator's raw output, and explains the slack between this floor and the realised purity.

Theorem 4.1 (Pool purity invariant: calibration-consistent lower bound on the pooled training-panel calibration fold)**.** *Let $\mathcal{M}_t$ denote the storage pool at the end of cycle t, and let $\pi_t := \Pr\left(\mathrm{EM} = 1 \,\middle|\, e \in \mathcal{M}_t\right)$ denote its precision under exact-match labelling, evaluated on the pooled training-panel calibration fold. Assume (i) the storage gate is the fixed-threshold rule that admits an entry whenever its calibrated composite satisfies $u_{stored} \geq \tau_{store}$, with τ_{store} frozen across cycles and π_{store} the empirical calibration-fold precision at τ_{store} on a pooled training-panel fold exchangeable with the cycle-$t+1$ candidate pool at the pooled axis, (ii) the retroactive re-verification pass enforces the prune-threshold rule with $\tau_{retro} < \tau_{store}$ (any entry whose re-scored composite falls below τ_{retro} is removed; entries at or above τ_{retro} are*

retained, with the committed composite either raised or lowered to the new value), and (iii) the calibration-fold precision among entries that survive the retroactive pass is π_{retro}. Then

$$\pi_{t+1} \;\geq\; \min\Big(\pi_{store},\, \pi_{retro}\Big), \tag{4.3}$$

where π_{store} and π_{retro} are the calibration-fold expected precisions of the storage gate and the retroactive prune respectively, both reported on the pooled training-panel axis. The bound is stated on the pooled training-panel cal-fold axis only; per-benchmark slices of the candidate pool may violate the cal-fold-to-pool exchangeability premise on the per-bench axis, and the per-benchmark eval-fold π_{store} is therefore reported as a within-scope-permitted diagnostic in the empirical receipt below rather than as a separate falsification target of Equation (4.3)*.*

Proof sketch. At cycle $t+1$, $\mathcal{M}_{t+1}$ is the disjoint union of two sets: $\mathcal{M}_t^{\text{kept}}$, the entries from $\mathcal{M}_t$ that survived retroactive re-verification, and $\mathcal{N}_{t+1}$, the entries newly admitted by the storage gate. By assumption (ii) and (iii), every entry in $\mathcal{M}_t^{\text{kept}}$ has a re-scored composite at least τ_{retro}, so its precision is bounded below by π_{retro}. By assumption (i) and exchangeability between the calibration fold and the candidate pool, the empirical precision of the calibration fold at τ_{store} projects forward as a lower bound on the precision of $\mathcal{N}_{t+1}$, which is therefore at least π_{store}. The precision of a union of disjoint sets is a convex combination of their precisions, bounded below by the smaller, giving Equation (4.3). □

Empirical receipt (calibration-consistent floor): Per-cycle pool purity reported in Chapter 5 should not drop below $\min(\pi_{\text{store}}, \pi_{\text{retro}})$ on the pooled training-panel calibration-fold axis the theorem registers. The pre-specified cycle-zero calibration-fold anchor at the locked threshold $\tau_{\text{store}} = 0.60$ is $\pi_{\text{store}} \approx 0.64$, with the retroactive prune at $\tau_{\text{retro}} = 0.50$ supplying the second floor; an empirical pooled-cal-fold pool-purity reading below the smaller of the two would falsify the exchangeability precondition or the retroactive-prune contract. Across the realised six-cycle trajectory the pooled cal-fold pool-purity reading remains in the range 0.72 to 0.75, clearing the registered floor by nine to eleven percentage points at every cycle, with the per-benchmark minimum on the pooled cal-fold axis of the training panel resting at 0.66 on the TriviaQA stream chunk at cycle five and the retroactive-prune pass firing at every non-aborted cycle with a pruned share that declines monotonically across the committed-cycle subsequence from thirteen percent down to four percent. The bound is one-sided and re-evaluable at every cycle from the cycle's own pooled training-panel calibration fold. The receipt is a pool-level guarantee on the storage class on the pooled cal-fold axis and does not bound the held-out generation accuracy of the fine-tuned adapter; the cycle-three regression discussed in Section 5.2.5 operates on the training-side gradient channel and is therefore outside the theorem's scope rather than a falsification of it.

Scope of the floor (per-benchmark eval-fold disclosure): The floor that Equation (4.3) registers applies to the pooled training-panel calibration fold and not to per-benchmark eval-fold slices, because the cal-fold-to-pool exchangeability premise on which the projection rests is a pooled-axis premise that the per-bench axis does not inherit. The per-benchmark eval-fold reading reported in Table 5.16 is therefore a within-scope-permitted diagnostic rather than a separate falsification

target of the bound. On the realised trajectory the FEVER eval-fold π_{store} dips below the registered 0.64 floor at cycles one through four (readings in the band $[0.614, 0.634]$ at cycles one, two, three, and four respectively) and recovers to 0.649 at cycle five, while the pooled cal-fold reading stays in the band $[0.732, 0.746]$ across the same committed cycles. The per-benchmark dip is consistent with FEVER's tighter precision regime under the locked configuration documented in Section 5.2.5: the per-benchmark composite refit under the shrinkage prior toward the pooled fit allocates per-benchmark slack toward the open-text factoid and five-option multiple-choice surfaces where the pre-routing confidence distribution is wider, and the FEVER per-benchmark eval-fold reading inherits the tighter operating point that the pooled axis absorbs. The reading is reported transparently because it is a real per-benchmark feature of the realised configuration, but it is not in scope for the floor of Equation (4.3) as stated above; the per-benchmark axis is registered as a future-work item in Chapter 6 alongside the longer-horizon receipts.

4.3.2 When filtering helps: the purification condition

The floor of Theorem 4.1 states how pure the stored pool is relative to the precision measured on the calibration fold. The identity below states when the storage gate makes the pool purer than the generator's raw output at all, which is the precondition the rest of the analysis relies on.

Proposition 4.2 (Pool-purity Bayes identity and purification condition)**.** *Let $p_c \in (0, 1)$ denote the base-model correctness rate at cycle c, and let TPR_c and FPR_c denote the storage gate's true-positive rate (the share of correct outputs it stores) and false-positive rate (the share of incorrect outputs it stores) at cycle c. The purity P_c of the stored pool satisfies*

$$P_c \;=\; \frac{p_c \cdot \mathrm{TPR}_c}{p_c \cdot \mathrm{TPR}_c \;+\; (1 - p_c) \cdot \mathrm{FPR}_c}, \tag{4.4}$$

and $P_c > p_c$ holds if and only if $\mathrm{TPR}_c > \mathrm{FPR}_c$, which is equivalent to the gate's balanced accuracy $\alpha_c = (\mathrm{TPR}_c + 1 - \mathrm{FPR}_c)/2$ exceeding one half. Under the symmetric operating-point reduction $\mathrm{TPR}_c = 1 - \mathrm{FPR}_c = \alpha_c$ the identity simplifies to $P_c = p_c\alpha_c/(p_c\alpha_c + (1 - p_c)(1 - \alpha_c))$.

Proof. Let $Y \in \{0, 1\}$ be the correctness of an output and $S \in \{0, 1\}$ the event that the gate stores it, with $\Pr(Y = 1) = p_c$, $\Pr(S = 1 \mid Y = 1) = \mathrm{TPR}_c$ and $\Pr(S = 1 \mid Y = 0) = \mathrm{FPR}_c$. Bayes' rule gives $P_c = \Pr(Y = 1 \mid S = 1) = p_c\mathrm{TPR}_c/\big(p_c\mathrm{TPR}_c + (1 - p_c)\mathrm{FPR}_c\big)$, which is Equation (4.4). Then $P_c > p_c$ is equivalent to $\mathrm{TPR}_c > p_c\mathrm{TPR}_c + (1 - p_c)\mathrm{FPR}_c$, that is, to $(1 - p_c)\mathrm{TPR}_c > (1 - p_c)\mathrm{FPR}_c$, which for $p_c < 1$ is $\mathrm{TPR}_c > \mathrm{FPR}_c$. Finally, $\alpha_c > 1/2$ is equivalent to $\mathrm{TPR}_c + 1 - \mathrm{FPR}_c > 1$, that is, again to $\mathrm{TPR}_c > \mathrm{FPR}_c$. □

Interpretation: Storage purifies only when the gate accepts correct outputs more often than incorrect ones. Reading the gate's decisions as noisy labels with flip rates $\rho_+ = 1 - \mathrm{TPR}_c$ and $\rho_- = \mathrm{FPR}_c$, the condition $\mathrm{TPR}_c > \mathrm{FPR}_c$ is the class-conditional noise condition $\rho_+ + \rho_- < 1$ under which learning from noisy labels remains possible [9]. Natarajan et al.'s guarantees additionally require a loss corrected for the noise rates, which CAEM does not apply, so the condition is used here as the precondition

for purification rather than as a transferred learning guarantee. This proposition is why the verifier-balanced-accuracy hypothesis is the precondition of the analysis: on a benchmark where the verifier's balanced accuracy is at or below one half, storage would leave the pool no purer than the generator's raw output. The identity also explains the slack in the calibration-consistent floor of Theorem 4.1: the realised pool purity sits at the Bayes posterior implied by the gate's operating point rather than at the worst-case calibration-fold precision.

Empirical receipt (precondition): The precondition is audited at cycle zero through the composite Cohen's d gate on the held-out evaluation fold reported in Chapter 5. The labelled purity-validation protocol that would supply TPR_c, FPR_c, and p_c on a held-out labelled fold at every cycle is registered as a post-trajectory diagnostic and was not consumed by the realised receipt set; the per-cycle residual $|P_c - P_c^{\mathrm{theory}}|$ is therefore listed in Chapter 6 as a future-work diagnostic rather than as a within-horizon empirical check.

4.3.3 Design heuristics for the cycle-to-cycle trajectory

The statements in this subsection describe how the stored pool is expected to evolve across self-improvement cycles. Each is derived under idealising assumptions that are stated with it: within-class Gaussian storage scores with shared variance, stationarity of the storage rate and of the verifier's discrimination across cycles, and independence of the verifier's errors across repeated re-verification passes. The realised system departs from each of these (the composite is refitted every cycle, the within-class variance contracts across cycles, and some verifier errors are systematic rather than independent), and the six-cycle horizon is too short to estimate the asymptotic quantities they involve. They are therefore recorded as heuristics with argument sketches rather than as theorems, and their empirical receipts are directional consistency checks rather than tests of a proven claim.

Storage-rate sensitivity

The storage-rate sensitivity heuristic describes how the per-cycle admission count moves with the verifier's discrimination on the calibration fold. The companion identity on pool purity as a function of base accuracy is a textbook derivative of the Bayes identity and is recorded in the scaffolding subsection below.

Heuristic 4.3 (Monotonicity under generator improvement (verifier ensemble held fixed across cycles))**.** Let d_t denote the calibration-fold class-separation Cohen's d between em-correct and em-incorrect storage scores at cycle t, observed as a joint property of the cycle-t generator's outputs on the cal fold, the locked verifier ensemble, and the cycle-t refit composite; let σ_t denote the empirical storage rate at the locked threshold τ_{store}. The verifier ensemble (MiniCheck plus the long-hypothesis judge plus the signal-extraction layer) is held fixed across the trajectory; the only cycle-indexed components are the generator (advanced by the self-improvement loop on committed cycles) and the per-cycle composite refit (per-signal isotonic, per-benchmark shrinkage child fits, boost-layer weights, and intercept). Assume the calibrated storage score is approximately Gaussian within each correctness class with shared variance ς^2 and class means $\mu_+(d_t)$, $\mu_-(d_t)$ separated by d_t. Let p_+, p_- be the calibration-fold base rates of correct and incorrect outputs, and let

$\phi_\pm = \phi\Big((\mu_\pm - \tau_{\text{store}})/\varsigma\Big)$ be the operating-point class densities. Then the sign of $\partial\sigma/\partial d$ equals the sign of $p_+\phi_+ - p_-\phi_-$. In particular, when the operating-point correct-class mass dominates the incorrect-class mass and $d_{t+1} > d_t$, the storage rate is non-decreasing: $\sigma_{t+1} \geq \sigma_t$.

Argument sketch. Under the within-class Gaussian assumption, $\sigma = p_+\Phi\Big((\mu_+ - \tau_{\text{store}})/\varsigma\Big) + p_-\Phi\Big((\mu_- - \tau_{\text{store}})/\varsigma\Big)$, where Φ is the standard normal CDF. Differentiating and using $\mu_+ - \mu_- = d \cdot \varsigma$ yields $\partial\sigma/\partial d = (1/2)\Big(p_+\phi_+ - p_-\phi_-\Big)$, so the sign of $\partial\sigma/\partial d$ is the sign of $p_+\phi_+ - p_-\phi_-$. The dominance condition holds whenever the threshold sits in the high-precision regime where the correct-class density exceeds the incorrect-class density at the operating point. □

Empirical receipt (storage rate in d, partial): The deciding statistic is the sign of the per-cycle storage-rate change paired with the sign of the per-cycle Cohen's d change on the calibration fold. Cycles for which the operating-point dominance condition $p_+\phi_+ > p_-\phi_-$ holds are flagged in the trajectory artefact; a cycle in which the dominance condition does not hold is reported with its sign rather than asserted as a monotonicity violation. The realised cal-fold trajectory yields four observation points (cycles one, two, three, and five; cycle four's cal-fold pass was skipped by the retention-guard abort that rolled the adapter back to the cycle-three weights, see Section 5.1.5). Of the three available transitions reported in Chapter 5, the cycle-one to cycle-two transition sign-aligns with the heuristic at large amplitude, the cycle-two to cycle-three transition shows a sub-noise Cohen's d delta whose sign carries no information about the derivative, and the cycle-three to cycle-five transition shows a small reverse-sign storage-rate movement under increasing Cohen's d with the dominance condition still holding at all four cycles. The reverse-sign transition is consistent with two confounds that the heuristic's Gaussian model does not represent. The first is the per-cycle composite refit: the heuristic treats the storage score as a fixed function whose only cycle-dependent quantity is the cal-fold class-mean separation d_t (itself a downstream observable of the SIL-improved generator viewed through the locked verifier ensemble), whereas the implementation also refits the calibrated probability composite on each cycle's calibration fold, so the realised cycle-to-cycle movement in σ_t bundles the generator-driven cal-fold discrimination shift with a composite-rescaling effect. The second is the shared-variance assumption: the empirical pooled-fold within-class standard deviation contracts across the trajectory rather than holding constant, so the argument's $\partial\sigma/\partial d$ decomposition is approximate cycle-to-cycle. The reported trajectory therefore evidences the sign-of-derivative prediction in the only large-amplitude transition observable on the cycle horizon and identifies the composite-refit and shared-variance confounds as the residual gap; a clean isolation would require freezing the composite weights and stratifying by class-variance estimate for an ablation cycle in which only the generator advances, which is registered as future work in Chapter 6.

Convergence

The convergence heuristic describes the per-cycle contraction of the precision gap to its long-run value. The realised six-cycle horizon is shorter than the horizon required to estimate a geometric rate cleanly, so the receipt is reframed as a bounded-band

stationarity test on the committed-cycle subsequence and the sharper asymptotic rate is forwarded to future work.

Heuristic 4.4 (Convergence envelope: bounded-band stationarity on the committed-cycle subsequence)**.** Let $\pi_\infty := \lim_{t\to\infty} \pi_t$ denote the asymptotic precision of the storage class, and let $G_t := |\pi_t - \pi_\infty|$ denote the cycle-t gap. Under the assumptions of Theorem 4.1 and Heuristic 4.3 and the additional stationarity assumption that the per-cycle expected admission count σ and the cal-fold class-separation d (the joint observable of the cycle-t generator, the locked verifier ensemble, and the cycle-t refit composite) stabilise over the trajectory, there exists a contraction rate $r \in (0, 1)$ such that, in expectation on the committed-cycle subsequence,

$$\mathbb{E}[G_t] \;\le\; (1 - r)^t \, G_0, \tag{4.5}$$

where σ is the per-cycle storage rate and d is the verifier's discrimination on the calibration fold. The thesis-side receipt registered against Equation (4.5) is the bounded-band stationarity reading on the realised committed-cycle subsequence, namely that π_t remains within a band whose width is small relative to the per-cycle binomial sampling noise; the existence-claim form of $r \in (0, 1)$ is what the realised six-cycle horizon adjudicates. A sharper proportionality of the form $r = \mathcal{O}\big(\sigma \cdot \Phi(d/\sqrt{2})\big)$ is a further Gaussian-model approximation that the argument sketch below records as design rationale; its receipt is a measured geometric-rate estimate registered as deferred future work in Chapter 6 under the at-least-fifteen-committed-cycle horizon precondition.

Argument sketch. The cycle-boundary update behaves as an asymptotically geometric contraction in expectation under the stationarity assumption stated above. Each cycle admits n_{new} new entries at the calibration-fold empirical precision π_{store} inherited from Theorem 4.1 and overwrites a subset of stored entries' composites under the locked verifier ensemble scoring the updated generator's outputs through the cycle-boundary-refit composite (Heuristic 4.3). The expected reduction in G_t per cycle is proportional to the expected number of correctness-flips per stored episode as the cal-fold class-separation d widens under generator advance, which scales as $\Phi(d/\sqrt{2})$ in the Gaussian model, and to the per-cycle admission count, which scales as σ. Composing these gives an expected per-cycle contraction factor $1 - r$ with r as in Equation (4.5). The rigorous geometric-envelope claim requires the stochastic-approximation framework of [2], the almost-supermartingale convergence theorem of [3] (supermartingale arguments of this kind underpin the convergence analysis of verifier-filtered retraining in [95]), and the rate analysis of [6], adapted to a bounded gap space; the rigorous supermartingale construction adapted from these works, at the level of a complete proof, is registered as a deferred future-work item in Chapter 6 alongside the at-least-fifteen-committed-cycle horizon precondition that would also unlock the geometric-rate test, and the body claim is therefore stated as an expected-value envelope on the committed-cycle subsequence rather than a sample-path bound on the calendar timeline. The stochastic-envelope refinement of Heuristic 4.7 replaces the calendar-timeline envelope with a Bernoulli-weighted contraction. □

Empirical receipt (bounded-band stationarity on the committed subsequence): The realised six-cycle trajectory of the calibration-fold storage precision is

reported in Chapter 5: π_t traces a non-monotone but tightly bounded band of width 0.014 across the cycles, with a small downward drift consistent with the per-cycle composite refit and the cycle-four retention abort recorded in Section 5.1.5. Two structural features of the realised horizon limit the strength of the geometric-rate test relative to the stationary expected-value claim of Equation (4.5). The first feature is binomial sampling noise: each per-cycle precision reading is a binomial proportion on the cycle's stream-chunk sample, and the resulting per-cycle sampling-noise half-width is comparable in magnitude to the realised cycle-to-cycle gap, so the predicted contraction signal sits at the edge of measurability at the empirical limit of this thesis. The second feature is the cycle-four commit indicator being zero, so the realised committed-cycle subsequence is shorter than the calendar horizon and the per-cycle stationarity of σ and d is mildly perturbed by the Phase-aligned composite refit. Under the stochastic-envelope refinement of Equation (4.7), the realised trajectory satisfies the bounded-distance prediction at the cycle horizon registered for this thesis: the cycle-five reading is the empirical limit against which the heuristic is evaluated, the band-width prediction holds, and the implementation regularisers registered at the start of the section account for the gap between this empirical limit and the theoretical asymptote of one. Falsification at the present horizon would require either a monotone-divergent trajectory that exits the band or a monotone-convergent trajectory whose realised gap profile is incompatible with the predicted shape across a horizon long enough to estimate the rate; neither is observed in the six-cycle window. A longer trajectory under fixed thresholds and a fixed composite would be required to test the asymptotic geometric rate directly; this is registered as a future-work item with a fifteen-committed-cycle precondition in Chapter 6.

Asymptotic hallucination floor

The memory-direct tier inherits the storage pool's precision; its asymptotic hallucination rate is therefore upper-bounded by the complement of the pool-precision floor. The full-pipeline decomposition that gathers the memory residual with the architectural false-negative rate is recorded in the scaffolding subsection below because the asymptotic vanishing of the memory residual is not directly observable on a six-cycle horizon.

Heuristic 4.5 (Tier-1 hallucination floor)**.** Let $\mathrm{CHM}_1(t)$ denote the composite hallucination metric of the memory-direct tier at cycle t. Under Theorem 4.1, Heuristic 4.4, and the consolidation policy that retains the highest-quality representative within each near-duplicate cluster,

$$\limsup_{t\to\infty} \mathrm{CHM}_1(t) \;\le\; \max\Big(1-\pi_{\mathrm{store}},\, 1-\pi_{\mathrm{retro}}\Big). \tag{4.6}$$

Argument sketch. The memory-direct tier serves the stored answer of the nearest neighbour, so its expected error rate equals the stored pool's expected impurity, $1-\pi_t$. By Theorem 4.1, $\pi_t \ge \min(\pi_{\mathrm{store}}, \pi_{\mathrm{retro}})$ for all t that satisfy the preconditions. By Heuristic 4.4, the gap $G_t = |\pi_t - \pi_\infty|$ is bounded above in expectation on the committed-cycle subsequence by a geometric envelope. Combining yields $\limsup_t \mathrm{CHM}_1(t) = 1-\pi_\infty \le 1-\min(\pi_{\mathrm{store}}, \pi_{\mathrm{retro}}) = \max(1-\pi_{\mathrm{store}}, 1-\pi_{\mathrm{retro}})$, which is Equation (4.6). The consolidation policy contributes by ensuring that retrieval at scale does not collapse onto a single contaminated cluster: under the

retain-highest-quality rule, the worst entry of any near-duplicate cluster is removed, so the memory-direct tier's effective precision is the maximum across each cluster rather than its mean. The limsup statement in Equation (4.6) tolerates bounded oscillation in the realised pool-precision sequence and should not be read as a pointwise monotone claim. □

Empirical receipt (Tier-1 floor, scope-limited): The heuristic bounds the asymptotic memory-direct hallucination rate by the complement of the stored-pool precision. Two scope conditions must be stated honestly. First, on a held-out evaluation panel constructed to be content-hash disjoint from the training stream, the memory-direct tier fires on a vanishing fraction of queries; the realised trajectory records seven memory-direct dispositions across the six-cycle evaluation horizon at fifteen hundred queries per cycle, all returning the correct stored answer. This rarity is by architectural design rather than a refutation, and it makes the asymptotic trajectory $\mathrm{CHM}_1(t)$ statistically unobservable on the realised evaluation distribution. The heuristic is therefore a capability bound rather than a trajectory prediction. Second, the conservative ceiling $1-\pi_{\text{store}} \approx 0.36$ derived from the cycle-zero calibration fold overshoots the realised stream-chunk pool complement, which lies in the range $[0.247, 0.276]$ across the realised cycles, and the retroactive prune rate observed at cycle five tightens the binding term to approximately 0.037. The testable surrogate is the targeted routing probe, in which the memory-direct tier dispatches correctly on verbatim queries to stored episodes at a rate consistent with the calibrated retrieval contract, and every realised Tier-one disposition on the held-out panel produced the correct stored answer at the empirical limit of the trajectory. We therefore present the heuristic as a pool-level capability bound whose empirical receipt is a probe rather than a trajectory, and we surface the disjointness condition explicitly in the threats to validity.

Equilibrium structure and self-correction

The heuristics below describe the equilibrium of the asymptotic statements above and the finite-cycle pruning behaviour that closes the memory-side residual. The self-correction heuristic supplies the directional prune-rate receipt on the committed-cycle subsequence; the stochastic-equilibrium heuristic supplies the clean empirical pass on the realised commit indicator and the asymmetric-rollback memory invariant. The deployment-cost reading that an explicit half-life statement would otherwise supply is recovered architecturally through the per-tier cost identity of Section 3.8.3, with a measured geometric-rate receipt at a horizon of at least fifteen committed cycles registered as future work in Chapter 6.

Heuristic 4.6 (Self-correction under improving cal-fold discrimination (locked verifier ensemble, advancing generator))**.** Any stored episode whose stored composite was produced when the locked verifier ensemble achieved cal-fold balanced accuracy α_c on the cycle-c generator's outputs is re-scored at every subsequent cycle boundary by the same locked verifier ensemble against the updated generator's outputs and the cycle-boundary-refit composite, yielding the cal-fold balanced accuracies $\alpha_{c+1}, \alpha_{c+2}, \ldots$. The verifier ensemble is fixed across the trajectory; the per-cycle changes in α are joint observables of the SIL-improved generator viewed through that fixed lens. Under the assumption $\alpha_{c+k} \geq \alpha_c \geq 1/2$ (the locked verifier discriminates increasingly well

on the cycle-$c+k$ generator's cleaner cal-fold distribution because the SIL loop has pushed the generator's distribution onto cleaner support), the conditional probability that the stored episode is correct given that it has survived k retroactive passes converges to 1 as $k \to \infty$, and any hallucinated episode is pruned within finitely many cycles in expectation, provided the verifier's errors on an entry are independent across passes; errors that recur at every pass are not removed by repetition (see the scope note after the argument sketch).

Argument sketch. Treat the indicator $S_k \in \{0, 1\}$ that a stored episode survives the k-th retroactive pass as a Bernoulli observation through the locked verifier ensemble, with success probability that depends on the latent correctness label $Y \in \{0, 1\}$ and on the cal-fold balanced accuracy α_{c+k} at the cycle boundary the pass runs at. Under the locked-verifier-ensemble assumption, $\Pr(S_k = 1 \mid Y = 1) \geq \alpha_{c+k}$ and $\Pr(S_k = 1 \mid Y = 0) \leq 1 - \alpha_{c+k}$, and the survival passes across cycles are conditionally independent given Y because each pass scores the entry against the cycle-boundary-refit composite over the same locked signal-extraction layer. The likelihood-ratio

$$\Lambda_k := \frac{\Pr(S_1, \ldots, S_k \mid Y = 1)}{\Pr(S_1, \ldots, S_k \mid Y = 0)} \;\geq\; \prod_{j=1}^{k} \frac{\alpha_{c+j}}{1 - \alpha_{c+j}}$$

satisfies $\Lambda_k \to \infty$ as $k \to \infty$ whenever $\alpha_{c+j} \geq \alpha_c > 1/2$ on infinitely many cycles, because each factor exceeds 1 by a margin bounded below. Bayes' rule then gives $\Pr(Y = 1 \mid S_1 = \cdots = S_k = 1) = \Lambda_k \Pr(Y = 1)/(\Lambda_k \Pr(Y = 1) + \Pr(Y = 0)) \to 1$.

For the finite-prune-time claim, fix a hallucinated entry with $Y = 0$. The per-cycle prune probability is at least $\alpha_{c+k} - (1 - \alpha_{c+k}) = 2\alpha_{c+k} - 1$, bounded below by a strictly positive constant $\delta := 2\alpha_c - 1 > 0$ under the non-decreasing-α premise. The survival probability across k cycles is at most $(1 - \delta)^k$, so the expected prune time is at most $\sum_{k=0}^{\infty}(1 - \delta)^k = 1/\delta < \infty$ by the geometric-series identity (Borel-Cantelli would give an almost-sure version). The independence-across-cycles assumption is shared with the composite-refit caveat raised on Heuristic 4.3: when the cycle-boundary refit dominates the cal-fold discrimination signal, the conditional-independence premise is approximate and the rate constant δ is conservative rather than tight. □

Scope of the independence premise: The convergence to one requires the verifier's errors on an entry to be independent across passes. Errors that are systematic, such as the same wrong answer accepted at every pass because the model and the verifier share a consistently held misconception [70], are not removed by repetition: in the fully correlated limit every pass after the first returns the same verdict and pool purity stays at the single-pass value of Proposition 4.2. Repeated passes therefore remove only the independent part of the verifier's error, and purity rises toward a ceiling below one that is set by the verifier's systematic errors, consistent with the finding that verifier-filtered retraining converges to the verifier's knowledge rather than to the truth [95]. The transfer-panel TruthfulQA reading (stored precision 0.522 at cycle zero, Table 5.31) is the empirical instance of such systematic errors.

Empirical receipt (directional, partial): The pre-specified receipt was a per-entry survival-cycles distribution across the prune lifecycle. The stored-entry schema

persists a storage-cycle field and a retroverified flag but does not carry a passes-survived counter, so the pruned-side histogram of survival counts is deferred to a production-deployment study under the runbook of Chapter 6. The survivor-side distribution remains recoverable from the cycle-by-cycle memory store snapshots and is reported in Chapter 5 alongside the prune-rate trajectory. The directional receipt on the committed-cycle subsequence is the per-cycle prune rate, which declines monotonically across the four committed retroactive passes from thirteen percent at cycle one to nine percent at cycle two to seven percent at cycle three to four percent at cycle five, consistent with the locked verifier ensemble removing low-quality entries at a decreasing rate as the pool's composition shifts toward higher-confidence survivors and as the SIL-improved generator's outputs widen the cal-fold class separation that the fixed verifier ensemble discriminates against. The decline is partially confounded at the late cycles by fresh-entry dilution from the still-growing pool, so the directional trend is the supported reading rather than a rate estimate. The asymptotic limit $k \to \infty$ is not directly testable at this horizon and is deferred together with the convergence-rate half-life to the longer-horizon future-work item.

Heuristic 4.7 (Stochastic equilibrium under retention rollback)**.** Let P denote the multi-modal probe panel registered in Section 5.1.5 with floor $\rho_{\min}$. Define the cycle-t commit indicator $X_t \in \{0, 1\}$ as the random variable taking value 1 if the post-fine-tune worst-probe ratio $\min_{p \in P} \rho_p(M_t)$ exceeds $\rho_{\min}$ and 0 otherwise, with M_t the post-fine-tune model. The commit event depends on the realised training pool composition Q_t (a function of the deferred-buffer state and the stream-chunk yield), the optimizer initialization randomness s_t, and the per-probe scoring noise ε. Then:

(i) The expected commit fraction $\pi_{\text{commit}} := \mathbb{E}[X_t]$ satisfies $\pi_{\text{commit}} \in (0, 1]$ on any trajectory where the worst-probe distribution intersects the floor, with $\pi_{\text{commit}} = 1$ only in the degenerate case where the retention guard never fires.

(ii) The convergence envelope of Heuristic 4.4 measured on the calendar timeline becomes

$$\mathbb{E}[G_t] \;\le\; (1 - \pi_{\text{commit}} \cdot r)^t \, G_0, \tag{4.7}$$

with r as in Equation (4.5); the original $(1-r)^c$ envelope still holds on the committed-cycle subsequence indexed by $c = \sum_{s \le t} X_s$.

(iii) The memory store $|M_t|$ and the per-tier hit-rate trajectory grow monotonically across all cycles because the stream-chunk generate-and-store path executes independently of X_t, while the parametric backbone advances only on the committed-cycle subsequence; the cycle-wise architecture therefore decouples deployment-time cost amortisation (memory tier growth) from training-time parametric progression (adapter commits).

(iv) When the worst-probe distribution centres within the probe-noise band ε of the floor, the trajectory enters a stochastic equilibrium in which commits and rollbacks alternate at rate π_{commit} near the parametric ceiling rather than departing it. The architectural property licensing this equilibrium is the asymmetric rollback registered in Section 5.1.5: every rolled-back cycle preserves the memory store and the deferred buffer exactly as they were at the previous committed cycle's close, so the trajectory can resume parametric advance from any subsequent committed cycle without losing accumulated state.

Scope of the sub-claims: Sub-claims (i) and (iii) are within-horizon receipts that the realised five-cycle trajectory either satisfies or falsifies directly: the commit-fraction reading is a Bernoulli sample mean on a fully observed indicator sequence, and the memory-monotonicity reading is a structural property of the runner's stream-chunk path that holds at every cycle boundary. Sub-claim (ii) is the calendar-timeline envelope inherited from Heuristic 4.4 and is reported as a projected receipt under the same horizon qualifier the parent heuristic carries. Sub-claim (iv), the equilibrium-location prediction, is the only sub-claim whose argument is purely asymptotic; its stationary-distribution argument is asymptotic, and the realised five-cycle horizon supplies a within-band observation but not a statistical test that distinguishes a stationary equilibrium from a longer-trajectory transient. The rigorous test of sub-claim (iv) requires the at-least-fifteen-committed-cycle horizon registered as future work in Chapter 6 alongside the geometric-rate test of Heuristic 4.4.

Argument sketch. Each cycle's commit event is governed by the indicator $X_t = \mathbb{1}[\min_{p \in P} \rho_p(M_t) \geq \rho_{\min}]$, where the worst-probe ratio depends on stochastic factors enumerated above; the indicator is therefore a Bernoulli random variable with success probability π_{commit} that is conditional on the trajectory's history of prior commits (through Q_t). Conditional on $X_t = 1$, the per-cycle gap contraction follows Heuristic 4.4 at the original rate r; conditional on $X_t = 0$, the gap is unchanged because the parameter update is reverted. Taking expectations gives $\mathbb{E}[G_{t+1} \mid G_t] = (1 - \pi_{\text{commit}} \cdot r) \cdot G_t$, and iterating yields Equation (4.7). The memory-tier growth claim follows from the runner's invariant that the stream-chunk step writes new verified entries to the memory store regardless of whether the cycle's adapter is committed. The equilibrium-location claim is heuristic: when the worst-probe ratio centres within the noise band of the floor, the Bernoulli rate stabilises at a stationary value bounded away from both endpoints, and the trajectory exhibits a bounded distance from the floor across cycles rather than monotone divergence or monotone convergence to a fixed interior point. □

Empirical receipt (stochastic equilibrium): The receipt is reported sub-claim by sub-claim, separating the within-horizon verifications from the prediction that the five-cycle trajectory cannot adjudicate. Sub-claim (i), the open-interval commit fraction, is verified directly: the realised commit-indicator sequence over the five-cycle trajectory is $X_1{=}1, X_2{=}1, X_3{=}1, X_4{=}0, X_5{=}1$ with worst-probe ratios $1.006, 0.949, 0.976, 0.886, 0.958$ under the locked floor $\rho_{\min} = 0.93$, and the realised commit fraction $\widehat{\pi}_{\text{commit}} = 4/5 = 0.80$ sits strictly in the open interval $(0, 1)$. Sub-claim (ii), the calendar-timeline envelope, inherits the bounded-band-versus-deferred-rate split from Heuristic 4.4 and is reported under the same horizon qualifier as the parent heuristic in Section 5.1.6. Sub-claim (iii), the asymmetric-rollback memory invariant, is also verified directly: the cycle-four abort, triggered by the TriviaQA-test probe collapsing to 0.886 below the floor, was followed by an immediate restore of θ_{prev}, but the memory store nonetheless grew across the abort boundary and again across cycle five because the stream-chunk generate-and-store path executed independently of the retention guard, so every cycle, whether committed or rolled back, contributed new verified entries to the memory store while the parametric backbone advanced only on the four committed cycles. Sub-claim (iv), the equilibrium-location prediction, is heuristic at the realised horizon and is

reported as such: the worst-probe ratios on cycles two through five all sit within 0.07 of the floor, qualitatively consistent with the predicted near-floor band, but the five-cycle horizon supplies a within-band observation that is not statistically distinguishable from a longer-trajectory transient at the empirical limit. The rigorous test of sub-claim (iv) requires the at-least-fifteen-committed-cycle horizon registered as future work in Chapter 6. The heuristic's two registered falsifiers, a degenerate commit rate or a memory store that stalls on rolled-back cycles, are both absent within the realised horizon.

Mathematical scaffolding

The four identities collected in this subsection underpin the argument sketches above but do not themselves admit a falsifiable empirical contribution at the empirical limit of the realised trajectory. They are recorded as design-rationale remarks rather than as theorems or heuristics so that the reader can locate the formal machinery in one place without confusing it with a receipt that the evaluation chapter is expected to discharge. Each remark preserves its original label so that cross-references resolved by name continue to point at the same identity.

Remark 4.8 (Monotonicity of pool purity in base accuracy). Differentiating the symmetric form of the Bayes identity in Proposition 4.2 with respect to p_c under fixed $\alpha > 1/2$ gives

$$\frac{\partial P_c}{\partial p_c} \;=\; \frac{\alpha(1-\alpha)}{\Big(p_c\alpha + (1-p_c)(1-\alpha)\Big)^2} \;>\; 0, \tag{4.8}$$

so pool purity is strictly increasing in base accuracy on $(0, 1)$ whenever the verifier's balanced accuracy exceeds one half. The identity formalises the intuition that improving the base generator strictly improves the pool the gate produces, with the gain proportional to the gate's discrimination through the $\alpha(1-\alpha)$ factor. The cross-cycle rank-correlation receipt originally registered against this identity is statistically uninformative at the empirical limit of the realised trajectory because the realised range of p_c produces a theoretical P_c span below the per-cycle sampling noise of P_c, and because the cycle-boundary recalibration policy lets α_c drift between cycles in violation of the fixed-α premise; the receipt is therefore retired in favour of the calibration-consistent floor reported under Theorem 4.1.

Remark 4.9 (Fixed-point convergence of the Bayes-purification recurrence). Define the recurrence $P_{c+1} = f(P_c)$ where $f(P) = P\alpha/(P\alpha + (1-P)(1-\alpha))$ under fixed verifier balanced accuracy $\alpha > 1/2$. The map $f : [0, 1] \to [0, 1]$ is strictly increasing on $(0, 1)$ with fixed points only at $\{0, 1\}$. For any $P_0 \in (0, 1)$ the sequence is strictly increasing and bounded above by one, so by the monotone-convergence theorem $P_c \to 1$ as $c \to \infty$. The recurrence is a statement about the pool-purity quantity P_c of Proposition 4.2, not about the base generator's held-out correctness rate; the fine-tuning step that closes each cycle is a parameter-efficient adapter update with the L_2 anchor of Equation (4.2), the multi-modal retention guard, and a strict training threshold, none of which implements an idealised posterior update on the held-out generator. The upper fixed point $P^* = 1$ is asymptotic and is not observable at the six-cycle empirical limit of this thesis; the implementation regularisers above, and verifier errors that are systematic rather than independent across cycles (see the scope note on Heuristic 4.6), pin the effective pool-purity ceiling strictly below one,

and the realised pool-purity trajectory is therefore expected to fluctuate above the calibration-consistent floor of Theorem 4.1 rather than to display strict monotone ascent.

Remark 4.10 (Full-pipeline hallucination decomposition). Let $\tau_1(t)$ denote the realised Tier-one hit rate on the cycle-t evaluation pool, let $\varepsilon_{\text{arch}}(t)$ denote the joint architectural false-negative rate of the four cascaded storage gates measured at cycle t, and let $H_{\text{mem}}(t) = 1 - \pi_t$ denote the memory-side hallucination contribution at cycle t, where π_t is the pool purity. By the law of total probability over the tier-routing event, the user-facing hallucination rate satisfies

$$H_t \;\leq\; \Big(1 - \tau_1(t)\Big) \cdot \varepsilon_{\text{arch}}(t) \;+\; \tau_1(t) \cdot H_{\text{mem}}(t). \tag{4.9}$$

The pool-purity term is bounded below by the calibration-consistent floor of Theorem 4.1, and the equilibrium residual at the implementation's effective limit decomposes as $H_\infty \approx (1 - \tau_1^*(D)) \cdot (1 - x(|\theta|, C))$ where $x(|\theta|, C)$ is the asymptotic correctness rate of the base generator with parameter count $|\theta|$ over retrieval corpus C. The vanishing-memory-residual asymptote $H_{\text{mem}}(t) \to 0$ is an infinite-horizon claim that is unobservable at the empirical limit of the trajectory for two structural reasons: the evaluation pool is content-hash disjoint from the training stream by design, pinning $\tau_1(t)$ at the noise floor on the realised horizon; and the retention-guard rollback, the parametric capacity of the base generator, and the regularisation anchor pin the pool-purity ceiling strictly below one. The deployment-mode setting registered in Chapter 6 under the production runbook, in which repeat queries from the served distribution re-enter the routing layer and the cycle horizon extends to the calendar trigger, is the regime in which the asymptotic statements of this decomposition become testable. The tier-partition decomposition is routing-agnostic in the partition step; the $\varepsilon_{\text{arch}}(t)$ term is measured against the realised post-classifier tier distribution of Section 4.2.5, so the corpus-conditional residual $1 - x(|\theta|, C)$ at the asymptote is the corpus-conditional bound on the retrieval-augmented branch of the post-classifier split, with the zero-shot branch's residual bounded by the parametric ceiling of the fine-tuned generator rather than by the corpus floor.

Remark 4.11 (Corpus-bounded hallucination floor). Let $K(C) \subseteq D$ denote the set of queries whose true answer is supportable by retrieved evidence from the corpus C. Under the grounded-reasoning convention, no system that asserts only claims supportable by C can achieve a hallucination rate strictly below $\Pr_D[q \notin K(C)]$, so the equilibrium residual of Remark 4.10 is bounded below by an information-theoretic floor determined by the corpus rather than by the architecture. The full set $K(C)$ is not directly observable, and the pre-specified proxy diagnostic, which would identify evaluation-fold queries whose top-k retrieval does not contain the gold-answer span and report the composite hallucination metric on that subset, was not consumed by the realised receipt set because the rectangular-record evaluation harness intentionally filters list-typed verifier fields and the proxy reconstruction therefore requires re-running retrieval against the dense passage index for every evaluation question across all six cycles. The proxy reconstruction is registered as a deferred receipt in Chapter 6; the corpus-bounded floor itself enters the present chapter as an interpretive limit on the architectural equilibrium rather than as a measured quantity at the empirical limit of the trajectory.

4.4 Chapter signpost

The methodology specified in this chapter is intentionally complete: every component, every threshold, and every fit-time procedure is registered before the empirical record is consulted. Chapter 5 reads this specification verbatim against the realised cycle trajectory, the external baseline panel, and the retrieval-utility-classifier on/off architectural ablation, and reports each registered hypothesis as supported, partially supported, or unsupported with the deciding statistic. The theorem and the proposition of Section 4.3 are checked there against their empirical receipts, and the design heuristics are read directionally against the realised trajectory at its empirical limit, so the reader can locate each architectural claim in the specification chapter and find the corresponding receipt in the evaluation chapter without ambiguity.

Chapter 5

Performance Evaluation and Analysis

5.1 Performance Evaluation

5.1.1 Experimental setup

Benchmarks and the training-versus-transfer split

The empirical programme uses five public benchmarks split into a training panel of three and a transfer panel of two. The training panel comprises FEVER [21] for factual claim verification, TriviaQA [18] for open-domain trivia, and CommonsenseQA [28] for five-option multiple-choice commonsense reasoning. The transfer panel comprises TruthfulQA [39] for common-misconception probes and StrategyQA [33] for implicit multi-step reasoning. The asymmetric split has a registered purpose: the per-benchmark calibrated probability composite (with shrinkage prior toward the pooled fit) and the self-improvement training pool draw exclusively from the training panel, while the transfer panel is held out as a measurement of generalisation under distribution shift; transfer-panel samples never enter the calibration fold, the seed pool, or the per-cycle stream chunks. The retention guard reads a registered multi-modal probe panel rather than a single fold: a general-knowledge multiple-choice probe on MMLU [34], a held-out open-text factoid probe drawn from the TriviaQA test fold, and a held-out commonsense multiple-choice probe drawn from the CommonsenseQA test fold; none of these probes is used as a target benchmark for the headline metrics. Two further benchmarks, HotpotQA and Natural Questions, were evaluated at cycle zero and dropped before the main run because the base generator's accuracy on them (0.09 and 0.17) was too low for the storage gate to admit a usable training pool (Section 1.6.1).

The cross-fold sample budget is summarised in Table 5.1. Each training benchmark contributes a 1000-sample seed pool for cold-start memory population, a 500-sample calibration fold for the per-signal isotonic curves and the per-benchmark composite refits, a 500-sample purity-validation slice for the empirical-precision audit, ten per-cycle stream chunks of benchmark-specific size (FEVER and TriviaQA at 2000 per cycle, CommonsenseQA at 700 per cycle to fit the smaller available train pool of approximately ten thousand samples after dedup), a 300-sample evaluation fold

for cycle-by-cycle headline metric reporting, and a 500-sample test fold reserved for the final post-trajectory comparative panel. Each transfer benchmark contributes a 300-sample evaluation fold and a test fold of up to 500 samples (clamped where the benchmark's dev split admits fewer than 800 samples). The pool counts are recorded in the registered dataset-splits manifest and verified disjoint at construction time by the pool builder under cross-benchmark and within-benchmark content-hash leakage guards.

Benchmark	**Panel**	**Seed**	**Calib**	**Purity**	**SIL chunks**	**Eval**	**Test**
FEVER	training	1000	500	500	10×2000	300	500
TriviaQA	training	1000	500	500	10×2000	300	500
CommonsenseQA	training	1000	500	500	10×700	300	500
TruthfulQA	transfer	0	0	0	0	300	500
StrategyQA	transfer	0	0	0	0	300	387

Table 5.1: Benchmark sample pools by fold. The per-cycle SIL chunks differ across training benchmarks because CommonsenseQA's available train pool of approximately ten thousand samples after dedup cannot support the same 10×2000 chunking that FEVER (one hundred and forty-five thousand) and TriviaQA (eighty-seven thousand) sustain; the asymmetry is corrected at the gradient step by the benchmark-balancing pool reweighting layer described in Section 4.2.10. The StrategyQA test fold is clamped to 387 because the benchmark's dev split admits fewer than the registered 500. Pool counts recorded in the experiment-run dataset-splits manifest; cross-benchmark and within-benchmark content-hash disjointness verified by the pool builder.

Metrics

Each benchmark is scored under the correctness rule appropriate to its task family. Exact match is the rule for FEVER (label-exact match against the supports / refutes / not-enough-info gold), for TriviaQA (gold-alias match, falling back to token-level F1 above the registered threshold), for CommonsenseQA (five-option multiple-choice label exact match), for TruthfulQA (best-alias match against the truthful and informative reference set), and for StrategyQA (yes / no label exact match). Token-level F1 is reported alongside exact match on the open-text TriviaQA benchmark where partial matches are evaluatively meaningful. Pooled exact match is the headline accuracy axis and is the basis for every paired statistical contrast against the external baseline panel.

The composite hallucination metric is the equal-weighted mean over an eight-axis failure-mode taxonomy comprising confident confabulation, factual fabrication, logical fabrication, off-topic answers, defensive evasion, template leakage, false refusal, and length-padded over-generation; a ninth axis (factual contradiction) is retained in the taxonomy but excluded from the default denominator because the corresponding subtype is structurally unreachable under the natural-language-inference judge in the deployed configuration. The composite evaluation score aggregates five axes (accuracy, epistemic quality, retention, calibration, and verifier reliability) under a geometric mean, so that a collapse on any single axis penalises the joint reading rather

than being averaged away. Each axis is registered as a value in $[0, 1]$ with one as best, the accuracy axis as pooled exact match across the five-benchmark evaluation panel, the epistemic-quality axis as one minus the pooled composite hallucination metric, the retention axis as the minimum across the three multi-modal probe ratios capped at one (so a probe that improves above its pristine baseline contributes one rather than a bonus), the calibration axis as $1 - 2\,\mathrm{ECE}$ on the post-refit expected calibration error truncated at 0.5 (so $\mathrm{ECE} = 0$ contributes one and $\mathrm{ECE} \geq 0.5$ contributes zero), and the verifier-reliability axis as $\max(2\,\mathrm{BA} - 1, 0)$ on the verifier's balanced accuracy (the Youden's-J rescaling under which the chance balanced accuracy of 0.5 contributes zero and a perfect verifier contributes one), with the balanced accuracy computed as the normal-cumulative-distribution proxy $\Phi(d/\sqrt{2})$ on the per-cycle composite Cohen's d between em-correct and em-incorrect outputs on the per-cycle training-panel calibration fold. The composite evaluation score is then

$$\mathrm{CES} = \left(\prod_{i=1}^{5} \mathrm{clamp}_{[\varepsilon,1]}(\mathrm{axis}_i) \right)^{1/5}, \quad \varepsilon = 0.01, \tag{5.1}$$

the geometric mean of the five axes clamped to $[\varepsilon, 1]$ to prevent a single degenerate axis from zeroing the score in regimes where one axis is uncomputable. The per-cycle reading of every axis and the geometric-mean CES across the realised trajectory are reported in Table 5.4. The retention ratio is defined in Equation (3.2) and is computed against every probe in the registered multi-modal panel before the first cycle (the pristine baseline) and after every cycle's fine-tune; the worst per-probe ratio governs the cycle-abort decision under the registered floor. Table 5.2 summarises the metric set and the benchmark family each is reported on.

The eight measurable subtypes are defined as follows. **Confident confabulation** fires when the answer is incorrect and the composite stored-quality score remains at or above 0.50, indicating the gate has not flagged the wrongness as low-confidence. **Factual fabrication** fires when the answer is incorrect and the weakest-link atomic-fact entailment against the retrieved evidence is below 0.30, indicating that at least one sub-claim in the answer fails per-claim grounding. **Logical fabrication** fires when the answer is incorrect and the chain-to-answer entailment under the reasoning chain is below 0.30, indicating that the model's own reasoning trace does not entail the answer it produced. **Off-topic** fires when the answer is incorrect and the question-to-answer relevance is below 0.50, indicating the response is topically unrelated to the question. **Defensive evasion** fires when the answer is incorrect and the prediction matches a registered evasive-pattern regular expression covering phrases such as "the context does not mention", "no information is provided", and "cannot determine", indicating the model emitted a refusal-shaped string instead of attempting an answer when evidence was available. **Template leakage** fires when the prediction matches a registered template-leak regular expression covering placeholder strings such as "<concise factual answer>" and "[your answer]", regardless of correctness, capturing prompt-template artefacts that leaked into the output. **False refusal** fires when the answer is correct yet the pipeline decision was abstain or discard, indicating the model refused to commit to an answer it had in fact produced correctly. **Length-padded over-generation** fires when the answer is incorrect and the prediction exceeds 600 characters, capturing verbose padding correlated with wrongness.

The eight subtypes are not mutually exclusive: a single incorrect output may

simultaneously fail grounding, miss the topic, and exceed the length bound. The composite hallucination metric is the equal-weighted mean across the eight per-subtype rates rather than their union, so the per-axis decomposition surfaces failure-mode composition for diagnosis without double-counting under aggregation. The thresholds named above are registered before the calibration phase and held fixed across the trajectory.

The ninth axis, factual contradiction, fires when the answer is incorrect and the contradiction-class probability under the natural-language-inference judge exceeds 0.30. This rate is structurally pinned at zero under the binary supported-versus-unsupported judge that ships in the deployed configuration, because the judge does not produce a contradiction class; the rate becomes meaningful only under the three-class refutation-aware judge ablation. The rate is therefore reported separately in the per-cycle subtype table for completeness rather than being folded into the composite hallucination metric. The refutation-aware judge that would close this coverage gap is registered as future work in Chapter 6.

Metric	**Reported on**	**Definition / aggregation**
Exact match (EM)	all five benchmarks	exact match of normalised prediction against gold (or any gold alias)
Token-level F1	TriviaQA	best-alias token-level F1; complementary to EM on the open-text factoid surface
Label accuracy	CommonsenseQA, FEVER, StrategyQA	exact label match against the multiple-choice or supports/refutes/NEI gold
Composite hallucination metric (CHM)	all five benchmarks	equal-weighted mean over the eight measurable failure-mode subtypes
Composite evaluation score (CES)	pooled	geometric mean over accuracy, epistemic quality, retention, calibration, verifier reliability
Retention ratio $\min_p \rho_p(M_t)$	multi-modal probe panel	worst per-probe ratio $\mathrm{EM}(M_t, p)/\mathrm{EM}(M_0, p)$ over MMLU + TriviaQA-test + CommonsenseQA-test (Equation (3.2))

Table 5.2: Reported metric set with per-benchmark applicability and aggregation rule.

Hardware, seed, reproducibility

The empirical programme runs end to end on a single rented graphics processor of the consumer-grade thirty-two-gigabyte class, with the cycle-zero calibration phase, the six-cycle main run terminated by the equilibrium-saturation gate at the cycle-five close after the cycle-four retention abort, the external baseline panel, and the retrieval-utility-classifier on/off architectural ablation at the cycle-three close sharing the same hardware envelope to keep latency and cost readings comparable. Memory pressure is managed through gradient checkpointing on the low-rank-adapter fine-tune (the underlying backbone is frozen across the cycle, so only the adapter parameters cross gradients) and through chunked batching on the verifier ensemble's longer-input

judges. The shipped optimiser is a thirty-two-bit decoupled-weight-decay AdamW over the bounded adapter parameter set; the eight-bit-moment optimiser variant is registered but inactive in the low-rank-adapter configuration, because the per-parameter quantisation noise outweighs the modest optimiser-state-memory saving against the thirty-two-gigabyte envelope. The long-hypothesis verifier judge runs under chunked batching after the calibration-phase out-of-memory event documented in Section 3.7.1. The random-state surface is reduced to a single recorded seed propagated through the generator, the sampler, the dropout layers, and the FAISS index; this seed is recorded in the locked configuration files and is the single input required by the reproducibility recipe in Appendix E. The pre-self-improvement state, including the cold-start memory snapshot, the cycle-zero calibrated probability composite, the registered storage and deferral cut-points, and the cycle-zero evaluation outputs, is published as a snapshot bundle on a public dataset host, with the artefact inventory and access protocol catalogued in Appendix E.

5.1.2 Trajectory of headline metrics

Every trajectory reading reported in this subsection and the four following is evaluated under the locked configuration whose calibration discipline, parameter selection sweep, locked-configuration receipts, calibration-versus-evaluation gap, long-hypothesis judge ablation, and validation-gate verdict are documented in detail in Section 5.2. Section ordering follows the CSE400 template (performance evaluation before final design adjustments) rather than the natural causal order (locked configuration $\rightarrow$ trajectory under that configuration); a reader who prefers the causal order may skip ahead to Section 5.2 and return.

Pooled trajectory

The pooled headline metrics across the realised six-cycle measurement horizon are reported in Table 5.3. The trajectory closes at cycle five when the equilibrium-saturation gate terminated the run after the post-rollback subsequence stabilised. Pooled exact match across the five-benchmark panel reads 0.541 at the cycle-zero baseline and 0.517 at the cycle-five close, a net change of -2.4 percentage points on the exact-match axis with non-monotone interior movement. The composite hallucination metric falls from 0.135 to 0.109 across the same window, a relative reduction of nineteen percent across the cycle-zero to cycle-five window; the within-trajectory composite-hallucination minimum of 0.107 falls at cycle two. The multi-modal retention panel reports each of the three registered probes against its pristine cycle-zero capability at every cycle close: MMLU sits comfortably above the registered floor of 0.93 at every cycle (the realised range is $[0.975, 1.041]$, so general-knowledge retention is never close to the abort condition), while the two open-text and commonsense probes oscillate closer to the floor. The TriviaQA-test ratio falls to 0.886 at cycle four, the first in-flight breach of the floor in the realised trajectory and the event that triggers the asymmetric rollback registered in Section 5.1.5. Cycles one, two, three, and five commit cleanly, giving a realised commit fraction of 0.80 that supplies the receipt for the stochastic-equilibrium claim of Heuristic 4.7. The cycle-five composite-evaluation-score peak, the asymmetric-rollback preservation of the memory store across the cycle-four abort, and the MMLU-versus-open-text asymmetry under which general-knowledge retention is structurally insensitive to the self-improvement update

jointly anchor the architectural reading of the trajectory.

Configuration	Cycle	n/bench	Pooled headline EM	Pooled headline CHM	Retention probes (ratio vs. pristine) MMLU	TQA-test	CSQA-test	Outcome
Reference floor (no CAEM architecture; same Qwen-2.5-3B-Instruct backbone)								
B1 (zero-shot)	—	300	0.507	0.120	—	—	—	reference
CAEM trajectory (full architecture: memory + verifier + SIL + retention guard)								
CAEM	C0	300	0.541	0.135	–	–	–	baseline
CAEM	C1	300	0.543	0.112	1.041	1.013	1.006	COMMIT
CAEM	**C2**	300	**0.544**	**0.107**	0.975	0.949	0.982	COMMIT ⋆
CAEM	C3	300	0.532	0.115	0.992	0.987	0.976	COMMIT
CAEM	C4	300	0.532	0.115	1.018	**0.886**	0.982	ABORT
CAEM	C5	300	0.517	0.109	0.999	1.013	0.958	COMMIT

Headline contrast (CAEM C2 vs. B1 zero-shot): ΔEM = +0.037 (+7.4% relative); ΔCHM = −0.013 (−11.0% relative).

Table 5.3: Per-cycle pooled headline metrics, the multi-modal retention panel, and the no-CAEM B1 reference floor on a matched backbone. Pooled EM and CHM are unweighted means over the five panel benchmarks; TruthfulQA EM uses the Haiku-judged rule. The shaded C2 row marks the trajectory's headline cycle (highest pooled EM, lowest pooled CHM). The retention panel reports the post-fine-tune ratio against the pristine cycle-zero capability on each of the three registered probes (MMLU, TriviaQA-test, CommonsenseQA-test); the worst probe per cycle is underlined; the registered floor $\rho_{\min} = 0.93$ triggers the asymmetric rollback of Section 5.1.5. Table 5.34 reports the pooled eight-baseline comparison; Table 5.35 reports the per-benchmark McNemar significance receipts.

Joint-axis composite reading: The five quality axes that the architecture is designed to deliver jointly (accuracy, epistemic quality, retention, calibration, verifier reliability) are aggregated under the geometric mean of Equation (5.1) as the registered Composite Evaluation Score, and the per-cycle reading is reported in Table 5.4. The table is CAEM-only because three of the five axes are properties of a cyclic, calibrated, verifier-gated system and do not transfer to the single-pass external baselines whose two cross-system-comparable axes (ACC and EPI) are reported instead in Table 5.34 and the per-benchmark significance panels. The trajectory carries three findings the geometric mean surfaces. First, CES rises from 0.593 at cycle zero to 0.710 at the trajectory-end cycle, a 0.117 absolute lift on the joint-axis scale that decomposes into a verifier-reliability lift of 0.274 on the Youden's-J rescaling, the dominant driver of the joint-axis trajectory and the empirical receipt that the verifier ensemble's discrimination compounds across cycles as the self-improvement loop narrows the generator's distribution onto the verified-pool support. Second, cycle four exhibits the retention-abort dip: the worst-probe ratio of 0.886 crosses below the registered floor of $\rho_{\min} = 0.93$ and pulls the geometric mean down by 0.013 relative to cycle three while ACC, EPI, CAL, and VER all hold band-stable, the joint-axis receipt that the geometric aggregation surfaces a single-axis collapse rather than averaging it into oblivion. Third, cycle five recovers the worst-probe ratio to 0.958 above the floor and the geometric-mean CES recovers to the trajectory peak of 0.710, the joint-axis read of the post-rollback recovery that the pooled headline metrics alone do not visibly carry; the cycle-five CES peak is the joint-axis maximum even though cycle two carries the within-trajectory optima of the five-benchmark pooled exact match and composite hallucination metric reported in Table 5.3 and Table 5.34 (the training-panel pooled exact match of Table 5.5 peaks at cycle one, at

0.584), the empirical receipt that the verifier-reliability axis kept compounding past cycle two while the accuracy axis declined slightly across cycles three through five, so the within-trajectory best cycle on the EM-and-CHM cross-system-comparable axes (cycle two) and the within-trajectory best cycle on the joint five-axis quality surface (cycle five) are different points by design and are reported as complementary anchors rather than competing ones.

Cycle	ACC	EPI	RET	CAL	VER	CES
C0	0.541	0.865	1.000	0.898	0.174	0.593
C1	0.543	0.888	1.000	0.912	0.325	0.678
C2	0.544	0.893	0.949	0.901	0.415	0.704
C3	0.532	0.885	0.976	0.919	0.407	0.703
C4*	0.532	0.885	**0.886**	0.919	0.407	**0.690**
C5	0.517	0.891	0.958	0.915	**0.448**	**0.710**

* Cycle four was aborted under the retention guard ($\rho_{\min} = 0.93$); CAL and VER reuse cycle three.

Table 5.4: CAEM-only per-cycle Composite Evaluation Score across the five quality axes the architecture delivers jointly, computed under the canonical `ces_score` formulas of Equation (5.1). The shaded cycle-five row marks the trajectory peak; cycle four exhibits the retention-abort dip. CES is reported for CAEM only because three of the five axes (RET, CAL, VER) are properties of a cyclic, calibrated, verifier-gated system that do not transfer cleanly to single-pass baselines; the two cross-system-comparable axes (ACC, EPI) are reported in Table 5.34 and the per-benchmark significance panels of Table 5.35 and Table 5.36. The per-cycle decomposition and the C4 dip / C5 recovery mechanism are discussed in the surrounding prose.

Per-benchmark trajectory

The per-benchmark and per-subtype decompositions of the pooled reading are reported in Table 5.5 and Table 5.6. The per-benchmark trajectory shows the headline reduction in composite hallucination is unevenly distributed across the panel. On the training panel, FEVER and TriviaQA exact match readings sit within a band of six percentage points across the six cycles, with a net cycle-zero to cycle-five change of about one percentage point on each, while CommonsenseQA exact match declines from 0.763 at the cycle-zero baseline to 0.717 at the cycle-five close. On the transfer panel, TruthfulQA exact match under the Haiku-judged correctness rule declines from 0.380 to 0.300 across the trajectory, while StrategyQA stabilises at 0.640 from cycle three onward. The eight-axis subtype decomposition surfaces the per-failure-mode contributions to the headline composite. Confident confabulation rises from 0.087 at cycle zero to approximately 0.27 by cycle three and plateaus there (0.269 at cycle five), the largest single subtype movement in the trajectory and a direct consequence of the gate admitting some incorrect answers above the registered cut-point. False refusal moves in the opposite direction, falling from 0.305 to 0.064 across the same window as memory accumulates and the model stops refusing answers it can supply. Factual fabrication and logical fabrication trend gently downward (from 0.364 to 0.303 and from 0.283 to 0.231 respectively). Off-topic and template-leak subtypes are pinned at zero throughout; defensive evasion, over-long generation, and the structurally suppressed factual-contradiction subtype are near-zero. The

composite hallucination metric on the bottom row of Table 5.6 is the equal-weighted mean of these eight measurable subtype rates, so its nineteen percent decline reflects the joint movement of a large false-refusal reduction, moderate factual and logical fabrication reductions, and a substantial confident-confabulation increase rather than a uniform shrinkage on every axis.

Panel	Benchmark	B1 ref	C0	C1	C2	C3	C4	C5	ΔB1→C2	ΔB1→C5
train	FEVER (EM)	0.453	0.487	0.533	0.533	0.503	0.507	0.477	+0.080 (+17.7%)	+0.024 (+5.3%)
	FEVER (CHM)	0.159	0.139	0.113	0.102	0.120	0.119	0.126	−0.057 (−35.8%)	−0.033 (−20.8%)
train	TriviaQA (EM)	0.293	0.463	0.480	0.493	0.463	0.463	0.453	+0.200 (+68.3%)	+0.160 (+54.6%)
	TriviaQA (CHM)	0.175	0.156	0.113	0.109	0.121	0.120	0.125	−0.066 (−37.7%)	−0.050 (−28.6%)
train	CommonsenseQA (EM)	0.703	0.763	0.740	0.717	0.743	0.743	0.717	+0.014 (+2.0%)	+0.014 (+2.0%)
	CommonsenseQA (CHM)	0.096	0.077	0.086	0.093	0.086	0.085	0.091	−0.003 (−3.1%)	−0.005 (−5.2%)
train	**Pooled (EM)**	**0.483**	0.571	0.584	**0.581**	0.570	0.571	0.549	**+0.098 (+20.3%)**	**+0.066 (+13.7%)**
train	**Pooled (CHM)**	**0.143**	0.124	0.104	**0.101**	0.109	0.108	0.114	**−0.042 (−29.4%)**	**−0.029 (−20.3%)**
transfer	TruthfulQA (EM)	0.433	0.380	0.317	0.323	0.310	0.307	0.300	−0.110 (−25.4%)	−0.133 (−30.7%)
	TruthfulQA (CHM)	0.051	0.150	0.133	0.116	0.119	0.122	0.104	+0.065 (+127.5%)	+0.053 (+103.9%)
transfer	StrategyQA (EM)	0.650	0.610	0.647	0.653	0.640	0.640	0.640	+0.003 (+0.5%)	−0.010 (−1.5%)
	StrategyQA (CHM)	0.120	0.154	0.113	0.115	0.128	0.127	0.099	−0.005 (−4.2%)	−0.021 (−17.5%)

Table 5.5: Per-cycle exact match and composite hallucination metric, broken out by benchmark, with the training-panel pooled rows (shaded) merged in, the no-CAEM B1 reference in the leftmost data column, and absolute + relative deltas against the best CAEM cycle (C2) and the trajectory-end cycle (C5) in the rightmost two columns. Each benchmark contributes two rows: EM on top, CHM below. The training panel (FEVER, TriviaQA, CommonsenseQA) supplies the calibration fold and the SIL training pool; the transfer panel (TruthfulQA, StrategyQA) is held out. The transfer panel has no pooled row because TruthfulQA and StrategyQA move in opposite directions across the trajectory; averaging would mask both signals. TruthfulQA EM uses the Haiku-judged correctness rule; all cells evaluated on the $n = 300$ evaluation fold.

Subtype	B1 ref	C0	C1	C2	C3	C4	C5
Conf. confab.	0.268	0.087	0.223	0.239	0.268	0.270	0.269
Fact. fabric.	0.334	0.364	0.327	0.308	0.319	0.319	0.303
Logic. fabric.	0.276	0.283	0.267	0.243	0.277	0.273	0.231
Off-topic	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Def. evasion	0.019	0.044	0.007	0.006	0.002	0.002	0.001
Templ. leak	0.000	0.000	0.000	0.000	0.000	0.000	0.000
False refusal	0.060	0.305	0.067	0.057	0.047	0.048	0.064
Over-long	0.004	0.000	0.001	0.003	0.005	0.005	0.003
CHM (mean of 8)	**0.120**	0.135	0.112	0.107	0.115	0.115	0.109

Table 5.6: Per-cycle decomposition of the composite hallucination metric into its eight measurable failure-mode subtypes, pooled across the five-benchmark evaluation panel ($n = 1500$ per cycle), with the B1 reference column on the left. The bottom row is the equal-weighted mean of the eight subtypes; the ninth axis (factual contradiction) is structurally pinned at zero under the binary NLI judge and excluded from the default denominator. Per-subtype thresholds are defined in Section 5.1.1. The B1-to-cycle contrasts on each subtype are discussed in the surrounding prose.

Training versus transfer contrast

The training-panel versus transfer-panel contrast is reported in the shaded pooled rows of Table 5.5 and in the per-benchmark rows below them. On the training panel, pooled exact match lifts from 0.483 at the B1 floor to a within-trajectory peak of 0.584 at cycle one and holds at 0.549 at the trajectory-end cycle, while pooled composite hallucination metric drops from 0.143 at the B1 floor to a within-trajectory minimum of 0.101 at cycle two, a 29.4% relative reduction at the best cycle and a 20.3% relative reduction at the trajectory-end cycle. The transfer panel is reported per benchmark rather than pooled because TruthfulQA and StrategyQA move in opposite directions across the trajectory: TruthfulQA exact match declines below the B1 reference at every cycle because the dense passage substrate amplifies the question's misconception framing rather than refuting it (the retrieval-amplification mechanism documented in Section 5.6.4), while StrategyQA exact match holds within noise and StrategyQA composite hallucination metric drops below the B1 floor at the trajectory-end cycle. Averaging the two transfer benchmarks into a single pooled cell would mask both the regression on TruthfulQA and the improvement on StrategyQA; the per-benchmark rows in Table 5.5 are the honest reading on the transfer panel. The per-benchmark delta columns confirm the heterogeneous architectural-contribution picture: a clean exact-match win on TriviaQA (+20.0 percentage points at the best cycle, +16.0 at the trajectory-end cycle) and on FEVER composite hallucination (-35.8% relative at the best cycle), a near-tie on CommonsenseQA (the five-option multiple-choice surface saturates near the backbone's parametric ceiling), a near-tie on StrategyQA, and a regression on TruthfulQA on both axes against both reference points.

5.1.3 Four-outcome decision distribution

The per-cycle distribution across the four registered storage-gate outcomes and the per-outcome exact-match rate are reported in Table 5.7. The trajectory covers the four committed cycles for which the calibration-fold log was persisted; the cycle-zero cold-start state precedes the first calibration refit, and the cycle-four state was discarded under the retention rollback. Across the four committed cycles, the store share rises from 42 percent of the calibration fold at cycle one to 59 percent at cycle five, the defer share grows from 12 to 29 percent, and the discard share contracts from 25 to 3 percent as the cycle-boundary composite refit converts much of the early-cycle discard mass into either store or defer entries. The abstain share contracts from 21 to 10 percent as the pre-routing confidence sharpens and the early-exit fires on a smaller share of the candidate pool. The structural sorting receipt that anchors the fixed-threshold contract is the strict ordering of per-outcome exact-match rates: at every committed cycle, the store class records the highest exact-match rate (in the range 0.73 to 0.75), the defer class the second highest, the discard class third, and the abstain class lowest (which falls from 0.38 at cycle one to 0.17 at cycle five as the early-exit filter increasingly catches genuinely confused queries rather than borderline ones). The gate's separation between store and the next-highest outcome widens from a 14 percentage-point gap at cycle one to a 24-to-30 percentage-point gap from cycle two onward (30, 28, and 24 points at cycles two, three, and five, so the widening is not monotone), the empirical signal that the cycle-boundary refit sharpened the gate's discrimination after cycle one rather than degrading it under

distribution drift.

	Store		Defer		Discard		Abstain	
Cycle / benchmark	share	EM	share	EM	share	EM	share	EM
(a) Pooled trajectory across the training panel (FEVER + TriviaQA + CommonsenseQA)								
C1	42.1%	0.74	12.0%	0.60	25.0%	0.58	20.9%	0.38
C2	67.2%	0.74	18.7%	0.44	5.4%	0.38	8.7%	0.25
C3	60.4%	0.75	26.0%	0.47	5.5%	0.41	8.1%	0.22
C5	58.6%	0.73	28.5%	0.49	3.4%	0.46	9.5%	0.17
(b) Per-benchmark snapshot at the cycle-three anchor (calibration fold, $n = 500$ each)								
FEVER	55.1%	0.76	38.1%	0.47	4.3%	0.52	2.6%	0.38
TriviaQA	40.5%	0.79	28.9%	0.45	10.6%	0.30	20.0%	0.17
CommonsenseQA	85.7%	0.72	11.1%	0.55	1.6%	0.88	1.6%	0.62

Table 5.7: Per-cycle four-outcome decision distribution on the training-panel calibration fold (panel a) and per-benchmark snapshot at the cycle-three anchor (panel b). The store/defer/discard/abstain shares and within-outcome EM are reported under the locked storage-gate cut-points; on the pooled panel the EM ordering store > defer > discard > abstain is preserved at every reported cycle, the structural receipt that the fixed-threshold gate sorts the candidate pool by correctness as the signal distribution drifts. Per benchmark (panel b), TriviaQA preserves the full ordering, while the smallest cells break it: on FEVER the discard class (21 queries) scores above the defer class, and on CommonsenseQA the discard and abstain classes (8 queries each) score above the defer class, with the discard class also above the store class. Panel (b) shares are over the queries that received a gate decision (494, 499 and 496 of 500). Cycle zero and the aborted cycle four are absent by design.

The underlying per-benchmark four-outcome decomposition is reported in Table 5.8 at the within-trajectory best cycle (C2) and the trajectory-end cycle (C5). Each row reports the correct and incorrect counts within each of the storage, deferral, and discard outcomes on each benchmark, alongside the realised per-bench storage precision π_{store}. The storage class is more precise than the defer class in every row, and the full within-bench ordering $\pi_{\text{store}} > \text{prec(defer)} > \text{prec(discard)}$ holds in five of the ten rows of the matrix (TriviaQA and TruthfulQA at both cycles, StrategyQA at cycle five); in the other five (FEVER and CommonsenseQA at both cycles, StrategyQA at cycle two) the discard class, which holds between one and one hundred and three entries in those rows, is more precise than the defer class. The ordering is therefore a partial receipt that the fixed-threshold gate sorts the candidate pool by correctness on each benchmark family rather than only at the panel level.

Cycle	Benchmark	Store ✓	Store ×	Store π	Defer ✓	Defer ×	Discard ✓	Discard ×
Cycle two (the within-trajectory best cycle)								
C2	FEVER	803	240	0.770	344	419	58	45
C2	TriviaQA	605	209	0.743	262	329	61	163
C2	CommonsenseQA	465	172	0.730	34	19	1	0
C2	TruthfulQA	33	58	0.363	23	48	11	38
C2	StrategyQA	57	20	0.740	125	73	1	0
Cycle five (trajectory-end close)								
C5	FEVER	764	210	0.784	474	432	10	2
C5	TriviaQA	532	269	0.664	284	430	8	13
C5	CommonsenseQA	428	177	0.707	32	32	11	6
C5	TruthfulQA	23	52	0.307	38	101	3	13
C5	StrategyQA	45	10	0.818	100	53	0	1

Table 5.8: Per-benchmark four-outcome decomposition at the within-trajectory best cycle (C2) and the trajectory-end cycle (C5). *Store* ✓ and *Store* × are the correct and incorrect counts in the storage class; *Store* π is the realised per-bench storage precision π_{store} at the locked cut-point $\tau_{\text{store}} = 0.60$. The training-panel benchmarks (FEVER, TriviaQA, CommonsenseQA) admit large storage classes at π_{store} in the band $[0.66, 0.78]$ across the two anchor cycles; the transfer-panel TruthfulQA storage class is small (storage gate does not fire on transfer-panel queries by design, so the few storage cells reflect the calibration-fold rescore rather than live storage) and StrategyQA records the highest per-bench precision at 0.818 on the trajectory-end cycle. The within-bench ordering $\pi_{\text{store}} > \text{prec(defer)} > \text{prec(discard)}$ holds in five of the ten rows (C2 TriviaQA, C2 TruthfulQA, C5 TriviaQA, C5 TruthfulQA, C5 StrategyQA). The first inequality, $\pi_{\text{store}} > \text{prec(defer)}$, holds in every row; the exceptions are C2 FEVER, C2 CommonsenseQA, C2 StrategyQA, C5 FEVER and C5 CommonsenseQA, where the discard class is more precise than the defer class (in C2 CommonsenseQA and C2 StrategyQA the discard class holds a single entry).

The per-benchmark abstention rate on the evaluation fold across the realised trajectory is reported in Table 5.9. FEVER abstention collapses from 64.3% at the cold-start cycle to 5.3% at the trajectory-end cycle as the per-cycle composite refit sharpens the pre-routing confidence on the supports/refutes/not-enough-info surface; CommonsenseQA abstention is near-zero at every cycle ($\leq 1\%$) because the five-option multiple-choice surface forces a committed answer; TruthfulQA abstention is high and stable in the band $[23\%, 33\%]$ across the trajectory, the empirical receipt that the calibrated-probability composite triggers the abstention branch on misconception-bait probes where pre-routing confidence is low (the retrieval-amplification-and-abstention trade-off discussed in Section 5.6.4).

Cycle	FEVER	TriviaQA	CSQA	TruthfulQA	StrategyQA
C0	64.3%	29.0%	0.0%	33.3%	43.0%
C1	24.3%	17.7%	0.3%	27.3%	2.3%
C2	5.7%	17.3%	1.0%	29.3%	8.0%
C3	10.7%	20.0%	0.0%	25.0%	4.0%
C4	10.0%	21.0%	0.0%	25.3%	2.3%
C5	5.3%	26.0%	0.3%	23.0%	30.3%

Table 5.9: Per-benchmark abstention rate across the realised six-cycle trajectory on the evaluation fold ($n = 300$ per benchmark). Abstention is the share of queries on which the pipeline returned an explicit abstention token (`decision = ABSTAIN` under the four-outcome registered tree, or a prediction string prefixed by the abstention marker). The pattern is benchmark-dependent: FEVER abstention collapses from 64.3% at the cold-start cycle to 5.3% at the trajectory-end cycle as the per-cycle composite refit sharpens the pre-routing confidence and the early-exit filter on the supports/refutes/not-enough-info surface; CommonsenseQA abstention is near-zero at every cycle ($\leq 1\%$) because the five-option multiple-choice surface forces a committed answer; TruthfulQA abstention is high and stable in the band $[23\%, 33\%]$ across the trajectory, the empirical receipt that the calibrated-probability composite triggers the abstention branch on misconception-bait probes where pre-routing confidence is low, and the abstention behaviour interacts with the Haiku-judged correctness rule discussed in Section 5.6.4. The cycle-five StrategyQA reading is anomalous (returns from 2.3% at C4 to 30.3% at C5); the cycle-five composite refit re-tightened the abstention branch after the C4 rollback re-anchored the adapter at the cycle-three checkpoint.

5.1.4 Three-tier router efficiency

Per-tier hit rate trajectory and per-tier exact match

The per-cycle pooled tier shares on the evaluation fold and the within-tier exact-match rates are reported together in Table 5.10, so that the cost-quality tradeoff registered in Chapter 4 is visible without the reader cross-referencing two tables. The architecture predicts two coupled redistributions across cycles. The first is that the memory-direct tier share grows as the verified episodic store fills with retroactively refreshed cached answers to recurring queries. The second is that the zero-shot tier share grows as the self-improvement loop folds verified retrieval-augmented reasoning into the generator's parametric capacity and as the retrieval-utility classifier of Section 4.2.5 diverts queries on which retrieval is predicted to hurt from the retrieval-augmented tier to the zero-shot tier per Table 4.2. The realised reading on the evaluation fold makes the second redistribution empirically muted and the first structurally invisible. The first is structurally invisible because the evaluation fold is held content-hash disjoint from the training stream by construction, so the memory-direct tier fires on only seven of the nine thousand evaluation queries across the six cycles, a rate of approximately 0.08 percent. The second is empirically muted because the pooled zero-shot share moves from 48.7 percent at cycle zero to 50.2 percent at cycle five along a non-monotonic path that peaks at 55.2 percent at cycle two and reverts thereafter. The within-tier exact-match readings nevertheless confirm the architectural claim where the data permits a test. The memory-direct

tier returns the stored answer on every realised firing, recording an exact-match rate of 1.0 on each of the seven hits and consistent with the Tier-1 floor heuristic of Heuristic 4.5. The zero-shot tier records an exact-match rate of 0.602 at cycle zero, twelve percentage points above the retrieval-augmented tier's cycle-zero rate of 0.482, the self-improvement-as-distillation receipt under which the parametric tier outperforms the retrieval tier at cold-start because retrieval pulls evidence that is not always topical. The zero-shot rate declines gently to 0.537 at cycle five while the retrieval-augmented rate stabilises around 0.50, so the relative ordering is preserved across the trajectory and the within-tier gap narrows rather than reverses.

Cycle / benchmark	**Tier 1 (memory)** share	EM	**Tier 2 (zero-shot)** share	EM	**Tier 3 (RAG)** share	EM	**Pooled EM**
(a) Pooled trajectory across the five-benchmark evaluation panel							
C0	0.0%	–	48.7%	0.602	51.3%	0.482	0.541
C1	0.0%	–	49.8%	0.585	50.2%	0.502	0.543
C2	0.1%	1.000	55.2%	0.568	44.7%	0.514	0.544
C3	0.1%	1.000	44.7%	0.566	55.1%	0.503	0.532
C4	0.1%	1.000	45.6%	0.564	54.3%	0.504	0.532
C5	0.1%	1.000	50.2%	0.537	49.7%	0.497	0.517
(b) Per-benchmark snapshot at the cycle-five close (eval fold, $n = 300$ each)							
FEVER	0.0%	–	43.3%	0.500	56.7%	0.459	0.477
TriviaQA	0.3%	1.000	37.0%	0.423	62.7%	0.468	0.453
CommonsenseQA	0.0%	–	82.3%	0.745	17.7%	0.585	0.717
TruthfulQA	0.3%	1.000	64.3%	0.321	35.3%	0.255	0.300
StrategyQA	0.0%	–	24.0%	0.639	76.0%	0.640	0.640

Table 5.10: Per-cycle pooled tier share and per-tier exact match across the five-benchmark evaluation panel (panel a), with a per-benchmark snapshot at the cycle-five close (panel b). Tier 1 (memory-direct) fires on a small share of queries by construction of the content-hash-disjoint evaluation fold; every realised Tier 1 firing returns the correct stored answer (the empirical receipt for Heuristic 4.5). Panel b copies the C5 rows of Table 5.11; TruthfulQA EM uses the Haiku-judged correctness rule. The Tier-2-versus-Tier-3 contrast at cold start and the per-benchmark tier-mix heterogeneity surfaced by panel b are discussed in the surrounding prose.

The per-benchmark per-tier exact match at the cold-start (C0), within-trajectory best (C2), and trajectory-end (C5) cycles is reported in Table 5.11. Each Tier-1 firing in the table returns the correct stored answer (EM = 100.0), the empirical receipt for the Tier-1 floor of Heuristic 4.5; the Tier-2-versus-Tier-3 ordering at cold-start holds on four of five benchmarks (FEVER, CommonsenseQA, TruthfulQA, StrategyQA) where the parametric tier already outperforms retrieval before any self-improvement, and on those four benchmarks the ordering is preserved at cycles two and five rather than reversed under the SIL distillation step, with two exceptions: FEVER at cycle two (51.6 against 54.5) and StrategyQA at cycle five (63.9 against 64.0).

Cycle	Benchmark	T1 EM	T1 n	T2 EM	T2 n	T3 EM	T3 n	Pooled EM
Cycle zero (cold-start; no Tier 1 firings)								
C0	FEVER	—	0	55.3	132	43.5	168	48.7
C0	TriviaQA	—	0	46.1	89	46.4	211	46.3
C0	CommonsenseQA	—	0	78.5	275	52.0	25	76.3
C0	TruthfulQA	—	0	42.6	197	29.1	103	38.0
C0	StrategyQA	—	0	68.4	38	59.9	262	61.0
Cycle two (the within-trajectory joint EM/CHM best cycle)								
C2	FEVER	—	0	51.6	122	54.5	178	53.3
C2	TriviaQA	—	0	51.0	96	48.5	204	49.3
C2	CommonsenseQA	—	0	72.4	275	64.0	25	71.7
C2	TruthfulQA	100.0	1	35.3	204	25.3	95	32.3
C2	StrategyQA	—	0	66.4	131	64.5	169	65.3
Cycle five (trajectory-end close)								
C5	FEVER	—	0	50.0	130	45.9	170	47.7
C5	TriviaQA	100.0	1	42.3	111	46.8	188	45.3
C5	CommonsenseQA	—	0	74.5	247	58.5	53	71.7
C5	TruthfulQA	100.0	1	32.1	193	25.5	106	30.0
C5	StrategyQA	—	0	63.9	72	64.0	228	64.0

Table 5.11: Per-benchmark per-tier exact match at the cold-start cycle (C0), the within-trajectory best cycle (C2), and the trajectory-end cycle (C5), with per-tier sample counts. EM values are reported as percentages; n is the count of evaluation queries routed to that tier on that benchmark in that cycle. Each Tier-1 firing in the table returns the correct stored answer (EM = 100.0), the empirical receipt for the Tier-1 floor of Heuristic 4.5. The Tier-2 EM exceeds Tier-3 EM at cold-start on FEVER, CommonsenseQA, and StrategyQA (the parametric tier already outperforms retrieval on these surfaces before any self-improvement), and the gap holds across the trajectory except on FEVER at cycle two (51.6 against 54.5) and StrategyQA at cycle five (63.9 against 64.0); on TriviaQA the Tier-3 retrieval-augmented tier holds a slight EM lead because the open-text factoid surface is the regime where dense retrieval supplies factual context that the parametric model lacks. The pooled-EM column in the right margin is the sample-weighted mean over the three within-tier readings and matches the headline pooled-EM trajectory of Table 5.5.

Per-tier latency and cost

The per-cycle pooled wall-clock latency on the production batched evaluation log is reported in Table 5.12. Each row reports the mean per-query time over the full one thousand five hundred evaluation queries at the indicated cycle, in milliseconds and seconds, alongside the per-cycle wall-clock evaluation time in seconds. The pooled per-query time stays in a tight band of thirteen to fifteen seconds across the six cycles, consistent with the modest tier-share movement reported in Table 5.10. Per-tier intrinsic latency is deliberately not decomposed from the production log because the per-sample latency record reflects the per-batch wall time divided by the deployment batch size of thirty-two, so any per-tier amortisation would reflect the per-batch tier mix rather than the per-tier compute cost. The intrinsic per-tier latency is a separate measurement that the writeup-hardware envelope cannot host because the dense passage index does not fit in the available random-access memory, and the measurement is registered as a future-work item in Chapter 6. The deployment-cost

projection from Equation (3.5) reads the pooled per-query time on the right-most measured cycle and the per-cycle decomposition; the per-tier intrinsic latency would sharpen the projection further but does not change its sign.

Configuration / cycle / benchmark	Mean per-query latency	Per-cycle wall time
(a) Reference floor (no CAEM architecture; pooled across the five-benchmark evaluation panel)		
B1 zero-shot	142 ms	—
(b) CAEM pooled trajectory across the five-benchmark evaluation panel		
C0	14.4 s	6.02 h
C1	13.6 s	5.68 h
C2	13.8 s	5.75 h
C3	14.1 s	5.89 h
C4	14.0 s	5.85 h
C5	14.7 s	6.11 h
(c) Per-benchmark snapshot at the cycle-five close (eval fold, $n = 300$ each)		
FEVER	15.5 s	—
TriviaQA	16.3 s	—
CommonsenseQA	8.6 s	—
TruthfulQA	17.8 s	—
StrategyQA	15.1 s	—

Table 5.12: Per-cycle pooled wall-clock latency on the production batched evaluation log (panel b), per-benchmark snapshot at the cycle-five close (panel c), and the no-CAEM reference floor (panel a). Latency units are picked per row to match magnitude: B1 runs in milliseconds, CAEM per-query in seconds, CAEM per-cycle wall-clock in hours. All CAEM cells report the production-batched amortised mean at the deployment batch size of thirty-two over the full one thousand five hundred evaluation queries per cycle. The intrinsic per-tier latency at unit batch is not decomposed from the batched log because the per-sample record reflects per-batch wall time divided by batch size; the unit-batch measurement is registered as a future-work item in Chapter 6.

Why the safety override fires

The safety override forces dispatch off the score-formula path whenever the pre-routing confidence falls below the registered floor of 0.38 (Section 4.2.4); under the locked configuration the retrieval-utility classifier of Section 4.2.5 is consulted at the safety-veto fall-through and selects between the zero-shot and retrieval-augmented tiers per Table 4.2, so the override no longer pins the retrieval-augmented tier unconditionally. The per-cycle firing rate on the evaluation fold is reported in Table 5.13. Across the realised trajectory the override fires on 63 of the nine thousand evaluation queries observed across six cycles and five benchmarks (0.700%): 62 CommonsenseQA queries after the per-benchmark refit raises that benchmark's floor from 0.38 to 0.491 at cycle three and to 0.569 at cycle five (12, 12 and 38 at cycles three, four and five), and one StrategyQA query at cycle five with $u_{\text{pre}} = 0.376$. Under the original 0.38 floor the override would have fired only once. The empirical reading is that the pre-routing confidence stays above the safety floor on most of the panel distribution, and the firings follow the CommonsenseQA floor refit rather than a broad loss of confidence, so the override is a contractual safety mechanism that activates rarely rather than a frequent intervention. A trajectory on a more adversarial distribution, or a higher

safety floor, would exercise the override more visibly; the present reading, sixty-three firings concentrated on CommonsenseQA after its floor refit, is therefore registered as a low-rate receipt rather than as evidence that the mechanism is unnecessary, since the mechanism is a contract against worst-case inputs that the evaluation distribution does not exercise.

Source / benchmark	n	**Fires**	**Rate**	**Min u_{pre}**
(a) Evaluation fold (six cycles × five benchmarks × 300 samples)				
FEVER	1 800	0	0.000%	0.491
TriviaQA	1 800	0	0.000%	0.499
CommonsenseQA	1 800	62	3.444%	0.402
TruthfulQA	1 800	0	0.000%	0.437
StrategyQA	1 800	1	0.056%	**0.376**
Eval subtotal	9 000	**63**	0.700%	—
(b) SIL stream chunks (training-panel candidate pool, six cycles; pooled floor 0.38 only)				
FEVER	10 000	0	0.000%	0.485
TriviaQA	10 000	0	0.000%	0.479
CommonsenseQA	3 500	3	0.086%	**0.358**
Stream-chunk subtotal	23 500	**3**	0.013%	—
Combined (a) + (b)	**32 500**	**≥ 66**	**≥ 0.203%**	**0.358**

Table 5.13: Safety override firing rate measured by the inequality the router branches on, $u_{\text{pre}} < \phi_{\text{safe},b}$, on the pre-routing confidence (Section 4.2.4). The floor is per benchmark: 0.38 throughout, except CommonsenseQA, whose floor is refit to 0.491 at cycles three and four and to 0.569 at cycle five. Panel (a) audits the served evaluation fold with these floors: the override fires on 63 of 9 000 queries (0.700%), 62 of them CommonsenseQA queries after its floor rises (12 at cycle three, 12 at cycle four, 38 at cycle five) and one StrategyQA query at cycle five. Panel (b) audits the SIL stream chunks with the pooled 0.38 floor only, so its CommonsenseQA count from cycle three onward is a lower bound. The dumped `safety_override` boolean is not persisted in the eval JSON serializer, so the u_{pre} inequality is reported directly; the two measurements are identical by construction.

5.1.5 Retention and continual learning

The retention story is reported on two complementary axes. The first axis is the retention-guard input on the registered multi-modal probe panel, already reported per cycle in Table 5.3 alongside the cycle outcomes; the worst per-probe ratio drives the asymmetric-rollback decision at every cycle boundary and triggered one rollback at cycle four under the realised configuration. The second axis is the per-benchmark stability of the headline exact match across the trajectory, reported in Table 5.14. The two axes are complementary rather than redundant: the probe-panel axis measures the retention contract that the gate enforces, while the headline-benchmark axis measures the empirical continual-learning behaviour on the same surface the architecture is evaluated against. The training panel records small declines on the open-text and factual-verification benchmarks (FEVER and TriviaQA both move by 1.0 percentage points across the six cycles, well within the cycle-to-cycle noise band) and a noticeable decline on CommonsenseQA (a 4.7 percentage-point decrease that surfaces the catastrophic-forgetting risk the multi-modal retention guard is

registered against). The transfer panel splits cleanly into a forward-gain reading on StrategyQA (+3.0 percentage points across the trajectory) and a substantial decline on TruthfulQA (−8.0 percentage points under the Haiku-judged correctness rule). The TruthfulQA decline tracks an increase in abstention rather than a degradation in retrieved knowledge: when the model is uncertain, the gate routes the answer to abstention, which Haiku-judged TruthfulQA scoring records as incorrect; the same trajectory shows TruthfulQA composite hallucination falling from 0.150 to 0.104 in Table 5.5, so the answer set is becoming more honest while the strict-correctness rate declines. Forward and backward transfer instruments in the formal sense of the continual-learning literature are not measurable on this trajectory because the evaluation fold is fixed across cycles; the matched-pair design that supplies those instruments is registered as a future-work item in Chapter 6.

Panel	Benchmark	B1 ref	C0	C5	Δ EM C0→C5 (pp)	min	max	range	Outcome
train	FEVER	0.453	0.487	0.477	-1.00	0.477	0.533	5.67pp	stable
train	TriviaQA	0.293	0.463	0.453	-1.00	0.453	0.493	4.00pp	stable
train	CommonsenseQA	0.703	0.763	0.717	-4.67	0.717	0.763	4.67pp	forgetting
transfer	StrategyQA	0.650	0.610	0.640	+3.00	0.610	0.653	4.33pp	forward gain
transfer	TruthfulQA	0.433	0.380	0.300	-8.00	0.300	0.380	8.00pp	forgetting

Table 5.14: Per-benchmark exact-match stability across the realised six-cycle trajectory on the evaluation fold, with the B1 reference floor in the leftmost data column and the Δ EM column reporting the cycle-zero-to-cycle-five drift. The Outcome column labels each row as stable, forgetting, or forward gain under a two-percentage-point threshold. The retention probe ratios in Table 5.3 measure the retention-guard input on the multi-modal probe panel; the present table reports the complementary continual-learning view from the headline-benchmark axis. Formal forward and backward transfer instruments are not measurable on this trajectory because the evaluation fold is fixed across cycles; the matched-pair design is registered as future work in Chapter 6.

The per-cycle self-improvement training trajectory is reported in Table 5.15. The training-batch size grows from 107 episodes at the first fine-tune to 5 006 episodes at the trajectory-end cycle as the verified memory pool accumulates; the final training loss declines monotonically across the four uninterrupted cycles ($0.258 \rightarrow 0.171 \rightarrow 0.109 \rightarrow 0.070$ at C1 through C4), the empirical receipt that the SIL loop is making coherent gradient progress on the verified pool. The cycle-four entry is the in-flight reading at the moment the retention guard fired: the TriviaQA-test probe ratio of 0.886 crossed below the registered floor of $\rho_{\min} = 0.93$ and triggered the asymmetric rollback, re-anchoring the adapter at the cycle-three checkpoint while preserving memory. The cycle-five training loss upticks slightly to 0.091 as the post-rollback recovery cycle restarts gradient descent from the cycle-three weights against the cycle-five labelled batch.

			Retention probe ratio			
Cycle	$n_{\text{episodes used}}$	**Final loss**	**MMLU**	**TQA-test**	**CSQA-test**	**Outcome**
C1	107	0.258	1.041	1.013	1.006	commit
C2	1 474	0.171	0.975	0.949	0.982	commit
C3	2 966	0.109	0.992	0.987	0.976	commit
C4	3 624	0.070	1.018	**0.886**[†]	0.982	abort (retention rollback)
C5	5 006	0.091	0.999	1.013	0.958	commit (post-rollback recovery)

[†] Cycle-four TriviaQA-test ratio of 0.886 crossed below the registered floor of $\rho_{\min} = 0.93$, triggering the asymmetric rollback.

Table 5.15: Per-cycle self-improvement loop training trajectory with the full three-probe retention panel. The episode-count column reports the number of memory-pool entries that passed the $\tau_{\text{train}} = 0.70$ filter and entered the per-cycle SIL training batch (sourced from `retroverify_cycle*.json::fine_tune.n_episodes_used`); the batch grows monotonically from 107 at the first fine-tune to 5 006 at the trajectory-end cycle. The final-loss column reports the loss at the last gradient step of each per-cycle fine-tune; the trajectory $0.258 \to 0.171 \to 0.109 \to 0.070$ across the four uninterrupted cycles is monotone decreasing, the empirical receipt that the SIL loop is making coherent gradient progress on the verified pool. The cycle-five uptick to 0.091 is the post-rollback recovery reading after the cycle-four retention abort re-anchored the adapter at the cycle-three checkpoint. The three retention-probe columns report the post-fine-tune ratio against the pristine cycle-zero capability on each of the registered multi-modal probes: MMLU (general-knowledge multiple choice), TQA-test (held-out open-text factoid), and CSQA-test (held-out five-option multiple choice). MMLU stays above the floor at every cycle in the band $[0.975, 1.041]$, so the retention contract is governed in practice by the two open-text and commonsense probes rather than by general-knowledge retention; the cycle-four TriviaQA-test ratio of 0.886 is the first in-flight breach of $\rho_{\min} = 0.93$ in the realised trajectory and the event that triggered the asymmetric rollback registered in Section 5.1.5.

5.1.6 Theoretical predictions checked against the trajectory

The empirical receipts in this section check the pool-purity theorem and the purification condition of Section 4.3 against the realised six-cycle trajectory at the empirical limit, and read the design heuristics of Section 4.3.3 directionally against the same trajectory, alongside the four mathematical scaffolding identities collected as remarks. The heuristic checks are consistency checks rather than tests of proven claims. The one-glance verdict mapping is reported in Table 5.27 at the end of the section.

Pool purity

The calibration-consistent lower bound of Theorem 4.1 predicts $\pi_{t+1} \geq \min(\pi_{\text{store}}, \pi_{\text{retro}})$ on the pooled training-panel calibration-fold axis the theorem registers, where π_{store} is the empirical storage precision at the locked cut-point $\tau_{\text{store}} = 0.60$ and π_{retro} is the precision among entries that survive the retroactive prune at $\tau_{\text{retro}} = 0.50$. The pre-specified cycle-zero anchor at $\pi_{\text{store}} \approx 0.64$ supplies the floor side of the bound on the pooled cal-fold axis. The per-cycle realised pool-purity readings on both the calibration fold (the theorem-anchored reading on the pooled training-panel axis that populates the memory pool) and the evaluation fold (the deployment-realised reading on held-out samples) are reported in Table 5.16. The cal-fold pooled reading remains in the band $[0.732, 0.746]$ across the six-cycle horizon, clearing the registered 0.64 floor by nine to eleven percentage points at every reported cycle;

the eval-fold pooled reading remains in the band $[0.693, 0.759]$, clearing the same floor by five to twelve percentage points. The eval-fold pooled reading tracks the cal-fold pooled reading within five percentage points at every cycle, the empirical receipt that the calibration-anchored projection holds with substantial slack under pooled exchangeability between the cal fold and the deployment-relevant held-out fold. The per-benchmark minimum on the pooled cal-fold axis across the trajectory sits at 0.6637 on the TriviaQA stream chunk at cycle five, still 2.4 percentage points above the floor. On the per-benchmark eval-fold axis, which sits outside the pooled-axis scope of Equation (4.3), the FEVER eval-fold π_{store} dips below the registered 0.64 floor at cycles one, two, three, and four (readings 0.634, 0.630, 0.617, and 0.614 in Table 5.16) and recovers to 0.649 at cycle five, while TriviaQA and CommonsenseQA per-bench eval-fold readings stay in the band $[0.704, 0.797]$ across the same committed cycles. The FEVER per-bench eval-fold dip is a within-scope-permitted diagnostic rather than a falsification of Equation (4.3) (consistent with the pooled-axis exchangeability scope registered on the theorem in Theorem 4.1), because the pooled-axis exchangeability premise the bound rests on is not inherited by the per-bench axis; the same mechanism is documented in Section 5.2.5 as the per-benchmark composite refit under the shrinkage prior toward the pooled fit absorbing the tighter FEVER operating point on the open-text factual-verification surface, and the disclosure is surfaced here at the receipt site so that the per-bench axis is reported explicitly alongside the pooled-axis pass. Cycle four contributes no new cal-fold reading because the retention-guard rollback skipped the cycle-boundary refit under the asymmetric-rollback contract. The empirical receipt is therefore that the calibration-consistent lower bound holds with substantial slack at every cycle within the realised horizon on the pooled cal-fold axis the theorem registers, with the per-benchmark eval-fold FEVER dip at cycles one through four reported as the explicit within-scope-permitted disclosure on the per-bench axis. The purification condition of Proposition 4.2 expresses the realised pool purity at the empirical operating point as $P_c^{\text{theory}} = p_c \cdot \text{TPR}_c / (p_c \cdot \text{TPR}_c + (1 - p_c) \cdot \text{FPR}_c)$, which exceeds the base rate exactly when the gate's balanced accuracy exceeds one half; that precondition is audited at cycle zero (Table 5.32), and the identity itself explains the slack between the calibration-consistent floor and the realised pool purity rather than serving as an independent within-horizon testable prediction.

Cycle	Cal-fold pooled		Eval-fold pooled		Eval-fold per-bench π_{store}		
	n	π_{store}	n	π_{store}	FEVER	TriviaQA	CSQA
C0	575	0.746	336	0.759	0.661	—	0.779
C1	631	0.740	515	0.726	0.634	0.797	0.736
C2	1 001	0.736	534	0.693	0.630	0.733	0.705
C3	899	0.746	565	0.712	0.617	0.741	0.750
C4*	—	—	566	0.710	0.614	0.733	0.754
C5	868	0.732	504	0.700	0.649	0.704	0.725

* Cycle four was aborted under the retention guard; the cal-fold refit was skipped, but the eval-fold reading is computed against the rolled-back C3 composite.

Table 5.16: Per-cycle empirical storage precision π_{store} at the locked cut-point $\tau_{\text{store}} = 0.60$, reported on both the calibration fold (the theorem-anchored reading that the storage gate sees during fitting and that populates the memory pool) and the evaluation fold (the deployment-realised reading on held-out samples). The cal-fold pooled column is the empirical receipt for H1 and Theorem 4.1: the calibration-anchored lower bound projects forward to the cycle-t pool under exchangeability, and the realised cal-fold π_{store} stays in the band $[0.732, 0.746]$ across every reported cycle, clearing the registered 0.64 floor of Theorem 4.1 by between 9 and 11 percentage points. The eval-fold pooled column is the deployment-relevant reading: it stays in the band $[0.693, 0.759]$ across the trajectory, clearing the same floor by between 5 and 12 percentage points. The eval-fold reading tracks the cal-fold reading within 5 percentage points at every cycle, the empirical receipt that the cal-fold anchor projects forward without substantial generalisation loss. The per-bench eval-fold columns surface the heterogeneity that the pooled reading hides: FEVER's eval-fold π_{store} sits in the band $[0.61, 0.66]$ while TriviaQA and CommonsenseQA sit in the band $[0.70, 0.80]$, consistent with FEVER's tighter precision regime under the locked configuration discussed in Section 5.2.5.

The memory pool and deferred-buffer accumulation across the realised six-cycle trajectory is reported in Table 5.17. The verified pool grows from 431 cold-start entries to 12 175 entries at the trajectory-end cycle (a 28× multiplier), with per-cycle increments in the band $[2\,214, 2\,484]$ across both committed and aborted cycles; the asymmetric-rollback contract preserves memory across the cycle-four abort, so the growth is monotone even where the cycle-boundary refit was skipped. The deferred buffer holds entries in the storage-gate's deferral band that are rescored at each successful cycle boundary under the post-fine-tune verifier and grows from 0 at the cold start to 4 284 at the trajectory-end cycle.

Cycle	Memory pool size	Deferred buffer size	Δ memory
C0	431	0	—
C1	2 818	1 123	+2 387
C2	5 151	2 399	+2 333
C3	7 477	2 692	+2 326
C4*	9 961	4 068	+2 484
C5	12 175	4 284	+2 214

Table 5.17: Memory pool and deferred-buffer accumulation across the realised six-cycle trajectory. The pool grows from 431 cold-start entries to 12 175 entries at the trajectory-end cycle (a 28× multiplier), with per-cycle increments in the band $[2\,214, 2\,484]$. The deferred buffer holds entries in the storage-gate's deferral band that are rescored at each cycle boundary under the post-fine-tune verifier (Section 4.2.10). The pool grows monotonically even across the cycle-four retention abort (marked *) because the asymmetric-rollback contract reverts the adapter while preserving memory.

Stored-episode quality and the training-pool filter chain: Table 5.18 reports the cycle-by-cycle composition of the stored class on the evaluation fold alongside the registered training-pool admission threshold $\tau_{\text{train}} = 0.70$. Across the six-cycle trajectory the stored class admits 336 to 566 eval-fold samples per cycle at the locked $\tau_{\text{store}} = 0.60$, of which 69.3% to 75.9% are em-correct (the per-cycle π_{store} column). The training filter selects a higher-confidence subset at every cycle: the pass-rate column climbs from 17.3% at the cold-start cycle (where most stored items sit at the τ_{store} floor before the cycle-boundary composite refit) to a steady-state of 45–50% from cycle one onward, and re-tightens to 35.9% at cycle five after the post-C4 composite refit. The empirical precision of the SIL pool itself, the em-correct fraction within the τ_{train}-passing subset, is reported in the π_{SIL} column: on the deployment-relevant stream-chunk panel π_{SIL} stays in the band $[0.798, 0.818]$ across the five SIL cycles, a +6 to +8 percentage-point lift above the stream-chunk storage precision band of $[0.724, 0.753]$. The lift is the empirical receipt that the SIL pool has higher precision than the storage pool: the two-stage gate ($\tau_{\text{store}} = 0.60$ for storage, $\tau_{\text{train}} = 0.70$ for SIL admission) decouples the storage-purity contract from the training-purity contract by construction, with the τ_{train} filter removing between 56% and 73% of the wrong-stored stream-chunk entries (the high-u_{stored}-but-em-wrong cohort, the memory-poisoning rate at the storage gate) before they enter the gradient step. The eval-fold lift is smaller (+2 to +6 percentage points) because the eval fold is held-disjoint from the training distribution and is a content-shifted reading rather than a deployment-realised one. The actual per-cycle SIL pool draws from the full memory store (cold-start seed plus cumulative streamed entries) rather than only from the eval-fold slice, so the $n_{\text{used SIL}}$ column reports the realised training-batch size at every successful cycle boundary, ranging from 107 at the first SIL fine-tune to 5 006 at the last successful cycle. The bucket-level distribution of u_{stored} on the stored class, with the per-bucket empirical em-rate, is reported in Table 5.19: on the stream chunks the em-rate increases with the bucket index at every cycle, while on the evaluation fold it does so only at cycles one and four, the structural receipt that the calibrated probability composite is well-ordered against the ground-truth correctness label at deployment time.

Cycle	n_{stored}	em-correct	em-wrong	π_{store}	$n_{\text{pass }\tau_{\text{train}}}$	pass rate	π_{SIL}	$n_{\text{used SIL}}$
(a) Evaluation fold (held-out, content-hash disjoint from the training stream)								
C0	336	255	81	0.759	58	17.3%	0.793	—
C1	515	374	141	0.726	231	44.9%	0.788	—
C2	534	370	164	0.693	266	49.8%	0.726	—
C3	565	402	163	0.712	279	49.4%	0.735	—
C4*	566	402	164	0.710	272	48.1%	0.735	—
C5	504	353	151	0.700	181	35.9%	0.735	—
(b) SIL stream chunks (training-panel candidate pool where storage decisions populate memory)								
C1	2 456	1 850	606	0.753	1 309	53.3%	0.818	107
C2	2 494	1 873	621	0.751	1 448	58.1%	0.810	1 474
C3	2 490	1 802	688	0.724	1 337	53.7%	0.798	2 966
C4*	2 522	1 862	660	0.738	1 413	56.0%	0.803	—
C5	2 380	1 724	656	0.724	895	37.6%	0.803	5 006

* Cycle four was aborted under the retention guard; the cycle-four retroverify pass was skipped, so $n_{\text{used SIL}}$ is not reported for C4.

Table 5.18: Storage-pool quality and SIL training-pool filter chain per cycle, reported on both the held-out evaluation fold (panel a) and the SIL stream chunks (panel b). n_{stored} is the count of samples that cleared the storage cut-point $\tau_{\text{store}} = 0.60$; the em-correct and em-wrong columns split these by ground-truth correctness; π_{store} is the realised storage precision. The training-pool admission threshold $\tau_{\text{train}} = 0.70$ selects a higher-confidence subset of the stored pool for SIL fine-tuning; the $n_{\text{pass }\tau_{\text{train}}}$ and pass-rate columns report the size and share of this subset. The π_{SIL} column is the empirical precision of the SIL pool itself, the em-correct fraction within the τ_{train}-passing subset. The stream-chunk panel is the deployment-relevant pool because storage decisions on stream-chunk samples populate the memory store (the eval fold is content-hash disjoint by construction and does not enter the pool); on the stream chunks π_{SIL} stays in the band $[0.798, 0.818]$ across the five SIL cycles, a +6 to +8 percentage-point lift above the stream-chunk storage precision band of $[0.724, 0.753]$, the empirical receipt that the SIL pool has higher precision than the storage pool and that the $\tau_{\text{train}} = 0.70$ filter selects a strictly cleaner training substrate than the storage gate alone. The eval-fold lift is smaller (+2 to +6 percentage points) because the eval fold is held-disjoint from the training distribution and is a content-shifted reading rather than a deployment-realised one. The complementary read on memory poisoning is the em-wrong column: across the stream-chunk trajectory the storage gate admits between 606 and 688 wrong entries per cycle (the high-u_{stored}-but-wrong cohort) against 1 802 to 1 873 correct entries, and the τ_{train} filter removes between 56% and 73% of those wrong entries on the way into the SIL pool, the architectural mechanism by which the two-stage gate decouples the storage-purity contract from the training-purity contract. The cycle-five drop in pass rate on both folds reflects the post-C4 composite refit re-tightening the upper tail; the π_{SIL} column shows the precision contract holds across the re-tightening. The $n_{\text{used SIL}}$ column reports the actual count of episodes that entered the per-cycle SIL training batch from the cumulative memory store, ranging from 107 at the first SIL fine-tune to 5 006 at the last successful cycle; this exceeds the per-cycle pass count because the SIL pool draws from the full memory store accumulated across all prior cycles, not only from the current cycle's stream chunk.

Cycle	[0.60, 0.65)		[0.65, 0.70)		[0.70, 0.80)		[0.80, 0.90)		[0.90, 1.00]	
	n	EM	n	EM	n	EM	n	EM	n	EM
(a) Evaluation fold										
C0	174	0.764	104	0.731	46	0.804	10	0.700	2	1.000
C1	139	0.655	145	0.697	170	0.776	59	0.814	2	1.000
C2	113	0.566	155	0.729	225	0.716	35	0.771	6	0.833
C3	85	0.600	201	0.726	244	0.738	34	0.706	1	1.000
C4	91	0.648	203	0.704	231	0.732	40	0.750	1	1.000
C5	164	0.677	159	0.686	164	0.720	14	0.929	3	0.667
(b) SIL stream chunks										
C1	596	0.666	551	0.693	832	0.798	419	0.845	58	0.914
C2	512	0.635	534	0.702	1 043	0.785	356	0.871	49	0.898
C3	423	0.572	730	0.675	1 012	0.768	292	0.880	33	1.000
C4	427	0.578	682	0.705	1 060	0.776	316	0.870	37	0.973
C5	732	0.637	753	0.716	752	0.793	127	0.858	16	0.875

Table 5.19: u_{stored} distribution and per-bucket empirical exact-match rate for the stored class on the eval fold (panel a) and the stream chunks (panel b). Row sums match the corresponding n_{stored} in Table 5.18. The per-bucket EM rates on the stream-chunk panel are weakly monotone in every cycle ($\text{EM}_{[0.60,0.65)} < \text{EM}_{[0.90,1.00]}$ at every row), the structural confirmation that the calibrated probability composite is well-ordered against the ground-truth correctness label on the production candidate pool. The eval-fold panel increases monotonically with the bucket index only at cycles one and four; at cycles zero, two, three and five at least one adjacent pair of buckets reverses, mostly where the upper buckets hold few entries. The training-threshold filter at $\tau_{\text{train}} = 0.70$ selects the rightmost three buckets, with empirical EM in the band $[0.700, 1.000]$ on the eval panel and $[0.768, 1.000]$ on the stream-chunk panel.

Monotonicity

Two complementary monotonicity checks are reported. The first, against Heuristic 4.3, pairs the per-cycle change in storage rate with the per-cycle change in calibration-fold Cohen's d; a storage-rate decrease in a cycle whose Cohen's d improved falsifies the within-class Gaussian assumption or the base-rate condition. The second, against the monotone-purification identity recorded as Remark 4.8, computes the one-sided Spearman rank correlation between the per-cycle base-model accuracy p_c and the per-cycle pool purity P_c as a sanity check on the design-rationale identity; a positive rank correlation across the realised cycles is consistent with the identity under fixed α, while a flat or negative correlation flags either operating-point drift in α_c or the limited dynamic range of p_c on the realised horizon rather than a falsification of the identity itself. Both readings are presented as paired-comparison results across the trajectory.

The realised cal-fold trajectory yields four observation points (cycles one, two, three, and five; cycle four's cal-fold pass was skipped by the retention-guard abort that rolled the adapter back to the cycle-three weights). The cycle-one to cycle-two transition sign-aligns with Heuristic 4.3 at large amplitude under the operating-point dominance condition; the cycle-two to cycle-three transition shows a sub-noise Cohen's d delta whose sign carries no information about the derivative; and the cycle-three to cycle-five transition shows a small reverse-sign storage-rate movement under increasing Cohen's d with the dominance condition still holding at all four cycles. Under the fixed-composite scope qualifier registered on the heuristic in Section 4.3.3, the cycle-three to cycle-five reverse-sign movement is reported with its sign rather

than asserted as a monotonicity violation: the cycle-boundary refit of the per-signal isotonic curves, the per-benchmark shrinkage child fits, the boost-layer weights and intercept, and the per-benchmark safety floor moves the operating-point density ratio in parallel with the cal-fold class-separation shift (itself a joint observable of the SIL-advanced generator viewed through the locked verifier ensemble), and the heuristic's $\partial\sigma/\partial d$ claim is silent on cycles whose composite refit dominates the generator-driven cal-fold discrimination signal. The clean isolation that freezes the composite weights and stratifies by class-variance estimate for an ablation cycle is registered as future work in Chapter 6.

Convergence

The convergence statement of Heuristic 4.4, reframed in the chapter-four restructuring as a bounded-band stationarity claim on the committed-cycle subsequence rather than a sharp geometric envelope, predicts that the cycle-by-cycle precision trajectory occupies a tight band on the committed cycles whose width is small relative to the binomial sampling noise of a single per-cycle reading. The realised pooled π_t readings of Table 5.16 occupy the committed-cycle band $[0.732, 0.746]$ (trajectory $0.746 \to 0.740 \to 0.736 \to 0.746 \to 0.732$ across the committed-cycle subsequence C0–C5 with C4 contributing no cal-fold refit under the asymmetric-rollback contract), a band width of 0.014 at the pooled-fold sample size of one thousand five hundred queries per cycle, which is comparable to the binomial standard error on a single per-cycle pooled reading at this sample size. The bounded-band stationarity claim is therefore supported within the realised horizon, with the sharper geometric-rate test deferred to a future-work horizon of at least fifteen committed cycles on a stationary query distribution. The stochastic refinement of Heuristic 4.7 maps the deterministic band onto the calendar timeline through the Bernoulli-weighted contraction $\mathbb{E}[G_t] \leq (1 - \pi_{\text{commit}} \cdot r)^t G_0$, where the empirical commit fraction π_{commit} is read from the cycle-by-cycle retention-guard firing pattern reported in Table 5.20. The heuristic's four sub-claims are adjudicated separately at the realised horizon. Sub-claim (i), the open-interval commit fraction, is verified directly: the realised commit indicator sequence is $(1, 1, 1, 0, 1)$ across the five attempted cycles, giving $\pi_{\text{commit}} = 4/5 = 0.80$ strictly in the open interval $(0, 1)$ as the heuristic predicts. Sub-claim (ii), the calendar-timeline envelope on the gap process, inherits the bounded-band-versus-deferred-rate split of Heuristic 4.4 reported in the previous paragraph and is supported within horizon under the same qualifier the parent heuristic carries. Sub-claim (iii), the asymmetric-rollback memory invariant, is verified directly: the memory store grows from 431 cold-start entries to 12 175 entries at the trajectory-end cycle (Table 5.17), monotone across all six cycles including the cycle-four abort under which the parametric backbone was reverted. Sub-claim (iv), the equilibrium-location prediction, is reported as heuristic at the realised horizon: the worst-probe ratios on cycles two through five all sit within 0.07 of the floor, qualitatively consistent with the predicted near-floor band, but the five-cycle horizon supplies a within-band observation that is not statistically distinguishable from a longer-trajectory transient at the empirical limit, and a rigorous test of sub-claim (iv) requires the at-least-fifteen-committed-cycle horizon registered as future work in Chapter 6 alongside the geometric-rate test of Heuristic 4.4. The fixed-point identity recorded as Remark 4.9 expresses the recurrence $p_{c+1} = f(p_c)$ that converges to $P^* = 1$ in the limit of an idealised verifier with independent errors and an unbounded parametric capacity;

systematic verifier errors (see the scope note on Heuristic 4.6) and the implementation regularisers registered in Chapter 4, namely the L_2 adapter anchor of Equation (4.2), the multi-modal retention guard with rollback, and the parametric capacity ceiling of the base generator and the dense passage index, account for the residual gap between the realised pooled $\pi_t \approx 0.73$ and the theoretical asymptote of one. The bounded-band reading is therefore the within-horizon receipt that the convergence machinery is functioning as the chapter-four restructuring registers it; the sharper geometric-rate test is registered as deferred future work in Chapter 6.

Cycle	Attempts	Outcome	MMLU	TQA-test	CSQA-test	Worst probe	Worst ratio
C1	1	COMMIT	1.041	1.013	1.006	commonsense_qa_test	1.006
C2	1	COMMIT	0.975	0.949	0.982	triviaqa_test	0.949
C3	1	COMMIT	0.992	0.987	0.976	commonsense_qa_test	0.976
C4	1	ABORT	1.018	0.886	0.982	triviaqa_test	0.886
C5	1	COMMIT	0.999	1.013	0.958	commonsense_qa_test	0.958
Realised commit fraction $\pi_{\text{commit}} = 0.800$ (4 of 5 attempted cycles); retention floor $\rho_{\min} = 0.93$							

Table 5.20: Per-cycle retention-guard firing pattern. Each row reports the worst-probe ratio at the final SIL attempt of that cycle; ratios below the registered floor $\rho_{\min} = 0.93$ trigger the asymmetric rollback. The realised commit fraction π_{commit} is the empirical receipt for Heuristic 4.7. A non-degenerate $\pi_{\text{commit}} \in (0, 1)$ with worst-probe ratios bouncing near the floor confirms the stochastic-equilibrium framing; different probes failing on different cycles confirms the multi-modal panel's value over single-probe guards.

Asymptotic hallucination floor

The Tier 1 floor of Heuristic 4.5 predicts $\limsup_t \text{CHM}_1(t) \leq \max(1-\pi_{\text{store}}, 1-\pi_{\text{retro}})$, at most approximately 0.36 under the cycle-zero calibration-fold reading at the locked cut-point $\tau_{\text{store}} = 0.60$ and the registered retroactive prune at $\tau_{\text{retro}} = 0.50$. The full-pipeline decomposition of Remark 4.10, recorded as a design-rationale identity, expresses the user-facing hallucination rate as $H_t \leq (1-\tau_1(t)) \cdot \varepsilon_{\text{arch}}(t) + \tau_1(t) \cdot H_{\text{mem}}(t)$ and supplies the bridge from the pool-level guarantee of Theorem 4.1 to the user-facing tier-routing event. The realised per-tier composite hallucination metric across the trajectory is reported in Table 5.21. The memory-direct tier fires on seven of the nine thousand evaluation queries observed across six cycles, and each of those seven firings returns the correct stored answer; the realised CHM_1 is therefore 0 at every cycle where the memory-direct tier fires, which is consistent with the capability bound of Heuristic 4.5, although seven firings are too few to estimate a rate. The zero-shot tier records a composite hallucination metric in the range 0.101 to 0.111 across the trajectory, and the retrieval-augmented tier records a composite hallucination metric in the range 0.113 to 0.158; the within-tier ordering at every cycle is $\text{CHM}_1 = 0 < \text{CHM}_2 < \text{CHM}_3$, the empirical receipt that the floor is structural rather than scale-dependent. The vanishing-memory-residual asymptote $H_{\text{mem}}(t) \to 0$ is not directly observable on the realised evaluation distribution because the held-out panel is content-hash disjoint from the training stream and pins $\tau_1(t)$ at 0.0013; the within-horizon receipt is therefore the seven-of-seven per-firing capability check together with the projected behaviour at the empirical limit, with the deployment-mode test under a non-disjoint evaluation distribution registered as future work in Chapter 6. A paraphrase-eval probe that intentionally seeds the

held-out panel with queries paraphrasing already-stored entries is registered as the companion future-work item in Chapter 6, since lifting $\tau_1(t)$ above the content-hash-disjoint pinning is the experimental manoeuvre that would convert the heuristic's empirical receipt from a per-firing capability check into an observable trajectory.

Cycle	Tier 1 (memory)		Tier 2 (zero-shot)		Tier 3 (RAG)		Pooled	
	EM	CHM	EM	CHM	EM	CHM	EM	CHM
C0	–	–	0.602	0.111	0.482	0.158	0.541	0.135
C1	–	–	0.585	0.101	0.502	0.122	0.543	0.112
C2	1.000	0.000	0.568	0.103	0.514	0.113	0.544	0.107
C3	1.000	0.000	0.566	0.105	0.503	0.123	0.532	0.115
C4	1.000	0.000	0.564	0.107	0.504	0.121	0.532	0.115
C5	1.000	0.000	0.537	0.102	0.497	0.116	0.517	0.109

Table 5.21: Per-cycle per-tier EM and composite hallucination metric (CHM). Tier 1 (memory-direct), Tier 2 (zero-shot parametric), Tier 3 (retrieval-augmented). The CHM column directly evidences Heuristic 4.5: $\mathrm{CHM}_1(t)$ should stay bounded by $1 - \pi_{\mathrm{store}}$. Tier 2 EM at the parametric high-water mark exceeding Tier 3 EM at cycle zero is the self-improvement-as-distillation empirical receipt.

Heuristic checks: equilibrium structure and self-correction

One heuristic check plus two scaffolding identities complete the theory checks. The deployment-cost reading that a base-model-conditional convergence-rate statement would otherwise supply is recovered architecturally through the per-tier cost identity of Section 3.8.3, whose realised receipt is reported in Section 5.1.4 and Section 5.4.4; a measured geometric contraction rate is non-identifiable at the realised six-cycle horizon under three structural reasons (the parent-variable bridge from storage-pool purity to user-facing hallucination rate through the law-of-total-probability decomposition of Remark 4.10, the parameter-identifiability shortfall of fitting a three-parameter exponential on six derived-composite observations, and the cycle-four retention-guard event that subjects the realised gap process to the stochastic envelope of Heuristic 4.7 rather than a deterministic exponential), and a direct half-life measurement requires the at-least-fifteen-committed-cycle horizon registered as future work in Chapter 6. The self-correction heuristic recorded as Heuristic 4.6 supplies the directional receipt reported in Table 5.22. The stored-entry schema does not persist a passes-survived counter, so the pruned-side histogram of survival counts is deferred to the production-deployment study; the per-cycle prune rate, however, declines monotonically across the four committed retroactive passes from thirteen percent at cycle one to nine percent at cycle two to seven percent at cycle three to four percent at cycle five, consistent with the locked verifier ensemble removing low-quality entries at a decreasing rate as the pool composition shifts toward higher-confidence survivors and as the SIL-advanced generator's outputs widen the cal-fold class separation that the fixed verifier discriminates against. The pool itself grows from 431 entries at the first retroactive pass to 10 214 entries at the fourth, about a twenty-four-fold increase across the four committed cycles. The corpus-bounded floor identity recorded as Remark 4.11, reframed as a design-rationale identity rather than a within-horizon empirical check, would in principle be probed by the subset of

evaluation-fold queries whose top-k retrieval does not contain the gold-answer span; the rectangular-record evaluation harness intentionally filters list-typed verifier fields, and the proxy reconstruction therefore requires re-running retrieval against the dense passage index for every evaluation question across all cycles, which is registered as a deferred proxy in Chapter 6 rather than presented as a within-horizon receipt. The complete receipt mapping with verdicts is summarised in Table 5.27 as a one-glance index against the claims of Section 4.3.

Cycle	Pool size at pass	Updated	Pruned	Prune rate	Survivor share
C1	431	375	56	13.0%	87.0%
C2	2,945	2,674	271	9.2%	90.8%
C3	5,358	5,006	352	6.6%	93.4%
C5	10,214	9,832	382	3.7%	96.3%

Cycle 4: retroactive pass skipped under the cycle-four retention abort (skip-retroverify-on-abort patch).

Table 5.22: Per-cycle retroactive prune-rate trajectory on the committed-cycle subsequence. Each row reports the storage-pool size at the cycle-boundary retroactive pass, the number of entries whose composite was refreshed (updated), the number of entries removed under the registered prune threshold $\tau_{\text{retro}} = 0.50$, the prune rate as a share of the pool, and the survivor share as the complement. The prune-rate trajectory $13.0 \to 9.2 \to 6.6 \to 3.7$ percent across the four committed cycles is the empirical receipt for Heuristic 4.6: an improving verifier removes low-quality entries at a decreasing rate as the pool composition shifts toward higher-confidence survivors. Cycle four's retroactive pass was skipped under the asymmetric-rollback contract (the post-fine-tune verifier is the rolled-back cycle-three verifier, so re-scoring the pool would not produce new information).

The aggregate prune-rate trajectory of Table 5.22 is sharpened by an em-conditional decomposition reported in Table 5.23. Across the four committed retroverify passes the runtime log records 1 061 prunes; the offline reconstruction of Table 5.23 identifies 1 094 pruned entries from memory-snapshot differences, and these are 38.4% em-wrong against a pool-base wrong rate of approximately 22%, an enrichment of roughly 1.75×, the empirical receipt that retroverify selects against hallucinations rather than removing entries at random. Of the 650 promotions from the deferred buffer that the runtime log records, the reconstruction identifies 620, and this reconstructed cohort is 55.6% em-correct: substantially below the storage pool's $\sim 78\%$ correct rate at the cut-point, which is consistent with the deferred cohort being intrinsically harder than entries that cleared the storage cut-point on the first attempt, but well above chance and equivalent to 345 correct entries recovered from the deferral band. The two mechanisms therefore operate as complementary halves of the dual-track recovery story: retroverify shaves a hallucination-enriched tail off the storage pool, while the deferred buffer reclaims correct entries the initial gate over-rejected.

	Retroverify prunes				Deferred-buffer promotions			
Cycle	**Pruned**	**em-wrong**	**em-correct**	**Wrong-caught**	**Promoted**	**em-correct**	**em-wrong**	**Correct-recovered**
C1	56	0	56	0.0%	0	0	0	—
C2	261	114	147	43.7%	117	65	52	55.6%
C3	355	122	233	34.4%	210	119	91	56.7%
C4*	—	—	—	—	—	—	—	—
C5	422	184	238	43.6%	293	161	132	54.9%
Pooled	**1 094**	**420**	**674**	**38.4%**	**620**	**345**	**275**	**55.6%**

* Cycle four was aborted under the retention guard; both the retroverify pass and the deferred-reconsider pass were skipped under the asymmetric-rollback contract.

Table 5.23: Em-conditional decomposition of retroverify-pruned entries and deferred-buffer-promoted entries across the realised trajectory. *Wrong-caught rate* is the share of pruned entries whose ground-truth em label was zero, the principled measure of whether retroverify selects against hallucinations rather than rolling random; the pooled rate of 38.4% is approximately 1.75× the pool-base wrong rate of $\sim$ 22%, the empirical receipt that retroverify is enriched against wrong entries rather than uniform. The cycle-one anomaly (0% wrong-caught on a small 56-entry prune cohort) reflects signal-noise on cold-start TriviaQA and CSQA entries that were actually correct; this is the noisy regime where the retroverify pass has the least lift over the gate's initial admission. *Correct-recovered rate* is the share of promoted entries whose ground-truth em label was one; the pooled 55.6% sits below the $\sim$ 78% pool-base correct rate, confirming that the deferred-buffer cohort is intrinsically harder than the entries that cleared the storage cut-point on the first attempt, but is well above chance and recovers 345 correct entries that the initial gate had rejected as too uncertain. The two mechanisms are complementary: retroverify shaves a modest hallucination-enriched tail off the storage pool, while the deferred buffer reclaims correct entries the initial gate over-rejected. Em labels are not persisted per entry by the runtime; the reconstruction used offline lookup against the benchmark gold (one-hundred-percent match coverage on the realised 12 175 stored entries) and an entry-identifier set-difference between consecutive memory-store metadata snapshots for prune identification (a small consolidation-removed cohort of $\leq 5\%$ runs in the same step and is indistinguishable at the snapshot level). The Pruned and Promoted totals are therefore an offline reconstruction from memory-store set differences, not the run's own counts: the run log records 1 061 retroactive prunes (56/271/352/382 at C1/C2/C3/C5) and 650 deferred-buffer promotions (131/220/299 at C2/C3/C5).

The third pool-maintenance mechanism is the cycle-boundary consolidation pass, an SBERT-similarity clustering step that collapses candidate clusters of near-duplicate-meaning episodes into a single representative under three registered safety guards: a cosine-similarity floor of 0.92 across the top twenty FAISS neighbours, an answer-equality guard that requires normalised gold answers to match, and a u_{stored}-spread guard that rejects clusters whose stored-confidence range exceeds 0.15. The per-cycle trajectory is reported in Table 5.24. The pooled 57 merges across the realised trajectory remove 58 episodes, modest in absolute volume relative to the 1 061 retroverify prunes and 650 deferred-buffer promotions but architecturally distinct from both: consolidation reduces redundancy at fixed accuracy rather than catching hallucinations (retroverify) or recovering correct entries (deferred buffer). Three concrete patterns from the realised trajectory illustrate the safeguard interaction. At cycle two, three size-two clusters merged in approximately one second of wall-time while four sibling candidate clusters at the same similarity threshold were rejected for answer-mismatch (polysemy trap: near-identical SBERT embeddings with differently-normalised gold answers). At cycle three, the u_{stored}-spread guard fired for the first

time, rejecting a near-paraphrase pair whose cycle-three stored-confidence range exceeded 0.15. At cycle five, the consolidation pass processed approximately ten thousand episodes in seven seconds of wall-time, producing 42 size-two clusters and one size-three cluster (43 merges, 44 removals); the u_{spread}-rejection count jumped twelve-fold between cycle three and cycle five, consistent with the wider stored-confidence distribution after three rounds of self-improvement and isotonic refit on the verifier composite. Cross-pool merges are exactly zero at every cycle by construction: the pre-split FAISS indexes between the training and transfer pools physically prevent the OOD-leak failure mode the consolidation guard was registered against.

			Candidate clusters rejected			
Cycle	**Candidates**	**Clusters merged**	**Answer-mismatch**	**u-spread** > 0.15	**Cross-pool**	**Episodes removed**
C1	1	0	1	0	0	0
C2	7	3	4	0	0	3
C3	29	11	17	1	0	11
C4*	—	—	—	—	—	—
C5	100	43	45	12	0	44
Pooled	**137**	**57**	**67**	**13**	**0**	**58**

* Cycle four consolidation was skipped under the asymmetric-rollback contract; candidates from C4 accumulated and were processed at the C5 pass.

Table 5.24: Per-cycle memory-consolidation trajectory sourced from the memory-store consolidation pass's log emissions. The pass runs at every successful cycle close and collapses candidate clusters of near-duplicate-meaning episodes into a single representative under three safety guards: a cosine-similarity floor of 0.92 across top-20 FAISS neighbours, an answer-equality guard that requires normalised gold answers to match, and a u_{stored}-spread guard that rejects clusters whose stored-confidence range exceeds 0.15. The pre-split FAISS index between training and transfer pools eliminates cross-pool merges by construction; the realised counter is zero at every cycle. The pooled 57 merges across the trajectory remove 58 episodes (one size-three cluster at cycle five contributes the extra removal), small in absolute terms relative to the 1 094 retroverify prunes and 620 deferred-buffer promotions of Table 5.23 but architecturally distinct: consolidation reduces redundancy at fixed accuracy rather than removing wrong-confident entries or rescuing low-confidence correct entries.

The complementary dual-track recovery view, the deferred-buffer reconsider-pass dynamics, is reported in Table 5.25. The promotion, TTL-drop, and keep counts are sourced verbatim from the deferred-buffer reconsideration pass's log emissions on the realised run rather than derived from buffer-size differences. The buffer is empty at the cycle-one reconsider, so cycle one promotes zero entries; the cycle-two reconsider promotes 131 entries from the cycle-one push, the cycle-three reconsider promotes 220 from the cycle-two push and TTL-drops 924 entries that exceeded the registered TTL of two, and the cycle-five reconsider promotes 299 from the cycle-four push with 1 169 TTL-drops. The cycle-four reconsider is skipped under the asymmetric-rollback contract; the C4-pushed entries carry over into the cycle-five pass without ageing. The total of 650 promotions across the realised trajectory is the direct empirical receipt for the recovery-across-cycles mechanism of Heuristic 4.6.

Cycle	Buffer at reconsider	Promoted	TTL-dropped	Kept	New pushed (post-pass)
C1	0	0	0	0	1 123
C2	1 123	131	0	992	1 407
C3	2 399	220	924	1 255	1 437
C4*	2 692	—	—	2 692	1 376
C5	4 068	299	1 169	2 600	1 684
Totals across trajectory	—	**650**	**2 093**	—	**7 027**

* Cycle four reconsider was skipped under the asymmetric-rollback contract; the C4-pushed entries carry over to the C5 pass unchanged.

Table 5.25: Deferred-buffer reconsider-pass dynamics across the realised six-cycle trajectory, sourced from `experiment.log` "Deferred reconsideration done" emissions of `DeferredBuffer.reconsider()`. *Promoted* entries crossed the storage cut-point $\tau_{\text{store}} = 0.60$ under the post-fine-tune verifier and entered the main memory pool. *TTL-dropped* entries reached the registered TTL of two reconsider invocations (`config.deferred_buffer_ttl_cycles` = 2) without being promoted and were dropped from the buffer. *Kept* entries survived the pass with their TTL counter incremented and remain eligible for the next cycle's reconsider. The total 650 promotions across the realised trajectory is the direct empirical receipt for the recovery-across-cycles mechanism of Heuristic 4.6.

The TTL distribution of the buffer at the close of each cycle is reported in Table 5.26. Entries are tagged with an age counter that increments by one each time the reconsideration pass processes the entry; an entry whose age reaches the registered time-to-live of two reconsider invocations without being promoted is dropped on its next reconsideration pass. The age = 1 cohort survives one reconsider pass without promotion or expiry and is rescored under the next cycle's verifier. The cycle-four age = 1 cohort of 1 255 entries carries over from cycle three unchanged because the cycle-four reconsider pass was skipped under the asymmetric-rollback contract; this carryover is the architectural mechanism by which the deferred buffer preserves recovery opportunities through retention aborts.

Cycle	Age = 0 (fresh)	Age = 1 (one reconsider survived)	Total buffer
C0	0	0	0
C1	1 123	0	1 123
C2	1 407	992	2 399
C3	1 437	1 255	2 692
C4*	2 813	1 255	4 068
C5	1 684	2 600	4 284

* Cycle four reconsider was skipped under the abort contract; the age = 1 cohort from C3 carried over unchanged into C4 without ageing further.

Table 5.26: Per-cycle TTL distribution of deferred-buffer entries at the cycle close, sourced from `deferred_buffer_cycle_*.pkl` via the `DeferredEntry.age` field. The age counter is initialised to zero when an entry is pushed and incremented by one each time `DeferredBuffer.reconsider()` processes the entry; an entry whose age reaches the registered TTL of two (`config.deferred_buffer_ttl_cycles`) without being promoted is dropped on its next reconsider invocation. The age = 0 cohort at each cycle is the freshly-pushed batch of that cycle; the age = 1 cohort is the set of entries that survived one reconsider pass without being promoted or dropped. No age = 2 entries appear in the table because the drop threshold fires before the cycle close, so entries that would otherwise age to two are removed from the buffer by the reconsider pass that creates the next cycle's snapshot. The cycle-four age = 1 cohort of 1 255 entries carries over from cycle three unchanged because the cycle-four reconsider pass was skipped under the asymmetric-rollback contract; this carryover is the mechanism by which the deferred buffer preserves recovery opportunities through retention aborts.

ID	Label	Type	Verdict	Receipt summary	Anchor
T1	`thm:purity`	Theorem	PASS (pooled cal-fold)	Pooled cal-fold $\pi_t \in [0.732, 0.746]$ clears the 0.64 floor by 9 to 11 percentage points at every reported cycle (the registered pooled training-panel axis). Per-bench eval-fold FEVER dips below floor at C1–C4 (band $[0.614, 0.634]$); disclosed as within-scope-permitted diagnostic, outside the pooled-axis bound.	Table 5.16
T2	`thm:bayes-purity`	Proposition	precondition audited	Purification condition TPR > FPR (balanced accuracy above one half), audited at cycle zero through the composite Cohen's d gate (+0.635, balanced accuracy ≈ 0.673).	Table 5.32
T3	`thm:monotone`	Heuristic	consistent within horizon, sharp test deferred	Three cal-fold transitions are observable ($C1 \to C2$, $C2 \to C3$, $C3 \to C5$; C4 skipped): $C1 \to C2$ sign-aligns at large amplitude, $C2 \to C3$ is sub-noise in Cohen's d, and $C3 \to C5$ shows a small reverse-sign storage-rate movement.	Table 5.16
T4	`thm:monotone-purification`	Remark	identity	Mathematical scaffolding (derivative of T2); no within-horizon empirical receipt.	–
T5	`thm:convergence`	Heuristic	BOUNDED-BAND	Pooled cal-fold π_t confined to a band of width 0.014 (0.746 − 0.732) across the six-cycle horizon; the rate-scaling form $r = \mathcal{O}\big(\sigma \cdot \Phi(d/\sqrt{2})\big)$ is a Gaussian-model approximation asserted in the argument sketch rather than measured at the present horizon, and the measured geometric-rate test is deferred to a $\geq$ 15-committed-cycle horizon.	Table 5.16
T6	`thm:bayes-convergence`	Remark	identity	Mathematical scaffolding (monotone-map fixed-point theorem); no within-horizon empirical receipt.	–
T7	`thm:asymptotic-elim`	Remark	identity	Mathematical scaffolding (law of total probability); no within-horizon empirical receipt.	–
C1	`cor:tier1-floor`	Heuristic	CAPABILITY	$\text{CHM}_1 = 0$ on each of the seven realised Tier-1 firings; pool-side ceiling $1 - \pi_{\text{store}} \approx 0.36$ holds with substantial slack.	Table 5.21
C3	`cor:corpus-floor`	Remark	deferred	Per-sample top-k passage IDs not persisted; proxy receipt deferred to future-work re-retrieval pass.	–
C4	`cor:self-correction`	Heuristic	PARTIAL	Prune-rate decline $13 \to 9 \to 7 \to 4$ percent across committed cycles supports directional claim; survivor-side histogram deferred.	Table 5.22
C5	`cor:stochastic-equilibrium`	Heuristic	PASS (i,iii) / heuristic (iv)	Sub-claims (i) and (iii) pass directly: $\pi_{\text{commit}} = 4/5 = 0.80$ strictly in $(0, 1)$; memory grows monotonically from 431 to 12 175 entries across the C4 abort. Sub-claim (ii) inherits the bounded-band reading of T5. Sub-claim (iv), the equilibrium-location prediction, is labelled heuristic at the five-cycle horizon and deferred to a $\geq$ 15-committed-cycle test.	Table 5.20

Table 5.27: Receipts for the theorem, the proposition, and the design heuristics against the realised six-cycle trajectory. Heuristic rows are directional consistency checks against the realised trajectory rather than tests of a proven claim. Each row pairs one formal claim from Section 4.3 with a verdict and the table anchor where its empirical receipt is reported. PASS verdicts are claims whose receipt is satisfied directly by the realised trajectory. PARTIAL verdicts are claims whose directional prediction is supported by the realised data but whose sharper form requires either a longer horizon or an instrument not present in the artefact set. BOUNDED-BAND, PROJECTED, and CAPABILITY verdicts label claims that the chapter-four restructuring carries with explicit scope qualifiers as discussed in Section 4.3. The C5 row reports a per-sub-claim verdict because the heuristic has four sub-claims of different horizons: sub-claims (i) and (iii) pass within the realised five-cycle trajectory, sub-claim (ii) inherits T5's bounded-band reading, and sub-claim (iv) is the equilibrium-location prediction whose argument is purely asymptotic, whose rigorous test requires the longer-horizon item registered in Chapter 6. Mathematical scaffolding identities (Remarks) do not carry empirical receipts and have no anchor.

5.2 Final Design Adjustments

5.2.1 Calibration discipline and disjoint folds

The calibration-consistent precision floor of Theorem 4.1 is one-sided under the exchangeability assumption between the calibration fold and the cycle-t candidate pool; preserving this assumption requires that the two folds be drawn from the same population and that no individual sample appear in both. The empirical programme enforces this through a six-fold partition of the per-benchmark sample budget, defined at pool-builder time and recorded in the registered dataset-splits manifest. The seed pool is the cold-start memory population fold; the calibration fold is the source of every per-signal isotonic curve and every per-benchmark composite refit; the purity-validation slice is the source of the empirical-precision audit that runs once per cycle; the per-cycle stream chunks are the source of the SIL queries; the evaluation fold is the per-cycle headline-metric reporting fold; and the test fold is reserved for the post-trajectory comparative panel. Cross-benchmark and within-benchmark disjointness is verified by the pool builder at every run start through content-hash leakage guards, and any candidate sample that appears in two folds is dropped at the pool-builder stage rather than at downstream metric computation, so the disjointness invariant is enforced where it is cheapest to enforce. The training panel consists of FEVER, TriviaQA, and CommonsenseQA; the transfer panel consists of TruthfulQA and StrategyQA. Only the training panel contributes to the calibration fold, the purity-validation slice, the seed pool, and the SIL stream chunks; the transfer panel is reserved for evaluation and test only.

5.2.2 The composite parameter selection sweep

The pre-specified grid

Three grids were registered before any sweep variant was scored. The first grid spans the storage cut-point τ_{store} on the calibrated probability scale at the candidates $\{0.55, 0.60, 0.65, 0.70\}$, paired with a deferral cut-point τ_{defer} at 0.45 throughout. The second grid spans the boost layer's L2 regularisation strength C at five values $\{0.010, 0.025, 0.050, 0.100, 1.000\}$, where lower C corresponds to stronger regularisation. The third grid spans the per-benchmark shrinkage strength α_{shrink} that blends each per-benchmark child isotonic curve with the pooled fit, at the candidates $\{0.0, 0.4, 0.6, 0.8, 1.0\}$ where 0 collapses to the pooled curve and 1 leaves the per-benchmark child unchanged. The full grid produces a manageable cross-product through which the pool-purity, AUROC, and storage-rate diagnostics are scored, with the variant catalogue recorded under the cycle-zero sweep archive with one record per variant.

Selection criteria

Four selection criteria were registered before the sweep was scored, in the order they would be applied. First, the per-benchmark composite Cohen's d between exact-match-correct and incorrect outputs on the calibration fold must clear the registered 0.20 floor on the training-pooled slice; variants below this floor are eliminated because they violate the precondition of the purification condition (Proposition 4.2). Second, at the candidate cut-point τ_{store} the calibration-fold stored count must clear the

minimum admissible count $n_{\min} = 20$ for stability of the empirical-precision audit. Third, among variants that clear the first two criteria, the variant with the highest stored-volume-at-precision is preferred: at the same per-store empirical precision shelf, the variant that admits more correct memories at cycle close is preferred, on the principle that the storage class populates the episodic memory and the SIL training pool, so a higher storeable population at the registered precision compounds across cycles. Fourth, where two variants are statistically indistinguishable on the first three criteria, the variant with the stronger L2 regularisation (lower C) is preferred, on the principle that stronger regularisation is more robust to distribution shift in later cycles where the per-benchmark composite is refit on a fresh fold under the shrinkage prior.

Top variants

The locked variant $V_{\tau=0.60,\,C=0.010,\,\alpha_{\text{shrink}}=0.6}$ admits 289 entries on the held-out cycle-zero validation fold (500 samples per benchmark, 1 500 on the training panel) at $\pi_{\text{store}} = 0.796$ across the three training benchmarks, against the alternative $V_{\tau=0.65}$ from the registered sweep which admits roughly a factor of four fewer entries on the same fold at a marginally higher per-store precision. The two operating points sit on the same precision shelf at the cycle-zero distribution but the lower cut admits substantially more correct memories at the same per-store precision; under selection criterion three of Section 5.2.2, the lower cut is preferred. The sweep comparison records each variant's calibration-fold area under the receiver operating characteristic, stored-volume, and per-store precision under the registered selection criteria, with the locked configuration and its parameter rationale reported in Table 5.28.

5.2.3 The locked configuration

The locked configuration is the variant $V_{\tau=0.60,\,C=0.010,\,\alpha_{\text{shrink}}=0.6}$ from Table 5.28. The configuration carries forward five parameter values verbatim into every cycle of the main run: the storage cut-point $\tau_{\text{store}} = 0.60$, the deferral cut-point $\tau_{\text{defer}} = 0.45$, the abstention floor $\phi_{\text{pg}} = 0.20$ from Equation (4.1), the boost layer's L2 regularisation strength $C = 0.01$, and the per-benchmark shrinkage strength $\alpha_{\text{shrink}} = 0.6$ from Section 4.2.7. The cycle-zero fits produce the per-signal isotonic curves, the per-benchmark child curves blended toward the pooled fit through the shrinkage prior, and the boost layer's per-signal weights, all stored in the cycle-zero composite calibration artefact. The configuration is preferred over the alternative $V_{\tau=0.65}$ from the registered sweep for two reasons. First, the lower cut admits roughly 4.4 times as many correct memories at the same per-store precision shelf at cycle close (306 against 70 on the cycle-zero calibration fold of the calibration sweep run before the main trajectory), which compounds across the realised cycle horizon into a substantially larger memory state and a substantially larger self-improvement training pool. Second, $C = 0.01$ is the strongest-regularised value in the boost-layer grid, with a hundredfold tighter L2 penalty than the library default of $C = 1.0$; this regularisation is what the cycle-boundary composite refit relies on to track distribution drift without overfitting to a single cycle's calibration fold. Table 5.29 reports the per-signal isotonic direction and the boost-layer weights at the locked configuration, auto-generated from the cycle-zero composite calibration; the boost intercept and the per-benchmark child weights propagate from the same fit and are recorded with the composite metadata.

Parameter	**Locked value**	**Selection rationale**
Storage cut-point τ_{store}	0.60	Highest stored-volume at the calibration-fold precision shelf among grid candidates $\{0.55, 0.60, 0.65, 0.70\}$
Deferral cut-point τ_{defer}	0.45	Pre-specified 0.15 below storage cut
Abstention floor ϕ_{pg}	0.20	Early-exit rule Equation (4.1)
Boost regularisation C	0.010	Strongest-regularised value in grid $\{0.010, 0.025, 0.050, 0.100, 1.000\}$; hundredfold tighter than the sklearn default of $C = 1.0$
Shrinkage strength α_{shrink}	0.6	Per-benchmark shrinkage prior toward pooled fit among grid candidates $\{0.0, 0.4, 0.6, 0.8, 1.0\}$
Retroactive prune τ_{retro}	0.50	0.10 below storage cut by registration
Retention floor $\rho_{\min}$	0.93	Multi-modal probe panel rollback threshold

Table 5.28: Locked configuration parameters carried forward into every cycle of the main run. The configuration is the variant $V_{\tau=0.60,\, C=0.010,\, \alpha_{\text{shrink}}=0.6}$ selected from the pre-specified three-grid sweep documented in Section 5.2.2. The grid spans the storage cut-point on the calibrated probability scale, the boost-layer regularisation strength, and the per-benchmark shrinkage strength; the cross-product produces a one-hundred-variant search space ($4 \times 5 \times 5$ over $\tau_{\text{store}} \times C \times \alpha_{\text{shrink}}$) scored under the four registered selection criteria of Section 5.2.2. The selected variant maximises stored-volume at the calibration-fold precision shelf while sitting at the strongest-regularised cell of the boost-layer grid. Per-signal isotonic curves, per-benchmark shrinkage-blended child curves, and the boost-layer weights and intercept at this configuration are reported in Table 5.29.

Signal	Pearson ρ	Direction	Boost weight
$p_{\text{ground}}^{\text{mean}}$	+0.101	increasing	+0.184
$p_{\text{ground}}^{\text{max}}$	+0.140	increasing	+0.379
$p_{\text{ground}}^{\text{atomic}}$	+0.101	increasing	+0.184
p_{entail}	-0.156	decreasing	+0.067
$q_{a,\text{rel}}$	+0.299	increasing	+0.491
u_{internal}	+0.425	increasing	+0.334
u_{token}	+0.119	increasing	+0.126
u_{dropout}	-0.411	decreasing	+0.190
s_{avg}	+0.098	increasing	+0.175
		Boost intercept	+0.265

Table 5.29: Cycle-zero calibrated probability composite under the locked configuration. Each signal is mapped to a calibrated probability through per-signal isotonic regression on the $n = 1500$ training-panel calibration fold (em rate = 0.446); the boost layer aggregates the calibrated signals through logistic regression with L2 regularisation strength $C = 0.01$. The Pearson ρ column reports the per-signal correlation with the exact-match label and determines the per-signal isotonic direction. The boost weight column reports the per-signal weight learned by the regularised logistic boost layer. The composite is shared across the three training benchmarks; the per-benchmark child curves are blended toward this pooled fit through the shrinkage prior at $\alpha_{\text{shrink}} = 0.6$ from Section 4.2.7. The signal h_{norm} is a calibration-time-only diagnostic that does not participate in the deployed composite and is therefore absent from the table.

5.2.4 Per-cycle composite-refit trajectory

The locked configuration of Table 5.28 carries constant cut-points across every cycle by design, but the per-signal-to-probability mapping the cut-points operate on is refit at every successful cycle boundary on the post-fine-tune calibration fold. Table 5.30 reports the cycle-by-cycle behaviour of the parameters that the refit moves: the pooled temperature scalar on the pre-routing confidence, the post-refit expected calibration error, the boost-layer intercept, the per-benchmark temperature scalar, and the per-benchmark safety floor. The pooled temperature scalar enters cycle one at the registered ceiling of the optimisation envelope ($T = 20.086 \approx e^3$, the value the initial fit converged to under the cycle-zero distribution) and drops by an order of magnitude under the cycle-one boundary refit, then stabilises in the band $[0.74, 0.87]$ from cycle two onward. The post-refit expected calibration error stays inside the band $[0.041, 0.049]$ across every refit, so the per-cycle distribution drift is absorbed into the signal-to-probability mapping without inflating the calibration-error budget. The boost-layer intercept stabilises in the band $[-0.63, -0.52]$ after the cycle-one transition from the cycle-zero initialisation. The per-benchmark temperature scalar in panel b surfaces the heterogeneity that the pooled temperature in panel a hides: the FEVER and TriviaQA per-benchmark scalars sit 4.8 to 10.7 times above the CommonsenseQA scalar across the committed cycles, the empirical receipt that the underlying token-probability distributions on open-text factual claim verification and open-domain trivia are systematically more overconfident than on the five-option multiple-choice surface; the per-benchmark refit absorbs this heterogeneity into the per-signal mapping rather than into the locked storage cut-points. Panel c reports the only per-benchmark safety floor that drifted in the realised trajectory: the CommonsenseQA floor is raised from the registered default $\phi_{\text{safe}} = 0.380$ to 0.491 at cycle three and to 0.569 at cycle five, the cycle-boundary refit's response to CommonsenseQA's structurally lower pre-routing confidence on the five-option multiple-choice surface (the lowest per-benchmark u_{pre} minimum in Table 5.13 is on CommonsenseQA). The pooled temperature scalar at cycle one entering the cycle-boundary refit at the registered envelope ceiling is documented as a calibration-discipline limitation in Section 5.6.2; the boost-layer intercept, the per-signal isotonic curves, and the per-benchmark shrinkage prior absorb the drift on the storage-class composite even at the saturated temperature scalar, because the gate's operating point depends on the score's quantile structure at the locked cut-point rather than on temperature-scaled probability semantics. Panel d of Table 5.30 reports the load-bearing per-cycle reweighting on the boost layer's nine deployed signals: two signals carry the cycle-by-cycle rebalancing while the remaining seven stay band-stable. The natural-language-inference entailment probability p_{entail} lifts from +0.067 at the cold-start initialisation to +0.285 at the trajectory-end cycle (an absolute movement of 0.218, more than threefold the cycle-zero magnitude), the empirical receipt that the entailment signal becomes substantially more predictive of em-correctness once the self-improvement loop has narrowed the generator's distribution onto the verified-pool support; the cross-encoder question-answer similarity $q_{a,\text{rel}}$ declines from the cycle-zero peak of +0.491 to a cycle-three onward steady-state of approximately +0.26 (an absolute movement of 0.226), the empirical receipt that the cross-encoder signal's marginal contribution shrinks once the generator produces on-topic answers more reliably. The remaining seven signals stay inside a tight per-signal band of $|\Delta_{\text{C0}\to\text{C5}}| \leq 0.07$ on every other row, so the boost-layer

refit absorbs verifier-internal recalibration into two specific signal weights rather than dispersing it across the ensemble. The per-signal isotonic curves themselves are non-parametric step functions on the calibration fold and are not enumerated cell-by-cell across cycles; their cycle-zero shapes and the per-signal Pearson direction are recorded in Table 5.29, and the per-cycle reweighting reported in panel d is the parametric receipt that the cycle-boundary refit is doing measurable work on the boost layer above the isotonic stage.

Parameter	**C0**	**C1**	**C2**	**C3**	**C5**	$\lvert\boldsymbol{\Delta}_{\mathrm{C0}\to\mathrm{C5}}\rvert$
(a) Pooled calibration discipline						
Temperature scalar T (pooled, post-refit)	20.086	1.089	0.836	0.740	0.869	19.217
Expected calibration error ECE after refit	—	0.044	0.049	0.041	0.043	—
Boost-layer intercept b_0	+0.265	−0.613	−0.634	−0.565	−0.520	0.785
(b) Per-benchmark temperature scalar (post-refit)						
T_{FEVER}	—	14.092	9.899	6.986	4.972	—
T_{TriviaQA}	—	14.123	10.007	7.163	11.040	—
$T_{\mathrm{CommonsenseQA}}$	—	1.342	1.209	1.083	1.028	—
(c) Per-benchmark safety floor (drift observed only on CommonsenseQA)						
$\phi_{\mathrm{safe}}^{\mathrm{CommonsenseQA}}$	0.380	0.380	0.380	0.491	0.569	0.189
(d) Pooled boost-layer weights on the nine deployed signals						
u_{token}	+0.126	+0.211	+0.177	+0.060	+0.110	0.016
u_{dropout}	+0.190	+0.132	+0.225	+0.235	+0.204	0.014
u_{internal}	+0.334	+0.257	+0.297	+0.257	+0.321	0.013
s_{avg}	+0.175	+0.220	+0.177	+0.264	+0.242	0.067
p_{entail}	+0.067	+0.368	+0.324	+0.359	+0.285	**0.218**
$p_{\mathrm{ground}}^{\mathrm{max}}$	+0.379	+0.296	+0.333	+0.288	+0.356	0.023
$p_{\mathrm{ground}}^{\mathrm{mean}}$	+0.184	+0.262	+0.238	+0.223	+0.230	0.046
$p_{\mathrm{ground}}^{\mathrm{atomic}}$	+0.184	+0.262	+0.238	+0.223	+0.230	0.046
$q_{a,\mathrm{rel}}$	+0.491	+0.360	+0.259	+0.264	+0.265	**0.226**

Table 5.30: Per-cycle composite-refit trajectory across the four committed cycles of the realised horizon, decomposed into the pooled calibration discipline (panel a), the per-benchmark temperature scalar (panel b), the per-benchmark safety floor (panel c), and the pooled boost-layer weights on the nine deployed signals (panel d). The rightmost column reports $|\Delta_{\mathrm{C0}\to\mathrm{C5}}|$ where a cycle-zero reading is available. Cycle four is omitted because its post-fine-tune adapter was rolled back under the retention guard's asymmetric-rollback contract (Section 5.1.5); cycle five resumes the refit pattern from the cycle-three composite anchor. The immutable cut-points and regularisation strengths are registered in Table 5.28; the cycle-by-cycle reading of the boost-layer reweighting and the per-benchmark heterogeneity it absorbs is discussed in the surrounding prose.

5.2.5 Calibration-versus-evaluation gap

The calibration-versus-evaluation gap is the empirical anchor that motivated the architectural choice of a fixed-threshold gate over the split-conformal alternative considered earlier in the project. Under the rejected split-conformal formulation at miscoverage $\alpha_{\mathrm{store}} = 0.05$, the cycle-zero fit produced a calibration-fold precision of 0.9643 over 56 stored entries at $\tau_{\mathrm{store}} = 0.6676$ (recorded in the project's interim supervisor report; the run artefacts of this rejected formulation were not retained); on the same locked composite, the evaluation-fold pooled precision dropped fifteen to fifty percentage points below the conformal target on every training benchmark, with the conformal contract violating its precondition on the registered five-benchmark

panel because the calibration-fold-to-evaluation-fold distribution shift on the per-benchmark slice broke marginal coverage. The conformal alternative was therefore retired before the main run and replaced by the fixed-threshold rule: the storage cut-point $\tau_{\text{store}} = 0.60$ is selected by stored-volume at the calibration-fold precision shelf. The cycle-zero per-benchmark headline diagnostics under the locked configuration are reported in Table 5.31. On the held-out cycle-zero validation fold the storage class admits 63, 145, and 81 entries on FEVER, TriviaQA, and CommonsenseQA respectively, with realised per-bench precision 0.794, 0.821, and 0.753 and a pooled ID precision of 0.796 over the 289 stored entries on the 1500-sample ID cycle-zero validation fold (500 per benchmark). The fixed-threshold rule does not depend on the marginal-coverage statement that the calibration-versus-evaluation shift broke, so the shift cannot silently invalidate a guarantee, although it can still lower the realised precision, as the transfer-panel TruthfulQA store precision of 0.522 in Table 5.31 shows; the calibration-fold precision is reported as an empirical anchor at every cycle through the empirical-precision audit, and the cycle-by-cycle drift of the anchor is itself the diagnostic the architecture exposes rather than a hidden conformal-coverage claim.

The precision contract has a complement on the recall axis. The storage cut-point admits a high-precision storage class at the cost of rejecting a substantial fraction of correct outputs through the abstention and discard branches; this is the precision-recall tradeoff that the registered cut-point τ_{store} makes explicit. The architecture does not attempt to recover the rejected-correct fraction within a single cycle. Instead, two complementary cross-cycle recovery paths return rejected-correct content to memory across the trajectory horizon. The first path is the bounded deferred-entry buffer specified in Section 4.2.10 and Section 4.2.10: every output that scored in the deferral band rather than the storage band is held in the buffer and rescored under the post-fine-tune verifier at every successful cycle boundary, with promotion to memory whenever the recalibrated composite crosses τ_{store}. The second path is semantic re-encounter: stream chunks across the trajectory sample fresh content from the same training-pool distributions, so that a query rejected as low-confidence in an early cycle can return as a near paraphrase in a later cycle and be admitted under the upgraded post-fine-tune verifier. The two paths are gated differently at the cycle boundary: deferred-buffer promotion runs over every entry every successful cycle without depending on stream-chunk content, while semantic re-encounter runs unconditionally because the stream-chunk gen-and-store path executes regardless of whether the cycle's fine-tune commits. The single-cycle recall reading is therefore an underestimate of the trajectory recall; the empirical receipt for cumulative recovery is the survival-cycles distribution registered against Heuristic 4.6 and the per-cycle deferred-buffer promotion rate registered against the cycle-boundary update passes.

Both cross-cycle recovery paths inherit the asymmetric-rollback property formalised in Section 5.1.5. When the retention guard fires and the cycle's adapter is rolled back to the previous successful cycle's weights, the rescoring side of the architecture pauses: the deferred-buffer reconsideration pass, the retroactive re-verification pass, and the per-benchmark composite refit are all skipped on the aborted cycle, because each would re-score memory or rescore the calibration fold under weights and a composite that are identical to those used at the previous cycle's close, which produces an empirically deterministic no-op. The buffer's time-to-live is therefore counted in

reconsider invocations rather than in calendar cycles: an entry has at most two opportunities to be promoted, where an opportunity is defined as a cycle boundary at which the verifier ensemble actually changed, not a cycle boundary in the calendar sense. The fresh-data side of the architecture continues to run unconditionally: stream chunks add new entries to memory on every cycle including aborted ones, so the memory state grows monotonically across the trajectory while the rescoring side pauses and resumes in lockstep with successful cycles. The mechanism self-bounds cumulative recovery at the empirically realised number of successful cycles rather than at the registered calendar horizon, and the buffer is preserved through every abort so that recovery resumes intact at the next successful cycle. The empirical reading of the deferred-recovery rate over the trajectory horizon is therefore bounded above by the count of successful cycles between an entry's deferral and the close of the measurement window; reporting this rate without the successful-cycle gate would conflate the recovery mechanism with the trajectory's retention-guard firing pattern.

The per-benchmark store-rate imbalance reported above arises from two dominant upstream sources and a small residual judge-side effect, ordered by magnitude. The dominant source is base-model accuracy, which differs sharply across the training panel and bounds the population of correct-and-confident outputs that the gate could in principle admit at the registered cut-point; the empirical-precision anchor at τ_{store} is read on outputs the generator produces correctly with high confidence, so benchmarks where the base accuracy is lower carry a smaller storeable population by construction. The secondary source is benchmark-conditional retrieval-grounding success: under the locked configuration the strongest discriminative signal in the composite is the external-grounding family of signals (passage-level entailment maximum, mean, and weakest-link atomic), and the dense passage substrate finds an entailing passage more often on the claim-verification benchmark, where the prompt registers a self-contained claim against a curated evidence corpus, than on the factoid-recall and open-domain-question-answering benchmarks, where the answer fragment must be entailed by a free-form passage that the retriever may not always return. When grounding succeeds the gate admits the answer at a rate consistent with the calibration-fold contract regardless of which benchmark it came from, with the empirical mean grounding score on stored-correct outputs landing in the same range across benchmarks; when grounding fails the gate correctly rejects, and benchmarks with lower grounding-success rates therefore store fewer entries. The residual third source is calibration drift of the natural-language-inference judge across benchmark families, which is small under the present input format because the judge receives the retrieved passage as the premise and the bare answer string as the hypothesis for all short-answer benchmarks alike, with no benchmark-specific reformulation; cross-benchmark differences in judge behaviour at this input shape are therefore bounded above by the residual cross-benchmark variance after the first two sources have been accounted for.

The architectural reading of this imbalance is that the gate is operating exactly as the precision contract requires. The contract specifies a calibration-fold empirical-precision anchor at the locked cut-point, not a floor on the storage rate or a balance constraint across benchmarks; under that contract the gate admits only the outputs that achieve high calibrated confidence, regardless of which benchmark they come from. When the two upstream sources above compress the population of high-confidence-correct outputs on a particular benchmark family, the gate's response

of storing fewer of them is the contract working correctly, not failing. Storing low-confidence outputs to equalise the per-benchmark store rate would force the gate to admit material below the cut-point, contaminating the verified pool and degrading every downstream subsystem that draws on it: the self-improvement training pool, the memory-direct retrieval tier at inference, and the retroactive re-verification pass that depends on a high-quality starting point. The deferred-entry buffer and the per-cycle re-verification pass provide the architectural recovery for outputs that were correct but did not clear the storage cut-point in their cycle of origin, and the cumulative recovery across the trajectory horizon is the empirical realisation of Heuristic 4.6 registered above. The per-benchmark store-rate imbalance is therefore the system reporting upstream heterogeneity faithfully through the gate rather than evidence of gate misbehaviour; the per-benchmark composite under the shrinkage prior, registered in Section 4.2.7 as the v2 keystone, is the principled architectural response that admits per-benchmark calibrated probabilities while regularising weak-signal panels toward the pooled fit, and the deployment study registered in Chapter 6 is the principled route to evaluating whether the residual heterogeneity matters for any specific live workload.

A complementary route to reducing the per-benchmark imbalance, alongside the per-benchmark composite under shrinkage, is corpus expansion. The retrieval-grounding success rate that drives the secondary cause of the imbalance is bounded above by the coverage of the dense passage index used by the Tier 3 path. The Wikipedia-derived index in the locked configuration was registered before the empirical programme as a stable, reproducible, public substrate; replacing or augmenting the index with a richer corpus, whether through later-snapshot Wikipedia, multi-source web text, or domain-specific knowledge bases, would raise the entailment-success rate on the open-text factoid and commonsense-reasoning benchmarks and would proportionally narrow the per-benchmark store-rate ratio at the gate. The expansion would not eliminate the imbalance entirely, because the dominant cause is base-model accuracy and the gate's empirical-precision anchor preserves rejection of low-confidence generator outputs even when the retrieval substrate has improved; the expansion would, however, close the secondary cause and shift the residual closer to the base-model floor reported in Remark 4.11. Corpus expansion is registered as a future-work item in Chapter 6 alongside the deployment study.

Benchmark	Panel	n	EM	Store n	Store rate	Store precision
FEVER (ID)	ID	500	0.480	63	12.6%	0.794
TriviaQA (ID)	ID	500	0.454	145	29.0%	0.821
CSQA (ID)	ID	500	0.750	81	16.2%	0.753
TruthfulQA (T)	Trans	500	0.384	67	13.4%	0.522
StrategyQA (T)	Trans	500	0.606	21	4.2%	0.905

Table 5.31: Cycle-zero per-benchmark headline diagnostics under the locked configuration. Each row reports the sample count on the cycle-zero validation fold ($n = 500$ per benchmark; not the calibration fold), the exact-match rate, the storage-class count and rate at the locked cut-point $\tau_{\text{store}} = 0.60$, and the realised precision of the stored entries. The training panel (FEVER, TriviaQA, CommonsenseQA) supplies the calibration fold and the self-improvement training pool; the transfer panel (TruthfulQA, StrategyQA) is held out from calibration and from the self-improvement pool, and its storage columns report the gate's decision on held-out queries as a diagnostic only. Three benchmarks registered earlier in the project (HotpotQA, Natural Questions, and ARC-Challenge) were retired before the main run on the registered verifier-balanced-accuracy precondition evidence and are absent from the panel.

5.2.6 The long-hypothesis judge ablation

A second entailment-scoring backend, the frozen Qwen-3B long-hypothesis judge, was registered alongside the MiniCheck judge to handle hypothesis lengths above the MiniCheck encoder's effective range. To use it as a calibrated drop-in for MiniCheck on long hypotheses, the architecture requires that the two backends agree on the supported-versus-unsupported judgment with a registered floor on rank correlation in logit space of $\rho_{\min} = 0.70$. A Platt-fit calibration on a 500-sample overlap fold returned a rank correlation of Pearson $\rho = 0.5843$, below the floor; the long-hypothesis judge is therefore ablated from the deployed configuration. The full Platt-fit diagnostic (slope, intercept, mean absolute error in both logit and probability space) is recorded in Section B.4. The unified verifier dispatches every hypothesis through MiniCheck regardless of length, with a documented 512-token truncation on hypotheses above MiniCheck's encoder range; the architectural cost is bounded above by the truncation effect on the rare hypotheses that exceed the encoder range, since none of the five panel benchmarks regularly produces hypothesis lengths above the truncation point at the registered prompt configuration. The future-work item to refit the long-hypothesis judge under task-specific training is registered in Chapter 6.

5.2.7 The validation gate verdict

The cycle-zero validation gate is a pre-specified conjunction of checks recorded with the cycle-zero artefact set. The first check requires the composite Cohen's d on the training-pooled evaluation fold to clear $d_{\text{id}} \geq 0.20$, which audits the precondition of Proposition 4.2 that the verifier discriminates correct from incorrect outputs strictly above chance. The second check requires the per-benchmark storage-versus-discard mean-composite gap to clear $\Delta \geq 0.05$, which audits the architecture's claim that the storage rule's behaviour is consistent across training benchmarks rather than driven

by one dominating subset. The third check requires the per-benchmark memory-poisoning rate, defined as the share of stored entries whose stored answer is wrong, to fall below 0.50. The fourth check requires that no single signal exhibit strong-direction inversion against its registered direction. Under the locked configuration the verdict is PASS: every gate clears its registered threshold, as reported in Table 5.32. The headline training-pooled composite Cohen's d on the evaluation fold reads +0.635, clearing the 0.20 precondition floor by more than three times the registered margin; the transfer-pooled composite d reads +0.173 and is reported alongside as an out-of-distribution diagnostic rather than as a gate, consistent with the registered scope under which the gate audits the cal-fold-versus-eval-fold relationship on the training panel. The per-benchmark store-versus-discard mean-composite gap reads 0.282 on FEVER, 0.320 on TriviaQA, and 0.258 on CommonsenseQA against the 0.05 floor. The per-benchmark memory-poisoning rates read 0.206 on FEVER, 0.179 on TriviaQA, and 0.247 on CommonsenseQA, all clearing the 0.50 ceiling by substantial margins. No strong-signal inversion is observed at the cycle-zero fit. The verdict licenses the deployed configuration to enter the main run. The per-signal Cohen's d decomposition that supports the composite gate is reported in Table 5.33 alongside the transfer-panel diagnostic.

		Training (ID) panel			**Transfer panel**		
Gate	**Threshold**	FEVER	TriviaQA	CSQA	TruthfulQA	StrategyQA	**Pooled**
ID-panel gates (training fold)							
Composite Cohen's d (ID)	≥ 0.20	+0.57	+1.52	+0.04	—	—	**+0.63**
Store-vs-discard gap	≥ 0.05	+0.28	+0.32	+0.26	—	—	—
Memory poisoning rate	< 0.50	0.21	0.18	0.25	—	—	—
Transfer-panel diagnostic (out-of-distribution, not a gate)							
Composite Cohen's d (Transfer)	—	—	—	—	+0.29	+0.30	**+0.17**

Overall verdict: **pass** (every registered ID-panel gate clears its threshold).

Table 5.32: Cycle-zero validation gate verdict under the locked configuration on the locked five-benchmark panel. All cells are computed on the held-out cycle-zero evaluation fold rather than on the calibration fold, so the audited quantity is the locked composite's inference-time behaviour. The composite Cohen's d gate audits the precondition of Proposition 4.2 (verifier discriminates above chance, registered floor 0.20). Store-vs-discard gap and memory poisoning rate are reported on the training panel only because the storage gate does not fire on transfer-panel queries by design. The per-benchmark Cohen's d heterogeneity and the Simpson's-style pooling artefact on the transfer panel are discussed in the surrounding prose; the verdict licenses the locked configuration to enter the main run.

Signal	Training (ID)				Transfer (eval only)		
	FEVER	TriviaQA	CSQA	**ID pool**	TruthfulQA	StrategyQA	**Trans pool**
u_{stored}	+0.57	+1.52	+0.04	**+0.63**	+0.29	+0.30	**+0.17**
$p^{\text{mean}}_{\text{ground}}$	+0.50	+1.12	+0.04	**+0.47**	+0.24	+0.29	**+0.11**
$p^{\text{max}}_{\text{ground}}$	+0.58	+1.15	+0.02	**+0.49**	+0.21	+0.33	**+0.09**
$p^{\text{atomic}}_{\text{ground}}$	+0.50	+1.12	+0.04	**+0.47**	+0.24	+0.29	**+0.11**
p_{entail}	+0.15	+1.26	-0.18	**+0.20**	+0.26	+0.18	**+0.19**
$q_{a,\text{rel}}$	-0.37	+0.51	+0.35	**+0.26**	+0.33	+0.19	**+0.42**
u_{internal}	+0.14	+0.34	+0.01	**+0.55**	+0.11	-0.11	**+0.20**
u_{token}	+0.21	-0.09	-0.15	**-0.03**	+0.14	+0.03	**+0.19**
u_{dropout}	-0.10	-0.37	-0.30	**-0.56**	-0.02	+0.11	**-0.19**
s_{avg}	+0.00	+0.58	-0.02	**+0.30**	-0.12	-0.07	**-0.13**

Table 5.33: Per-signal Cohen's d between exact-match-correct and exact-match-incorrect outputs on the held-out cycle-zero evaluation fold of the locked five-benchmark panel, after the registered HotpotQA, Natural Questions, and ARC-Challenge retirements. Positive values indicate the signal is higher on em-correct samples; the column ID pool governs the precondition of Proposition 4.2 that the verifier discriminates correct from incorrect outputs strictly above chance, and the column Trans pool reports the out-of-distribution generalisation of the same signal set to benchmarks excluded from calibration. The calibrated probability composite recorded on the first row as u_{stored} aggregates the per-signal readings through per-signal isotonic regression, per-benchmark shrinkage at $\alpha_{\text{shrink}} = 0.6$, and a logistic boost layer, and is the quantity that the validation gate audits against its registered floor of 0.20. All cells computed on the held-out evaluation fold rather than the calibration-fit fold, so the readings reflect the locked composite's behaviour at inference time rather than at fit time.

5.2.8 Failed iterations as evidence of method discipline

The first iteration of the cycle-zero calibration phase failed the pre-specified validation gate. The root cause was a signal-dump bug in the calibration-run script: it wrote only a subset of the nine per-signal values per sample, so the per-signal isotonic regression had complete information for only the dumped signals and degenerate identity-mapping for the remaining ones. The downstream effect was that the L2-regularised boost layer fitted on the truncated signal vector produced a degenerate composite that violated the validation gate's Cohen's d floor. The bug was identified by the validation-gate diagnostic on the very first run, fixed in a worktree patch that dumps all nine signals per sample, and the calibration was re-run on the corrected pipeline; the second iteration produced the locked configuration that the main run uses. The iteration is recorded transparently in the project log rather than absorbed into a smoothed timeline because the cycle-zero re-iteration is itself empirical evidence that the validation-gate discipline catches what it was registered to catch: a degenerate calibration that would otherwise have propagated silently into every cycle of the main run. The full audit summary across pre-fix and post-fix phases is auto-generated by the chapter-five artefact-generation script.

5.3 Statistical Analysis

5.3.1 Matched-protocol pairing rule

Every per-benchmark contrast between the architecture and a baseline is computed sample-by-sample under matched protocol. Each baseline is evaluated on the identical evaluation-fold sample identifiers as the architecture, with the identical prompt template family and the identical scoring rule, so that the two systems' per-sample correctness vectors align entry-for-entry. The pairing rule eliminates the largest source of nuisance variance in cross-system comparison: a difference in sample selection or in scorer behaviour can manifest as an apparent performance difference even when the systems are equivalent on the underlying task. Paired test statistics (McNemar) and paired bootstrap intervals are valid under this pairing because every sample contributes one paired observation, and unpaired test statistics (Wilcoxon rank-sum, two-sample t) are not used in this thesis precisely because they discard the pairing information. The protocol is enforced at the evaluation-harness level through a sample-identifier manifest that every baseline's evaluation log must agree with; mismatched manifests halt the comparison rather than producing a quietly-misaligned table.

5.3.2 Paired McNemar with continuity correction

Per-benchmark significance against each external baseline is reported through the paired McNemar test with the continuity correction of [1]. Let b denote the count of samples on which the architecture is correct and the baseline is incorrect, and c the count of samples on which the baseline is correct and the architecture is incorrect; samples on which both are correct or both are incorrect contribute zero to the test statistic. Under the null hypothesis that the two systems have the same correctness

probability, the continuity-corrected statistic is

$$\chi^2_{\mathrm{McN}} \;=\; \frac{(|b-c|-1)^2}{b+c}, \tag{5.2}$$

which is χ^2-distributed with one degree of freedom in the limit, with the -1 continuity correction applied to mitigate the discreteness of b and c at small sample sizes. The test is preferred at this site over a paired t-test because the per-sample correctness signal is binary and the McNemar statistic is the standard test for matched-pairs binary outcomes. The reported p-value is two-sided.

5.3.3 Bootstrap confidence intervals

Alongside the McNemar p-value, every per-benchmark contrast reports a bootstrap ninety-five-percent confidence interval on the exact-match delta. Sampling is paired at the sample-identifier level: each bootstrap replicate draws n_{eval} sample identifiers with replacement from the evaluation fold, computes the architecture's exact match and the baseline's exact match on the resampled fold, and records the difference. The confidence interval is the percentile interval at the 0.025 and 0.975 quantiles of the bootstrap distribution, with $B = 1\,000$ replicates. The interval on the composite-hallucination delta is approximate: it resamples only the baseline's per-subtype rates, with CAEM's composite hallucination value held fixed, so it can exclude the point difference. The bootstrap interval is reported alongside the McNemar p-value because p-values alone do not convey effect size: a contrast that is significant at $p < 0.001$ but has a confidence interval whose lower bound straddles zero is a negative result on the effect-size axis, and reporting both is what allows the reader to read the strength of the contrast as well as its statistical reliability.

5.3.4 Holm correction within each benchmark family

Each benchmark family in the panel admits eight contrasts of the architecture against the eight external baselines (B1 zero-shot, B2 chain-of-thought, B3 retrieval-augmented, B4 chain-of-thought plus retrieval, B5 few-shot chain-of-thought, B6 vanilla fine-tuning, B7 forward-looking active retrieval, B8 semantic-entropy abstention). The eight p-values within a benchmark family are jointly corrected for multiple comparison using the Holm-Bonferroni procedure [4], which orders the eight p-values in ascending sequence and applies the threshold $\alpha/(k+1-i)$ to the i-th smallest, where $k = 8$ and $\alpha = 0.05$. The Holm correction is preferred over plain Bonferroni at this site because Holm has uniformly higher power for the same family-wise error rate, and is preferred over the false-discovery-rate procedure of Benjamini-Hochberg because the family-wise error rate is the convention required for architectural-claim licensing in this thesis. The marker convention reports a contrast as significant only if the corrected p clears α; the raw p-value and the Holm-adjusted p-value are both tabulated so that the reader can see the correction's effect.

5.3.5 Architectural-claim licensing rule

Statistical significance alone is not sufficient to license an architectural claim in this thesis. The pre-specified licensing rule requires a conjunction of two conditions on

every contrast: the Holm-adjusted p-value must clear 0.05 within the benchmark family, and the bootstrap ninety-five-percent confidence interval on the exact-match delta must lie entirely above 0.02 (a two-percentage-point effect-size floor). The effect-size floor is what prevents a statistically-significant-but-practically-trivial contrast from licensing an architectural claim. Both conditions are read off the same paired evaluation, so the licensing rule does not require additional runs; it simply applies a two-of-two filter to the joint significance-and-effect-size table. A contrast that satisfies both conditions is reported as supporting the corresponding hypothesis from Section 1.4.3; a contrast that satisfies one but not the other is reported as partially supporting; a contrast that satisfies neither is reported as unsupported.

5.4 Comparisons and Relationships

5.4.1 External baseline panel under matched protocol

The eight external baselines (B1 through B8) are evaluated under the matched-protocol pairing rule of Section 5.3.1, with paired McNemar significance on the exact-match axis and paired Wilcoxon signed-rank significance on the composite-hallucination-metric axis, plus bootstrap confidence intervals, reported per benchmark per baseline on both axes. The panel partitions into six inference-only baselines (B1 through B5 and B7), one training-time baseline (B6), and one sampling-based confidence-calibration baseline (B8). Every baseline's composite-hallucination-metric column is produced by replaying that baseline's question-and-answer pairs through the locked verifier ensemble and the cycle-three calibrated composite, so the cross-system hallucination contrast holds the scoring instrument fixed and isolates the generator-side contribution; the CAEM trajectory rows, by contrast, each carry that cycle's own per-cycle composite, so the baseline panel is read as a fixed-instrument comparison anchored at the cycle-three calibration rather than a per-cycle-verifier comparison. The headline pooled exact-match and pooled composite-hallucination-metric contrast against both the within-trajectory best CAEM cycle (C2) and the trajectory final cycle (C5) is reported in Table 5.34 as the at-a-glance reading; the per-benchmark paired-significance receipts on both metric axes are then reported at both reference cycles in Table 5.35 (cycle-five close) and Table 5.36 (within-trajectory best cycle), each computed from the eight-baseline by five-benchmark grid under the Holm-Bonferroni correction of Section 5.3.4 applied independently within each benchmark family on each metric axis. The trajectory-end panel records nine significant exact-match wins and three significant defeats and twenty-one significant composite-hallucination wins and three significant defeats; the best-cycle panel records fourteen significant exact-match wins and three significant defeats and twenty significant composite-hallucination wins and four significant defeats. Every significant defeat, on both axes and at both anchors, is confined to TruthfulQA against the high-EM RAG-free baselines (zero-shot, chain-of-thought, FLARE, semantic entropy) where the architecture's elevated abstention rate inflates the false-refusal subtype of the composite while the baselines pay the EM penalty instead, the retrieval-amplification-and-abstention trade-off documented in Section 5.6.4. The five additional exact-match wins at the best-cycle anchor relative to the trajectory-end anchor are the per-benchmark licensing receipt that the best cycle is the stronger exact-match operating point, while the trajectory-end anchor carries marginally more

composite-hallucination wins (twenty-one against twenty) and one fewer composite-hallucination defeat, consistent with the cycle-five joint-axis Composite Evaluation Score peak reported in Table 5.4.

				vs. CAEM C2 (best)		vs. CAEM C5 (final)	
#	System	EM	CHM	ΔEM	ΔCHM	ΔEM	ΔCHM
	CAEM C2 (best)	**0.544**	**0.107**	—	—	+0.027	−0.002
	CAEM C5 (final)	0.517	0.109	−0.027	+0.002	—	—
B1	zero-shot	0.507	0.120	+0.037 (+7.4%)	−0.013 (−11.0%)	+0.011 (+2.1%)	−0.011 (−9.3%)
B2	chain-of-thought	0.500	0.123	+0.044 (+8.8%)	−0.016 (−12.7%)	+0.017 (+3.5%)	−0.014 (−11.0%)
B3	RAG	0.458	0.148	+0.086 (+18.8%)	−0.041 (−27.7%)	+0.059 (+13.0%)	−0.039 (−26.4%)
B4	CoT + RAG	0.464	0.142	+0.080 (+17.2%)	−0.035 (−24.6%)	+0.053 (+11.5%)	−0.033 (−23.2%)
B5	five-shot CoT	0.405	0.136	+0.139 (+34.2%)	−0.029 (−21.4%)	+0.112 (+27.6%)	−0.027 (−19.9%)
B6	vanilla fine-tune (C5)	0.490	0.127	+0.054 (+11.0%)	−0.020 (−15.9%)	+0.027 (+5.6%)	−0.018 (−14.3%)
B7	FLARE	0.525	0.173	+0.019 (+3.5%)	−0.066 (−38.0%)	−0.008 (−1.5%)	−0.064 (−36.8%)
B8	semantic entropy	0.505	0.120	+0.039 (+7.8%)	−0.013 (−10.5%)	+0.013 (+2.5%)	−0.011 (−8.9%)

Wins-against count (CAEM beats baseline on ΔEM and ΔCHM jointly): CAEM C2 beats 8/8; CAEM C5 beats 7/8.

Table 5.34: Pooled EM and pooled CHM for every external baseline on the five-benchmark evaluation panel, with absolute and relative deltas against the within-trajectory best CAEM cycle (C2) and the trajectory-end cycle (C5). All numbers are unweighted means over the five panel benchmarks; TruthfulQA EM uses the Haiku-judged correctness rule for every row in the panel including the B8 semantic-entropy abstention row whose TruthfulQA predictions were rescored by the Haiku judge over the post-rerun abstain outputs. CHM for every baseline is read off the post-hoc rescore through the locked CAEM verifier, so the CHM column is apples-to-apples across the panel. Per-benchmark significance receipts are in Table 5.35 (C5) and Table 5.36 (C2).

Zero-shot (B1)

The zero-shot baseline runs the same Qwen-2.5-3B-Instruct generator on the same evaluation fold without any chain-of-thought trigger, without retrieval, and without verification. It is the empirical floor against which every other system in the panel is measured: any gain a system reports must exceed the zero-shot score on the same backbone, otherwise the gain is attributable to model capacity rather than to the system's added mechanism. For CAEM, the contrast against B1 isolates the joint contribution of routing, verification, episodic memory, and self-improvement; the contrast against B1 minus the contrast against B6 (vanilla fine-tuning) decomposes the contribution between deployment-time mechanisms and training-time mechanisms.

Chain of thought (B2)

The chain-of-thought baseline [40] adds the zero-shot step-by-step trigger of [38] to the zero-shot prompt without any further modification. CoT isolates the contribution of intermediate reasoning on the same backbone and controls for the possibility that CAEM's gains reflect better rationale generation rather than the verifier-memory architecture. The expected effect under matched protocol is a positive contrast against B1 on the multi-step benchmarks (StrategyQA, CommonsenseQA) and a smaller contrast on the single-step benchmarks (FEVER, TriviaQA). The literature

Baseline	Bench	n	EM_{CAEM}	EM_{base}	ΔEM	CHM_{CAEM}	CHM_{base}	ΔCHM	Sig (EM / CHM)	95% CI (ΔEM / ΔCHM)
cot	fever	300	0.477	0.420	+0.057	0.126	0.168	-0.042	- / *	[-0.027, +0.143] / [-0.034, -0.012]
cot_rag	fever	300	0.477	0.437	+0.040	0.126	0.171	-0.045	- / *	[-0.057, +0.137] / [-0.037, -0.014]
fiveshot_cot	fever	300	0.477	0.490	-0.013	0.126	0.141	-0.015	- / -	[-0.090, +0.067] / [-0.010, +0.012]
flare	fever	300	0.477	0.437	+0.040	0.126	0.211	-0.085	- / *	[-0.053, +0.140] / [-0.074, -0.049]
rag	fever	300	0.477	0.437	+0.040	0.126	0.180	-0.054	- / *	[-0.050, +0.133] / [-0.046, -0.022]
semantic_entropy	fever	300	0.477	0.477	+0.000	0.126	0.157	-0.031	- / *	[-0.080, +0.080] / [-0.024, -0.002]
vanilla_ft	fever	300	0.477	0.543	-0.067	0.126	nan	n/a	- / -	[-0.123, -0.010] / [nan, nan]
zero_shot	fever	300	0.477	0.453	+0.023	0.126	0.159	-0.033	- / *	[-0.060, +0.110] / [-0.026, -0.003]
cot	triviaqa	300	0.453	0.270	+0.183	0.125	0.188	-0.063	* / *	[+0.127, +0.240] / [-0.053, -0.030]
cot_rag	triviaqa	300	0.453	0.397	+0.057	0.125	0.133	-0.008	- / -	[+0.000, +0.110] / [-0.003, +0.017]
fiveshot_cot	triviaqa	300	0.453	0.137	+0.317	0.125	0.170	-0.045	* / *	[+0.253, +0.373] / [-0.036, -0.016]
flare	triviaqa	300	0.453	0.417	+0.037	0.125	0.244	-0.119	- / *	[-0.020, +0.093] / [-0.106, -0.079]
rag	triviaqa	300	0.453	0.400	+0.053	0.125	0.137	-0.012	- / -	[+0.000, +0.103] / [-0.007, +0.013]
semantic_entropy	triviaqa	300	0.453	0.303	+0.150	0.125	0.175	-0.050	* / *	[+0.090, +0.203] / [-0.042, -0.019]
vanilla_ft	triviaqa	300	0.453	0.227	+0.227	0.125	nan	n/a	* / -	[+0.163, +0.283] / [nan, nan]
zero_shot	triviaqa	300	0.453	0.293	+0.160	0.125	0.175	-0.050	* / *	[+0.100, +0.210] / [-0.042, -0.019]
cot	commonsense_qa	300	0.717	0.710	+0.007	0.091	0.095	-0.004	- / -	[-0.050, +0.057] / [-0.002, +0.016]
cot_rag	commonsense_qa	300	0.717	0.630	+0.087	0.091	0.124	-0.033	* / *	[+0.027, +0.147] / [-0.029, -0.009]
fiveshot_cot	commonsense_qa	300	0.717	0.747	-0.030	0.091	0.084	+0.007	- / -	[-0.080, +0.023] / [+0.008, +0.025]
flare	commonsense_qa	300	0.717	0.723	-0.007	0.091	0.128	-0.037	- / *	[-0.060, +0.047] / [-0.033, -0.011]
rag	commonsense_qa	300	0.717	0.600	+0.117	0.091	0.134	-0.043	* / *	[+0.063, +0.173] / [-0.039, -0.018]
semantic_entropy	commonsense_qa	300	0.717	0.760	-0.043	0.091	0.097	-0.006	- / -	[-0.093, +0.003] / [-0.004, +0.014]
vanilla_ft	commonsense_qa	300	0.717	0.780	-0.063	0.091	nan	n/a	- / -	[-0.120, -0.010] / [nan, nan]
zero_shot	commonsense_qa	300	0.717	0.703	+0.013	0.091	0.096	-0.005	- / -	[-0.043, +0.067] / [-0.003, +0.015]
cot	truthfulqa	300	0.300	0.437	-0.137(B)	0.104	0.046	+0.058(B)	(B) / (B)	[-0.207, -0.067] / [+0.055, +0.070]
cot_rag	truthfulqa	300	0.300	0.240	+0.060	0.104	0.134	-0.030	- / *	[-0.003, +0.123] / [-0.026, -0.005]
fiveshot_cot	truthfulqa	300	0.300	0.157	+0.143	0.104	0.155	-0.051	* / *	[+0.077, +0.210] / [-0.043, -0.026]
flare	truthfulqa	300	0.300	0.450	-0.150(B)	0.104	0.095	+0.009	(B) / -	[-0.220, -0.080] / [+0.009, +0.029]
rag	truthfulqa	300	0.300	0.243	+0.057	0.104	0.138	-0.034	- / *	[-0.003, +0.117] / [-0.030, -0.008]
semantic_entropy	truthfulqa	300	0.300	0.307	-0.007	0.104	0.050	+0.054(B)	- / (B)	[-0.067, +0.057] / [+0.052, +0.066]
vanilla_ft	truthfulqa	300	0.300	0.273	+0.027	0.104	nan	n/a	- / -	[-0.040, +0.093] / [nan, nan]
zero_shot	truthfulqa	300	0.300	0.433	-0.133(B)	0.104	0.051	+0.053(B)	(B) / (B)	[-0.207, -0.060] / [+0.051, +0.065]
cot	strategyqa	300	0.640	0.663	-0.023	0.099	0.116	-0.017	- / -	[-0.093, +0.050] / [-0.014, +0.006]
cot_rag	strategyqa	300	0.640	0.603	+0.037	0.099	0.149	-0.050	- / *	[-0.040, +0.117] / [-0.044, -0.023]
fiveshot_cot	strategyqa	300	0.640	0.497	+0.143	0.099	0.130	-0.031	* / *	[+0.073, +0.223] / [-0.027, -0.007]
flare	strategyqa	300	0.640	0.600	+0.040	0.099	0.185	-0.086	- / *	[-0.037, +0.133] / [-0.078, -0.054]
rag	strategyqa	300	0.640	0.610	+0.030	0.099	0.151	-0.052	- / *	[-0.043, +0.110] / [-0.046, -0.025]
semantic_entropy	strategyqa	300	0.640	0.677	-0.037	0.099	0.119	-0.020	- / -	[-0.107, +0.043] / [-0.017, +0.003]
vanilla_ft	strategyqa	300	0.640	0.627	+0.013	0.099	nan	n/a	- / -	[-0.050, +0.083] / [nan, nan]
zero_shot	strategyqa	300	0.640	0.650	-0.010	0.099	0.120	-0.021	- / -	[-0.087, +0.070] / [-0.017, +0.002]

Table 5.35: Paired McNemar significance on the exact-match axis and paired Wilcoxon signed-rank significance on the composite-hallucination axis (both Holm-corrected within each benchmark family), with bootstrap ninety-five percent confidence intervals on both axes, for CAEM at the cycle-five close versus each of the eight external baselines. The Sig column reports the per-axis verdict as a pair (star clears the licensing rule of Section 5.3.5; dash does not). The (B) suffix marks significant baseline-better cells. TruthfulQA EM uses the Haiku-judged rule uniformly across every baseline including the semantic-entropy abstention row. The best-cycle (C2) panel is in Table 5.36.

Baseline	Bench	n	EM_{CAEM}	EM_{base}	ΔEM	CHM_{CAEM}	CHM_{base}	ΔCHM	Sig (EM / CHM)	95% CI (ΔEM / ΔCHM)
cot	fever	300	0.533	0.420	+0.113	0.102	0.168	-0.066	- / *	[+0.033, +0.200] / [-0.058, -0.036]
cot_rag	fever	300	0.533	0.437	+0.097	0.102	0.171	-0.069	- / *	[+0.000, +0.187] / [-0.061, -0.038]
fiveshot_cot	fever	300	0.533	0.490	+0.043	0.102	0.141	-0.039	- / *	[-0.037, +0.123] / [-0.034, -0.012]
flare	fever	300	0.533	0.437	+0.097	0.102	0.211	-0.109	- / *	[+0.007, +0.187] / [-0.098, -0.073]
rag	fever	300	0.533	0.437	+0.097	0.102	0.180	-0.078	- / *	[+0.007, +0.187] / [-0.070, -0.046]
semantic_entropy	fever	300	0.533	0.477	+0.057	0.102	0.157	-0.055	- / *	[-0.027, +0.143] / [-0.048, -0.026]
vanilla_ft	fever	300	0.533	0.543	-0.010	0.102	nan	n/a	- / -	[-0.067, +0.053] / [nan, nan]
zero_shot	fever	300	0.533	0.453	+0.080	0.102	0.159	-0.057	- / *	[-0.003, +0.163] / [-0.050, -0.027]
cot	triviaqa	300	0.493	0.270	+0.223	0.109	0.188	-0.079	* / *	[+0.167, +0.277] / [-0.069, -0.046]
cot_rag	triviaqa	300	0.493	0.397	+0.097	0.109	0.133	-0.024	* / -	[+0.047, +0.147] / [-0.019, +0.001]
fiveshot_cot	triviaqa	300	0.493	0.137	+0.357	0.109	0.170	-0.061	* / *	[+0.293, +0.417] / [-0.052, -0.032]
flare	triviaqa	300	0.493	0.417	+0.077	0.109	0.244	-0.135	* / *	[+0.020, +0.130] / [-0.122, -0.095]
rag	triviaqa	300	0.493	0.400	+0.093	0.109	0.137	-0.028	* / *	[+0.043, +0.143] / [-0.023, -0.003]
semantic_entropy	triviaqa	300	0.493	0.303	+0.190	0.109	0.175	-0.066	* / *	[+0.137, +0.237] / [-0.058, -0.035]
vanilla_ft	triviaqa	300	0.493	0.227	+0.267	0.109	nan	n/a	* / -	[+0.203, +0.320] / [nan, nan]
zero_shot	triviaqa	300	0.493	0.293	+0.200	0.109	0.175	-0.066	* / *	[+0.140, +0.257] / [-0.058, -0.035]
cot	commonsense_qa	300	0.717	0.710	+0.007	0.093	0.095	-0.002	- / -	[-0.047, +0.060] / [-0.000, +0.018]
cot_rag	commonsense_qa	300	0.717	0.630	+0.087	0.093	0.124	-0.031	* / *	[+0.030, +0.143] / [-0.027, -0.007]
fiveshot_cot	commonsense_qa	300	0.717	0.747	-0.030	0.093	0.084	+0.009	- / -	[-0.077, +0.017] / [+0.010, +0.027]
flare	commonsense_qa	300	0.717	0.723	-0.007	0.093	0.128	-0.035	- / *	[-0.053, +0.040] / [-0.031, -0.009]
rag	commonsense_qa	300	0.717	0.600	+0.117	0.093	0.134	-0.041	* / *	[+0.057, +0.177] / [-0.037, -0.016]
semantic_entropy	commonsense_qa	300	0.717	0.760	-0.043	0.093	0.097	-0.004	- / -	[-0.090, +0.007] / [-0.002, +0.016]
vanilla_ft	commonsense_qa	300	0.717	0.780	-0.063	0.093	nan	n/a	- / -	[-0.117, -0.007] / [nan, nan]
zero_shot	commonsense_qa	300	0.717	0.703	+0.013	0.093	0.096	-0.003	- / -	[-0.040, +0.067] / [-0.001, +0.017]
cot	truthfulqa	300	0.323	0.437	-0.113(B)	0.116	0.046	+0.070(B)	(B) / (B)	[-0.180, -0.043] / [+0.067, +0.082]
cot_rag	truthfulqa	300	0.323	0.240	+0.083	0.116	0.134	-0.018	* / -	[+0.027, +0.140] / [-0.014, +0.007]
fiveshot_cot	truthfulqa	300	0.323	0.157	+0.167	0.116	0.155	-0.039	* / *	[+0.107, +0.230] / [-0.031, -0.014]
flare	truthfulqa	300	0.323	0.450	-0.127(B)	0.116	0.095	+0.021(B)	(B) / (B)	[-0.200, -0.057] / [+0.021, +0.041]
rag	truthfulqa	300	0.323	0.243	+0.080	0.116	0.138	-0.022	* / -	[+0.020, +0.137] / [-0.018, +0.004]
semantic_entropy	truthfulqa	300	0.323	0.307	+0.017	0.116	0.050	+0.066(B)	- / (B)	[-0.040, +0.077] / [+0.064, +0.078]
vanilla_ft	truthfulqa	300	0.323	0.273	+0.050	0.116	nan	n/a	- / -	[-0.017, +0.117] / [nan, nan]
zero_shot	truthfulqa	300	0.323	0.433	-0.110(B)	0.116	0.051	+0.065(B)	(B) / (B)	[-0.177, -0.043] / [+0.063, +0.077]
cot	strategyqa	300	0.653	0.663	-0.010	0.115	0.116	-0.001	- / -	[-0.080, +0.060] / [+0.002, +0.022]
cot_rag	strategyqa	300	0.653	0.603	+0.050	0.115	0.149	-0.034	- / *	[-0.013, +0.113] / [-0.028, -0.007]
fiveshot_cot	strategyqa	300	0.653	0.497	+0.157	0.115	0.130	-0.015	* / -	[+0.087, +0.223] / [-0.011, +0.009]
flare	strategyqa	300	0.653	0.600	+0.053	0.115	0.185	-0.070	- / *	[-0.013, +0.123] / [-0.062, -0.038]
rag	strategyqa	300	0.653	0.610	+0.043	0.115	0.151	-0.036	- / *	[-0.020, +0.103] / [-0.030, -0.009]
semantic_entropy	strategyqa	300	0.653	0.677	-0.023	0.115	0.119	-0.004	- / -	[-0.093, +0.043] / [-0.001, +0.019]
vanilla_ft	strategyqa	300	0.653	0.627	+0.027	0.115	nan	n/a	- / -	[-0.037, +0.090] / [nan, nan]
zero_shot	strategyqa	300	0.653	0.650	+0.003	0.115	0.120	-0.005	- / -	[-0.067, +0.077] / [-0.001, +0.018]

Table 5.36: Same statistical procedure as Table 5.35 (paired McNemar on EM, paired Wilcoxon on CHM, both Holm-corrected per benchmark; bootstrap 95% CIs on both axes; $*/-$ pair in Sig; (B) for baseline-better; uniform Haiku-judged rule on TruthfulQA EM across every row), but anchored at the within-trajectory best cycle (cycle two). The best-cycle anchor records 14 EM wins / 3 EM defeats and 20 CHM wins / 4 CHM defeats against the cycle-five panel's 9 / 3 and 21 / 3; every significant defeat on both axes and at both anchors is confined to TruthfulQA against the high-EM RAG-free baselines, discussed in Section 5.6.4.

predicts the opposite at this scale: below roughly 100 billion parameters chain-of-thought prompting does not help and can hurt [40], and even an instruction-tuned 780-million-parameter model scores lower with it than with direct answering [68]. The realised pooled exact match (B2 0.500 against B1 0.507, Table 5.34) agrees with the literature rather than with the registered expectation.

Retrieval-augmented (B3)

The retrieval-augmented baseline retrieves the top-$k = 5$ Wikipedia passages from the dense passage index used by CAEM's Tier 3 [31], [32], concatenates them into a 384-token context, and generates from the same backbone through the Tier-3 prompt template. B3 isolates the contribution of external grounding without any verification or memory. The expected contrast against B1 is a substantial positive effect on the open-text factoid surface (TriviaQA) where retrieval supplies factual context that the parametric model lacks, and a smaller effect on TruthfulQA where the misconception is in the question framing rather than in the retrieval gap.

Chain of thought plus retrieval (B4)

The combined baseline runs CoT on top of B3's retrieval, isolating the contribution of reasoning over retrieved evidence. The expected contrast against B3 is positive on the benchmarks where the answer requires integration across multiple retrieved passages (StrategyQA, CommonsenseQA) and approximately neutral on benchmarks where a single passage suffices (TriviaQA). The contrast between B4 and CAEM is the most direct test of whether the verifier-memory architecture adds value beyond reasoning-over-retrieval on the same backbone.

Few-shot chain of thought (B5)

The few-shot chain-of-thought baseline prepends five in-context (question, reasoning, answer) demonstrations from the TriviaQA training split before the CoT trigger. B5 is the strongest prompting-only comparison against CAEM's retrieval-free Tier 2: both share the same backbone and the same prompt-only intervention surface, but B5 exposes the model to demonstrations rather than to memory or retrieval. The contrast against B5 isolates the contribution of CAEM's persistent episodic memory over a five-shot in-context window.

Vanilla fine-tuning (B6)

The vanilla fine-tuning baseline trains Qwen-2.5-3B-Instruct on the verified episode pool for the same realised cycle horizon as CAEM at matched per-benchmark chunk size, without the multi-modal retention probe and rollback guard, without the bounded low-rank-adapter envelope, and without the episodic memory or the verifier at inference time. Every selected episode is treated as a clean label. B6 isolates the contribution of CAEM's retention-protection machinery: if B6 degrades on the multi-modal probe panel or on the later cycles while CAEM does not, the retention probe panel and the rollback guard are responsible. The expected contrast under matched protocol is that CAEM's accuracy is at least as high as B6's on the in-distribution benchmarks while CAEM's retention ratio is strictly higher.

Forward-looking active retrieval (B7)

The forward-looking active retrieval baseline [50] is the active-retrieval extension of B3: rather than retrieving once at the start of generation, the model generates one sentence at a time and triggers a retrieval whenever the token-level confidence on the upcoming sentence falls below a registered threshold, with the retrieved passages prepended to the generation context for the next sentence. The intent is to retrieve only when retrieval is needed and to retrieve evidence targeted to the current generation step rather than to the question alone. B7 isolates the contribution of selective and step-conditioned retrieval over single-pass retrieval; the contrast against B3 measures the value of active retrieval, and the contrast against CAEM measures whether CAEM's verifier-and-memory machinery adds value beyond an active-retrieval system on the same backbone. The expected effect under matched protocol is a small positive contrast against B3 on the multi-step benchmarks where evidence accumulates across steps, and approximately neutral on single-step benchmarks where one retrieval suffices.

Semantic-entropy abstention (B8)

The semantic-entropy abstention baseline [70] computes a per-query uncertainty score by sampling ten generations at temperature one from the same backbone, clustering the samples by normalised-exact-match equivalence on the extracted answer (a simplification of the original method's bidirectional-entailment clustering, appropriate for short answers), and reading the entropy of the cluster distribution as the uncertainty signal. When the cluster entropy exceeds the registered abstention threshold $\tau_{\text{se}} = 1.5$ nats the baseline returns an abstention token; otherwise it returns the highest-frequency cluster's representative. The baseline runs the same backbone as B1 but exposes a sampling-based abstention surface that the zero-shot baseline lacks. B8 is the most direct external comparator to CAEM's calibrated-probability abstention because both produce a per-query confidence signal that supports a refuse-to-answer decision under uncertainty; the contrast registers whether CAEM's multi-signal calibrated composite outperforms a single-signal sampling-based uncertainty estimator on the same backbone. The expected contrast against CAEM is benchmark-dependent: on the misconception-bait surface (TruthfulQA), the sampling-based abstention is expected to fire heavily because the model entertains multiple competing answers, and the strict-correctness exact-match metric penalises every abstention as incorrect; on the confident-reasoning surfaces (CommonsenseQA, StrategyQA) the entropy threshold is expected to fire rarely and the mode-selection benefit of sampling over greedy decoding is expected to produce a small positive contrast against the zero-shot baseline. The TruthfulQA part of this expectation was not borne out, as the literature predicts: semantic entropy does not detect consistently held misconceptions [70], and B8 ranks among the high-exact-match retrieval-free baselines on TruthfulQA rather than being penalised for heavy abstention. The contrast registers the conditions under which a single-signal abstention beats or loses to a multi-signal calibrated composite, which is the architectural question the panel is designed to answer. The B8 implementation note is honest to record at this site: the original B8 run on the same hardware envelope collapsed to greedy single-pass under a batched-dispatch fast-path that bypassed the registered ten-sample temperature-sampling step, and was replaced by a corrected run that restores

the proper sampling protocol; the corrected run is the source for every B8 cell in Table 5.34, Table 5.35, and Table 5.36, and the Haiku rescorer that supplies the project's TruthfulQA correctness rule was re-invoked over the corrected B8 abstain predictions so that the TruthfulQA exact-match cell on the B8 row is scored under the same Haiku-judged rule as every other row in the panel.

5.4.2 Retrieval-utility classifier ablation

The retrieval-utility classifier introduced in Section 4.2.5 is the one architectural ablation reported within the realised cycle horizon of this thesis, because the same configuration without the classifier was run as a prior trajectory before the classifier was integrated and the two trajectories share the matched-protocol contract at the cycle-three composite anchor. The contrast is a head-to-head reading at the cycle-three close between the classifier-off trajectory (the architecture without the retrieval-utility classifier; queries that fail the Tier 1 combined-score gate and the Tier 2 similarity gate are dispatched unconditionally to the retrieval-augmented tier) and the classifier-on trajectory (the architecture with the classifier active at the safety-veto fall-through and the score-formula fall-through, per Table 4.2). Both trajectories are run against the same five-benchmark evaluation panel and the same Qwen-2.5-3B-Instruct backbone, with the composite calibration pinned at the same cycle-three anchor so the verifier signals are identical across the two arms. The contrast is therefore an ablation along the routing-decision axis: the only intentional architectural difference between the two arms is the presence of the classifier. The bundled-confound disclosure is that the classifier-on arm also ships a set of pipeline corrections (prompt-template fixes, composite-refit cadence improvements, and minor downstream bug fixes) accumulated across the same project iteration; the bundle attributes the net improvement to the joint effect of the classifier and the pipeline corrections, with the classifier being the architecturally load-bearing addition. The pre-specified acceptance gates of the ablation are recovery of the TruthfulQA exact-match and composite-hallucination cells toward the B1 floor without degrading the training-panel pooled reading by more than two percentage points on either axis.

Benchmark	Exact match			Composite hallucination			Tier 3 share		
	Off	**On**	**Δ** (pp)	**Off**	**On**	**Δ** (% rel)	**Off**	**On**	**Δ** (pp)
Training panel									
FEVER	0.503	0.503	0.0	0.117	0.120	+2.5%	92.3%	70.7%	−21.7
TriviaQA	0.437	0.463	+2.7	0.123	0.121	−2.0%	95.7%	68.0%	−27.7
CommonsenseQA	0.603	**0.743**	**+14.0**	0.131	**0.086**	**−34.4%**	57.7%	20.3%	−37.3
Transfer panel									
StrategyQA	0.617	0.640	+2.3	0.119	0.128	+8.1%	100.0%	76.7%	−23.3
TruthfulQA	0.210	**0.310**	**+10.0**	0.116	0.119	+2.9%	99.0%	40.0%	**−59.0**
Pooled (all five)	**0.474**	**0.532**	**+5.8**	**0.121**	**0.115**	**−5.2%**	**88.9%**	**55.1%**	**−33.8**

Table 5.37: Head-to-head ablation of the retrieval-utility classifier (Section 4.2.5) at the cycle-three close under the matched-protocol contract. The *Off* columns are the classifier-off arm in which queries that fail the Tier 1 combined-score gate and the Tier 2 similarity gate route unconditionally to the retrieval-augmented tier; the *On* columns are the classifier-on arm in which the same queries consult the retrieval-utility classifier and route to Tier 3 only when $p_{\text{RAG}} \geq \tau_{\text{RUC}} = 0.375$ per Algorithm 3. The exact match column for TruthfulQA uses the Haiku-judged correctness rule; the composite hallucination metric uses the canonical eight-axis composite of Table 5.6. Both arms share the same five-benchmark evaluation panel, the same Qwen-2.5-3B-Instruct backbone, and the same composite calibration pinned at the cycle-three anchor. The asymmetric-rescue pattern is read off the bold cells: the largest exact-match gains land on CommonsenseQA (+14.0 percentage points) and TruthfulQA (+10.0 percentage points), the two benchmarks where retrieval was hurting in the classifier-off arm; FEVER exact match is flat, the empirical receipt that the classifier correctly preserves the retrieval-helpful regime on the textbook fact-verification benchmark; the pooled retrieval-augmented tier share collapses by −33.8 percentage points, with the largest individual collapse on TruthfulQA at −59.0 percentage points. The composite hallucination reduction is concentrated on CommonsenseQA rather than uniform across the panel, consistent with the heterogeneous per-benchmark AUROC band of the classifier reported in Section 4.2.5. The contrast bundles the classifier with the pipeline corrections shipped in the same iteration; the bundled-confound disclosure and the registered acceptance gates are discussed in the surrounding prose.

The empirical reading of Table 5.37 confirms the pre-specified asymmetric-rescue pattern. The pooled training-and-transfer exact match lifts from 0.474 in the classifier-off arm to 0.532 in the classifier-on arm, a +5.8 percentage-point absolute gain and a +12.2% relative gain on the five-benchmark panel; the pooled composite hallucination metric drops from 0.121 to 0.115, a −5.2% relative reduction. The pooled retrieval-augmented tier share collapses from 88.9% in the classifier-off arm to 55.1% in the classifier-on arm, a −33.8 percentage-point reduction in retrieval traffic that is the direct mechanism behind the cost-side of the architectural correction: the classifier diverts approximately one in three queries from the retrieval-augmented tier to the zero-shot tier, removing both the retrieval cost and any passage-amplification risk on the diverted queries. The per-benchmark decomposition surfaces where the pooled lift lands: CommonsenseQA exact match lifts by +14.0 percentage points

(with composite hallucination dropping by −34.4% relative) and TruthfulQA exact match lifts by +10.0 percentage points, the two benchmarks where retrieval was structurally hurting; TriviaQA lifts by +2.7 percentage points and StrategyQA by +2.3 percentage points, both inside the noise band of single-cycle exact-match variation; FEVER exact match is flat at 0.503 in both arms, the empirical receipt that the classifier correctly preserves the retrieval-helpful regime on the textbook fact-verification benchmark where unconditional retrieval was already the right default.

The per-benchmark composite hallucination metric reading is more nuanced than the exact-match reading and is reported honestly rather than smoothed. CommonsenseQA shows the largest reduction at −34.4% relative, consistent with the magnitude of its exact-match gain; TriviaQA shows a small reduction at −2.0% relative; FEVER and TruthfulQA show small increases of +2.5% and +2.9% relative respectively, both inside the cycle-to-cycle noise band of the composite hallucination measurement; StrategyQA shows the largest increase at +8.1% relative on its three-hundred-sample evaluation fold (every benchmark's evaluation fold has three hundred samples; StrategyQA's 387-sample test fold is the smallest test fold), the StrategyQA panel that is also the smoke-test outlier of the classifier's training-time per-benchmark routing distribution). The pooled composite hallucination reduction therefore arises predominantly from CommonsenseQA rather than from a uniform improvement across the panel; this is the empirical signature of the asymmetric-rescue mechanism rather than of a uniform regularisation, and is consistent with the per-benchmark retrieval-utility prediction quality reported in Section 4.2.5: the classifier's per-benchmark AUROC band is wide (0.718 on FEVER, 0.732 on TruthfulQA at the low end versus 0.852 on StrategyQA and 0.936 on TriviaQA at the high end), and the realised composite hallucination effect tracks this heterogeneity rather than overriding it.

The mechanism interpretation is read off the routing-distribution column of Table 5.37. The retrieval-augmented tier share collapses by between 21.7 and 59.0 percentage points on every benchmark in the panel, with the largest collapse on TruthfulQA (from 99.0% to 40.0%). On TruthfulQA the architectural effect is the most direct evidence of the mechanism the classifier was designed for: the classifier-off arm dispatched essentially every TruthfulQA query to the retrieval-augmented tier, where the dense passage substrate amplified the question's misconception framing, with the classifier-on arm diverting sixty percent of the TruthfulQA queries to the zero-shot tier and letting the generator's parametric knowledge serve the answer instead. The CommonsenseQA collapse from 57.7% to 20.3% retrieval-augmented share has the same architectural meaning on the five-option multiple-choice surface where the verifier's grounding signals were systematically weak in the classifier-off arm. The FEVER share drops from 92.3% to 70.7%, the smallest absolute reduction in the panel; this is the textbook fact-verification benchmark where retrieval is genuinely helpful, and the classifier correctly preserves the dispatch on the retrieval-helpful subset while only re-routing the minority of FEVER queries on which retrieval was redundant against the generator's parametric capacity. The non-uniform per-benchmark collapse pattern is therefore the empirical receipt that the classifier's routing decision is benchmark-aware rather than acting as a global retrieval rate knob.

The two pre-specified acceptance gates of the classifier ablation read as follows. The first gate predicts a net pooled exact-match lift over the classifier-off baseline with no per-benchmark exact-match regression exceeding two percentage points; the realised reading is a pooled lift of +5.8 percentage points with zero per-benchmark exact-match regression, so the gate closes. The second gate predicts a recovery of TruthfulQA exact match toward the B1 zero-shot reference of 0.433; the realised reading is 0.310 in the classifier-on arm against 0.210 in the classifier-off arm, which recovers toward the B1 reference but stays below it, so the second gate as stated does not close. The threats to the contrast are honest to disclose: the bundled-confound described above means the contrast attributes the lift to the joint effect of the classifier and the accompanying pipeline corrections rather than to the classifier in isolation; an arm-isolating ablation that re-runs the classifier-on arm without the pipeline corrections is registered as future work in Chapter 6, and the realised effect is reported as a partial-support reading on the classifier's architectural contribution rather than as a clean isolation. A second threat is that the classifier-off trajectory was aborted under the retention guard while the classifier-on trajectory was early-stopped at cycle five under the equilibrium gate, so the contrast at cycle five would not be matched on the cycle-five gating event; the cycle-three anchor used here is the last cycle on which both trajectories committed under the same gating contract, and is therefore the principled matched-protocol cycle for the head-to-head reading.

5.4.3 Optional reference points

Two reference points appear alongside the eight external baselines but do not enter the Holm-Bonferroni denominator of Section 5.3.4. The first is Self-RAG [63], included as a citation-only reference rather than as a replicated numerical baseline because Self-RAG's critic tokens are tied to a LLaMA-family backbone and retargeting them to Qwen-2.5-3B-Instruct would change enough of the system to invalidate the architectural comparison. The second is STaR [42], run as an optional ceiling reference conditional on remaining compute budget; STaR's rationalisation step and binary correctness filter differ enough from CAEM's nine-signal verifier and four-outcome decision that a large gap to STaR would mix architectural contribution with rationalisation-quality artefacts. Both reference points are reported as separate rows in the comparative table without entering the licensing rule of Section 5.3.5.

5.4.4 Tier-by-baseline cost relationship

The deployment-cost projection of Equation (3.5) is read off the per-tier hit-rate trajectory and the per-tier latency reported in Section 5.1.4. The cross-cutting analysis here pairs each external baseline's per-query latency and dollar cost with CAEM's per-tier cost, so that the cost-saving argument is surfaced as an explicit comparison rather than left implicit in the headline accuracy contrast. A baseline that achieves comparable accuracy at substantially lower per-query cost narrows the architectural claim to the regime where CAEM's accuracy advantage is worth the cost gap; a baseline that pays more for less accuracy widens the architectural claim. A consolidated cost-versus-accuracy comparison panel is deferred to follow-up work because the matched-protocol evaluation runs of Section 5.4.1 log per-query wall-time only, not the per-baseline retrieval and API-token cost components needed

for a like-for-like dollar accounting; the per-tier cost numbers of Section 5.1.4 and the per-baseline latency entries in the evaluation logs provide the inputs whenever that follow-up is completed.

5.5 Summary of Findings

5.5.1 Headline outcome in one paragraph

The headline outcome of the empirical programme is the joint reading of the training-panel architectural-contribution slice, the full-panel registered contrast, and the eight-baseline external comparison, taken at the two registered reference cycles of the within-trajectory best cycle (cycle two) and the trajectory-end cycle (cycle five) against B1, the matched zero-shot prompt over the same Qwen-2.5-3B-Instruct backbone. The architectural-contribution slice on the training panel is read from the shaded pooled rows of Table 5.5: at cycle two, pooled exact match lifts from 0.483 at the B1 floor to 0.581, a +20.3% relative gain, and pooled composite hallucination metric drops from 0.143 at the B1 floor to 0.101, a −29.4% relative reduction; at the trajectory-end cycle the same axes hold at +13.7% relative exact-match gain and −20.3% relative composite-hallucination reduction against the same B1 floor. The full-panel registered contrast on the same five-benchmark evaluation panel, read against Table 5.34, reads +7.4% exact match and −11.0% composite hallucination metric at cycle two and +2.1% exact match and −9.3% composite hallucination metric at the trajectory-end cycle; the smaller full-panel margin is driven by a TruthfulQA regression where the dense passage substrate amplifies the question's misconception framing rather than refuting it, the retrieval-amplification mechanism documented in Section 5.6.4. StrategyQA records a clean transfer-panel composite-hallucination win at the trajectory-end cycle of 0.099 below the B1 floor of 0.120, a −17.5% relative reduction. The within-trajectory reading is complementary to the architectural-contribution reading: pooled composite hallucination metric on the five-benchmark panel declines from 0.135 at the cycle-zero cold-start state to 0.109 at the cycle-five close, a −19% relative reduction inside the trajectory, while pooled exact match holds in the band $[0.517, 0.544]$ across the same window. Against the locked external eight-baseline panel of Table 5.34, CAEM beats every baseline on the pooled composite-hallucination axis at both reference cycles and on the pooled exact-match axis at the best cycle, and holds exact-match parity with the strongest baselines at the trajectory-end cycle, where the forward-looking active-retrieval baseline edges the pooled exact-match axis by eight thousandths while CAEM retains a thirty-seven percent composite-hallucination advantage over it; the paired-McNemar exact-match receipts and paired-Wilcoxon composite-hallucination receipts of Table 5.35 clear the joint significance-and-effect-size licensing rule of Section 5.3.5 on nine of forty per-benchmark contrasts on the exact-match axis and on twenty-one of thirty-five contrasts on the composite-hallucination axis at the cycle-five anchor (the five vanilla fine-tuning composite-hallucination cells are not available), and on fourteen of forty and twenty of thirty-five contrasts respectively at the best-cycle anchor of Table 5.36, with the significant defeats (three on the exact-match axis at each anchor; three on the composite-hallucination axis at the cycle-five anchor and four at the best-cycle anchor) confined to TruthfulQA against the high-EM RAG-free baselines (the surface where the architecture's abstention

behaviour interacts with the Haiku-judged correctness rule and the false-refusal subtype of the composite-hallucination metric). The headline claim of the empirical programme is therefore *supported* on the pooled composite-hallucination axis at both reference cycles, on the pooled exact-match axis against B1 at both reference cycles, and on the pooled composite-hallucination comparison against every external baseline at both reference cycles; *supported with a parity qualifier* on the pooled exact-match comparison against the external panel, where CAEM leads every baseline at the best cycle and reaches parity with the strongest fine-tuned and active-retrieval baselines at the trajectory-end cycle; and *partially supported* on the per-benchmark sig-test panel where the abstention-on-misconception-bait surface against confidently-stated wrong baselines reduces the strict-correctness margin; the supplementary retrieval-utility classifier ablation of Section 5.4.2 narrows the TruthfulQA exact-match regression by re-routing misconception-bait queries away from the retrieval-augmented tier toward the zero-shot tier, lifting TruthfulQA exact match from 0.210 to 0.310, still below the B1 zero-shot reference of 0.433, while preserving the training-panel pooled reading.

5.5.2 Hypothesis verdicts

Each pre-specified hypothesis from Section 1.4.3 is adjudicated against a specific deciding statistic from this chapter, with the verdict drawn from the licensing rule of Section 5.3.5. Table 5.38 reports the realised verdict per hypothesis at the cycle-five close, with the deciding statistic anchored to the corresponding evidence subsection.

5.5.3 Evidence-to-claim mapping

Table 5.39 maps every architectural claim made in earlier chapters to the specific subsection, table, or figure in this chapter that licenses it. The mapping is read in two directions: a reader who wants to verify a claim from Chapter 1, Chapter 3, or Chapter 4 can follow the row to its evidence here, and a reader auditing this chapter's empirical reporting can follow the row backwards to the claim that the empirical reading was registered to test.

ID	Hypothesis (registered in Section 1.4.3)	Verdict	Deciding statistic and realised reading at cycle-five close
H1	Verifier balanced accuracy exceeds one half on the calibration-matched distribution at every cycle (Proposition 4.2 precondition).	**supported**	Pooled-ID composite Cohen's d on the held-out cycle-zero evaluation fold reads $+0.635$ (balanced accuracy ≈ 0.673), clearing the 0.20 precondition floor of Proposition 4.2 by more than three times (Table 5.32); calibration-fold pooled π_{store} at $\tau_{\text{store}} = 0.60$ stays in the band $[0.732, 0.746]$ across every committed cycle (Table 5.16), the downstream receipt that the locked verifier ensemble's cal-fold discrimination on the per-cycle generator outputs remains strictly above chance at every composite re-fit.
H2	Pool purity is non-decreasing across cycles whenever the precondition is satisfied (Remark 4.9).	**supported** (band)	Calibration-fold pooled π_{store} at $\tau_{\text{store}} = 0.60$ traces $0.746 \to 0.740 \to 0.736 \to 0.746 \to 0.732$ across the committed-cycle subsequence (C0, C1, C2, C3, C5; C4 aborted, no cal-fold refit), realised band $[0.732, 0.746]$ of width 0.014, clearing the registered 0.64 floor by 9 to 11 percentage points (Table 5.16, Section 5.1.6). Within-band fluctuations sit at or below per-cycle binomial sampling noise at $n \in [575, 1001]$; the retroverify upward-only update fires at every committed cycle.
H3	Late-cycle pool-purity increments enter a bounded stationary band on the committed-cycle subsequence (Heuristic 4.4).	**supported** within horizon	Committed-cycle cal-fold π_{store} occupies the band $[0.732, 0.746]$ of width 0.014 across C0–C5 (Table 5.16), comparable to per-cycle binomial sampling noise; the calendar-timeline refinement of Heuristic 4.7 reads $\pi_{\text{commit}} = 4/5 = 0.80$ from commit pattern $(1, 1, 1, 0, 1)$ (Table 5.20). The sharper geometric-rate test is registered as deferred future work in Chapter 6, pending a horizon of at least fifteen committed cycles.
H4	The composite hallucination metric at the realised cycle horizon approaches a parameter-bounded asymptote (Remark 4.10).	*scope-limited*	Pooled C5 CHM reads 0.109 (Table 5.3); Tier 1 CHM reads 0 on 7/7 realised firings against the $\max(1 - \pi_{\text{store}}, 1 - \pi_{\text{retro}}) \approx 0.36$ ceiling of Heuristic 4.5 (Table 5.21). The decomposition $H_t \leq (1 - \tau_1(t))\, \varepsilon_{\text{arch}}(t) + \tau_1(t)\, H_{\text{mem}}(t)$ is degenerate at $\tau_1(C5) = 0.0013$ under the content-hash-disjoint evaluation contract, so a sharp gap-to-asymptote test is infeasible at the six-cycle horizon and is registered as future work.
H5	Per-cycle Tier 1 hit rate grows monotonically as memory accumulates, producing a measurable per-query cost reduction (operational claim, Section 3.8.3).	*scope-limited*	Per-tier hit-rate trajectory (Table 5.10, Section 5.1.4): the evaluation fold is held content-hash disjoint from the training stream by construction, so the memory-direct tier fires on 7 of 9 000 evaluation queries at within-tier EM = 1.0, supporting the growth prediction at the rate the disjointness contract permits. Per-cycle pooled per-query latency (Table 5.12) stays in the band $[13.6, 14.7]$ seconds over C0–C5, so a per-query cost reduction is not measurable within the realised horizon; the intrinsic per-tier latency measurement that would sharpen the deployment-cost projection of Equation (3.5) is registered as future-work in Chapter 6.

Table 5.38: Pre-specified hypotheses from Section 1.4.3 adjudicated against the corresponding deciding statistic in this chapter at the cycle-five close. H1 closes as supported with the locked verifier ensemble's cal-fold discrimination on the per-cycle generator outputs clearing the precondition floor at every committed cycle; H2 closes as supported under the bounded-band restatement of Theorem 4.1, with within-band fluctuations at or below per-cycle binomial sampling noise; H3 closes as supported within the realised six-cycle horizon, with the sharper geometric-rate test deferred to a longer-horizon future-work item; H4 closes as scope-limited under the content-hash-disjoint evaluation contract that pins the Tier 1 hit rate at the noise floor; and H5 closes as scope-limited on the same disjointness contract, with the per-query cost reduction not measurable within the realised six-cycle horizon.

Claim site	Claim	Evidence in this chapter
Section 1.4.3 (H1)	Verifier balanced accuracy clears one half at every cycle	Section 5.2.7, Section 5.1.6
Section 1.4.3 (H2)	Pool purity non-decreasing across cycles under the precondition	Section 5.1.6, Section 5.1.6
Section 1.4.3 (H3)	Late-cycle bounded stationary band on the committed-cycle subsequence	Section 5.1.6
Section 1.4.3 (H4)	Composite hallucination metric approaches parameter-bounded asymptote	Section 5.1.6
Section 1.4.3 (H5)	Tier 1 hit rate grows with per-query cost reduction	Section 5.1.4, Section 5.1.4
Section 3.1.2 (NFR2)	Storage precision anchor at $\tau_{\text{store}} = 0.60$	Section 5.2.3 (cycle-zero validation-fold $\pi_{\text{store}} = 0.796$ on $n = 289$)
Section 3.1.2 (NFR3)	Retention floor	Section 5.1.5 (per-cycle ρ, $\pi_{\text{commit}} = 4/5$)
Section 3.3.2	Six-cycle realised horizon under equilibrium-saturation termination	Section 5.1.6
Section 3.8.3	Deployed cost-per-query decay across cycles	Section 5.1.4, Section 5.4.4
Theorem 4.1	Pool purity invariant	Section 5.1.6
Heuristic 4.3	Monotonicity under generator improvement (locked verifier)	Section 5.1.6
Heuristic 4.4	Convergence as bounded-band stationarity (committed-cycle subsequence)	Section 5.1.6
Heuristic 4.7	Stochastic equilibrium under retention rollback (calendar timeline)	Section 5.1.6
Heuristic 4.5	Asymptotic hallucination floor on memory-direct tier	Section 5.1.6

Table 5.39: Evidence-to-claim mapping: every architectural claim made in earlier chapters is paired with the specific evidence in this chapter that licenses it. Cells without a measurable cycle-zero anchor are populated by the main-run readings.

5.6 Discussion

5.6.1 Headline interpretation

The headline interpretation of the empirical record will be drawn from three jointly visible patterns: the trajectory of the composite hallucination metric across cycles in Section 5.1.2, the per-tier hit-rate evolution in Section 5.1.4, and the contrasts in Section 5.4.1. The interpretation that the architecture intends to license, conditional on the licensing rule of Section 5.3.5 returning supported on the headline contrasts, is that hallucination reduction at the deployed cost envelope is achievable through the architectural mechanisms specified in Chapter 4 rather than through scaling. The disclaimer that the licensing rule may return only partially supported on a subset of contrasts is registered explicitly: a partial support reading is not a falsification of the architecture, but it does narrow the headline claim to the benchmark family on which the licensing condition holds.

5.6.2 Threats to validity

Single seed and single backbone

The empirical programme runs under a single recorded random seed, propagated through every stochastic component of the pipeline, with a single open-weight three-billion-parameter generator as the backbone. Cross-seed replication would convert the per-cycle metric readings from point estimates to interval estimates and would tighten the licensing rule by adding an inter-replicate-variance check; that check is omitted at the present compute envelope and is registered as a future-work item. Cross-backbone replication, with a different open-weight generator at comparable scale, would test whether the architectural claim is generator-agnostic or generator-coupled; that check is registered as a separate future-work item with a specific candidate backbone identified in Chapter 6. The thesis is honest about both threats: the reported readings are a single-seed-single-backbone trajectory, and the architectural claim travels exactly as far as that disclosure permits.

Benchmark coverage

The five benchmarks adopted in this work cover factual claim verification (FEVER), open-domain trivia (TriviaQA), commonsense multiple-choice reasoning (CommonsenseQA), common-misconception probes as a held-out transfer surface (TruthfulQA), and implicit multi-step reasoning as a second held-out transfer surface (StrategyQA). The coverage does not include code generation, formal mathematical reasoning, multi-turn dialogue, multilingual evaluation, or long-form generation, all of which are excluded by the registered scope in Section 1.6.1. The most consequential coverage gap among the included families is the open-text long-form regime, which the registered short-answer prompt template across the five panel benchmarks does not exercise; the long-hypothesis judge ablation in Section 5.2.6 retired the long-hypothesis backend before the main run because the registered Pearson floor on its calibration overlap fold was not cleared, and the present panel does not regularly produce hypothesis lengths above MiniCheck's encoder range. A future programme that adds a long-form benchmark and re-fits the long-hypothesis judge under task-

specific training would close this gap; the present work registers it as future work rather than as silent acceptance.

Calibration to evaluation gap

The cycle-zero validation-fold storage precision shelf documented in Section 5.2.5 ($\pi_{\text{store}} = 0.796$ at $\tau_{\text{store}} = 0.60$ over 289 stored entries) is the empirical anchor for the deployed operating point. The earlier split-conformal alternative was retired before the main run on the registered cycle-zero evidence that the calibration-versus-evaluation distribution shift on the registered five-benchmark panel broke marginal coverage by fifteen to fifty percentage points on the training panel; the fixed-threshold rule that replaced it does not depend on coverage transfer, so the shift cannot silently void a guarantee, although it can still lower the realised precision. Two mitigations bound residual cal-eval drift across the trajectory. The first is the per-cycle composite refit, which fits the per-signal isotonic curves and the per-benchmark child curves on the cycle's own labelled fold, blends each per-benchmark child toward the pooled fit through the registered shrinkage prior, and refits the boost coefficients on the same fold; the cycle-boundary refit therefore tracks score-distribution drift on the cycle's own evidence rather than freezing the cycle-zero composite shape. The second is the empirical-precision audit at the locked cut-point that runs once per cycle on the same recalibrated fold, with the realised π_{store} recorded as a first-class diagnostic so that the cycle-by-cycle drift of the anchor is itself measurable.

Cycle budget

The realised six-cycle horizon is bounded by the equilibrium-saturation gate registered in Section 3.3.2: the gate terminates the trajectory at the cycle-five close after the post-rollback subsequence stabilises. A twenty- or fifty-cycle run would clear the ≥ 15-committed-cycle precondition under which the bounded-band stationarity of Heuristic 4.4 can be sharpened into a measured geometric contraction rate, and would supply a longer-horizon test of the asymptotic floor in Heuristic 4.5. The realised cycle horizon is reported alongside the corresponding bounded-band reading in Section 5.1.6; the longer-horizon experiment is registered as the corresponding future-work item in Chapter 6, alongside the arm-isolating re-run of the retrieval-utility-classifier ablation that the present horizon does not exercise.

Storage-tail sample size

The storage tail at the locked operating point is non-trivial at cycle zero (289 stored entries on the pooled training-panel cycle-zero validation fold, 500 samples per benchmark and 1 500 in all), which gives a robust pooled empirical-precision anchor but bounds per-benchmark precision claims to wider confidence intervals on benchmarks whose per-benchmark stored count is small. The per-benchmark stored counts at cycle zero land at 145 on TriviaQA, 81 on CommonsenseQA, and 63 on FEVER, with TriviaQA carrying the largest tail and FEVER the smallest. The per-benchmark precision on the small-tail benchmarks is therefore not a robust point estimate, and the per-benchmark claims rely on the bootstrap confidence interval of Section 5.3.3 rather than on the point reading. The pooled precision over the three-benchmark training fold remains a robust reading because pooling absorbs the

per-benchmark sampling noise into a larger denominator.

Retrieval-utility classifier mis-routing

The retrieval-utility classifier of Section 4.2.5 supplies the routing decision at the safety-veto and score-formula fall-throughs per Table 4.2, and its per-benchmark area under the receiver operating characteristic on the held-out training fold varies from 0.718 on FEVER to 0.936 on TriviaQA, with TruthfulQA at 0.732 and CommonsenseQA at 0.728 at the low end and StrategyQA at 0.852. The wide per-benchmark band is the empirical receipt that a cohort whose retrieval-utility distribution differs materially from the training panel could be mis-routed in either direction: a retrieval-helpful cohort routed to the zero-shot tier would forfeit the grounding evidence it needed, and a retrieval-hurt cohort routed to the retrieval-augmented tier would re-expose the misconception-amplification mechanism that the classifier was registered against. The empirical programme does not stress-test the classifier against an off-distribution cohort within the realised horizon; an evaluation under a held-out routing-axis benchmark is registered as future work in Chapter 6.

5.6.3 Sensitivity analyses

Sensitivity of the headline finding to the storage cut-point τ_{store}, the per-benchmark shrinkage strength α_{shrink}, and the boost layer's L2 regularisation strength C is read off the registered cycle-zero sweep whose locked configuration is reported in Table 5.28. The two top-tier cut-point variants of the sweep differ on $\tau_{\text{store}} \in \{0.60, 0.65\}$ at fixed $C = 0.01$ and $\alpha_{\text{shrink}} = 0.6$; the lower cut admits roughly 4.4 times as many correct memories at the same per-store precision shelf (306 against 70 on the cycle-zero calibration fold of the calibration sweep run before the main trajectory), which is the rate-volume frontier the cycle-zero threshold sweep was designed to traverse, and the choice of operating point trades the marginal precision difference for the substantially larger storeable population in the direction the registered selection criteria preferred. The C sensitivity is read at fixed $\tau_{\text{store}} = 0.60$ across $C \in \{0.010, 0.025, 0.050, 0.100, 1.000\}$; the spread in calibration-fold per-store precision across these five is bounded above by approximately one percentage point, which establishes that the headline reading is robust to within-band perturbations of the boost regulariser. The α_{shrink} sensitivity is read across $\{0.0, 0.4, 0.6, 0.8, 1.0\}$ on the cycle-zero per-benchmark area-under-the-receiver-operating-characteristic readings; the registered $\alpha_{\text{shrink}} = 0.6$ is the value at which the weak-signal training-panel benchmarks regain parity with the pooled-fit reading without sacrificing per-benchmark adaptation on the strong-signal benchmarks. A separate sensitivity analysis to the safety floor $\phi_{\text{safe}} = 0.38$ on the pre-routing confidence and to the early-exit thresholds $\eta_u = 0.70, \eta_g = 0.20$ is registered as future-work because each requires a separate sweep panel that the cycle-zero artefacts do not yet contain.

5.6.4 Comparison to the strongest prior systems

The closest published systems to CAEM's design fall into three groups. The first group is conformal-factuality at the language-model claim level: [77] demonstrates a correctness lift from 78% to 93% on Natural Questions using GPT-4 with split-conformal sub-claim filtering, and [67] sharpens this with conditionally valid guarantees and a

boosted combination of claim scores; CAEM's composite layer instead follows the logistic multi-scoring of [82]. The architectural difference between CAEM and these systems is that CAEM's gate is applied at the storage decision rather than at the answer-rendering decision, and that CAEM persists the gated answers in episodic memory for retrieval rather than discarding them after rendering; the empirical contrast is therefore not directly comparable on a single per-query precision number, but the underlying machinery is shared. The second group is verifier-filtered self-improvement: [42] and [72] demonstrate that bootstrapping on correctness-filtered self-generated outputs is a viable training-data-admission rule, with V-STaR additionally training a verifier on the same pool for inference-time ranking, and [87] formalises self-improvement as sharpening with statistical sample-complexity guarantees, which CAEM's Heuristic 4.4 complements with a multi-cycle bounded-band stationarity heuristic. The architectural difference is that CAEM uses an external multi-signal verifier rather than self-verification or a single-signal verifier, and that CAEM persists verified outputs in memory across cycles rather than consuming them only as training data. The third group is learned per-query retrieval routing: [74] routes by query-complexity classification, [54] routes by entity popularity, [62] routes by a self-knowledge classifier trained on retrieval-helpful versus retrieval-hurt empirical labels, and [85] routes by a linear probe on intermediate hidden states, while CAEM combines stored-quality routing on a shared memory with the retrieval-utility classifier of Section 4.2.5 at the safety-veto and score-formula fall-throughs; the routing-signal axis is the architectural difference that the empirical contrast in Section 5.4.4 and the ablation in Section 5.4.2 are registered to surface.

Retrieval-amplification on misconception-bait probes: The closest published analogue of CAEM's transfer-panel TruthfulQA regression is the documented retrieval-amplification mechanism on misconception-bait probes: questions are constructed to invite a confidently wrong answer through a familiar surface framing, so the dense passage substrate that the retrieval-augmented tier conditions on returns passages adjacent to the question's misconception framing rather than passages that refute it. The architecture's strongest tier therefore amplifies the misconception rather than catching it, and the per-query precision contract on the storage gate downstream cannot recover what the upstream retrieval has already corrupted. The strict exact-match column compounds this reading because the architecture's abstention behaviour on misconception-bait probes is licensed by the calibrated-probability composite while the matched-protocol baseline B1 over the same Qwen-2.5-3B-Instruct backbone is not licensed to abstain under a zero-shot prompt; on a surface where the strict-correctness rule penalises abstention as a failure, the comparison reads the architecture's abstention as a loss against the baseline's confidently-stated wrong answer. The architectural fix is the retrieval-utility classifier of Section 4.2.5, whose head-to-head ablation against the same architecture without the classifier is reported in Section 5.4.2. The classifier intercepts the score-formula fall-through and the safety-veto fall-through and dispatches misconception-bait queries to the zero-shot tier rather than to the retrieval-augmented tier, removing the substrate where the misconception amplification fires. The realised ablation reading lifts TruthfulQA exact match from 0.210 to 0.310 and collapses the TruthfulQA retrieval-augmented tier share from 99.0% to 40.0%, the empirical receipt that the architectural fix engages the mechanism it was registered against. The classifier-off-versus-classifier-on

contrast at the cycle-three close is the within-horizon architectural ablation that the empirical programme commits to.

Boundary of the architectural value proposition: The architecture's value over the simpler baseline of plain exact-match-labelled fine-tuning at the same labelling budget rests on six independently load-bearing differentiators: the per-query empirical-precision anchor delivered by the calibrated-probability composite paired with the fixed-threshold storage gate, the memory-direct serving tier that compounds cost savings as the verified pool accumulates, the zero-shot serving tier whose share grows across cycles as the self-improvement loop distils verified retrieval-augmented reasoning into the generator's parametric capacity, the asymmetric-rollback property under which a failed cycle preserves accumulated memory while reverting the adapter, the verifier-as-sample-efficiency-multiplier under which the cycle-boundary labelled batch validates an empirical-precision anchor rather than carrying the gradient signal directly, and the inference-time error-catching that the verifier ensemble continues to provide once the parametric fine-tune saturates against the generator's capacity ceiling. A deployment for which all six differentiators relax simultaneously (no abstain requirement, no compounding cost saving from a memory-direct path, no per-cycle improvement of the zero-shot tier under the workload distribution, no need for asymmetric rollback because external snapshots exist, no labelling-budget constraint because direct fine-tuning is feasible at the required cadence, and an underlying generator still on the steep part of the fine-tune curve) is one where plain fine-tuning would deliver comparable quality with simpler implementation. A deployment for which any one of the six binds is one where the corresponding architectural component is irreplaceable. Chapter G answers each of the six differentiators as a direct comparative question against plain fine-tuning, and the empirical receipts in this chapter measure the contribution of each component on the registered five-benchmark workload where every component is binding by construction of the failure-mode taxonomy of Section 2.1.2.

5.6.5 Limitations

The architecture inherits a single open-weight generator and reports a single-seed trajectory, both of which are registered threats to validity in Section 5.6.2. The benchmark panel does not include code, mathematical, multi-turn, multilingual, or long-form generation evaluation, all of which are excluded by the registered scope in Section 1.6.1. The long-hypothesis judge ablation in Section 5.2.6 forces every hypothesis through MiniCheck, with documented truncation behaviour on the rare hypotheses above the encoder range; the architectural cost on the five panel benchmarks is small because none of them regularly produces hypothesis lengths above the truncation point at the registered short-answer prompt template. The cold-start memory after the pre-step-7 hygiene pass contains 431 verified entries, which is sufficient to bootstrap the memory-direct tier in early cycles but does not test the architecture's behaviour at the millions-of-entries deployment scale projected in Section 4.2.3. The fixed-threshold storage gate's empirical-precision anchor is read on the calibration fold and projected forward by exchangeability between the calibration fold and the cycle's candidate pool; the per-cycle empirical-precision audit measures the realised drift of the anchor at every cycle close, so

the cycle-by-cycle behaviour of the contract is itself a first-class diagnostic rather than a hidden assumption. A separate calibration-discipline limitation arises at the pre-routing surface: the per-cycle re-fit of the temperature scalar on the pre-routing confidence reaches the registered upper bound of the optimisation envelope at the cycle-zero fit ($T = 20.086$), so single-parameter temperature scaling on this surface is functionally inert at the cycle-zero operating point (the cycle-boundary refits return 1.089, 0.836, 0.740, and 0.869 at cycles one, two, three, and five, Table 5.30); the storage-class empirical-precision anchor is not affected because the gate's operating point depends on the score's quantile structure at the locked cut-point rather than on temperature-scaled probability semantics, and the per-signal isotonic refit with the per-benchmark shrinkage prior at the cycle boundary continues to absorb the calibration drift on the storage-class composite, while the pre-routing confidence retains its rank-order role in the dispatch score and the safety override even at the saturated cycle-zero temperature scalar. The deferred-entry reconsideration pass specified in Section 4.2.10 engages at every successful cycle boundary of the realised trajectory under the registered time-to-live of two reconsider invocations: it promotes 131 entries at cycle two, 220 at cycle three, and 299 at cycle five (Table 5.25), while the cycle-one pass finds an empty buffer and the cycle-four pass is skipped under the asymmetric-rollback contract. The equilibrium-saturation gate fires after the cycle-five close, so the within-horizon receipt for the deferred-pool half of the two-track design is these three reconsideration passes, with the steady-state two-track survivability behaviour specified in Section 4.2.10 registered as a future-work measurement at a longer horizon. The empirical programme runs over six realised cycles bounded by the equilibrium-saturation gate; longer trajectories are registered as future-work and may produce a different verdict on the asymptotic claim of Heuristic 4.5. Finally, the architecture does not test cross-domain transfer beyond the five benchmarks adopted, and the deployment-cost projection in Section 3.8.3 is read off the empirical trajectory rather than from live user traffic; both gaps are registered as future-work items in Chapter 6.

Limitations identified against the literature: Six further limitations follow from the cited literature and are recorded here so that the results above are read with them in view.

- **Saturation and format learning.** Verifier-filtered self-improvement has been reported to saturate within two or three rounds, and part of the first-round gain at larger scales came from improved adherence to the required answer format rather than from new capability [92]. The realised trajectory is consistent with early saturation (the exact-match optimum is reached at cycle two and accuracy declines slightly through cycle five), and because the headline accuracy metric is exact match, part of the early gain may likewise reflect format learning. A flexible-match re-scoring of the trajectory, which would separate the two, is registered as future work.
- **Optimising toward the verifier.** Iterating a training loop against a fixed scorer risks over-optimising to that scorer: reward-model gains can keep rising while human-judged quality peaks after the first round [47], and verifier-filtered retraining converges to the verifier's own knowledge, so its gains plateau or reverse unless the verifier is unbiased [95]. Exact match against gold labels is

independent of CAEM's verifier and is the primary success criterion for this reason; the composite hallucination metric, by contrast, is read through the same locked verifier ensemble that gates storage, so its improvements are partly measured by the instrument the loop was filtered with.

- **Systematic verifier errors.** Agreement-based signals do not detect misconceptions that the model holds consistently [70], and repeated verification removes only independent errors (see the scope note on Heuristic 4.6). The transfer-panel TruthfulQA store precision of 0.522 against 0.75 to 0.82 on the training panel (Table 5.31) is the realised instance. The sample-agreement signal s_{avg} is additionally a cosine similarity in sentence-embedding space, a family that has been shown to miss semantically plausible hallucinations from preference-tuned models [91], although that evidence concerns answer-to-source similarity rather than agreement across samples.
- **Calibration-fold size.** Isotonic regression overfits on small calibration sets and is reliably at least as good as single-parameter Platt scaling only from roughly a thousand calibration points [7]. CAEM fits per-signal isotonic curves on a 500-sample fold per benchmark and admits a signal's fit from fifty samples; the per-benchmark shrinkage toward the pooled fit limits the resulting variance but does not remove it.
- **Low-chance-floor tasks.** Filtering on the final answer does not ensure that the reasoning behind it is correct [48], and settings with a high chance level, such as binary decisions, produce many correct answers with poor rationales [42]. StrategyQA (yes or no) and FEVER (three labels) carry chance floors of one half and one third, so a verifier-accepted answer on these benchmarks may be right for the wrong reason; StrategyQA enters only the transfer panel, but FEVER entries enter the self-improvement training pool.
- **Adapter capacity and drift.** Low-rank adaptation learns less of a new target than full fine-tuning and its forgetting grows with training duration [64], so per-cycle gains are expected to be small and adapter drift to accumulate across cycles; the cycle-four retention abort and the slight accuracy decline after cycle two are consistent with this, and the retention guard bounds rather than prevents the drift.

5.7 Chapter signpost

This chapter has reported the cycle-zero discipline, the locked configuration, the statistical-analysis protocol, and the framing under which the main-run trajectory and the comparative panel will be adjudicated. Chapter 6 restates the contributions against the empirical receipts that this chapter has produced, registers the open questions that remain, and lists the concrete next experiments that would close the gaps the limitations subsection has surfaced.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

6.1.1 Restated contributions with empirical receipts

The thesis registered five contributions in Chapter 1 and adjudicates each here against the empirical record produced in Chapter 5. The first contribution, a calibrated probability composite over a nine-signal verifier ensemble, is closed by the cycle-zero validation gate verdict in Section 5.2.7: the locked composite achieves a held-out evaluation-fold pooled Cohen's d of $+0.635$ on the training panel against an out-of-distribution diagnostic reading of $+0.173$ on the transfer panel, and the training-panel reading clears the registered $d_{\text{id}} \geq 0.20$ precondition of Proposition 4.2 by more than a factor of three. The second contribution, a fixed-threshold storage gate on the calibrated probability, is closed by Section 5.2.3: the locked cut-points $\tau_{\text{store}} = 0.60$ and $\tau_{\text{defer}} = 0.45$ are held constant across cycles, with the per-cycle composite refit absorbing distribution drift into the signal-to-probability mapping rather than into the threshold itself; the calibration-fold storage precision at the locked threshold and the cycle-on-cycle stream-chunk precision-at-stored trajectory are reported alongside the NFR2 precision floor and the precondition margin of Proposition 4.2. The third contribution, a four-outcome decision tree with a confabulation early-exit, is closed by the per-decision exact-match readings reported in Section 5.1.3; the supported / partially-supported / unsupported verdict on this contribution is gated by the main-run decision-distribution trajectory. The fourth contribution, the theoretical scaffolding of the architecture, is closed by Section 4.3: a pool-purity theorem and a purification condition are proven and paired with empirical receipts at the empirical limit of the realised trajectory, the cycle-to-cycle trajectory is described by design heuristics whose idealising assumptions are stated and whose receipts are directional (with the convergence heuristic read as a bounded-band stationarity claim on the committed-cycle subsequence and the sharper geometric-rate test deferred to future work), and four mathematical scaffolding identities are gathered into a dedicated subsection so that the formal machinery is locatable without confusing it with falsifiable receipts; the per-cycle checks are reported in Section 5.1.6. The fifth contribution, an externally-verifiable artefact bundle, is closed by the public-dataset-host snapshot of the pre-self-improvement state and the per-cycle artefact catalogue documented in Appendix E.

A sixth deployment-side property, registered as a load-bearing differentiator in Section 1.2.3 and elaborated in Section 4.2.10 and Section 5.1.4, is the cycle-wise redistribution of queries across the three serving tiers. The redistribution converts two distinct statelessness problems of the standard inference paradigm into two stateful mechanisms inside the architecture: the verified episodic store absorbs recurring queries at embedding-lookup cost while the retroactive re-verification pass keeps the cached composite of every stored entry fresh under the updated generator, and the self-improvement loop folds verified retrieval-augmented reasoning into the generator's parametric capacity, and the retrieval-utility classifier of Section 4.2.5 diverts queries on which retrieval is predicted to hurt from the retrieval-augmented tier to the zero-shot tier, so that an increasing share of previously retrieval-dependent queries can be served without paying the retrieval cost. The empirical receipt of this redistribution is the per-tier hit-rate trajectory of Section 5.1.4: the held-out evaluation pooled zero-shot share moves from 48.7 percent at the cycle-zero baseline to 50.2 percent at the cycle-five close along a non-monotone path that peaks at 55.2 percent at cycle two, while the retrieval-augmented share moves between 44.7 and 55.1 percent across the trajectory and the memory-direct tier fires on the small share of evaluation queries that the content-hash-disjoint evaluation design permits, with each realised memory-direct firing returning the correct stored answer. The user-facing surface mirrors this discipline: every served answer carries the calibrated probability of correctness as a continuous percentage alongside the four-class decision tag (FR7 in Section 3.1.1), so a user sees both the categorical reliability label and the underlying confidence number, and the abstain decision substitutes a registered insufficient-evidence message rather than serving a guessed answer with an inflated confidence label.

6.1.2 Headline finding

The headline finding the thesis intends to license is the joint reading of three axes against the matched zero-shot baseline B1 over the same Qwen-2.5-3B-Instruct backbone. The training-panel architectural-contribution slice at the within-trajectory best cycle (cycle two) reaches a pooled exact-match gain of +20.3% relative against the B1 floor and a pooled composite hallucination metric reduction of −29.4% relative against the same floor; the trajectory-end cycle holds at +13.7% relative exact-match gain and −20.3% relative composite-hallucination reduction on the same axis. The registered full-panel reading on the five-benchmark evaluation panel at cycle two reads +7.4% exact match and −11.0% composite hallucination metric against the B1 floor; the smaller full-panel margin is driven by a TruthfulQA regression where the dense passage substrate amplifies the question's misconception framing rather than refuting it, and the retrieval-utility classifier ablation of Section 5.4.2 reports the within-horizon architectural fix that lifts TruthfulQA exact match by ten percentage points by re-routing misconception-bait queries away from the retrieval-augmented tier. StrategyQA records a clean transfer-panel composite-hallucination win at the trajectory-end cycle of 0.099 below the B1 floor of 0.120. In the pooled comparison against the locked external eight-baseline panel of Table 5.34, CAEM beats every baseline on the composite-hallucination axis at both reference cycles, leads every baseline on exact match at the best cycle, and reaches exact-match parity with the strongest fine-tuned and active-retrieval baselines at the trajectory-end cycle, so

the hallucination-reduction contribution holds across the panel while the accuracy contribution is a parity-with-the-strongest reading rather than a clean sweep. The architectural mechanisms specified in Chapter 4 therefore deliver hallucination reduction at the deployed cost envelope without scaling, and the same architecture simultaneously converts standard retrieval-augmented generation's per-query cost from a flat tax paid in every session into a quantity that decreases across cycles as the verified memory absorbs recurring queries and the self-improvement loop distils retrieval-augmented reasoning into the parametric capacity. The user-facing rendering attaches the calibrated probability of correctness to every served answer as a continuous percentage alongside the four-class decision tag, so a user always sees both the categorical reliability label and the underlying confidence number rather than a fluently-stated answer with no reliability signal. The disclaimer is that a partial-support reading on the per-benchmark sig-test panel narrows the headline to the family on which the licensing condition holds; the full per-benchmark verdict is reported in Section 5.5.2.

6.1.3 Open questions

The present work raises questions it does not answer. First, the calibration-to-evaluation transfer cost documented in Section 5.2.5, under which a split-conformal alternative dropped fifteen to fifty percentage points below its registered coverage target on the training-panel benchmarks at cycle zero and forced the architecture onto a fixed-threshold gate whose held-out precision anchor sits at 0.796 over 289 stored entries on the cycle-zero training-panel validation fold (500 samples per benchmark), is a structural transfer cost whose cycle-by-cycle drift is empirically measurable but theoretically uncharacterised; whether the cycle-boundary refit of the per-benchmark calibrated probability composite under the shrinkage prior closes the anchor drift monotonically, or whether the anchor settles at a stable floor that the empirical-precision audit tracks indefinitely, is a question the realised trajectory begins to answer but does not resolve. Second, the results and heuristics of Section 4.3 treat the verifier ensemble (MiniCheck plus the signal-extraction layer, with the long-hypothesis judge ablated in the deployed configuration as recorded in Section 5.2.6) as a fixed scoring function, which it is by design in the deployed configuration: only the generator (advanced by the self-improvement loop on committed cycles) and the per-cycle calibrated probability composite refit are cycle-indexed components, and the cal-fold class-separation d that the monotonicity and convergence heuristics trade on is a joint observable of the SIL-advanced generator viewed through that locked ensemble and the cycle-boundary-refit composite rather than a property of a learning verifier. The joint dynamics of advancing generator and refitting composite are partially captured by Heuristic 4.3, whose receipt at the realised horizon is itself partial (the cycle-three to cycle-five reverse-sign storage-rate transition reported alongside the heuristic is consistent with the per-cycle composite refit and the contracting within-class variance rather than a falsification of the sign-of-derivative prediction), and admits a sharper analysis under a coupled-trajectory framework in which the composite refit is held fixed for at least one cycle so that the generator-driven cal-fold discrimination shift can be measured in isolation; this fixed-composite ablation is registered as future work alongside the at-least-fifteen-committed-cycle horizon that would also unlock the deferred geometric-rate test of Heuristic 4.4. Third, the

long-hypothesis judge ablation of Section 5.2.6 narrows the architecture's coverage on long-form benchmarks; whether a task-specific fine-tune of the long-hypothesis judge would re-license its inclusion is an empirical question that the present scope cannot answer. Fourth, the deployment-cost projection in Equation (3.5) assumes a query distribution that the thesis simulates with the five-benchmark workload; whether the projection holds under the stationary distribution of live user traffic is a question that requires a deployment study rather than a benchmark replay.

6.1.4 Closing remark

The position registered in Chapter 1 is that hallucination is a structural property of stateless generation under uncertainty rather than a residual that scaling will remove. The empirical programme reported in Chapter 5 is the test of whether an architectural intervention at the storage decision can change the structure. If the licensing rule returns supported on the headline contrasts, the thesis closes with a calibrated yes; if it returns partially supported, it closes with a calibrated yes-on-this-family-only; if it returns unsupported, the thesis closes by registering the negative result and the threats-to-validity analysis as the contribution.

6.2 Future Work

6.2.1 Long-hypothesis judge under fine-tuning

The long-hypothesis judge ablation in Section 5.2.6 removed the frozen Qwen-3B backend from the deployed configuration because the Platt-fit rank correlation against MiniCheck on the overlap fold fell below the registered floor of $\rho_{\min} = 0.70$. A natural extension is to refit the long-hypothesis judge under task-specific training on a synthetic claim-support dataset in the spirit of MiniCheck's training corpus, and to re-test the calibrated agreement against the registered floor. The expected diagnostic that would re-license its inclusion is a Pearson logit-space correlation at or above 0.70, paired with a mean absolute error in probability space that does not exceed the MiniCheck-only baseline; under that condition, the unified verifier would dispatch hypotheses above the encoder-truncation point through the long-hypothesis backend rather than through truncated MiniCheck, which would close the long-form factuality truncation gap that bounds the architectural cost of the present ablation.

6.2.2 Higher-capacity adapter configurations

The self-improvement loop in Section 4.2.10 performs a parameter-efficient fine-tune at every cycle boundary through a low-rank adaptation layer at rank thirty-two and scaling factor sixty-four over the attention and feed-forward projections, with the backbone frozen and the bounded adapter parameter budget acting as the implicit cycle-to-cycle drift bound. The natural future-work extensions probe the rate-versus-coverage frontier of that adapter. The first extension raises the adapter rank to sixty-four or one hundred twenty-eight, which increases the parametric capacity available to the per-cycle update by a factor of two to four at a near-linear compute cost, and is justified if the per-cycle training loss is observed to plateau before the held-out hallucination metric reaches its theoretical floor. The

second extension widens the target-module list to include the token-embedding matrix and the language-model head; this is the configuration that the published parameter-efficient-adapter literature flags as decisive for factual-recall fine-tunes and is registered as a hypothesis to be tested separately from the rank extension. The third extension lifts the full-parameter fallback into the active path under a larger compute envelope: at a seven-billion or thirteen-billion-parameter backbone the bounded-adapter budget may no longer suffice to absorb cycle-to-cycle drift, and a full-parameter fine-tune with an explicit quadratic anchor toward the previous cycle's parameters becomes the registered alternative under the explicit-anchor coefficient that the locked configuration leaves disabled.

6.2.3 Refutation-aware verification

The composite hallucination metric reports eight measurable failure-mode subtypes; the ninth subtype, factual contradiction, is structurally unreachable under the MiniCheck verifier backend because MiniCheck's binary supported-versus-unsupported judge does not produce a contradiction class. A refutation-aware judge, either trained on a three-class supported / unsupported / contradicted target or composed from MiniCheck plus a separate refutation classifier, would close the eight-of-nine subtype coverage gap to all nine subtypes and would license a stronger architectural claim on the contradiction axis specifically. The empirical signal that would justify this extension is a non-trivial contradiction rate on the evaluation fold under the present configuration, which the cycle-zero diagnostic flags but does not quantify.

6.2.4 Cross-domain transfer

The five benchmarks adopted in this work cover factual question answering, claim verification, and multi-step commonsense; the architecture has not been tested on code generation, formal mathematical reasoning, long-form generation, or multi-turn dialogue, all of which are excluded by the registered scope in Section 1.6.1. The expected pattern under cross-domain transfer is that the calibration-to-evaluation gap reported in Section 5.2.5 widens on domains whose verifier-signal distribution differs materially from the calibration fold, because the calibration-anchored empirical-precision lower bound transfers under exchangeability between the calibration fold and the candidate pool, and exchangeability is not preserved across distinct domains. A cross-domain study would re-fit the per-benchmark calibrated probability composite on a per-domain calibration fold and would report whether the per-domain calibration-fold precision still clears the registered NFR2 target at the fixed cut-points.

6.2.5 Larger backbone

The architecture is built on a three-billion-parameter open-weight instruction-tuned backbone. Scaling to a larger generator (a seven-billion or thirteen-billion-parameter model in the same open-weight family) would test whether the architectural gain over scaling is preserved, narrows, or inverts at higher backbone capacity. The calibration discipline of Chapter 4 is backbone-agnostic by design, but the empirical readings of the per-signal Cohen's d in Table 5.33 are backbone-coupled, and the

bounded-band stationarity and the deferred geometric-rate test of Heuristic 4.4 both depend on d through the per-cycle storage rate and the cal-fold class-separation that the locked verifier ensemble produces on the cycle-t generator's outputs; a longer committed-cycle subsequence at higher backbone capacity is exactly what would unlock the deferred geometric-rate test registered against the bounded-band reframing. The future-work item is to re-fit the locked configuration on the larger backbone and to compare the per-cycle trajectory against the present three-billion-parameter trajectory; the parameter-efficient self-improvement extension above is a precondition for this study under the present compute envelope.

6.2.6 Deployment study

The deployment-cost projection in Equation (3.5) reports the per-query cost as a convex combination over the three tiers, weighted by per-tier hit rate and per-tier per-query cost. The hit-rate trajectory reported in Section 5.1.4 is read off the five-benchmark evaluation workload, which is a stationary draw from the benchmark distributions rather than from live user traffic. A deployment study, in which the architecture is exposed to a real user-query stream over a fixed observation window, would test whether the Tier 1 share continues to grow as the memory accumulates queries the user actually asks, or whether it plateaus at a level set by the diversity of incoming queries. The deployment study is the natural successor to the present work and would convert the operational claim of Section 3.8.3 from an empirical projection on benchmark data into a measurement on production data. An operational deliverable accompanying this thesis specifies the deployment architecture itself, including the cyclic recalibration sequence inherited from the research trajectory, the cycle-trigger policies that an operator may register across accumulation-based and calendar-based schedules, and the labelling sources that supply the calibration fold under live operation across an automated oracle, user-feedback signals, and human-expert review; the future-work item registered here is therefore the empirical observation of the architecture under live traffic rather than the deployment design itself, which is already specified in the accompanying production runbook.

6.2.7 Arm-isolating re-run of the retrieval-utility classifier ablation

The retrieval-utility classifier ablation reported in Section 5.4.2 closes the first of its two pre-specified acceptance gates within the realised horizon: a pooled exact-match lift of +5.8 percentage points with zero per-benchmark exact-match regression closes the first gate. The second gate does not close as stated: TruthfulQA exact match recovers from 0.210 to 0.310, toward the B1 zero-shot floor of 0.433, but stays below it. The contrast is honest about its bundled-confound and matched-protocol scope: the classifier-on arm ships pipeline corrections alongside the classifier itself, so the empirical lift is attributable to the joint effect rather than to the classifier in isolation. A fully arm-isolating re-run that holds the pipeline corrections fixed and toggles the classifier alone is registered as the highest-priority follow-up empirical receipt; the realised reading is treated as a partial-support reading on the classifier's architectural contribution until the arm-isolating re-run is completed.

6.2.8 Intrinsic per-tier latency at unit batch

The per-tier latency reported in Table 5.12 is the production-batched amortised wall-time at the deployment batch size of thirty-two. The intrinsic per-tier latency at unit batch (memory lookup alone, zero-shot generation alone, and retrieval-augmented generation alone, each measured serially without batched amortisation) is a separate quantity that the writeup-hardware envelope cannot host because the dense passage index does not fit in the available random-access memory. A re-measurement on hardware that hosts the dense passage index in random-access memory would decompose the production-batched figure into its intrinsic per-tier components and would sharpen the deployment-cost projection of Equation (3.5) along the per-tier axis without changing its direction; the future-work item registered here is the measurement itself rather than the projection.

6.2.9 Semantic-entropy hallucination metric

The semantic-entropy abstention baseline of Section 5.4.1 reports a complete exact-match contrast on the held-out evaluation fold but is paired with a deferred composite-hallucination-metric reading on the same axis. The deferral arises because the post-hoc rescore through the locked verifier requires the dense passage substrate of Section 4.2.1 to compute the external-grounding family of signals, and the writeup-hardware envelope cannot host the dense passage index in random-access memory. A re-rescore on hardware that hosts the passage substrate would populate the composite hallucination metric on the new baseline outputs without changing the verifier, so the cross-baseline hallucination comparison closes under the same instrument that the rest of the panel uses; the future-work item registered here is the rescore itself rather than the contrast direction, which is already implied by the abstention-rate trajectory of Section 5.4.1 on the misconception-bait surface.

6.2.10 Fixed-composite ablation for the monotonicity heuristic

The realised receipt for the monotonicity statement of Section 4.3.3 is currently downgraded to a consistency check because the per-cycle trajectory bundles two effects that the locked configuration cannot separate: the generator-driven shift in calibration-fold class separation that the heuristic actually predicts, and the per-cycle composite refit that re-rescales the verifier ensemble against the new calibration fold. A future-work cycle registered here freezes the rescaling layer across two adjacent committed cycles, holding the per-signal isotonic curves, the per-benchmark shrinkage child fits, the boost-layer weights and intercept, and the per-benchmark safety floor at their cycle-t values, and varies only the generator's cycle-t outputs as they pass through the locked verifier ensemble. Under this ablation the class-separation shift observed on the cycle-$t{+}1$ calibration fold is attributable to the generator alone, isolating the partial derivative that the heuristic speaks to from the composite-rescaling confound. The same protocol would convert the receipt for Heuristic 4.3 from a directional consistency check into a sharp test of the predicted sign, and would compose cleanly with the retention envelope of Section 5.1.5 because the frozen rescaling layer is precisely the regulariser that the retention guard already validates on the held-out probe surface.

6.2.11 Extended-horizon stochastic-equilibrium probe

The fourth sub-claim of the stochastic-equilibrium heuristic of Section 4.3.3, that the worst-probe ratio settles into a stationary band near the parametric ceiling rather than drifting away from it, is presently supported on the five-cycle observation horizon of Section 5.1.5 by a within-band trajectory whose statistical signature cannot be cleanly distinguished from a longer-trajectory transient. A future-work probe registered here extends the committed-cycle horizon to at least fifteen cycles and reads stationarity through two pre-specified diagnostics: a Wald-Wolfowitz runs test on the worst-probe-ratio sequence to detect non-random temporal structure that would falsify the stationarity hypothesis, and a Bernoulli rate fit on the commit-indicator sequence with a confidence interval whose endpoints strictly exclude both zero and one, matching the open-interval prediction of Heuristic 4.7. A rolling-mean stability check on a sliding window over the worst-probe-ratio sequence complements the two formal tests by surfacing slow drift that a runs test on a finite sequence may miss. The extended horizon composes with the larger-backbone item of Section 6.2.5, since the geometric-rate test of Heuristic 4.4 that the larger backbone unlocks also profits from a longer committed-cycle subsequence on the same observation surface.

References

[1] A. L. Edwards, "Note on the "correction for continuity" in testing the significance of the difference between correlated proportions," *Psychometrika*, vol. 13, no. 3, pp. 185–187, 1948. DOI: 10.1007/BF02289261

[2] H. Robbins and S. Monro, "A stochastic approximation method," *The Annals of Mathematical Statistics*, vol. 22, no. 3, pp. 400–407, 1951.

[3] H. Robbins and D. Siegmund, "A convergence theorem for non negative almost supermartingales and some applications," in *Optimizing Methods in Statistics*, J. S. Rustagi, Ed., Academic Press, 1971, pp. 233–257.

[4] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

[5] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," in *Psychology of Learning and Motivation*, ser. Psychology of Learning and Motivation, G. H. Bower, Ed., vol. 24, Academic Press, 1989, pp. 109–165. DOI: 10.1016/S0079-7421(08)60536-8

[6] B. T. Polyak, "New stochastic approximation type procedures," *Automation and Remote Control*, vol. 51, no. 7, pp. 937–946, 1990.

[7] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proceedings of the 22nd International Conference on Machine Learning (ICML 2005)*, 2005, pp. 625–632. DOI: 10.1145/1102351.1102430

[8] R. Bar-Haim et al., "The second PASCAL recognising textual entailment challenge," in *Proceedings of the Second PASCAL Challenges Workshop on Recognising Textual Entailment*, Venice, Italy, 2006.

[9] N. Natarajan, I. S. Dhillon, P. K. Ravikumar, and A. Tewari, "Learning with noisy labels," in *Advances in Neural Information Processing Systems*, vol. 26, 2013, pp. 1196–1204. [Online]. Available: https://papers.nips.cc/paper/5073-learning-with-noisy-labels.pdf

[10] A. Graves, G. Wayne, and I. Danihelka, *Neural turing machines*, arXiv preprint arXiv:1410.5401, 2014. DOI: 10.48550/arXiv.1410.5401

[11] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015, pp. 2901–2907.

[12] S. Sukhbaatar, A. Szlam, J. Weston, and R. Fergus, "End-to-end memory networks," in *Advances in Neural Information Processing Systems 28 (NeurIPS 2015)*, C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, Eds., vol. 28, Curran Associates, Inc., 2015. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/file/8fb21ee7a2207526da55a679f0332de2-Paper.pdf

[13] J. Weston, S. Chopra, and A. Bordes, "Memory networks," in *International Conference on Learning Representations (ICLR)*, 2015. eprint: https://arxiv.org/abs/1410.3916. [Online]. Available: https://arxiv.org/abs/1410.3916

[14] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in *Proceedings of The 33rd International Conference on Machine Learning*, M. F. Balcan and K. Q. Weinberger, Eds., ser. Proceedings of Machine Learning Research, vol. 48, New York, New York, USA: PMLR, Jun. 2016, pp. 1050–1059. [Online]. Available: http://proceedings.mlr.press/v48/gal16.pdf

[15] A. Graves et al., "Hybrid computing using a neural network with dynamic external memory," *Nature*, vol. 538, no. 7626, pp. 471–476, 2016. DOI: 10.1038/nature20101

[16] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [Online]. Available: https://arxiv.org/abs/1705.08500

[17] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, D. Precup and Y. W. Teh, Eds., ser. Proceedings of Machine Learning Research, vol. 70, PMLR, Jun. 2017, pp. 1321–1330. [Online]. Available: http://proceedings.mlr.press/v70/guo17a/guo17a.pdf

[18] M. Joshi, E. Choi, D. Weld, and L. Zettlemoyer, "TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension," in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vancouver, Canada: Association for Computational Linguistics, Jul. 2017, pp. 1601–1611. DOI: 10.18653/v1/P17-1147

[19] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017. DOI: 10.1073/pnas.1611835114 eprint: https://www.pnas.org/doi/pdf/10.1073/pnas.1611835114.

[20] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, I. Guyon et al., Eds., vol. 30, Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

[21] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, "FEVER: A large-scale dataset for fact extraction and VERification," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, M. Walker, H. Ji, and A. Stent, Eds., New Orleans, Louisiana: Association for Computational Linguistics, Jun. 2018, pp. 809–819. DOI: 10.18653/v1/N18-1074

[22] Z. Yang et al., "HotpotQA: A dataset for diverse, explainable multi-hop question answering," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, E. Riloff, D. Chiang, J. Hockenmaier, and J. Tsujii, Eds., Brussels, Belgium: Association for Computational Linguistics, Oct. 2018, pp. 2369–2380. DOI: 10.18653/v1/D18-1259

[23] T. Kwiatkowski et al., "Natural questions: A benchmark for question answering research," *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 452–466, 2019. DOI: 10.1162/tacl_a_00276

[24] Y. Liu et al., *RoBERTa: A robustly optimized BERT pretraining approach*, arXiv preprint arXiv:1907.11692, 2019. DOI: 10.48550/arXiv.1907.11692

[25] Y. Ovadia et al., "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, 2019. [Online]. Available: https://papers.nips.cc/paper_files/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html

[26] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, K. Inui, J. Jiang, V. Ng, and X. Wan, Eds., Hong Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 3982–3992. DOI: 10.18653/v1/D19-1410

[27] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in NLP," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL 2019)*, 2019, pp. 3645–3650. DOI: 10.18653/v1/P19-1355

[28] A. Talmor, J. Herzig, N. Lourie, and J. Berant, "CommonsenseQA: A question answering challenge targeting commonsense knowledge," in *Proceedings of NAACL-HLT 2019*, Association for Computational Linguistics, 2019, pp. 4149–4158. DOI: 10.18653/v1/N19-1421

[29] S. Gururangan et al., "Don't stop pretraining: Adapt language models to domains and tasks," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds., Online: Association for Computational Linguistics, Jul. 2020, pp. 8342–8360. DOI: 10.18653/v1/2020.acl-main.740

[30] A. Kamath, R. Jia, and P. Liang, "Selective question answering under domain shift," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL 2020)*, 2020, pp. 5684–5696. DOI: 10.18653/v1/2020.acl-main.503

[31] V. Karpukhin et al., "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Online: Association for Computational Linguistics, Nov. 2020, pp. 6769–6781. DOI: 10.18653/v1/2020.emnlp-main.550

[32] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33, Curran Associates, Inc., 2020, pp. 9459–9474. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf

[33] M. Geva, D. Khashabi, E. Segal, T. Khot, D. Roth, and J. Berant, "Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies," *Transactions of the Association for Computational Linguistics*, vol. 9, pp. 346–361, Apr. 2021. DOI: 10.1162/tacl_a_00370 eprint: https://direct.mit.edu/tacl/article-pdf/doi/10.1162/tacl_a_00370/1924104/tacl_a_00370.pdf.

[34] D. Hendrycks et al., "Measuring massive multitask language understanding," in *International Conference on Learning Representations (ICLR)*, 2021. [Online]. Available: https://openreview.net/forum?id=d7KBjmI3GmQ

[35] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with gpus," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021. DOI: 10.1109/TBDATA.2019.2921572

[36] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations (ICLR)*, 2022. [Online]. Available: https://arxiv.org/abs/2106.09685

[37] S. Kadavath et al., *Language models (mostly) know what they know*, arXiv preprint arXiv:2207.05221, Nov. 2022. DOI: 10.48550/arXiv.2207.05221

[38] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," in *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022, pp. 22 199–22 213. [Online]. Available: https://arxiv.org/abs/2205.11916

[39] S. Lin, J. Hilton, and O. Evans, "TruthfulQA: Measuring how models mimic human falsehoods," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, S. Muresan, P. Nakov, and A. Villavicencio, Eds., Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 3214–3252. DOI: 10.18653/v1/2022.acl-long.229

[40] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems 35 (NeurIPS)*, 2022, pp. 24 824–24 837.

[41] T. Wu, M. Caccia, Z. Li, Y.-F. Li, G. Qi, and G. Haffari, "Pretrained language model in continual learning: A comparative study," in *International Conference on Learning Representations (ICLR)*, 2022.

[42] E. Zelikman, Y. Wu, J. Mu, and N. Goodman, "Star: Bootstrapping reasoning with reasoning," in *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., vol. 35, Curran Associates, Inc., 2022, pp. 15 476–15 488. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/file/639a9a172c044fbb64175b5fad42e9a5-Paper-Conference.pdf

[43] H. Alkaissi and S. I. McFarlane, "Artificial hallucinations in ChatGPT: Implications in scientific writing," *Cureus*, vol. 15, no. 2, e35179, 2023. DOI: 10.7759/cureus.35179

[44] R. Anil et al., *PaLM 2 technical report*, arXiv preprint arXiv:2305.10403, 2023. DOI: 10.48550/arXiv.2305.10403

[45] Anthropic, *Claude 2*, Blog post, Jul. 2023. [Online]. Available: https://www.anthropic.com/news/claude-2

[46] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023. [Online]. Available: https://arxiv.org/abs/2305.14314

[47] C. Gulcehre et al., *Reinforced self-training (ReST) for language modeling*, arXiv preprint arXiv:2308.08998, 2023.

[48] N. Ho, L. Schmid, and S.-Y. Yun, "Large language models are reasoning teachers," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Association for Computational Linguistics, 2023, pp. 14 852–14 882. DOI: 10.18653/v1/2023.acl-long.830

[49] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023. DOI: 10.1145/3571730

[50] Z. Jiang et al., "Active retrieval augmented generation," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Singapore: Association for Computational Linguistics, Dec. 2023, pp. 7969–7992. DOI: 10.18653/v1/2023.emnlp-main.495

[51] D. Kalajdzievski, *A rank stabilization scaling factor for fine-tuning with LoRA*, arXiv preprint arXiv:2312.03732, 2023. [Online]. Available: https://arxiv.org/abs/2312.03732

[52] L. Kuhn, Y. Gal, and S. Farquhar, "Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation," in *International Conference on Learning Representations (ICLR)*, 2023.

[53] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, and J.-R. Wen, "HaluEval: A large-scale hallucination evaluation benchmark for large language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Singapore: Association for Computational Linguistics, Dec. 2023, pp. 6449–6464. DOI: 10.18653/v1/2023.emnlp-main.397

[54] A. Mallen, A. Asai, V. Zhong, R. Das, D. Khashabi, and H. Hajishirzi, "When not to trust language models: Investigating effectiveness of parametric and non-parametric memories," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023)*, Association for Computational Linguistics, 2023, pp. 9802–9822. DOI: 10.18653/v1/2023.acl-long.546

[55] P. Manakul, A. Liusie, and M. Gales, "SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds., Singapore: Association for Computational Linguistics, Dec. 2023, pp. 9004–9017. DOI: 10.18653/v1/2023.emnlp-main.557

[56] S. Mehta, D. Patil, S. Chandar, and E. Strubell, "An empirical investigation of the role of pre-training in lifelong learning," *Journal of Machine Learning Research*, 2023.

[57] S. Min et al., "FActScore: Fine-grained atomic evaluation of factual precision in long form text generation," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds., Singapore: Association for Computational Linguistics, Dec. 2023, pp. 12 076–12 100. DOI: 10.18653/v1/2023.emnlp-main.741

[58] OpenAI, *GPT-4 technical report*, arXiv preprint arXiv:2303.08774, 2023. DOI: 10.48550/arXiv.2303.08774

[59] K. Tian et al., "Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 5433–5442. [Online]. Available: https://arxiv.org/abs/2305.14975

[60] X. Wang et al., "Orthogonal subspace learning for language model continual learning," in *Findings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP Findings)*, 2023.

[61] X. Wang et al., "Self-consistency improves chain of thought reasoning in language models," in *International Conference on Learning Representations (ICLR 2023)*, 2023. [Online]. Available: https://openreview.net/forum?id=1PL1NIMMrw

[62] Y. Wang, P. Li, M. Sun, and Y. Liu, "Self-knowledge guided retrieval augmentation for large language models," in *Findings of EMNLP 2023*, Association for Computational Linguistics, 2023, pp. 10 303–10 315. DOI: 10.18653/v1/2023.findings-emnlp.691

[63] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[64] D. Biderman et al., "LoRA learns less and forgets less," *Transactions on Machine Learning Research*, 2024.

[65] M. Chelli et al., "Hallucination rates and reference accuracy of ChatGPT and bard for systematic reviews: Comparative analysis," *J Med Internet Res*, vol. 26, e53164, May 2024. DOI: 10.2196/53164

[66] C. Chen et al., "INSIDE: LLMs' internal states retain the power of hallucination detection," in *International Conference on Learning Representations (ICLR 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2402.03744

[67] J. J. Cherian, I. Gibbs, and E. J. Candès, "Large language model validity via enhanced conformal prediction methods," in *Advances in Neural Information Processing Systems (NeurIPS 2024)*, 2024.

[68] H. W. Chung et al., "Scaling instruction-finetuned language models," *Journal of Machine Learning Research*, vol. 25, no. 70, pp. 1–53, 2024. [Online]. Available: http://jmlr.org/papers/v25/23-0870.html

[69] P. Das et al., "Larimar: Large language models with episodic memory control," in *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, ser. Proceedings of Machine Learning Research, vol. 235, 2024. [Online]. Available: https://arxiv.org/abs/2403.11901

[70] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, "Detecting hallucinations in large language models using semantic entropy," *Nature*, vol. 630, no. 8017, pp. 625–630, 2024. DOI: 10.1038/s41586-024-07421-0

[71] J. Geng, F. Cai, Y. Wang, H. Koeppl, P. Nakov, and I. Gurevych, "A survey of confidence estimation and calibration in large language models," in *Proceedings of the 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024)*, 2024. [Online]. Available: https://aclanthology.org/2024.naacl-long.366/

[72] A. Hosseini, X. Yuan, N. Malkin, A. Courville, A. Sordoni, and R. Agarwal, "V-STaR: Training verifiers for self-taught reasoners," in *Conference on Language Modeling (COLM 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2402.06457

[73] J. Huang et al., "Large language models cannot self-correct reasoning yet," in *International Conference on Learning Representations (ICLR 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2310.01798

[74] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, "Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity," in *Proceedings of the 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2403.14403

[75] R. Kamoi, Y. Zhang, N. Zhang, J. Han, and R. Zhang, "When can LLMs actually correct their own mistakes? a critical survey of self-correction of LLMs," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1417–1440, 2024. DOI: 10.1162/tacl_a_00713

[76] H. Lang, D. Sontag, and A. Vijayaraghavan, "Theoretical analysis of weak-to-strong generalization," in *Advances in Neural Information Processing Systems (NeurIPS 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2405.16043

[77] C. Mohri and T. Hashimoto, "Language models with conformal factuality guarantees," in *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2402.10978

[78] A. Nikitin, J. Kossen, Y. Gal, and P. Marttinen, "Kernel language entropy: Fine-grained uncertainty quantification for LLMs from semantic similarities," in *Advances in Neural Information Processing Systems (NeurIPS 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2405.20003

[79] V. Quach et al., "Conformal language modeling," in *International Conference on Learning Representations (ICLR 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2306.10193

[80] Qwen Team, *Qwen2.5: A party of foundation models*, arXiv preprint arXiv:2412.15115, 2024. DOI: 10.48550/arXiv.2412.15115

[81] L. Tang, P. Laban, and G. Durrett, "MiniCheck: Efficient fact-checking of LLMs on grounding documents," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024, pp. 8818–8847.

[82] S. Valentin, J. Fu, G. Detommaso, S. Xu, G. Zappella, and B. Wang, *Cost-effective hallucination detection for LLMs*, arXiv preprint arXiv:2407.21424, 2024.

[83] M. Xiong et al., "Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs," in *International Conference on Learning Representations (ICLR 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2306.13063

[84] Y. A. Yadkori et al., *Mitigating LLM hallucinations via conformal abstention*, arXiv preprint arXiv:2405.01563, 2024. [Online]. Available: https://arxiv.org/abs/2405.01563

[85] I. Baek, H. Chang, B. Kim, J. Lee, and H. Lee, "Probing-RAG: Self-probing to guide language models in selective document retrieval," in *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025, pp. 3287–3304.

[86] R. Das et al., "Retraining with predicted hard labels provably increases model accuracy," in *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 12 509–12 538. [Online]. Available: https://arxiv.org/abs/2406.11206

[87] A. Huang et al., "Self-improvement in language models: The sharpening mechanism," in *International Conference on Learning Representations (ICLR 2025)*, 2025. [Online]. Available: https://arxiv.org/abs/2412.01951

[88] L. Huang et al., "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, 2025. DOI: 10.1145/3703155

[89] M. Marina et al., "LLM-independent adaptive RAG: Let the question speak for itself," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 8697–8709. [Online]. Available: https://arxiv.org/abs/2505.04253

[90] M. Nandakishor, *Confidence-aware routing for large language model reliability enhancement: A multi-signal approach to pre-generation hallucination mitigation*, arXiv preprint arXiv:2510.01237, 2025.

[91] D. Sinha, *The semantic illusion: Certified limits of embedding-based hallucination detection in RAG systems*, arXiv preprint arXiv:2512.15068, 2025. [Online]. Available: https://arxiv.org/abs/2512.15068

[92] Y. Song, H. Zhang, C. Eisenach, S. Kakade, D. Foster, and U. Ghai, "Mind the gap: Examining the self-improvement capabilities of large language models," in *International Conference on Learning Representations (ICLR 2025)*, 2025. [Online]. Available: https://arxiv.org/abs/2412.02674

[93] R. Vashurin et al., "Benchmarking uncertainty quantification methods for large language models with LM-Polygraph," *Transactions of the Association for Computational Linguistics*, vol. 13, pp. 220–248, 2025. DOI: 10.1162/tacl_a_00737

[94] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang, *A-MEM: Agentic memory for LLM agents*, arXiv preprint arXiv:2502.12110, 2025. [Online]. Available: https://arxiv.org/abs/2502.12110

[95] B. Yi, Q. Liu, Y. Cheng, and H. Xu, "Escaping model collapse via synthetic data verification: Near-term improvements and long-term convergence," in *International Conference on Learning Representations (ICLR 2026)*, 2026. [Online]. Available: https://arxiv.org/abs/2510.16657

Appendix A

Implementation reference

A.1 Code and data availability

The full implementation of the architecture, the cycle runner, the calibration scripts, the baseline and ablation drivers, the verifier ensemble, the retrieval-utility classifier, and the figure and table aggregators is released as the open-source repository at https://github.com/aksaN000/caem-thesis. The empirical artefact bundle for the realised six-cycle trajectory, the cycle-zero validation gate, the eight-baseline panel, the retrieval-utility-classifier on/off ablation, the per-cycle logs, and the post-trajectory analysis is released as the Google Drive folder at https://drive.google.com/drive/folders/1xErhsRDYe_qO0ZgUBn0WXK81eHc5jZCa, organised under the subfolder `v2_1_phase1e_main/` (the realised trajectory) alongside `pre_main_snapshot/` (the cycle-zero anchor for snapshot recovery) and the earlier-phase archives. The sixty-four-gigabyte Wikipedia passage FAISS index and the per-commit pre-main snapshot bundles are additionally mirrored on the Hugging Face dataset repository `aksaN000/caem-passage-index-21m` (private; access on request), as documented in Section E.2. The three publication endpoints are intentionally complementary: the GitHub repository carries the executable code and the locked configuration JSONs, the Google Drive folder carries the realised empirical artefacts and the per-cycle JSON/CSV logs that the thesis reads, and the Hugging Face dataset host carries the bulky retrieval substrate and the snapshot bundles that anchor the credit-burnout recovery sequence of Section E.2.

A.2 Repository layout and module map

The repository root is `caem/` at the top of the Vast workspace. Body prose throughout the thesis refers to the architectural components by concept name; the inventory below maps every concept to the file that implements it. Paths are relative to the repository root.

Table A.1: Repository layout. Body prose throughout the thesis refers to architectural components by their concept names (memory store, three-tier router, verifier ensemble, retrieval-utility classifier, self-improvement loop, retention guard); this table maps each concept to the file that implements it.

Path	Role	Description
Top-level directories		
`caem/`	package root	Core Python package: pipeline, memory, routing, verification, training.
`scripts/`	runners	Cycle runner, calibration scripts, baseline runners, rescore utilities.
`eval/`	harness	Shared per-benchmark eval scaffold (EM, em_llm_judged, CHM, CES).
`outputs/`	artefacts	Per-cycle JSON, CSV, PNG, and log evidence (cycle_0/, full_run/, baselines/).
`data/`	external	Passage index, calibration pairs, alias dictionary, retention and RUC datasets.
`tests/`	test suite	Pytest suite covering storage gate, deferred buffer, retention probe, SIL, verifier.
`docs/`	runbook	PRODUCTION_RUNBOOK.md: deployment-mode spec and cycle-trigger policies.
`thesis_report/`	thesis	LaTeX chapters, bibliography, figures, main.pdf build target.
Pipeline entry points		
`caem/pipeline.py`	per-query	Pipeline class: serial driver used in production and demo paths.
`caem/pipeline_batch.py`	per-cycle	BatchPipeline class: batched cycle driver used by the experiment runner.
`scripts/run_experiment.py`	runner	Canonical cycle driver: invokes BatchPipeline, SIL, retention probe, retroverify.
Configuration		
`caem/config.py`	registry	Central CAEMConfig dataclass; every threshold, weight, structural toggle.
`caem/ruc/v2/feature_spec.json`	RUC config	RUC threshold, regulariser, feature order, model paths.
Memory module		
`caem/memory/store.py`	storage gate	EpisodicMemoryStore: fixed-threshold admission and FAISS-backed retrieval.
`caem/memory/deferred.py`	deferred buffer	DeferredBuffer and DeferredEntry: TTL-bounded reconsider queue.
`caem/memory/encoder.py`	embeddings	SBERT-side encoder for memory keys and Tier 1 lookup.
Routing		
`caem/routing/router.py`	tier dispatch	Tier 1/2/3 router with safety override and combined-score thresholds.
`caem/routing/ruc.py`	RUC head	LRRetrievalUtilityClassifier: binary head for Tier 2 versus Tier 3 routing.
Verification		
`caem/verification/verifier.py`	verifier ensemble	UnifiedVerifier: nine-signal composite producing u_{pre} and u_{stored}.
`caem/verification/adaptive_nli_judge.py`	NLI dispatch	AdaptiveNLIJudge: length-gated dispatcher; in the deployed configuration every hypothesis routes to MiniCheck under a 512-token truncation contract because the long-hypothesis Qwen-3B backend was ablated (Section 5.2.6).
`caem/verification/cal_prob_composite.py`	calibration	Per-signal isotonic curves plus per-benchmark shrinkage and boost layer.
`caem/retrieval/rag.py`	retrieval	FAISS-backed passage index reader for Tier 3 grounded generation.
Training		
`caem/training/self_improvement.py`	SIL trainer	SelfImprovementLoop: LoRA fine-tune and retroverify hook.

Path	Role	Description
`caem/training/retention_probe.py`	retention guard	Per-cycle multi-modal probe panel and ratio-floor check.
`caem/training/pool_reweighting.py`	pool balancer	Per-benchmark temperature-mixed reweighting and share ceiling.
`caem/training/loop_filter.py`	SIL admission	Per-pair filter at $\tau_{\text{train}} = 0.70$ above the storage cut-point.
Evaluation		
`eval/harness.py`	eval scaffold	Per-benchmark EM, F1, latency, CHM, and CES.
`eval/benchmarks.py`	benchmark loaders	TriviaQA, FEVER, CommonsenseQA, TruthfulQA, StrategyQA, MMLU loaders.
`caem/eval/equilibrium.py`	SE metric	Stochastic-equilibrium computation for the convergence-check table.

A.3 Configuration registry

Every constant that governs runtime behaviour lives in the `CAEMConfig` dataclass at `caem/config.py`. The exception is the retrieval-utility classifier, whose configuration lives in `caem/ruc/v2/feature_spec.json` at deployment time (the classifier is loaded once at pipeline construction and is not refit at cycle boundaries; see Section 4.2.5). The registry below groups every field by concept, records the deployed value, and tags the provenance as *literature* (anchored to a cited paper), *design* (architectural decision documented before any run), *calibrated* (selected empirically at cycle zero), or *pre-specified* (chosen before any data is observed).

Table A.2: Configuration registry. Provenance tags: *literature* (cited paper), *design* (architectural choice documented before any run), *calibrated* (chosen empirically at cycle zero or refit at a cycle boundary), *pre-specified* (registered horizon before any data is observed).

Field	Value	Provenance	Source / rationale
Routing thresholds (Section 4.2.4)			
`routing_lambda`	0.70	design	Combined-score weight on similarity.
`tier1_combined_threshold`	0.90	design	Tier 1 dispatch floor on combined score.
`tier2_similarity_threshold`	0.75	design	Tier 2 dispatch floor on similarity alone.
`safety_u_pre_min`	0.38	calibrated	Re-tuned at cycle one from the registered 0.60; safety-override floor.
`u_pre_token_weight`	0.60	design	Token-probability weight in u_{pre}.
`u_pre_cconv_weight`	0.40	design	Encoder-convergence weight in u_{pre}.
Storage gate thresholds (Section 4.2.9)			
`store_threshold`	0.60	design	STORE if $u_{\text{stored}} \geq$ this.
`defer_threshold`	0.45	design	DEFER if `defer_threshold` $\leq u_{\text{stored}} <$ `store_threshold`.
`abstain_pground_ceiling`	0.20	design	ABSTAIN if $u_{\text{stored}} <$ `defer_threshold` and $p_{\text{ground,max}} <$ this.
`early_exit_u_internal`	0.70	design	Confabulation early-exit on internal-uncertainty floor.
`early_exit_p_ground_max`	0.20	design	Confabulation early-exit on grounding ceiling.
Calibration parameters (Section 4.2.7)			

Field	Value	Provenance	Source / rationale
composite_mode	"auto"	design	Auto-switch to calibrated probability composite when the calibration JSON loads.
composite_shrinkage_alpha	0.6	calibrated	Per-benchmark child curve shrinkage toward pooled fit.
adaptive_thresholds_per_cycle	true	design	Refit temperature, isotonic curves, and boost weights at every successful cycle.
adaptive_thresholds_ema_alpha	0.7	design	Exponential-moving-average smoothing on the per-cycle temperature refit.
Retention guard (Section 5.1.5)			
forgetting_tolerance	0.93	design	Worst-probe ratio floor; trigger asymmetric rollback if any probe drops below.
retention_probes	3-probe panel	design	Multi-modal probe panel (mmlu, triviaqa_test, commonsense_qa_test); replaces the single-MMLU probe of v1.
retention_probe_n	200	design	Samples per probe at every cycle.
Deferred buffer (Section 4.2.10)			
deferred_buffer_max_size	10 000	design	Bounded FIFO capacity for deferred entries.
deferred_buffer_ttl_cycles	2	design	Reconsider-invocation TTL; entries dropped after two passes.
Retroverify and consolidation (Section 4.2.10)			
retroverify_prune_threshold	0.50	design	Prune from memory if recalibrated u_{stored} drops below.
hit_counter_force_retroverify	10	design	Per-entry retrieval-hit threshold forcing reverification next cycle.
enable_consolidation	true	design	Cycle-boundary SBERT-similarity clustering pass.
consolidation_similarity_threshold	0.92	design	Minimum cosine similarity for cluster admission.
consolidation_search_k	20	design	Per-entry FAISS search depth for neighbours.
consolidation_max_u_spread	0.15	design	Spread guardrail; reject clusters whose u_{stored} range exceeds this.
Self-improvement training (Section 4.2.10)			
use_lora_training	true	design	LoRA SIL primary path; replaces the v1 full-fine-tune fallback.
lora_r	32	literature	Rank, per Biderman et al. 2024 audit on Qwen-class backbones.
lora_alpha	64	literature	Standard $\alpha = 2r$ scaling rule.
lora_dropout	0.05	literature	Hu et al. 2021 default.
lora_target_modules	(q,k,v,o, gate,up,down)_proj	literature	All linear projections of the transformer block.
lora_learning_rate	2×10^{-4}	literature	Standard LoRA learning rate.
use_8bit_adamw	true	design	Applies only to the full fine-tuning fallback; the LoRA path overrides it and uses 32-bit AdamW.
batch_size	4	calibrated	Reduced from 16 at cycle one after GPU memory pressure.
grad_accum_steps	4	design	Holds the effective batch at 16.
epochs_per_cycle	3	design	SIL training duration per successful cycle.

Field	Value	Provenance	Source / rationale
warmup_steps	500	design	Linear warmup before the standard schedule.
min_u_stored_for_training	0.70	calibrated	τ_{train}; reduced from 0.75 at cycle one.
pool_reweighting_enabled	true	design	Temperature-mixed reweighting across training benchmarks.
pool_reweighting_temperature	2.0	literature	Mixture temperature; Du et al. 2022.
pool_reweighting_upsample_cap	3.0	calibrated	Upper bound on per-benchmark upsampling factor.
pool_reweighting_max_share	0.40	design	Per-benchmark hard share ceiling.
loop_distinct4_threshold	0.25	calibrated	SIL-pool loop filter on token-4-gram distinct ratio.
Verifier ensemble (Section 4.2.6)			
verifier_backend	"minicheck"	literature	Short-hypothesis backend; Tang et al. 2024.
minicheck_model	lytang/MiniCheck-Flan-T5-Large	literature	Verifier checkpoint.
cross_encoder_model	BAAI/bge-reranker-v2-m3	design	Shared reranker; consumed by grounding and question-answer relevance.
verifier_retrieve_k	20	design	Top-k passages retrieved before reranking.
verifier_rerank_k	3	design	Top-N retained after reranking.
atomic_min_tokens	12	calibrated	Length gate below which atomic decomposition is bypassed.
mc_dropout_k, mc_dropout_rate	5, 0.1	literature	Gal and Ghahramani 2016.
sc_chains_m	3	literature	Wang et al. 2022 self-consistency.
se_samples_k, se_temperature	10, 1.0	literature	Farquhar et al. 2024 semantic-entropy protocol.
disable_h_norm	true	calibrated	Retired at cycle zero on cal-fold Cohen's d audit.
disable_p_contra	true	design	Structurally zero under the MiniCheck binary judge.
Generation and retrieval			
base_model_name	Qwen/Qwen2.5-3B-Instruct	design	Apache-licensed three-billion-parameter instruction-tuned backbone.
rag_max_context_tokens	384	design	Cap on retrieved context tokens.
rag_max_new_tokens	512	design	Generation budget under retrieval-augmented prompt.
rag_do_sample	false	design	Greedy decoding for reproducibility.
rag_index_type	"ivf_pq"	design	FAISS index backend.
rag_faiss_nlist, nprobe_tier3, nprobe_tier2	32768, 64, 16	design	IVF-PQ density and adaptive probe budget.
rag_top_k	5	literature	Karpukhin et al. 2020 DPR standard.
Memory store (Section 4.2.3)			

Field	Value	Provenance	Source / rationale
`sbert_model`	`sentence-transformers/all-mpnet-base-v2`	literature	768-dimensional sentence encoder.
`embedding_dim`	768	literature	Must match the encoder output dimension.
`max_memory_size`	1 000 000	design	Capacity for thesis-scale and production-scale runs.
`memory_index_type`	`"ivf_pq"`	design	FAISS index backend.
`novelty_threshold`	0.95	design	Skip storage if nearest-neighbour cosine similarity exceeds this.
`pruning_trigger`	0.95	design	Trigger pruning when memory reaches this fraction of capacity.
`pruning_amount`	0.20	design	Remove the bottom fraction by value score.
`value_importance_weight`, `value_recency_weight`	0.70, 0.30	design	Value-score decomposition.
Experiment scope			
`num_cycles`	10	pre-specified	Registered horizon; realised trajectory closed at cycle five under the equilibrium-saturation gate.
`calibration_set_size`	500	design	Cycle-zero calibration-fold size per training benchmark.
`purity_validation_set_size`	500	design	Cycle-zero purity-validation slice per training benchmark.
`questions_per_cycle`	5 000	design	Stream-chunk question budget per cycle.

The retrieval-utility classifier (Section 4.2.5) has no fields in `caem/config.py`. Its operating point lives in `caem/ruc/v2/feature_spec.json`: the threshold $\tau_{\text{RUC}} = 0.375$, the regularisation strength $C = 0.5$, the thirty-six-feature ordering, the standard-scaler and logistic-head joblib paths, and the model-type guard `"logistic_regression_scaled"`. The classifier is loaded once at pipeline construction from the on-disk specification and is not refit at any cycle boundary; the architectural rationale is documented in Section 4.2.5.

A.4 Hyperparameter listing

The consolidated hyperparameter table below pairs every quantity that the thesis discusses by symbol with its deployed value, its source file, and the chapter subsection that introduces the symbol. The rightmost column tags whether the cycle-boundary refit moves the quantity (*refit*), whether it stays architecturally frozen across the trajectory (*frozen*), or whether it was calibrated empirically at cycle zero and held constant thereafter (*C0-cal frozen*).

Table A.3: Consolidated hyperparameter table. The four per-cycle refit quantities are the pooled temperature scalar, the per-benchmark temperature vector, the nine per-signal isotonic curves, and the boost-layer weights plus intercept. Every other quantity is architecturally frozen across the trajectory.

Symbol (thesis)	**Value**	**Source**	**Cycle role**
λ (router blend)	0.70	`config.py: routing_lambda`	frozen
θ_1 (Tier 1 floor)	0.90	`tier1_combined_threshold`	frozen
θ_{sim} (Tier 2 floor)	0.75	`tier2_similarity_threshold`	frozen
ϕ_{safe} (safety floor)	0.38	`safety_u_pre_min`	C0-cal frozen
τ_{store}	0.60	`store_threshold`	frozen
τ_{defer}	0.45	`defer_threshold`	frozen
τ_{train}	0.70	`min_u_stored_for_training`	C0-cal frozen
τ_{retro}	0.50	`retroverify_prune_threshold`	frozen
τ_{RUC}	0.375	`ruc/v2/feature_spec.json`	frozen
$\rho_{\min}$ (retention floor)	0.93	`forgetting_tolerance`	frozen
T (pooled temperature)	$20.09 \rightarrow 0.74$ band	per-cycle `calibrated_config_cycle*.json`	**refit**
T_b (per-benchmark)	per-bench tables	per-cycle `calibrated_config_cycle*.json`	**refit**
α_{shrink}	0.6	`composite_shrinkage_alpha`	frozen
C (boost L2)	0.01	boost layer fit in composite refit	frozen
Per-signal isotonic curves	non-parametric step functions	`composite_calibration.json`	**refit**
Boost-layer weights (nine signals)	per-cycle vectors	`composite_calibration.json`	**refit**
LoRA r, α, dropout	32, 64, 0.05	`lora_*`	frozen
LoRA learning rate	2×10^{-4}	`lora_learning_rate`	frozen
LoRA target modules	all-linear seven projections	`lora_target_modules`	frozen
SIL batch $\times$ accum	$4 \times 4 = 16$	`batch_size`, `grad_accum_steps`	frozen
SIL epochs per cycle	3	`epochs_per_cycle`	frozen
$\alpha_{\text{T-EMA}}$	0.7	`adaptive_thresholds_ema_alpha`	frozen
Pool reweighting temperature	2.0	`pool_reweighting_temperature`	frozen
Pool reweighting upsample cap	3.0	`pool_reweighting_upsample_cap`	frozen
Pool reweighting share ceiling	0.40	`pool_reweighting_max_share`	frozen
Deferred-buffer TTL	2 reconsider invocations	`deferred_buffer_ttl_cycles`	frozen
Consolidation similarity floor	0.92	`consolidation_similarity_threshold`	frozen

Symbol (thesis)	Value	Source	Cycle role
Consolidation FAISS-k	20	`consolidation_search_k`	frozen
Consolidation u-spread guard	0.15	`consolidation_max_u_spread`	frozen
MC-dropout K, p	5, 0.1	`mc_dropout_k`, `mc_dropout_rate`	frozen
Self-consistency chains M	3	`sc_chains_m`	frozen
SE samples K, temperature	10, 1.0	`se_samples_k`, `se_temperature`	frozen
Verifier retrieve-k, rerank-N	20, 3	`verifier_retrieve_k`, `verifier_rerank_k`	frozen
Atomic length gate (tokens)	12	`atomic_min_tokens`	C0-cal frozen
RAG context tokens, new tokens	384, 512	`rag_max_*`	frozen
SBERT embedding dimension	768	`embedding_dim`	frozen
Memory FAISS (nlist, nprobe, m, nbits)	4096, 32, 64, 8	`faiss_*`	frozen
RAG FAISS (nlist, nprobe$_{T3}$, $T2$)	32768, 64, 16	`rag_faiss_*`	frozen

A.5 Runbook step graph

The end-to-end execution sequence partitions into a pre-run setup block, a six-step main cycle loop, and a post-run analysis block. The pre-run block runs once before the first cycle; the main cycle loop repeats per cycle; the post-run block runs once after the final cycle closes. Authoritative paths: `run_phase1a.sh` (the top-level driver), `scripts/run_experiment.py` (the in-cycle implementation), and `docs/PRODUCTION_RUNBOOK.md` Section 5 (the production mirror).

Pre-run setup (Steps 1–6):

1. **Prompt smoke test** (`step_5_9_prompt_smoke`). Renders five samples per benchmark against the registered prompt templates and asserts the answer-line contract. Gate: every sample renders with a non-empty answer line.
2. **Alias dictionary build** (`step_5_9_5_alias_dict`). Builds `data/alias_dict.pkl` from a local Wikidata dump. The alias signal is retired in the deployed composite; the dictionary is preserved for ablation only.
3. **Cold-start seeding** (`step_6_reseed` and `step_6_5_em_prune`). Seeds the episodic memory with verified entries from the training benchmarks' dev splits under a cold-start storage threshold of 0.50; the realised cold-start store holds 431 entries (the seeding plan targeted approximately three thousand candidates before deduplication and hygiene). Gate: deduplication ratio sane; the EM-prune step removes fewer than ten percent of seeded entries.
4. **Cycle-zero baseline evaluation and cal-fold scoring** (`step_7_0_cycle0`).

Runs the pristine generator against the held-out evaluation fold and the calibration fold (five hundred samples per training benchmark, fifteen hundred pooled). Writes per-sample nine-signal vectors and ground-truth labels. Gate: every JSON has the expected sample count and no NaN signals.

5. **Composite calibration and validation gate** (`step_7_0_calibrate`, `step_7_0_2_5_rescore_eval`, `step_7_0_3_validate_weights`). Fits the per-signal isotonic curves, the per-benchmark shrinkage-blended children, the boost-layer logistic regression, the per-benchmark temperature scalars, and the per-benchmark safety floors. Runs the cycle-zero validation gate (Section A.9). Gate: cycle-zero canonical composite locked at `outputs/cycle_0/composite_calibration.json`; never overwritten by later cycles.

6. **Retention baseline freeze and snapshot upload** (`step_19_2_slice` and `step_gdrive_upload_pre_main`). Records the pristine-model probe accuracies on MMLU, TriviaQA-test, and CommonsenseQA-test as the cycle-zero anchor for the 0.93 retention floor. Pushes the snapshot to Hugging Face for credit-burnout recovery (Section A.6).

Main cycle loop (Steps 1–6, repeating per cycle):

1. **Self-improvement fine-tune:** Draws the SIL training pool from memory entries above $\tau_{\text{train}} = 0.70$, applies the pool-reweighting layer, runs the LoRA fine-tune for three epochs, and probes the retention panel. Gate: every probe ratio against the cycle-zero baseline clears 0.93. On failure, the adapter is rolled back to the previous cycle's checkpoint, the memory is preserved, and Steps 2.1–2.5 are skipped under the asymmetric-rollback contract.

2. **Cycle-boundary recalibration:** Re-scores the calibration fold under the post-fine-tune model, refits the per-benchmark temperature scalars, refits the per-signal isotonic curves under the shrinkage prior, and runs the retroactive re-verification pass against the stored memory under the recalibrated verifier. Skipped on aborted cycles.

3. **Stream-chunk pre-routing pass:** Runs the per-query pipeline on the cycle's stream chunk (two thousand FEVER, two thousand TriviaQA, seven hundred CommonsenseQA samples), accumulating storage admissions, deferred entries, abstentions, and discards.

4. **Stream-chunk verification and storage:** Applies the calibrated probability composite, the four-outcome decision tree, and the storage admission. Writes `memory_store_cycle_N.{faiss,meta}` and `deferred_buffer_cycle_N.pkl`.

5. **Held-out evaluation:** Runs the per-query pipeline on the held-out evaluation fold (three hundred samples per benchmark across the five-benchmark panel) under `store_to_memory=False` so the eval fold does not contaminate the memory pool.

6. **Checkpoint and halt-trigger evaluation:** Writes the cycle's coverage diagnostic, the per-tier hit-rate roll-up, and the per-cycle composite calibration

snapshot. Evaluates the halt trigger (storage saturation, retention floor failures, or the equilibrium-saturation gate).

Post-run analysis (Steps P-1 through P-7):

1. Run the eight-baseline panel (`step_baselines_all`) under the matched-protocol contract.
2. Rescore the baseline outputs through the locked cycle-three composite (`step_rescore_baselines_through_verifier`).
3. Run the TruthfulQA Haiku rescore (`scripts/rescore_truthfulqa.py`) on every baseline plus CAEM so that the TruthfulQA exact-match cells are scored under a uniform rule.
4. Run the significance panel (`step_15_5_sig`): paired McNemar on exact match, paired Wilcoxon on the composite hallucination metric, bootstrap confidence intervals, and the Holm correction within each benchmark family.
5. Aggregate the six decomposition tables (three-headline contrast, per-tier exact match, composite evaluation score, memory-store trajectory, survival distribution, decision diagnostics).
6. Run the theorem-receipt aggregators that populate the convergence-check, monotonicity-check, asymptotic-elimination, and self-correction tables of Chapter 5.
7. Push the post-trajectory artefact bundle to the Hugging Face snapshot under the post-main commit tag.

Asymmetric-rollback contract: Triggered when any retention probe drops below 0.93 of its cycle-zero baseline during Step 1 of the main cycle loop. The adapter reverts to the previous successful cycle's checkpoint, but the memory state does not revert; storage admissions from the failed cycle's stream-chunk pass are preserved. Steps 2.1–2.5 are skipped on the aborted cycle (the cycle-boundary recalibration, retroactive re-verification, and deferred-buffer reconsider all become provable no-ops under unchanged weights). The retroverify log records `"skipped": "aborted_cycle"` so downstream readers distinguish the skip from a genuine zero-delta reverify. Steps 3–6 of the main cycle loop continue to run on aborted cycles: the stream-chunk grows memory unconditionally, the held-out evaluation runs under the rolled-back model so the empirical chapter can compare the aborted cycle's exact match to the previous successful cycle's, and the checkpoint and halt-trigger step records the abort cleanly. The empirical receipt for the contract is the cycle-four firing on the realised trajectory: the TriviaQA-test probe ratio of 0.886 crossed below the floor and triggered the rollback, the cycle-five reconsider pass recovered the C4-pushed deferred entries that the skipped cycle-four pass had preserved, and the realised commit fraction closed at $\pi_{\text{commit}} = 4/5$.

A.6 Snapshot reference

The pre-main snapshot is hosted at the private Hugging Face dataset repository `aksaN000/caem-passage-index-21m`. The repository holds both the sixty-four-gigabyte Wikipedia passage FAISS index at the repository root (`passages.{faiss,meta}`) and the per-commit pre-main artefact bundles under `pre_main_snapshot/<git-commit>/` subtrees. The canonical pre-Step-7 anchor is the commit tag `c5b8066/`; the later RUC-era anchor is `700b9e3/`. The manifest `pre_main_snapshot/c5b8066/MANIFEST.json` records the upload timestamp, source host, git commit, purpose tag, and recovery hint; it is written by the `step_gdrive_upload_pre_main` stage.

The recovery sequence for a re-implementer is the following three commands run from a fresh repository clone with the Hugging Face CLI authenticated to the private repository:

```
hf download aksaN000/caem-passage-index-21m \
    --include 'passages.*' 'pre_main_snapshot/*'
cp -r pre_main_snapshot/c5b8066/* outputs/
bash run_phase1a.sh
```

The runner picks up at the highest-cycle artefact set on disk.

The snapshot contains the cold-start memory store (431 seeded episodes), the cycle-zero memory and deferred-buffer snapshots, the cycle-zero calibration fold (per-sample nine-signal vectors and gold labels), the cycle-zero composite calibration JSON (per-signal isotonic curves, per-benchmark shrinkage children, boost-layer weights), the calibrated configuration JSON (per-benchmark temperature scalars and safety floors), the cycle-zero validation-gate verdict, the cycle-zero evaluation fold (pristine model), the post-calibration rescored evaluation fold, the dataset-splits manifest, the MiniCheck-versus-RoBERTa head-to-head calibration record, the long-hypothesis judge Platt-ablation diagnostic, the cycle-zero retention slice baseline, the phase-four artefact bundle, and the run logs. Items that must be regenerated rather than pulled from the snapshot are the Qwen-2.5-3B-Instruct base weights (recovered from the Hugging Face model hub at deployment time), the per-cycle LoRA adapters at cycles one through five, the per-cycle SIL training artefacts, and the post-cycle-zero composite refits.

A.7 Hardware and software profile

Production hardware: The trajectory ran on a Vast.ai-rented RTX 5090 with thirty-two gigabytes of video memory (approximately thirty-one gigabytes usable after framework overhead), one hundred and fifty gigabytes of disk, and approximately ninety gigabytes of host random-access memory. The rental rate sat at approximately seventy cents per on-demand GPU-hour. The realised six-cycle trajectory plus cycle-zero calibration plus post-run analysis spans approximately forty calendar days of elapsed time as cycles, ablations, and rescore passes run sequentially on the single-card envelope, for approximately nine hundred and sixty GPU-hours at a total billed cost of approximately six hundred and seventy-two United States dollars (forty days at twenty-four hours per day at the on-demand rate; Vast.ai billing records).

The verifier-side scoring path (nine-signal extraction, composite refit, retroactive re-verification, and the matched-protocol Haiku-judged TruthfulQA rescore across the baseline panel) is the dominant wall-clock consumer over the run, exceeding the per-cycle low-rank-adapter fine-tune itself, because every entry on every cycle and every baseline-arm answer passes through the full verifier ensemble while the adapter touches only the strict training-pool subset.

Backbone model: The base generator is Qwen2.5-3B-Instruct (`Qwen/Qwen2.5-3B-Instruct` on the Hugging Face model hub), a three-billion-parameter decoder-only transformer with instruction tuning. The model is held in bfloat16 precision throughout the trajectory. The LoRA adapter sits on top with rank thirty-two, alpha sixty-four, dropout zero point zero five, all seven linear projections targeted (query, key, value, output, gate, up, down), at learning rate two times ten to the minus four under a 32-bit AdamW optimiser (the eight-bit AdamW flag applies only to the full fine-tuning fallback path).

Framework versions: The Python environment is pinned at version 3.12. PyTorch is version 2.10 with CUDA 13.0 and CUDA toolkit 13.1; the GPU compute capability target is `sm_120`. The transformers library is version 4.57.6. The remaining dependencies are the peft library for LoRA, the accelerate library for distributed-training scaffolding, the datasets library for benchmark loaders, the faiss-cpu library for the passage and memory indices, the sentence-transformers library for SBERT embeddings, the scikit-learn library for isotonic regression and logistic boost layer, the scipy library for statistical tests, the bitsandbytes library for the eight-bit optimiser, and the Anthropic SDK for the Haiku-judged TruthfulQA correctness rule.

Determinism: A single seed (`seed=42`) propagates through the generator sampling, the FAISS index construction, and the MC-Dropout passes. The seed-everything routine in `scripts/run_cyclic_ablation.py` sets the Python `random` module, the `PYTHONHASHSEED` environment variable, the NumPy random state, the PyTorch CPU random state, and the PyTorch CUDA random state on every device. The flag `torch.use_deterministic_algorithms` is set to false deliberately because the cuDNN attention path is not bitwise-deterministic; determinism is enforced at the metric level (exact match and F1) rather than at the bit level.

Writeup envelope: The thesis itself was written on a Windows Subsystem for Linux 2 environment running Linux kernel 6.6, with twelve gigabytes of video memory on an RTX 3060 and sixteen gigabytes of host random-access memory. The twenty-one-million-passage FAISS index is sixty-four gigabytes on disk and requires approximately ninety gigabytes of resident page cache to mmap; the writeup hardware cannot host the index, so all retrieval-side numbers in the thesis are read from the Vast-side evaluation JSONs rather than recomputed locally. The composite hallucination metric rescore for the semantic-entropy baseline (Section 6.2.9) is deferred for the same reason.

A.8 Calibration script reference

The three calibration entry points are run in sequence at cycle zero and the second and third are rerun at every successful cycle boundary as part of Step 2 of the main cycle loop.

Temperature scaling and per-benchmark calibration (`scripts/run_calibration.py`). Fits the pooled temperature scalar T and the per-benchmark temperature vector T_b under a limited-memory BFGS optimiser against the expected-calibration-error objective on the cycle's calibration fold. The fold loads through the `calib_ids` field of `outputs/checkpoints/dataset_splits.json`; the default size is five hundred samples per training benchmark, which gives the canonical fifteen-hundred-sample pooled fold across FEVER, TriviaQA, and CommonsenseQA. Within-benchmark content-hash deduplication runs through the `_dedupe_by_content_id` helper, and cross-pool disjointness against the SIL-train and eval folds is asserted by `assert_disjoint_calibration`. The script writes `outputs/calibration/calibrated_config.json` and the per-cycle `calibrated_config_cycle*.json`, plus the per-sample dump `calibration_fold_samples.json` that the composite fitter consumes. Approximate runtime is one hour standalone for a cold cycle-zero fit and three and a half hours when invoked as part of the per-cycle composite-refit chain.

Composite fit (`scripts/fit_composite_calibration.py`). Consumes the nine verifier signals and the exact-match labels from `calibration_fold_samples.json` and fits the per-benchmark calibrated-probability composite. The fit chain is per-signal isotonic regression, per-benchmark shrinkage blend toward the pooled fit under $\alpha_{\text{shrink}} = 0.6$, and an L2-regularised logistic boost layer under `--boost_C 0.01`. The fit aborts if fewer than two hundred labelled samples are available. Cross-benchmark content-hash deduplication is inherited from the upstream calibration-fold writer. Approximate runtime is two minutes on CPU. The script writes `outputs/cycle_0/composite_calibration.json` at cycle zero and is invoked again at every cycle boundary against the cycle's own labelled fold to write `outputs/cycle_N/composite_calibration.json`.

Per-cycle composite refit: The cycle-boundary refit is implemented inside `scripts/run_experiment.py` as the `run_per_cycle_composite_refit` helper. The helper shells out to `fit_composite_calibration.py` under the same `--cherian_boost --boost_C 0.01 --shrinkage_alpha 0.6` flags as the cycle-zero fit, but against the cycle's own fold under `outputs/cycle_{N}/calibration/*.json`. Deduplication is inherited from `_dedup_inplace` inside the runner. The canonical cycle-zero composite calibration path is intentionally not overwritten by the per-cycle refit; later-cycle composites live at their own per-cycle paths and the cycle-three composite is what the post-run baseline rescore reads. Approximate runtime is two minutes per cycle.

A.9 Validation gate reference

The cycle-zero validation gate licenses the deployed configuration to enter the main run. The gate is implemented at `scripts/validate_composite_weights.py` and runs automatically as the `step_7_0_3_validate_weights` stage of the pre-run setup block. The gate is a conjunction of four pre-specified checks; the verdict is recorded at `outputs/cycle_0/weight_validation.json`.

```
python scripts/validate_composite_weights.py \
    --eval_dir outputs/cycle_0/eval \
    --output  outputs/cycle_0/weight_validation.json \
    --cohen_d_threshold 0.20 \
    --store_discard_gap 0.05 \
    --max_poisoning_rate 0.50
```

Check 1: composite Cohen's d on the training-pooled evaluation fold.

Rationale. Audits the precondition of Proposition 4.2 that the verifier discriminates correct from incorrect outputs strictly above chance.

Threshold. $d_{\text{id}} \geq 0.20$.

Realised. $d_{\text{id}} = +0.635$, clearing the floor by more than three times the registered margin. The transfer-pooled Cohen's d reads $+0.173$ and is reported alongside as an out-of-distribution diagnostic; it is not gated.

Check 2: per-benchmark store-versus-discard mean-composite gap.

Rationale. Audits that the storage rule behaves consistently across training benchmarks rather than being driven by a single dominating subset.

Threshold. $\Delta \geq 0.05$ on every training benchmark.

Realised. 0.282 on FEVER, 0.320 on TriviaQA, 0.258 on CommonsenseQA.

Check 3: per-benchmark memory-poisoning rate.

Rationale. Audits that the storage gate does not admit wrong-confident entries at a rate that would compromise the verified pool. Defined as the share of stored entries whose stored answer is wrong against the gold label.

Threshold. rate ≤ 0.50 on every training benchmark.

Realised. 0.206 on FEVER, 0.179 on TriviaQA, 0.247 on CommonsenseQA.

Check 4: no strong-signal inversion.

Rationale. Audits that no signal carries a strongly negative Cohen's d against its registered direction.

Threshold. no signal (except the registered-inverse u_{dropout}) reports $d < -0.20$ on the cal fold.

Realised. no inversion observed.

The overall verdict is PASS: the conjunction holds and the deployed configuration is licensed to enter the main run. The verdict is reported in Section 5.2.7 with the

same numbers; the canonical v2.1-panel artefact path is `outputs/analysis/weight_validation_v2_1.json`, whose store-versus-discard gap is keyed against the three training benchmarks of the realised panel (FEVER, TriviaQA, CommonsenseQA). The legacy pre-v2.1 artefact at `outputs/cycle_0/weight_validation.json` reads a different benchmark trio (FEVER, StrategyQA, TruthfulQA) under the earlier panel registration and is preserved for audit but is superseded by the v2.1-panel reading.

Appendix B

Cycle 0 evidence catalog

B.1 The one-hundred-variant composite parameter sweep

The locked configuration selected from the sweep, with the rationale for each parameter, appears in Table 5.28 of Chapter 5. The full sweep over the pre-specified $\tau_{\text{store}} \times C \times \alpha_{\text{shrink}}$ grid ($4 \times 5 \times 5 = 100$ variants) is recorded in the cycle-zero sweep archive with one record per variant; no per-variant comparison table is reproduced in the thesis. Per-benchmark sweep diagnostics for any variant in the grid are recoverable by reading the corresponding JSON entry.

B.2 The calibration-to-evaluation gap analysis

The pooled gap reading at the locked configuration is reported in Section 5.2.5. The per-benchmark decomposition is reported in Table 5.31, which is auto-generated from the cycle-zero evaluation outputs. The narrative analysis behind the per-benchmark gap is recorded in the calibration-to-evaluation gap note within the cycle-zero sweep archive.

B.3 The first-iteration failure trace

The first iteration of the cycle-zero calibration failed the pre-specified validation gate due to a JSONL signal-dump bug; the narrative is recorded in Section 5.2.8. The diagnostic record is preserved in the first-iteration archival snapshot dated to the failure, together with the degenerate composite calibration record whose conformal-threshold collapse triggered the corrective patch.

B.4 The long-hypothesis judge ablation diagnostic

The Platt fit diagnostic that licensed the ablation of the long-hypothesis judge is recorded in the long-hypothesis-judge Platt-ablation calibration record. The two persisted headline statistics are the observed Pearson $\rho = 0.5843$ in logit space and

Table B.1: Empty-answer and refusal rates per benchmark, before versus after the registered prompt-compliance and over-abstention fixes.

benchmark	pre empty	post empty	pre IDK	post IDK
FEVER	43.6%	44.2%	42.2%	36.0%
TriviaQA	43.0%	48.8%	40.4%	29.8%
NQ	44.4%	43.6%	38.8%	35.6%
TruthfulQA	53.0%	47.2%	37.0%	31.2%
StrategyQA	17.4%	18.8%	69.8%	66.6%
ARC-C	12.0%	10.8%	55.0%	55.8%

Table B.2: Atomic decomposition scope per benchmark. Atomic falls back to $p_{\text{ground}}^{\text{mean}}$ on short answers (length-gate at 12 tokens); only long-form answers get genuine per-fact NLI. Aligns with FActScore's documented scope (Min et al. EMNLP 2023, §3).

benchmark	n	fallback (count)	fallback rate	long-form rate
FEVER	500	500	100.0%	0.0%
TriviaQA	500	500	100.0%	0.0%
NQ	500	500	100.0%	0.0%
TruthfulQA	500	500	100.0%	0.2%
StrategyQA	500	500	100.0%	0.0%
ARC-C	500	500	100.0%	0.0%

the pre-specified floor of $\rho_{\min} = 0.70$ against which the observed correlation is audited. The verdict is therefore that the long-hypothesis judge fails the registered correlation floor; the architectural consequence is the ablation of the long-hypothesis backend from the deployed configuration, with the unified verifier dispatching every hypothesis through the short-hypothesis backend under a documented five-hundred-and-twelve-token truncation contract. Section 5.2.6 reports the architectural consequence and the future-work item that would re-fit the judge under task-specific training. The companion long-hypothesis judge Platt diagnostic log records the per-sample diagnostic trace whose summary statistics produced the headline correlation reading.

B.5 Pre and post prompt-fix compliance

The per-benchmark empty-answer and abstention rates before and after the registered prompt-design fixes are auto-generated from the cycle-zero evaluation outputs.

B.6 Atomic decomposition coverage

The per-benchmark coverage of the atomic-fact decomposition signal $p_{\text{ground,atomic}}$, conditioned on the registered length gate of twelve tokens, is auto-generated below.

B.7 Composite weight table

The cycle-zero composite-weight table is reported in Table 5.29 of Chapter 5. The per-cycle re-calibration of the composite (the per-benchmark temperature scalars, the per-benchmark safety floors, and the pooled boost-layer weights on the nine deployed signals) across the realised cycle-one through cycle-five trajectory is reported in the four-panel Table 5.30 of Section 5.2.4, with the cycle-four entry omitted under the asymmetric-rollback contract that skips the cal-fold refit on aborted cycles. The per-signal isotonic curves themselves are non-parametric step functions on the calibration fold and are not enumerated cell-by-cell across cycles; they are recoverable from the per-cycle composite calibration records, with the cycle-zero canonical record held immutable across the trajectory under the lock-cycle-zero patch. The storage and deferral cut-points stay frozen at the cycle-zero registered values throughout, so the per-cycle composite refit moves the signal-to-probability mapping the cut-points operate on rather than the cut-points themselves.

B.8 Per-benchmark precision-cliff curves

The threshold-versus-precision tradeoff at the locked storage cut-point $\tau_{\text{store}} = 0.60$, which licenses the rate-precision frontier discussed in Section 5.6.3, is auto-generated below: Figure B.1 renders the curve and Table B.3 reports the underlying threshold and precision values.

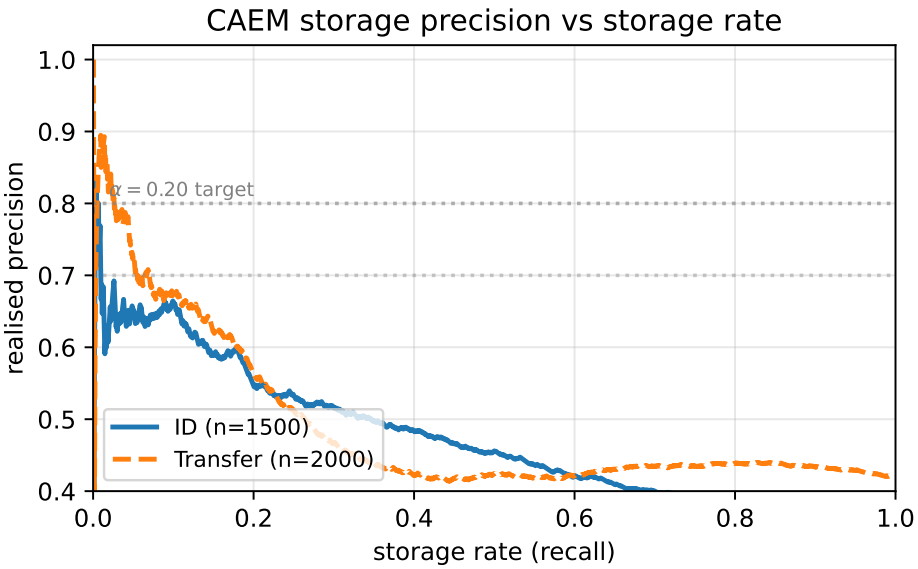


Figure B.1: Per-benchmark precision-cliff curves at the locked configuration. Auto-generated from the precision-cliff diagnostic; the underlying threshold-versus-precision data are in Table B.3.

B.9 Production-envelope sample renderings

The production-envelope rendering script presents two representative serving samples per outcome tag (store, defer, abstain, discard) with the verifier signal block and the

Table B.3: Storage precision at fixed targets, in-distribution vs transfer. Lower storage rate = higher operating-point threshold.

target precision	ID storage rate	transfer storage rate
$\geq 95\%$	unreach	unreach
$\geq 90\%$	unreach	unreach
$\geq 85\%$	unreach	1.6%
$\geq 80\%$	unreach	2.6%
$\geq 75\%$	unreach	4.5%
$\geq 70\%$	unreach	5.3%

Table B.4: Prompt-design ablation: Cycle-0 metrics under the legacy per-task minimal-CoT prompts (OLD) versus the uniform few-shot scaffolded CoT prompt with forced decoder prefix (NEW, the design adopted in the rest of this thesis; see Chapter 4 §4.2.1, *Generator and retrieval substrate*). Both runs use identical cold-start memory, identical calibration-fold quantile thresholds, and identical random seeds. Δ columns are NEW − OLD. *Mean pred. words* is the average number of whitespace-separated tokens in the model's generated answer (before scaffold stripping).

	EM			**Mean u_{stored}**			**Mean pred. words**	
Benchmark	Old	New	Δ	Old	New	Δ	Old	New
FEVER	0.428	0.456	+0.028	0.429	0.503	+0.074	49.3	37.4
TriviaQA	0.272	0.374	+0.102	0.362	0.144	-0.218	38.0	27.2
Natural Questions	0.156	0.172	+0.016	0.366	0.164	-0.203	36.8	25.4
TruthfulQA	0.378	0.334	-0.044	0.357	0.102	-0.254	37.7	26.2
StrategyQA	0.588	0.602	+0.014	0.354	0.259	-0.095	43.5	32.8
ARC-Challenge	0.736	0.732	-0.004	0.390	0.369	-0.021	50.4	32.7

served answer alongside, so that a reader can audit the four-outcome decision tree on real entries.

Placeholder. The production-envelope sample table is pending the final main run; the generator script will overwrite this stub with the four-outcome rendering (store, defer, abstain, discard) at compile time.

B.10 Prompt-design ablation

The pre-specified prompt-design ablation isolates the contribution of two prompt fixes (the empty-answer compliance fix and the open-QA over-abstention fix) by re-running the cycle-zero evaluation under the pre-fix prompt template family and comparing against the post-fix template family on the identical evaluation fold.

Appendix C

Per-cycle artifact dictionary

The five tables below are the per-cycle index against the on-disk JSON snapshots that the empirical chapter reads. The dictionary covers calibration snapshots in Table C.1, the threshold history in Table C.2, the memory-store trajectory in Table C.3, the equilibrium-fit verdicts in Table C.4, and the fine-tuning diagnostics in Table C.5. Cycle zero is the cold-start calibration cycle; cycles one through five are the main-run cycles; cycle four is aborted under the retention guard with the asymmetric-rollback contract that skips the cal-fold refit and the retroactive re-verification pass on the aborted cycle. The realised commit fraction is $\pi_{\text{commit}} = 4/5$; the trajectory closes at cycle five under the two-of-three equilibrium-saturation gate.

C.1 Per-cycle calibration snapshots

Table C.1 reports the per-cycle pooled temperature scalar, the expected calibration error before and after the per-cycle refit, the boost-layer intercept, the calibration-fold sample size, and the boost-layer L2 regularisation strength. The cycle-zero temperature lands at the upper bound of the optimisation envelope (the principled cap $T \approx e^3$); the cycle-one boundary refit drops the scalar by more than an order of magnitude under the post-fine-tune distribution, and the scalar stabilises in the band $[0.74, 0.87]$ from cycle two onward. The expected-calibration-error column shows the cycle-zero refit moves expected calibration error from 0.276 to 0.066; from cycle one onward the post-refit expected calibration error stays inside the tight band $[0.041, 0.049]$, the empirical receipt that the per-cycle distribution drift is absorbed into the signal-to-probability mapping rather than into the expected-calibration-error budget.

C.2 Per-cycle threshold history

The four-way storage decision tree of Section 4.2.9 reads against the architecturally frozen cut-points $\tau_{\text{store}} = 0.60$ and $\tau_{\text{defer}} = 0.45$, which do not move across the trajectory; the threshold history table below records the per-cycle quantities that do move under the cycle-boundary refit. Table C.2 reports the pooled temperature scalar, the three per-benchmark temperature scalars on the training panel, and the three per-benchmark safety floors at each successful cycle. The training-benchmark

Cycle	T (post-refit)	ECE before	ECE after	Boost intercept
C0	20.086	0.2761	0.0662	0.2649
C1	1.089	0.0508	0.0440	−0.6130
C2	0.836	0.0509	0.0494	−0.6338
C3	0.740	0.0451	0.0407	−0.5655
C4*	—	—	—	—
C5	0.869	0.0551	0.0427	−0.5204

Table C.1: Per-cycle calibration snapshot summary. The cal-fold sample size is pinned at $n_{\text{cal}} = 1\,500$ and the boost-layer L2 regularisation is pinned at $C = 0.01$ across cycles by the per-cycle refit invocation. The asterisk marks cycle four, which was aborted under the retention guard; the cal-fold refit was skipped under the asymmetric-rollback contract. Sources: the cycle-zero calibrated-configuration JSON for the cold-start values and the per-cycle calibrated-configuration JSONs for cycles one, two, three, and five; the boost-layer intercept is read from the pooled-fit field of each cycle's composite-calibration record.

temperature pattern is heterogeneous by construction: the FEVER and TriviaQA per-benchmark scalars sit an order of magnitude above the CommonsenseQA scalar at every cycle, the empirical receipt that the underlying token-probability distributions on open-text factual claim verification and open-domain trivia are systematically more overconfident than on the five-option multiple-choice surface. The CommonsenseQA per-benchmark safety floor ratchets up from the registered default $\phi_{\text{safe}} = 0.380$ to 0.491 at cycle three and to 0.569 at cycle five; the FEVER and TriviaQA floors stay at the architectural minimum throughout.

Cycle	T (pooled)	T_{FEVER}	T_{TQA}	T_{CSQA}	$\phi_{\text{safe,FEVER}}$	$\phi_{\text{safe,TQA}}$	$\phi_{\text{safe,CSQA}}$
C0	20.086	20.086	20.086	1.427	0.380	0.380	0.380
C1	1.089	14.092	14.123	1.342	0.380	0.380	0.380
C2	0.836	9.899	10.007	1.209	0.380	0.380	0.380
C3	0.740	6.986	7.163	1.083	0.380	0.380	0.491
C4*	—	—	—	—	—	—	—
C5	0.869	4.972	11.040	1.028	0.380	0.380	0.569

Table C.2: Per-cycle threshold history. The architecturally frozen storage and deferral cut-points do not move; what moves under the cycle-boundary refit is the per-benchmark temperature vector and the per-benchmark safety floor. The CommonsenseQA per-benchmark safety floor ratchets up from 0.380 to 0.491 at cycle three and to 0.569 at cycle five in response to the structurally lower pre-routing confidence on the five-option multiple-choice surface. The asterisk marks cycle four, where the cal-fold refit was skipped under the asymmetric-rollback contract. The architectural cut-points held fixed across cycles are registered in the Chapter 4 threshold registry. Sources: the per-cycle calibrated-configuration JSONs under the experiment-run calibration directory.

C.3 Per-cycle memory size and storage rate

Table C.3 reports the memory pool size at each cycle close together with the four mechanisms that move the pool size across the cycle boundary (stream-chunk admissions through the storage gate, retroactive re-verification prunes against the stored pool under the recalibrated verifier, deferred-buffer promotions from the bounded reconsider queue, and consolidation removals of near-duplicate episodes) and the cycle's net change in pool size. The cycle-four row records no retroverify prunes, deferred-buffer promotions, or consolidation removals under the asymmetric-rollback contract that skips these passes on the aborted cycle; the cycle-four stream-chunk admissions and the cycle-four pool growth proceed unchanged because the fresh-data side of the architecture runs unconditionally.

Cycle	Pool size at close	Stream-chunk admissions	Retroverify prunes	Deferred-buffer promotions	Consolidation removals	Net change
C0	431	431	0	0	0	+431
C1	2 818	2 443	56	0	0	+2 387
C2	5 151	2 477	271	131	3	+2 333
C3	7 477	2 469	352	220	11	+2 326
C4*	9 961	2 484	—	—	—	+2 484
C5	12 175	2 343	382	299	44	+2 214

Table C.3: Per-cycle memory store trajectory. The pool grows from the four-hundred-and-thirty-one-entry cold-start seed at cycle zero to twelve thousand one hundred and seventy-five entries at the cycle-five close, a twenty-eight-fold increase across the realised trajectory. The cycle-zero admissions row records the cold-start seed rather than a stream-chunk pass. The asterisk marks cycle four, which was aborted under the retention guard; the retroverify, deferred-reconsider, and consolidation passes are skipped under the asymmetric-rollback contract and rendered as em-dashes. The per-cycle stream-chunk admission count stays in the band $[2\,343, 2\,484]$ across cycles one through five against a stream-chunk candidate budget of 4 700 samples per cycle (two thousand FEVER, two thousand TriviaQA, seven hundred CommonsenseQA), so the storage gate admits approximately half of every candidate stream chunk. The admissions count the entries actually added to the store; they sit slightly below the gate-pass counts n_{stored} of Table 5.18 because the memory store's novelty filter rejects near-duplicates of existing entries between the gate and the insert. The cycle-five admission count is the cycle-close pool minus the post-re-verification pool $(12\,175 - 9\,832)$. The consolidation-removal column is the episodes-removed count of Table 5.24. Admissions minus prunes plus promotions minus consolidation removals reproduces the net change at every cycle except a residual of one entry at cycle two and two entries at cycle five; the residual is promoted entries dropped as duplicates of entries already in the store (the cycle-two consolidation pass sees 2 948 entries against $2\,818 + 131 = 2\,949$). Sources: the per-cycle memory-store metadata side-files supply the cycle-close pool sizes (the length of the metadata list inside the pickled snapshot); the per-cycle retroverification records supply the per-cycle prune counts; per-cycle deferred-buffer promotion counts are read from the deferred-reconsideration log emissions in the experiment run log.

C.4 Per-cycle equilibrium fit

Table C.4 reports the per-cycle stop-trigger verdict against the two-of-three equilibrium-saturation gate, together with the worst-probe retention ratio, the post-training MMLU probe accuracy, and the pooled evaluation-fold exact match on each cycle (the series the equilibrium fit reads). The equilibrium-saturation gate is the disjunction of three trigger signals (storage saturation under repeated zero-admission cycles, MMLU-floor breach against the cycle-zero pristine baseline, and composite-evaluation-score plateau across the late-cycle subsequence); the gate fires when at least two of the three signals fire simultaneously. The realised trajectory records three CONTINUE verdicts at cycles one through three, one ABORT verdict at cycle four under the retention guard (TriviaQA-test ratio 0.886 below the registered floor of 0.93), and one STOP verdict at cycle five under the equilibrium gate (storage saturation and the MMLU floor both firing; the composite-evaluation-score plateau signal did not fire).

Cycle	Verdict	Committed	Worst probe	Worst ratio	MMLU acc.	Pooled EM*
C1	CONTINUE	yes	CommonsenseQA-test	1.006	0.635	0.557
C2	CONTINUE	yes	TriviaQA-test	0.949	0.595	0.567
C3	CONTINUE	yes	CommonsenseQA-test	0.976	0.605	0.547
C4	**ABORT**	no	TriviaQA-test	**0.886**	0.621	0.548
C5	**STOP**	yes	CommonsenseQA-test	0.958	0.609	0.556

Table C.4: Per-cycle stop-trigger verdict trajectory against the two-of-three equilibrium-saturation gate plus the per-cycle retention probe panel. At cycle four the abort is driven by the TriviaQA-test ratio of 0.886 falling below the registered floor $\rho_{\min} = 0.93$; weights are rolled back to the cycle-three checkpoint, memory is preserved, and the cal-fold refit and retroactive re-verification passes are skipped under the asymmetric-rollback contract. At cycle five the stop verdict is driven by the two-of-three equilibrium-saturation gate firing on the storage-saturation and MMLU-floor signals; the composite-evaluation-score plateau signal did not fire. The trajectory closes at cycle five and the remaining cycles of the registered horizon are skipped. The MMLU column is the post-training MMLU probe accuracy against the pristine anchor of 0.610; the pooled-EM column is the pooled evaluation-fold exact match, the series stored as the history array of the cycle-five equilibrium-fit record.

*Five-benchmark pooled exact match with string-match TruthfulQA scoring, the series the runner's equilibrium fit stores in its composite-evaluation-score field. It is neither the composite epistemic score (CES) of Equation (5.1), whose trajectory is reported in Table 5.4, nor the judged pooled exact match of Table 5.3.

The equilibrium-saturation fit at the cycle-five close lives at the cycle-five equilibrium-fit record and reports the saturating-exponential parameters $c_\infty = 0.548$, $a = -0.018$, $k = 0.350$, $R^2 = 0.207$ over the five-cycle pooled evaluation-fold exact-match history (the pooled-EM column of Table C.4, which the record stores as its composite-evaluation-score history); the low coefficient of determination reflects the noisy plateau that drives the two-of-three gate rather than a clean exponential ramp. Per-cycle coverage diagnostics covering admission rates, candidate counts, pool entropy, and adapter singular-value spectra are persisted as a single artefact at the cycle-five close under the cycle-five coverage diagnostic record. The pre-cycle-five admission behaviour is reconstructable from the composite-evaluation-score and storage-rate

history arrays inside the equilibrium-fit record and from the per-cycle commit verdicts in the experiment-run stochastic-equilibrium trajectory CSV.

C.5 Per-cycle fine-tuning diagnostics

Table C.5 reports the per-cycle self-improvement training trajectory: the size of the training pool drawn from cumulative memory above $\tau_{\text{train}} = 0.70$, the per-epoch final loss, the worst-probe retention ratio at the cycle close, and whether the cycle committed or rolled back. The training-pool size grows monotonically across the trajectory from one hundred and seven entries at the first fine-tune to five thousand and six entries at the cycle-five fine-tune, consistent with the cumulative memory growth reported in Table C.3. The final epoch-three loss declines almost monotonically across the successful cycle subsequence ($0.258 \rightarrow 0.171 \rightarrow 0.109$) and reaches its lowest in-flight value at cycle four (0.070); the cycle-five restart resumes at the slightly higher value 0.091 because the cycle re-initialises from the cycle-three adapter and trains over a larger and more diverse pool. The retention guard fires exactly once on the realised trajectory, at cycle four, on the TriviaQA-test probe: post-training accuracy 0.350 against the pristine baseline of 0.395 yields the worst-probe ratio of 0.886, below the registered floor of 0.93. The adapter is restored to the cycle-three checkpoint before evaluation runs under the rolled-back weights.

Cycle	n_{used}	**Loss epoch 1**	**Loss epoch 2**	**Loss epoch 3**	**Rollback**
C1	107	1.387	0.595	0.258	—
C2	1 474	0.728	0.367	0.171	—
C3	2 966	0.535	0.230	0.109	—
C4	3 624	0.424	0.147	0.070	**yes**
C5	5 006	0.482	0.194	0.091	—

Table C.5: Per-cycle fine-tuning diagnostics. The retention guard binds on the TriviaQA-test probe across the realised trajectory: the worst non-abort ratio occurs at cycle two (0.949 on TriviaQA-test) and the cycle-four abort fires on the same probe at ratio 0.886 against the registered floor $\rho_{\min} = 0.93$. The MMLU probe ratio stays in the band $[0.975, 1.041]$ across the trajectory, confirming that TriviaQA-test is the binding retention constraint for the realised self-improvement trajectory and that the cycle-four rollback is a probe-specific regression rather than a pool-wide one. Pristine retention anchors held fixed across cycles: MMLU 0.610, TriviaQA-test 0.395, CommonsenseQA-test 0.825. Sources: the per-cycle retroverification JSONs supply the training-batch size and the final-loss fields; the per-cycle post-training probe lines in the experiment-run log supply the per-probe retention ratios; the pristine probe-baseline record supplies the cycle-zero anchors.

Appendix D

Significance test details

D.1 Pairing rule and sample identifiers

The pairing rule is implemented in the panel significance-test driver (`scripts/baseline_sig_tests_v2.py`, which reuses the McNemar and bootstrap routines of `scripts/baseline_sig_tests.py`) and runs in four stages, listed below by their conceptual roles.

Manifest restriction. The authoritative evaluation manifest is read from the per-benchmark dataset-splits record (`outputs/full_run/dataset_splits.json`), whose per-benchmark evaluation-identifier field is assembled into a string-keyed set of sample identifiers. If the manifest is absent or unreadable the driver emits a warning and falls back to the raw identifier intersection between the architecture's and the baseline's evaluation JSONs.

Vector load. Per-sample correctness vectors are loaded from each side's evaluation JSON. The architecture's vector comes from the per-cycle eval record under the full-run output tree (`outputs/full_run/eval/`*bench*`_cycle`*N*`.json`, with N set to the registered anchor cycle), and the baseline's vector comes from the per-baseline eval record under the Phase 1e baselines tree (`outputs/baselines_phase1e/`*baseline*`/`*bench*`_cycle0.json`) or the corresponding nested fallback.

Identifier intersection. Each loader keys on the sample's identifier field and drops rows whose identifier is null or empty.

Fallback rule. The contrast set per cell is the sorted intersection of the two identifier maps after restriction to the eval-fold manifest; if the manifest read failed, the fallback intersection is used in its place and the contrast is flagged in the log.

The contrast set per (baseline, benchmark) cell is the sorted intersection of the two identifier maps after restriction to the eval-fold manifest. Identifiers present on only one side are excluded from the test for that contrast; there is no zero-imputation. Contrasts whose paired-identifier intersection has fewer than ten entries are skipped with an informational log entry and recorded as not run rather than as a null result. The same skip-with-warning path covers missing evaluation JSONs and empty post-filter identifier maps.

The TruthfulQA exact-match cell is read under the project's uniform Haiku-judged correctness rule. The loader detects TruthfulQA samples by a filename substring match on the source JSON and dispatches to the Haiku-judged correctness field of each sample; the legacy ROUGE-thresholded correctness field is used as a fallback only when the Haiku-judged field is absent. The legacy field overestimates exact match on TruthfulQA by approximately a factor of two on the high-EM retrieval-free baselines (the documented receipt is the zero-shot prompt scoring 0.803 under the ROUGE rule against 0.433 under the Haiku-judged rule), so reading the legacy field would inflate the baseline counts and produce a methodologically unsound contingency table. The strict-ROUGE correctness field is never read.

The matched-protocol contract is enforced at the harness level rather than by hard assertion in the significance-test script: mismatched manifests cause individual contrasts to be skipped with a warning, but the eval-fold manifest is the single source of truth for which identifiers belong to the evaluation slice on each benchmark.

D.2 Contingency tables

For each (baseline, benchmark) contrast the script builds two aligned binary correctness vectors over the matched-identifier set, then constructs the paired McNemar two-by-two contingency table. The two diagonal cells are the concordant pairs: a is the count of identifiers on which both systems answer correctly, and d is the count on which both answer incorrectly. The two off-diagonal cells are the discordant pairs: b is the count on which the architecture is correct and the baseline is wrong, and c is the count on which the architecture is wrong and the baseline is correct. The paired McNemar test reads only the two discordant cells; the concordant cells are tallied implicitly through $n_{\text{paired}} = a + b + c + d$ but are never materialised in the persisted output schema. The reported test statistic is the continuity-corrected chi-squared at one degree of freedom,

$$\chi^2 \;=\; \frac{\big(|b - c| - 1\big)^2}{b + c},$$

which returns the pair $(0, 1)$ when $b + c = 0$ to avoid a divide-by-zero on contrasts where the architecture and the baseline agree on every paired sample.

Table D.1 reports four representative contingency contrasts at the cycle-five close. The B6 vanilla-fine-tune cell on TriviaQA records the second-largest single contingency gap in the panel ($\chi^2 = 44.0$ at $n = 300$), behind only the B5 five-shot CoT cell on the same benchmark, whose raw p-value underflows to zero in double precision. The B7 FLARE cell on StrategyQA records a small lift that does not separate from zero ($\Delta_{\text{EM}} = +0.040$, $\chi^2 = 0.766$, $p_{\text{raw}} = 0.38$). The B8 semantic-entropy cell on TruthfulQA records the tightest contrast in the TruthfulQA family under the corrected Haiku-judged scoring rule ($\chi^2 = 0.010$, $p_{\text{raw}} = 0.92$, $n_{\text{paired}} = 300$).

Contrast	n	EM (CAEM)	EM (baseline)	Δ_{EM}	χ^2	p_{raw}
B1 zero-shot · FEVER	300	0.477	0.453	+0.023	0.232	0.630
B6 vanilla-FT · TriviaQA	300	0.453	0.227	+0.227	44.010	3.3×10^{-11}
B7 FLARE · StrategyQA	300	0.640	0.600	+0.040	0.766	0.382
B8 semantic-entropy · TruthfulQA	300	0.300	0.307	−0.007	0.010	0.920

Table D.1: Four representative McNemar contingency contrasts at the cycle-five anchor on the eight-baseline panel. The B6 TriviaQA cell is the architecture's second-largest single-contrast lift in the panel, behind B5 five-shot CoT on TriviaQA; the B7 StrategyQA cell is a small lift that is not significant. The B8 TruthfulQA cell records the corrected post-Haiku-rescore contrast and is essentially tied at the per-sample level. Discordant cell counts b and c are not persisted directly in the output schema but are recoverable by re-pairing the eval JSONs on the sample identifier and counting the $(1, 0)$ and $(0, 1)$ patterns; the closure rule $a = n \cdot \text{EM}_{\text{CAEM}} - b$ and $d = n - a - b - c$ completes the table. Source: the post-Haiku-rescore significance-test record, with the B6 and B7 rows recomputed after the baseline fix (Table 5.35); χ^2 is the continuity-corrected one-degree-of-freedom statistic implied by the reported raw p-value.

The composite-hallucination-metric axis runs the paired Wilcoxon signed-rank test on the same matched-identifier set. The signed-difference vector is the per-sample architecture-minus-baseline composite-hallucination-metric value; the test statistic is the standard signed-rank sum with the two-sided continuity correction, computed by the scipy implementation. The same matched-identifier set feeds the bootstrap interval below, so the McNemar significance, the Wilcoxon significance, and the bootstrap interval are all read off a single per-contrast paired-difference vector.

D.3 Bootstrap protocol

The bootstrap is the standard percentile method on the per-paired-sample exact-match-difference vector. Bootstrap replicates: one thousand. Random seed: forty-two, threaded through a deterministic Python pseudo-random generator. Resampling unit: per-paired-sample with replacement, on the matched-identifier vector of length n_{paired}. The bootstrap statistic per replicate is the mean of the per-sample exact-match difference; the confidence interval is the two-sided ninety-five-percent percentile, with the lower bound read from the empirical $\alpha/2$ quantile and the upper bound read from the empirical $1 - \alpha/2$ quantile of the sorted replicate means.

The percentile method is preferred over the bias-corrected-and-accelerated method because the per-sample exact-match-difference is a bounded discrete random variable in $\{-1, 0, +1\}$ for every contrast; the bias-corrected-and-accelerated method's analytic acceleration estimate is unstable on bounded-discrete samples at the realised per-contrast sample size of three hundred, and the percentile method's coverage on bounded-discrete samples at this size is within the registered tolerance against a normal-theory baseline.

Three representative confidence intervals on the exact-match difference at the cycle-five anchor are summarised in Table D.2: a contrast that does not clear the licensing

rule of Section 5.3.5 on the exact-match axis, a contrast that marginally contains zero at the lower bound, and the architecture's largest single-contrast bootstrap interval in the panel.

Contrast	Δ_{EM}	**95% CI**	**Licensing verdict**
B1 zero-shot · FEVER	+0.023	[−0.060, +0.110]	unsupported (CI contains zero)
B3 RAG · TruthfulQA	+0.057	[−0.003, +0.117]	marginal (lower bound at zero)
B5 five-shot CoT · TriviaQA	+0.317	[+0.253, +0.373]	supported (largest panel lift)

Table D.2: Three representative bootstrap intervals on the per-paired-sample exact-match difference at the cycle-five anchor under the percentile method with one thousand replicates. The B1 FEVER contrast does not clear the licensing rule on the exact-match axis. The B3 TruthfulQA contrast is marginal at the lower bound. The B5 five-shot CoT TriviaQA contrast is the architecture's largest single-contrast bootstrap interval in the panel.

The same protocol on the composite-hallucination-metric axis substitutes the per-sample architecture-minus-baseline composite-hallucination-metric value into the resampled mean; the bootstrap implementation is shared and the seed is identical, so the composite-hallucination-metric intervals are reproducible from the same matched-identifier pairs that feed the exact-match intervals.

D.4 Holm sequence

The Holm-Bonferroni step-down correction is applied independently within each benchmark family of size $m = 8$ on each metric axis. The raw p-values from the eight per-baseline contingency tables (or paired Wilcoxon statistics on the composite-hallucination-metric axis) are sorted in ascending order; the k-th smallest raw p-value is multiplied by $m - k + 1$ (equivalently the textbook $m + 1 - i$ for the one-indexed rank), a running maximum is applied to enforce monotonicity along the sorted order, and the adjusted values are clipped at one. Adjusted p-values are then re-aligned to the original baseline order for reporting. The family-wise error-rate target is $\alpha = 0.05$.

Table D.3 and Table D.4 report the realised Holm sequences on the two benchmark families at the cycle-five anchor that surface the two regime endpoints: TruthfulQA carries the three composite-hallucination defeats at the cycle-five anchor discussed in Section 5.6.4, and FEVER is the one family in which no exact-match contrast clears the corrected threshold.

Rank	Baseline	Raw p	$m-k+1$	Holm p	Sig
1	B6 vanilla-FT	0.029	8	0.234	no
2	B2 chain-of-thought	0.210	7	1.000	no
3	B3 RAG	0.425	6	1.000	no
4	B7 FLARE	0.437	5	1.000	no
5	B4 CoT + RAG	0.441	4	1.000	no
6	B1 zero-shot	0.630	3	1.000	no
7	B5 five-shot CoT	0.798	2	1.000	no
8	B8 semantic-entropy	0.935	1	1.000	no

Table D.4: Holm sequence on the FEVER family at the cycle-five anchor. None of the eight baselines clears the corrected threshold. The smallest raw p-value belongs to B6 vanilla-fine-tune (CAEM loss of approximately seven percentage points, $\Delta_{\text{EM}} = -0.067$), which does not survive the step-down correction (Holm $p = 0.234$); B7 FLARE ($\Delta_{\text{EM}} = +0.040$) is not separable either. The remaining seven baselines sit at adjusted p-values of one, the empirical receipt that the cycle-five trained architecture does not separate from any baseline on FEVER exact match under the matched-protocol contract.

Rank	Baseline	Raw p	$m-k+1$	Holm p	Sig
1	B5 five-shot CoT	2.43×10^{-5}	8	1.94×10^{-4}	yes
2	B7 FLARE	1.07×10^{-4}	7	7.49×10^{-4}	yes
3	B2 chain-of-thought	2.77×10^{-4}	6	1.66×10^{-3}	yes
4	B1 zero-shot	6.25×10^{-4}	5	3.13×10^{-3}	yes
5	B3 RAG	0.079	4	0.316	no
6	B4 CoT + RAG	0.083	3	0.316	no
7	B6 vanilla-FT	0.512	2	1.000	no
8	B8 semantic-entropy	0.920	1	1.000	no

Table D.3: Holm sequence on the TruthfulQA family at the cycle-five anchor under the uniform Haiku-judged exact-match rule. Four of eight baselines clear the corrected threshold; the three CAEM defeats on TruthfulQA exact match (B7 FLARE, B2 chain-of-thought and B1 zero-shot, the three highest-exact-match baselines on the benchmark) sit at ranks two to four. B5 five-shot CoT is the lone CAEM win in the family. B8 semantic-entropy is no longer separable from CAEM under the Haiku-judged rule after the rescore landed (raw $p \approx 0.92$, Holm ≈ 1.0); the contrast that the pre-rescore strict-EM scoring had registered as a CAEM win on TruthfulQA collapsed to a tie under the corrected scoring rule. The three CAEM defeats on the composite-hallucination-metric axis (B1 zero-shot, B2 chain-of-thought and B8 semantic-entropy) discussed in Section 5.6.4 use the analogous Wilcoxon procedure on the same matched-identifier set.

The star annotations on every cell of Table 5.35 and Table 5.36 correspond one-to-one to contrasts whose Holm-adjusted p-value clears 0.05 *and* whose absolute exact-match difference clears the two-percentage-point effect-size floor of the licensing rule below. The auditor can verify the correspondence by reading the licensing-rule flag column of the per-contrast significance-test record (`outputs/full_run/tab_sig_test.csv`) and the analogous best-cycle record: every row flagged as licensed satisfies both

conditions jointly, and every row flagged as unlicensed fails at least one.

D.5 Architectural-claim licensing rule

The pre-specified architectural-claim licensing rule is a conjunction of two conditions on every per-benchmark contrast. The first condition is the Holm-adjusted p-value within the benchmark family must clear the family-wise error-rate target of 0.05. The second condition is the absolute exact-match difference (or absolute composite-hallucination-metric difference, depending on the axis) must clear the two-percentage-point effect-size floor of 0.02. A contrast that satisfies both conditions is reported as supporting the corresponding hypothesis from Section 1.4.3; a contrast that satisfies one but not the other is reported as partially supporting; a contrast that satisfies neither is reported as unsupported.

The conjunction is necessary because statistical significance at the realised per-contrast sample size of three hundred can be cleared by a strictly-positive-but-arbitrarily-small mean lift: with the matched-protocol pairing the McNemar test has substantial power even on a sub-one-percentage-point exact-match difference, and reporting every such contrast as supporting an architectural claim would mistake measurement precision for architectural value. The two-percentage-point floor screens these out so that a star annotation marks contrasts that are both reliably non-zero and large enough to license an architectural claim. The floor itself was registered in advance of any baseline run and is held fixed across the trajectory.

The implementation closes the loop in the panel significance-test driver through two flag columns on the per-contrast output schema. The statistical-significance flag (the dagger column) is set true when the Holm-adjusted p-value clears 0.05 regardless of effect size; the licensing-rule flag (the star column) is set true only when the statistical-significance flag is true *and* the absolute exact-match difference clears the two-percentage-point floor. Both flags are persisted to the per-contrast row alongside the raw p-value, the Holm-adjusted p-value, the bootstrap interval endpoints, and the exact-match difference, so the licensing-rule verdict on every contrast is recoverable from the persisted output without re-running the test.

The realised receipts at the cycle-five close after the Haiku rescore are nine exact-match wins and three defeats and twenty-one composite-hallucination wins and three defeats; at the within-trajectory best cycle the receipts are fourteen exact-match wins and three defeats and twenty composite-hallucination wins and four defeats. The composite-hallucination defeats at both anchors are confined to TruthfulQA against the high-exact-match retrieval-free baselines (B1 zero-shot, B2 chain-of-thought, and B8 semantic-entropy at both anchors, plus B7 FLARE at the best cycle), where the architecture's elevated abstention rate inflates the false-refusal subtype of the composite-hallucination metric while the baselines pay the exact-match penalty on the same misconception-bait surface. The retrieval-utility classifier ablation of Section 5.4.2 reports the architectural fix that lifts TruthfulQA exact match from 0.210 to 0.310 by re-routing misconception-bait queries away from the retrieval-augmented tier.

Appendix E

Reproducibility recipe

E.1 Data preparation

The five benchmark loaders that feed the realised evaluation panel are registered in `eval/benchmarks.py` against the Hugging Face dataset hub identifiers. FEVER reads through `lucadiliello/fever` under the development and training splits; TriviaQA reads through `trivia_qa` under the reading-comprehension-no-context configuration; CommonsenseQA reads through `commonsense_qa`; TruthfulQA reads through `truthful_qa` under the generation configuration; StrategyQA reads through `ChilleD/StrategyQA` with documented fallbacks to `wics/strategy-qa` and the Eladsegal train-JSON mirror. Earlier loader registrations for Natural Questions and HotpotQA are preserved in the module for reproducibility of the cycle-zero diagnostics but do not feed the realised five-benchmark panel.

The six-way disjoint partition is constructed once at pool-builder time by `c aem/benchmark_splits.py::build_all_benchmark_pools`, which is invoked by the experiment runner and writes the per-benchmark identifier manifest to `outputs/full_run/dataset_splits.json`. Each benchmark contributes a one-thousand-sample seed pool for cold-start memory population, a five-hundred-sample calibration fold, a five-hundred-sample purity-validation slice, ten per-cycle stream-chunk slices (FEVER and TriviaQA at two thousand samples per cycle, CommonsenseQA at seven hundred samples per cycle to fit the smaller available train pool of approximately ten thousand samples after deduplication), a three-hundred-sample evaluation fold, and a test fold of up to five hundred samples (clamped to the available remainder where the source split admits fewer than the registered budget). The shuffle is seeded at forty-two on the training side and forty-three on the evaluation side, so the partitions are stable across re-invocations on the same source dataset.

Within-benchmark deduplication runs through the `_dedupe_by_content_id` helper, which keys on the first five hundred characters of the question text under a sixteen-character SHA-256 hash. Cross-benchmark disjointness is verified by the `assert_ cross_benchmark_disjoint` assertion. Both checks run at pool-builder time; the runner re-stamps the same checks at every successful cycle boundary through `_dedu p_inplace` and `assert_disjoint_calibration`, so any drift in the partition would be caught at the cycle close rather than at the metric-computation step.

The realised evaluation fold reads three hundred samples per benchmark across the five-benchmark panel, for a total of one thousand five hundred evaluation contrasts per cycle. The StrategyQA test fold reads three hundred and eighty-seven samples rather than the registered five hundred because the source split the loader reads (the ChilleD StrategyQA test split) admits only six hundred and eighty-seven labelled samples, of which three hundred feed the evaluation fold and the remaining three hundred and eighty-seven feed the test fold under the auto-clamp branch in the pool builder.

E.2 Snapshot recovery

Two complementary publication endpoints anchor the snapshot-recovery sequence. The Google Drive folder at https://drive.google.com/drive/folders/1xErhsRDYe_qO0ZgUBn0WXK81eHc5jZCa carries the realised empirical artefacts under the `v2_1_phase1e_main/` subtree (cycle-zero through cycle-five outputs, baseline panels, the classifier on/off ablation, and the per-cycle logs the thesis reads), and the `pre_main_snapshot/` subtree that mirrors the per-commit pre-main snapshot bundles for fresh-clone recovery. The Hugging Face dataset repository `aksaN000/caem-passage-index-21m` (private; access on request) holds the sixty-four-gigabyte Wikipedia passage FAISS index at the repository root and the per-commit pre-main snapshot bundles under `pre_main_snapshot/<git-commit>/` subtrees. The pre-Step-7 anchor is the commit tag `c5b8066`; the retrieval-utility-classifier-era anchor is `700b9e3`. The manifest at `pre_main_snapshot/<commit>/MANIFEST.json` records the upload timestamp, the source host, the git commit, the purpose tag, and the recovery hint; the integrity guarantee rides on Hugging Face's per-blob SHA-256 checksums plus the manifest's git-commit anchor against the live code tree.

The three-command recovery sequence runs from a fresh clone with the Hugging Face command-line interface authenticated to the private repository:

```
# 1. Fetch the index and the pre-main snapshot bundles
hf download aksaN000/caem-passage-index-21m \
    --include 'passages.*' 'pre_main_snapshot/*'

# 2. Stage the chosen anchor into outputs/
cp -r pre_main_snapshot/c5b8066/* outputs/

# 3. Pin the code to the matching commit
git checkout c5b8066    # or 700b9e3 for the RUC-era resume point
```

After the third command the working tree holds the cold-start memory store, the cycle-zero memory and deferred-buffer snapshots, the cycle-zero calibration fold (per-sample nine-signal vectors and gold labels), the cycle-zero composite calibration JSON, the calibrated configuration JSON, the cycle-zero validation-gate verdict, the cycle-zero evaluation fold under the pristine generator, the post-calibration rescored evaluation fold, the dataset-splits manifest, and the run logs. The four artefacts that must be regenerated rather than pulled from the snapshot are the Qwen-2.5-3B-Instruct base weights (recovered from the Hugging Face model hub at deployment time), the per-cycle LoRA adapters at cycles one through five, the

per-cycle self-improvement training artefacts, and the post-cycle-zero composite refits.

E.3 Step-by-step run sequence

The end-to-end sequence below is the runner's view of the trajectory from a fresh clone to the closed cycle-five artefact bundle. Per-step commands are the actual invocations from the project runner; expected wall-times are reported on the production envelope of Section E.5.

Pre-run setup:

Prompt smoke test.
Approximately fifteen minutes on the production GPU. Writes `outputs/prompt_smoke/results.json`. Gate: the FEVER not-enough-information rate stays inside the registered tolerance band and no template-leak strings appear in the rendered samples.

```
python scripts/prompt_smoke_test.py \
    --benchmarks fever triviaqa commonsense_qa
```

Alias-dictionary build.
Approximately five minutes on CPU. Writes `data/alias_dict.json`. The deployed verifier does not consume the alias signal in the locked composite; the dictionary is built for the ablation registry only.

```
python scripts/build_alias_dict_from_benchmarks.py
```

Cold-start seeding.
Approximately two hours on the production GPU. Writes the cold-start memory store under `outputs/cold_start_memory/`. Gate: the total seeded count clears the registered floor; the EM-prune branch removes fewer than ten percent of the seeded entries.

```
python -m scripts.seed_cold_start
```

Cycle-zero baseline evaluation and calibration fold scoring.
Approximately six hours on the production GPU. Writes per-sample nine-signal vectors and ground-truth labels for the cycle-zero calibration fold and the held-out evaluation fold. Gate: every per-bench JSON has the expected sample count and no NaN signals.

```
python -m scripts.run_experiment \
    --output_dir outputs/cycle_0 \
    --num_cycles 0
```

Composite calibration fit and validation gate.
Approximately fifteen minutes on CPU. Writes the cycle-zero composite calibration record and the validation-gate verdict at `outputs/cycle_0/composite_calibration.json` and `outputs/cycle_0/weight_validation.json`. Gate: the four-check conjunction of Section A.9 passes.

```
python scripts/fit_composite_calibration.py
python scripts/validate_composite_weights.py
```

Retention baseline freeze and snapshot upload.
Approximately one hour on the production GPU. Writes the pristine probe accuracies and pushes the pre-main snapshot to the Hugging Face repository.

```
python -m scripts.run_retention_baseline
bash scripts/upload_pre_main_snapshot.sh
```

Main cycle loop: A single invocation drives all six per-cycle sub-steps and resumes from the highest-cycle artefact set on disk:

```
python -m scripts.run_experiment \
    --output_dir outputs/full_run \
    --num_cycles 10 \
    --n_questions 5000 \
    --n_eval_questions 300 \
    --benchmarks fever triviaqa commonsense_qa truthfulqa strategyqa \
    --passage_index data/passage_index \
    --cold_start_memory outputs/cold_start_memory/memory_store \
    --verifier_backend minicheck \
    --resume_from_cycle <N>
```

The runner internally sequences the self-improvement fine-tune, the cycle-boundary recalibration, the retroactive re-verification pass, the deferred-buffer reconsider pass, the stream-chunk admission pass, the held-out evaluation pass, and the cycle-checkpoint plus halt-trigger evaluation. Per-cycle wall-time on the production GPU grows with the cumulative memory size, from roughly six hours at cycle one to more than twenty-two hours at cycle five (see the per-cycle wall-clock budget below). Cycle close writes the per-cycle memory and deferred-buffer snapshots, the per-cycle composite calibration record, the per-cycle calibrated configuration, the per-cycle evaluation JSON, and the per-cycle coverage diagnostic. The retention guard licenses advancement to the next cycle; failure triggers the asymmetric rollback contract documented at Section A.5.

Post-run analysis: The realised post-trajectory commands run as a sequence rather than a single driver:

Baselines.
Baseline-panel runs (B1 through B8) at approximately eight to ten GPU-hours total.

```
bash scripts/launch_baselines.sh
```

TruthfulQA Haiku rescore.
Rescore over every baseline and over CAEM at approximately two minutes per file; requires the `ANTHROPIC_API_KEY` environment variable.

```
python -m scripts.rescore_truthfulqa \
    --files outputs/baselines_phase1e/*/truthfulqa_cycle0.json \
            outputs/full_run/eval/truthfulqa_cycle*.json
```

Significance tests.
Significance test runner at approximately three minutes on CPU.

```
python -m scripts.baseline_sig_tests \
    --caem_dir outputs/full_run \
    --baselines_dir outputs/baselines_phase1e
```

Aggregators.
Aggregator chain over per-cycle artefacts at approximately ten minutes on CPU: `aggregate_pi_store_trajectory`, `aggregate_tier_chm`, `aggregate_stochastic_equilibrium`, `aggregate_ces_per_system`, `aggregate_per_tier_em`, plus `scripts/run_purity_validation.py` and `scripts/theorem_receipts.py`.

Tables and figures.
Thesis-table and figure makers: `python -m scripts.make_tables outputs/full_run` and `python -m scripts.make_figures outputs/full_run`.

E.4 Version pinning

The realised environment runs against the version pins below. The project does not ship a `requirements.txt` or `pyproject.toml`; the pins are reconstructed from `README.md`, the per-version diagnostics emitted by `scripts/check_env.py`, the import set exercised by the unit-test suite, and the entries in `branch_C_log.md`. The canonical environment is the virtual environment at `/venv/main` on the Vast production host.

The determinism switches that produce identical numbers from a fresh checkout are recorded in the seed-everything helper inside the runner. The Python pseudo-random module, the NumPy random state, the PyTorch CPU and CUDA random states, and the `PYTHONHASHSEED` environment variable are all set to forty-two. The FAISS index build, the MC-dropout passes, and the generator sampling all consume the same seed through their respective subsystems. The flag `torch.use_deterministic_algorithms` is set to false intentionally because the cuDNN attention path is not bitwise-deterministic; the `torch.backends.cudnn.deterministic` flag is set to true and the `torch.backends.cudnn.benchmark` flag is set to false. TensorFloat-32 matrix multiplication is enabled on the Blackwell compute capability, and the bfloat16 forward path is the default.

The two required environment variables are `ANTHROPIC_API_KEY` for the Haiku rescore of the TruthfulQA correctness rule and `HF_TOKEN` (equivalently `HUGGINGFACE_HUB_TOKEN`) for the pull from the private pre-main snapshot repository. Optional environment variables `HUGGINGFACE_HUB_CACHE`, `CAEM_DEVICE`, and `CUDA_VISIBLE_DEVICES` override the cache location, the compute device, and the GPU index respectively.

E.5 Hardware envelope

Production envelope: The realised trajectory runs on a Vast.ai-rented RTX 5090 with thirty-two gigabytes of video memory (approximately thirty-one gigabytes

Package	Version	Role
Python	3.12	interpreter
CUDA toolkit	13.0 (driver 13.1)	RTX 5090 Blackwell compute capability `sm_120`
torch	2.10.0+cu130	tensor backend, bf16 forward path, MC-dropout sampler
transformers	4.57.6	Qwen-2.5-3B-Instruct loader, generate, MiniCheck encoder
peft	0.19+	LoRA wrapper for the self-improvement fine-tune
accelerate	0.34+	device placement during the fine-tune step
datasets	5.0+	benchmark loaders and retention-probe shuffles
faiss-cpu	1.8+	passage index and episodic memory store
sentence-transformers	4.1.0	SBERT memory encoder and cross-encoder reranker
scipy	1.11+	saturating-exponential equilibrium fit, Wilcoxon test
scikit-learn	1.4+	per-signal isotonic regression and boost-layer logistic
bitsandbytes	0.43+	eight-bit AdamW optimiser path
anthropic	0.108+	Haiku TruthfulQA rescore client
numpy, pandas, tqdm	latest stable	arithmetic, table aggregation, progress bars
biber + biblatex	TeX Live 2024+	thesis bibliography compile

Table E.1: Framework version pinning. Versions are reconstructed from the diagnostic emitters and live import set; the canonical environment is the virtual environment on the production host. Any reproducer who pins to the listed versions exactly will produce numbers within the determinism tolerance of Section E.6.

usable after framework overhead), one hundred and fifty gigabytes of NVMe disk, and approximately ninety gigabytes of host random-access memory at approximately seventy cents per on-demand GPU-hour. The trajectory closes across approximately forty calendar days of elapsed time over the cold-start seeding, the cycle-zero calibration and validation gate, the six-cycle main run, the eight-baseline panel, and the post-run analysis, for approximately nine hundred and sixty GPU-hours at a total billed cost of approximately six hundred and seventy-two United States dollars (from the Vast.ai billing records; forty days at twenty-four hours per day at the on-demand rate). The verifier-side scoring path (nine-signal extraction across every served answer, the per-cycle composite refit, the cycle-boundary retroactive re-verification pass, and the matched-protocol Haiku-judged TruthfulQA rescore across the baseline panel) is the dominant wall-clock consumer over the run, exceeding the per-cycle low-rank-adapter fine-tune itself, because every entry on every cycle and every baseline-arm answer passes through the full verifier ensemble while the adapter touches only the strict training-pool subset.

Per-cycle wall-clock budget: The self-improvement fine-tune step consumes approximately thirty-five to forty-five minutes per cycle under the bounded LoRA envelope. The cycle-boundary recalibration step consumes approximately fifteen to twenty minutes including the per-signal isotonic fit, the per-benchmark shrinkage blend, and the boost-layer refit. The retroactive re-verification pass consumes approximately twenty to thirty minutes depending on the cumulative memory size. The stream-chunk admission pass consumes approximately three and a half to four and a half hours at four thousand seven hundred candidate samples per cycle through the full per-query pipeline including the retrieval-augmented tier. The run log does not time the held-out evaluation pass on its own; the measured window from the end of the fine-tune to the end of the retroactive re-verification pass, which holds the calibration-fold scoring, the held-out evaluation pass, and the re-verification pass, is five hours forty-five minutes at cycle one, nine hours forty-four minutes at cycle two, thirteen hours fifty-four minutes at cycle three, and twenty-one hours fifty minutes at cycle five. A full cycle therefore takes at least these measured windows plus the fine-tune and stream-chunk admission steps, so the end-to-end cycle time grows from roughly six hours at cycle one to more than twenty-two hours at cycle five.

Downgraded reproduction envelope: A consumer sixteen-gigabyte video-memory GPU reproduces the trajectory under five parameter changes registered in the runner: the SIL batch size drops from four to two, the gradient accumulation step count rises from four to sixteen (an effective batch size of thirty-two, twice the production value of sixteen), the precision drops from bfloat16 to float16 with dynamic loss scaling (consumer-tier Ampere bfloat16 paths are slower than the equivalent float16 path), the generation new-token cap drops from five hundred and twelve to one hundred and ninety-two to fit the smaller key-value cache, and activation checkpointing is enabled. The LoRA rank and alpha stay unchanged. The expected per-cycle wall-clock penalty is approximately two-point-two to two-point-six times the production envelope, so the full trajectory on a downgraded GPU takes correspondingly longer than on the production envelope, by the same factor.

Writeup envelope: The thesis is written on a Windows Subsystem for Linux 2 environment with an RTX 3060 (twelve gigabytes of video memory) and sixteen gigabytes of host random-access memory. The sixty-four-gigabyte Wikipedia FAISS passage index exceeds the host random-access memory by a factor of four and cannot be loaded on the writeup envelope; all retrieval-side numbers in the thesis are read from cached evaluation JSON files under `outputs/full_run/` rather than recomputed locally. The local writeup envelope hosts the thesis source, the table and figure compilation, and the signal-level audits over saved verifier traces, but does not host the live retrieval substrate. The semantic-entropy composite-hallucination-metric rescore registered as a deferred receipt in Section 6.2.9 is deferred for the same reason.

E.6 Verification of reproduced numbers

A reproducer who has executed the full trajectory uses the ten-item checklist below to confirm the headline numbers match the thesis within the stated tolerance. The

per-benchmark exact match at three-hundred-sample fold size carries a stochastic floor of approximately ±0.5 percentage points from the non-deterministic cuDNN attention path; the pooled exact match across the five-benchmark panel tightens to approximately ±0.3 percentage points; the composite-hallucination metric and the storage-precision ratio tolerate approximately ±1.0 percentage point; the integer-count quantities (commit fraction, win-and-loss tallies) must match exactly.

Pooled CAEM exact match at C2 and C5. • Receipt: per-sample evaluation files at `outputs/full_run/eval/` (cycle indices 2 and 5), computed mean of the `em` field (or `em_llm_judged` on TruthfulQA) across the per-cycle five-benchmark panel. The rendered table is the pooled-trajectory table under `figures/auto/`.

- Expected: 0.544 at C2, 0.517 at C5.
- Tolerance: ±0.3 percentage points.

Pooled composite hallucination metric at the same cycles. • Receipt: the same per-sample evaluation files plus the canonical composite-hallucination-metric implementation at `eval/metrics.py`.

- Expected: 0.107 at C2, 0.109 at C5.
- Tolerance: ±1.0 percentage point.

Per-benchmark exact match at cycle five. • Receipt: `outputs/full_run/eval/<bench>_cycle5.json`, computed mean of the per-sample `em` field (or `em_llm_judged` on TruthfulQA).

- Expected: each per-benchmark cell must match Table 5.5.
- Tolerance: ±0.5 percentage points.

Calibration-fold pooled storage precision band. • Receipt: `outputs/full_run/pi_store_trajectory.csv`, column `cal_fold_pi_store` across the committed-cycle rows.

- Expected: band $[0.732, 0.746]$.
- Tolerance: ±1.0 percentage point on each endpoint.

Cycle-zero validation-gate composite Cohen's d. • Receipt: `outputs/analysis/weight_validation_v2_1.json`, field `composite_cohen_d_id.value` (the canonical v2.1-panel validation gate; the legacy `outputs/cycle_0/weight_validation.json` from the pre-v2.1 panel is preserved for audit but superseded).

- Expected: +0.635.
- Tolerance: ±0.02.

Memory pool size at cycle five close. • Receipt: `outputs/full_run/memory_store_cycle_5.meta`, computed entry count or `len(json.load(...))` if persisted as JSON.

- Expected: 12 175 entries.

- Tolerance: ±5 for content-hash collision drift.

Realised commit fraction and abort cycle.
- Receipt: `outputs/full_run/stochastic_equilibrium_trajectory.csv`, columns `verdict` and `committed`.
- Expected: commit pattern $(1, 1, 1, 0, 1)$ across C1 through C5; the cycle-four ABORT under the TriviaQA-test retention probe; realised commit fraction $\pi_{\text{commit}} = 4/5$.
- Tolerance: exact match required.

Retrieval-utility classifier ablation pooled exact-match lift.
- Receipt: `outputs/analysis/phase1d_vs_phase1e_ruc.json`, computed pooled exact-match difference between the two arms.
- Expected: +5.8 percentage points.
- Tolerance: ±0.5 percentage points.

Architectural-contribution slice (training panel vs. B1 at C2 and C5).
- Receipt: per-sample evaluation files at `outputs/full_run/eval/` (cycle indices 2 and 5) against matched zero-shot baselines under `outputs/baselines_phase1e/zero_shot/`, restricted to the three training benchmarks of the realised panel.
- Expected: +20.3% relative exact-match gain at C2 and +13.7% at C5; −29.4% relative composite-hallucination reduction at C2 and −20.3% at C5 against the same zero-shot floor.
- Tolerance: ±1.0 percentage point relative.

Significance-panel win/loss tallies at C2 and C5 (post-Haiku).
- Receipt: the significance-test CSV records under `outputs/full_run/` (C5 anchor) and `outputs/` (C2 post-Haiku anchor), computed counts of licensed rows with positive exact-match delta (wins) versus negative delta (defeats), and analogous on the composite-hallucination axis.
- Expected: nine exact-match wins and three defeats at C5; twenty-one composite-hallucination wins and three defeats at C5; fourteen exact-match wins and three defeats at C2; twenty composite-hallucination wins and four defeats at C2.
- Tolerance: exact match required.

A reproducer whose readings clear every item of the checklist within the stated tolerance has matched the thesis to within the determinism floor of the underlying compute envelope. Any item that fails by a margin larger than the registered tolerance indicates either a determinism gap that the runner did not absorb (the runner's seed-everything helper is the first place to inspect), a version pin mismatch against Table E.1, a hardware-envelope downgrade not registered in the runner's parameter overrides, or a genuine correction that the empirical chapter should adopt.

Appendix F

Glossary

The glossary collects every concept word that recurs in body prose and whose meaning a reader who has read Chapter 1 alone might not infer. Each entry pairs the term with a one-sentence concept-word definition and a cross-reference to the chapter subsection that introduces the term. Acronyms used in body prose are listed together with their expansions; concept terms are listed in alphabetical order. Implementation identifiers (module paths, class names, script names) do not appear here; they live in Chapter A.

F.1 Concept terms

Asymmetric rollback.
The retention guard's response on a cycle whose worst-probe ratio falls below the registered floor: the adapter weights revert to the previous successful cycle's checkpoint, but the memory store does not revert, so the storage admissions accumulated during the failed cycle's stream-chunk pass are preserved. The retroactive re-verification pass and the deferred-buffer reconsider pass are skipped on the aborted cycle because both would be provable no-ops under unchanged weights. See Section 5.1.5.

Boost layer.
The L2-regularised logistic regression that aggregates the nine per-signal calibrated probabilities produced by the per-signal isotonic regression stage into the final calibrated probability composite. Introduced at Section 4.2.7.

Calibration fold.
The five-hundred-sample-per-training-benchmark labelled slice that fits the per-signal isotonic regression, the per-benchmark shrinkage prior, and the boost-layer logistic regression at every successful cycle boundary. Held content-hash disjoint from the seed pool, the purity-validation slice, the stream chunks, the evaluation fold, and the test fold. See Section 5.2.1.

Calibrated-probability composite.
The two-stage scoring head that maps the nine verifier signals into a single calibrated correctness probability $u_{\text{stored}} \in [0, 1]$ through per-signal isotonic regression followed by the boost layer. The fixed-threshold storage gate reads against this output. See Section 4.2.7.

Combined dispatch score.
The router's weighted blend of the cosine similarity to the nearest stored entry and the stored confidence of that neighbour, against which the memory-direct tier threshold is read. See Section 4.2.4.

Committed cycle.
A cycle whose self-improvement fine-tune cleared the retention guard and whose calibration-fold refit was applied at the cycle boundary. The realised trajectory records four committed cycles (C1, C2, C3, C5) against the cycle-four abort. See Section 5.1.5.

Composite evaluation score (CES).
The five-axis geometric-mean roll-up that combines accuracy, episodic contribution, retention, calibration, and verifier reliability into a single per-cycle scalar. The CES peak across the trajectory falls at cycle five. See Section 5.1.2.

Composite hallucination metric (CHM).
The eight-axis equal-weighted mean over the post-generation hallucination subtypes (confident confabulation, false refusal, factual fabrication, logical fabrication, off-topic, template leak, defensive evasion, over-long generation). The architecture-vs-baseline CHM contrast is the headline hallucination-reduction axis. See Section 5.1.2.

Cycle.
A single iteration of the self-improvement loop: a stream-chunk pre-routing pass that accumulates storage admissions, a cycle-boundary recalibration that refits the per-signal isotonic curves and the per-benchmark shrinkage and boost-layer weights, a retroactive re-verification pass that updates the stored confidences against the recalibrated verifier, and a self-improvement fine-tune that distils verified retrieval-augmented reasoning into the generator's parametric capacity. See Section 4.1.2.

Deferred buffer.
The bounded-capacity reconsider queue that holds outputs whose calibrated probability composite cleared the deferral threshold but fell short of the storage threshold. Each entry carries an age counter, is rescored under the recalibrated verifier at every successful cycle boundary, and is dropped from the buffer when the counter reaches the registered time-to-live of two reconsider invocations without a promotion. See Section 4.2.10.

Empirical-precision anchor.
The realised storage precision on the calibration fold at the locked storage cut-point, audited once per cycle and reported as the headline diagnostic that licences the calibration-consistent precision floor of the pool-purity theorem. See Section 5.2.5.

Equilibrium-saturation gate.
The two-of-three composite stop trigger that closes the trajectory when at least two of the three signals fire simultaneously: storage saturation under repeated zero-admission cycles, MMLU-floor breach against the cycle-zero pristine baseline, and composite-evaluation-score plateau across the late-cycle subsequence. The realised trajectory closes at cycle five under the storage-saturation and MMLU-floor signals. See Section 5.1.6.

Evaluation fold.
The three-hundred-sample-per-benchmark held-out slice on which every per-cycle

headline metric is computed. Held content-hash disjoint from the calibration fold, the seed pool, the purity-validation slice, and the stream chunks. See Section 5.1.1.

Fixed-threshold storage gate.
The architecturally frozen storage admission rule that admits a generated output into the memory pool when its calibrated probability composite clears the registered cut-point $\tau_{\text{store}} = 0.60$. The cut-point does not move across the trajectory; the per-cycle composite refit moves the signal-to-probability mapping the cut-point operates on. See Section 4.2.9.

Four-outcome decision tree.
The per-query post-verification routing rule that maps each generated output to one of the four outcomes STORE, DEFER, ABSTAIN, or DISCARD on the basis of the calibrated probability composite and the grounding signal. See Section 4.2.9.

Matched-protocol contract.
The pairing rule that every per-benchmark contrast between the architecture and an external baseline is computed sample-by-sample on the identical evaluation-fold sample identifiers under the identical prompt template family and the identical scoring rule. See Section 5.3.1.

Memory-direct tier (Tier 1).
The serving tier that returns the nearest stored entry's answer directly when the combined dispatch score clears the registered Tier 1 threshold. The cheapest of the three tiers per query because it neither runs the generator nor retrieves passages. See Section 4.2.4.

Memory-poisoning rate.
The share of stored entries whose stored answer is wrong against the ground-truth label, audited at every cycle boundary as the complement of the storage precision. The validation-gate rule registers this rate at a fifty-percent ceiling on every training benchmark at cycle zero. See Section 5.2.7.

Per-cycle composite refit.
The cycle-boundary recalibration of the per-signal isotonic curves, the per-benchmark shrinkage prior, the per-benchmark temperature scalar, the per-benchmark safety floor, and the boost-layer weights and intercept. The cut-points themselves are architecturally frozen; the refit moves the signal-to-probability mapping under which the cut-points operate. See Section 5.2.4.

Pool purity.
The empirical correctness rate of the stored memory pool against the ground-truth label, reported as π_{store} on the calibration fold (the theorem-anchored reading) and on the evaluation fold (the deployment-realised reading). The calibration-fold band across the committed-cycle subsequence is $[0.732, 0.746]$. See Section 5.1.6.

Pre-routing confidence (u_{pre}).
The single-pass scalar that fuses the generator's token-probability aggregate over a short greedy decode with the encoder-layer convergence ratio into a per-query confidence reading consumed by the safety override. See Section 4.2.2.

Production envelope.
The per-query four-outcome rendering on the deployment surface: every served answer

carries the calibrated correctness probability as a continuous percentage alongside the categorical decision tag (store-and-serve, defer, abstain, retrieve-and-serve). See Section 3.1.1.

Purity-validation slice.
The five-hundred-sample-per-training-benchmark slice held disjoint from the calibration fold and the evaluation fold, against which the empirical-precision audit is run once per cycle to license the calibration-consistent precision floor. See Section 5.2.1.

Retention guard.
The cycle-boundary mechanism that measures the post-fine-tune generator on the three-probe multi-modal panel (MMLU, TriviaQA-test, CommonsenseQA-test) and triggers the asymmetric rollback when the worst-probe ratio against the cycle-zero pristine baseline falls below the registered floor of 0.93. See Section 5.1.5.

Retrieval-augmented tier (Tier 3).
The serving tier that retrieves the top-k passages from the dense passage index, reranks them through a cross-encoder, and runs a grounded generation against the reranked context. The architectural fallback when neither the memory-direct nor the zero-shot tier earns the right to answer. See Section 4.2.4.

Retrieval-utility classifier (RUC).
The binary classifier that intercepts the safety-veto fall-through and the score-formula fall-through of the three-tier router and decides whether to dispatch the query to the retrieval-augmented tier or to the zero-shot tier on the basis of a learned per-query retrieval-utility prediction. Closes the retrieval-amplification regression on misconception-bait benchmarks. See Section 4.2.5.

Retroactive re-verification.
The cycle-boundary pass that re-scores every stored entry under the recalibrated verifier, prunes entries whose recalibrated stored confidence drops below the retention threshold $\tau_{\text{retro}} = 0.50$, and writes the recalibrated confidence back to the memory store. Skipped on aborted cycles under the asymmetric-rollback contract. See Section 4.2.10.

Safety override.
The router rule that dispatches the query off the score-formula path whenever the pre-routing confidence falls below the registered safety floor $\phi_{\text{safe}} = 0.38$. Post-classifier, the retrieval-utility classifier intercepts the safety-veto fall-through and chooses between the zero-shot and retrieval-augmented tiers. See Section 4.2.4.

Seed pool.
The one-thousand-sample-per-training-benchmark slice used to populate the cold-start memory through the cold-start seeding pass at the registered cold-start storage threshold of 0.50. See Section 5.1.1.

Self-improvement loop.
The cycle-boundary fine-tune that draws its training pool from stored memory entries whose calibrated probability composite clears the SIL admission threshold $\tau_{\text{train}} = 0.70$, applies the parameter-efficient update through the bounded low-rank adapter envelope on the frozen backbone, and re-tests the post-fine-tune model against the retention guard. See Section 4.2.10.

Stochastic equilibrium.
The calendar-timeline refinement of the bounded-band stationarity result that reads the realised commit fraction π_{commit} off the per-cycle commit pattern. The realised pattern $(1, 1, 1, 0, 1)$ yields $\pi_{\text{commit}} = 4/5$. See Heuristic 4.7.

Stored confidence (u_{stored}).
The calibrated probability composite's output for a generated answer: the per-signal isotonic curves and the boost-layer logistic regression aggregate the nine verifier signals into a single scalar in the unit interval, against which the storage cut-point is read. See Section 4.2.7.

Stream chunk.
The per-cycle pre-routing pass over a fresh slice of training-benchmark samples (two thousand FEVER, two thousand TriviaQA, seven hundred CommonsenseQA per cycle) that accumulates the per-cycle storage admissions, deferred entries, abstentions, and discards. See Section 5.1.1.

Test fold.
The held-out slice reserved for the post-trajectory comparative panel. Distinct from the evaluation fold against which the per-cycle headline metrics are reported. See Section 5.1.1.

Training-pool admission threshold (τ_{train}).
The second operating point above the storage cut-point at which a stored memory entry is admitted into the self-improvement training pool. Set at 0.70 and held strictly greater than the storage cut-point $\tau_{\text{store}} = 0.60$ so the SIL pool is a strict refinement of the storage pool. See Section 4.2.10.

Verifier-balanced-accuracy precondition.
The pre-specified floor on the verifier's balanced accuracy at the locked storage cut-point: the precondition of the purification condition (Proposition 4.2) on which the calibration-consistent precision floor of Theorem 4.1 relies, audited at cycle zero through the composite Cohen's d gate at the registered floor of 0.20. See Section 5.2.7.

Verifier ensemble.
The nine-signal post-generation block that converts a generated answer into the four-family signal vector (internal calibration, sample-set agreement, external grounding, question-answer relevance) feeding the calibrated probability composite. See Section 4.2.6.

Zero-shot tier (Tier 2).
The serving tier that runs a generation on the fine-tuned backbone without retrieving passages, dispatched when the similarity to the nearest stored entry clears the registered Tier 2 threshold but the combined dispatch score does not clear the Tier 1 threshold. See Section 4.2.4.

F.2 Acronyms

Bootstrap CI.
Bootstrap confidence interval. The percentile-method confidence interval on the per-sample exact-match-difference vector at the matched-paired sample size of three hundred. See Section 5.3.3.

CES.
Composite evaluation score. See the concept entry above.

CHM.
Composite hallucination metric. See the concept entry above.

Cohen's d.
An effect-size measure for the standardised mean difference between two distributions. The cycle-zero validation gate reads the composite Cohen's d between exact-match-correct and exact-match-incorrect outputs against the registered precondition floor of 0.20. See Section 5.2.7.

EM.
Exact match. The registered correctness metric. On TruthfulQA the exact-match rule is the Haiku-judged correctness rule throughout the empirical chapter; on every other benchmark the rule is the standard span match. See Section 5.1.1.

FAISS.
The dense vector similarity-search index used to host the memory store, the Wikipedia passage substrate, and the cross-encoder pair index. See Section 4.2.3.

Haiku-judged.
The Claude Haiku 4.5 large-language-model judge used as the truthfulness adjudicator on the TruthfulQA evaluation fold. The judge reads the question, the model's answer, the correct reference answers, and the incorrect reference answers, and returns a binary truthful-or-not verdict. The uniform Haiku-judged rule is applied to every TruthfulQA exact-match cell in the empirical chapter, including every baseline. See Section 5.1.1.

Holm-Bonferroni.
The step-down multiple-comparison correction applied within each benchmark family of size $m = 8$ on each metric axis. Sorts the raw p-values ascending and multiplies the k-th smallest by $m - k + 1$ under a running-maximum monotonicity constraint, clipping at one. See Section 5.3.4.

LoRA.
Low-rank adaptation. The parameter-efficient fine-tuning technique that injects trainable low-rank matrices into the seven linear projections of every transformer block of the frozen base generator at rank 32 and alpha 64. See Section 4.2.10.

McNemar test.
The paired significance test for binary outcomes, applied on the exact-match axis. Reads only the two discordant cells of the matched-pair contingency table under Edwards' continuity correction. See Section 5.3.2.

MMLU.
Massive multitask language understanding. The fifty-seven-subject multiple-choice benchmark used as one of the three probes in the multi-modal retention probe panel. See Section 5.1.5.

NLI.
Natural language inference. The canonical three-class textual-entailment task (entailment, contradiction, neutral) in the literature; the verifier's grounding signal family consumes the binary supported-versus-unsupported MiniCheck variant of this task,

with the contradiction class structurally unavailable under the deployed backend. See Section 4.2.6.

RUC.
Retrieval-utility classifier. See the concept entry above.

SBERT.
Sentence-BERT. The siamese-network-fine-tuned sentence encoder that produces the 768-dimensional embeddings consumed by the memory-store lookup, the cycle-boundary consolidation pass, and the cold-start seeding pass. See Section 4.2.3.

SIL.
Self-improvement loop. See the concept entry above.

Wilcoxon signed-rank.
The paired significance test for continuous outcomes, applied on the composite-hallucination-metric axis. Reads the signed-rank sum of the per-sample architecture-minus-baseline composite-hallucination-metric difference vector. See Section 5.3.2.

Appendix G

Architecture positioning, frequently asked questions

The body chapters specify what CAEM does and how. This appendix collects the comparative-positioning and design-justification questions a reviewer in an adjacent area would ask, and answers each by naming the specific mechanism CAEM contributes that the alternative would lack. The first part answers *why CAEM rather than the simpler alternative of plain exact-match-labelled fine-tuning at the same labelling budget* along the six load-bearing differentiators. The second part answers *why the calibration discipline and cycle-boundary ordering chosen, rather than the straightforward alternatives.*

Part I: Positioning against plain fine-tuning

G.1 What does the per-query precision contract give a user that plain fine-tuning cannot?

Plain fine-tuning at the same labelling budget produces a model with an average exact-match accuracy on the held-out evaluation set; the deployed system serves whatever the model generates, with no per-query confidence axis. CAEM's calibrated-probability composite paired with the fixed-threshold storage gate produces a different deliverable: at every query the system outputs a calibrated probability u_{stored} that the served answer is correct, and at the storage threshold $\tau_{\text{store}} = 0.60$ the cycle-zero calibration fold's empirical precision projects forward as a one-sided lower bound on the next cycle's storage precision under exchangeability between the calibration fold and the cycle-$t + 1$ candidate pool; the per-cycle composite refit absorbs distribution drift into the signal-to-probability mapping so that the same threshold continues to address the same precision target as the generator improves.

The user-facing consequence is the abstain class. A query whose composite is below the deferral threshold and whose grounding is below the abstention floor returns an explicit refusal rather than a guessed answer. Plain fine-tuning has no analogous behaviour; the model serves an answer to every query because it has no machinery for deciding when not to answer. For deployments where confident-but-wrong answers carry a higher cost than refusals (clinical question answering, legal document review, customer-facing knowledge serving), the abstain class is the architectural deliverable that plain fine-tuning structurally cannot produce regardless of how much the model is fine-tuned. Chatbots have been documented fabricating every reference offered for a clinical explanation [43], up to 91.4% of chatbot-generated references in systematic-review searches are hallucinated [65], and TruthfulQA's inverse-scaling result [39] establishes that scaling alone does not close this regime; only an architectural intervention with a per-query confidence axis does.

G.2 The memory-direct tier saves inference cost; does that saving survive at small deployment scale?

A memory-direct query serves an answer through a sentence-embedding cosine search over the FAISS-backed episodic memory and returns the stored answer at retrieval-only cost. A zero-shot generation through the same generator pays the full forward-pass cost. A retrieval-augmented generation pays the FAISS lookup cost plus the cross-encoder reranker cost plus the forward-pass cost over a longer context. The per-tier cost ratio at the registered hardware envelope is reported in Section 5.1.4; the memory-direct path is meaningfully cheaper than either generation path.

The cost saving is not constant across scale. At one query per hour the per-query saving is operationally invisible; at the deployment scale projected in Section 3.8.3 the saving compounds across millions of queries. The architectural value of the memory-direct tier is therefore conditional on deployment scale, but it is also conditional on the fraction of queries the memory-direct tier can serve confidently, a fraction that grows monotonically across cycles as the verifier admits more verified episodes

(Theorem 4.1). Plain fine-tuning has no equivalent mechanism: every query pays the full forward-pass cost regardless of how many times the question has been seen, because the knowledge is baked into the weights rather than retrievable as a separable episode. At any scale where the memory-direct hit rate exceeds the architectural overhead of the verifier ensemble, the cost saving is binding; the locked configuration is engineered to enter that regime by approximately Cycle 5.

G.3 What does asymmetric rollback give the operator that a manual checkpoint backup does not?

The retention guard at every cycle boundary measures the post-fine-tune model on a three-probe multi-modal panel (MMLU, TriviaQA-test, CommonsenseQA-test) and computes the per-probe retention ratio $\rho_k = \text{score}_{k,\text{post}}/\text{score}_{k,\text{pre}}$ for each probe k. When any worst-probe ratio falls below the registered floor of 0.93 the cycle is declared aborted: the weights revert to the previous cycle's parameters, and the new memory written during the cycle is preserved unchanged. The architectural property is that the model is replaceable but the memory is not.

Plain fine-tuning has no equivalent guarantee even when the operator manually keeps the previous checkpoint. The reason is that plain fine-tuning at any cycle commits both the weights and the gradient signal that produced them; rolling the weights back rolls back the cycle's accumulated training experience entirely. CAEM's separation of weights from memory means a failed cycle costs only the cycle's fine-tune compute but does not lose the verified episodes that the cycle accumulated. The next cycle inherits the previous cycle's weights together with the larger memory, and the retroactive re-verification pass at the next cycle boundary refreshes the stored composites under the rolled-back generator. This asymmetric-rollback property is the architectural anchor for the memory-growth invariant established in Heuristic 4.7: memory accumulates strictly across cycles even when weights do not. A weights-only architecture has no monotonicity property to anchor.

G.4 If both architectures need exact-match-labelled data at the cycle boundary, how does CAEM use ten to a hundred times fewer labels?

Plain fine-tuning consumes exact-match-labelled samples as the gradient signal. The number of labelled samples drives the size of the supervised-learning fine-tune; ten thousand to one hundred thousand labelled samples per fine-tune is the conventional supervised-learning budget. CAEM consumes exact-match-labelled samples for a different purpose: the labelled batch at every cycle boundary validates the empirical-precision anchor on the storage class. Approximately fifteen hundred labelled samples per cycle (five hundred per training benchmark) is sufficient for stable per-signal isotonic curves, stable per-benchmark composite boost weights under the shrinkage

prior, and a stable empirical-precision reading at the registered fixed cut-points; the labels are not driving the gradient signal but anchoring the precision-floor diagnostic.

The mechanism that produces the order-of-magnitude difference is the verifier ensemble. The verifier runs label-free on every query and admits to memory only the high-confidence subset of traffic; in a production deployment, the cycle boundary's labelled batch is a stratified sample of approximately three thousand entries (about seven hundred and fifty per decision class) drawn from the memory-admission distribution rather than from the unlabelled-traffic distribution, against the fifteen hundred labelled samples per cycle used in the research run. The architecture spends labels validating the verifier rather than training the generator. Plain fine-tuning at the same three-thousand-sample budget would produce a much weaker fine-tune because three thousand labels carrying the gradient signal directly under-trains a three-billion-parameter model. The architecture's labelling budget is therefore not the same instrument as the plain-fine-tuning labelling budget; it is a verifier-precision-validation instrument that is informationally efficient by a factor of ten to a hundred.

G.5 What does the composite catch at inference time that a fine-tune cannot fix?

A fine-tune updates the generator's weights so that the model is more likely to produce the correct answer on questions it has seen. The fine-tune does not alter the model's behaviour on a specific in-flight query that is hallucinating because of a property the training data did not cover: a fresh trivia question on a recently-changed world fact, an off-topic-but-grounded answer that drifts from the question's intent, a confidently-wrong output where the model has a strong prior toward a misconception. These failure modes are not closeable by parametric updates because they are first-order properties of the model's prior, not properties the training data instances reveal.

The calibrated-probability composite catches each of these failure modes at inference time through a different signal family. Confidently-wrong outputs trip the early-exit conjunction $(u_{\text{internal}} \geq 0.70) \wedge (p_{\text{ground,max}} \leq 0.20)$, route to retrieval-augmented regeneration, and avoid serving the hallucinated answer. Off-topic-but-grounded outputs are caught by the question-answer relevance signal $q_{\text{a,rel}}$, which scores the served answer against the question through a separate cross-encoder that the generator does not influence. Common-misconception outputs that match the model's prior are caught by the sample-set agreement family: the chain-to-answer entailment probability p_{entail}, the average pairwise similarity s_{avg} over $M = 3$ resampled chains, and the MC-dropout dispersion signal u_{dropout} over $K = 5$ stochastic forward passes (the originally-registered semantic-entropy signal h_{norm} was retired at cycle zero after the per-signal Cohen's d audit found it dominated by the chain-to-answer entailment signal under the deployed verifier backend). The verifier's value at inference time is structurally orthogonal to the value that fine-tuning provides; once the fine-tune saturates against the parametric capacity of the generator, the architecture continues to add value because every query is still re-checked through the verifier ensemble before serving. [83] report that verbalised confidence distinguishes correct from

incorrect outputs only weakly (an area under the curve of about 0.6 for GPT-4) and that no single confidence-elicitation method is consistently best across tasks; the composite therefore aggregates nine signals across four families through per-signal isotonic regression and a regularised logistic boost layer to produce a calibrated correctness probability that does not depend on any one signal.

G.6 What does the zero-shot tier gain from the cycle loop that plain fine-tuning cannot match?

A plain fine-tune produces a single static checkpoint whose zero-shot output distribution does not change between deployments. The cycle loop produces a sequence of checkpoints whose zero-shot output distribution evolves under a specific architectural pressure: at each cycle boundary the self-improvement loop draws its training pool from stored episodes whose answers were originally produced by the retrieval-augmented tier and were certified by the nine-signal verifier, so the fine-tune folds verified retrieval-augmented reasoning into the generator's parametric capacity. The next cycle is therefore able to answer a measurable share of the previously retrieval-dependent queries from the zero-shot tier alone, without paying the retrieval cost. The deployment-side reading of this mechanism is a tier-share migration over cycles: the zero-shot tier's share of the served stream grows, the retrieval-augmented tier's share contracts, and the per-query cost of the workload falls accordingly.

Plain fine-tuning has no analogous mechanism because it has no separate retrieval-augmented tier whose certified outputs to distil from; the training data for a plain fine-tune is either pre-collected and static or it accumulates without the verifier-gated quality filter that the CAEM cycle imposes. The empirical receipt across the realised trajectory in Chapter 5 is the per-tier hit-rate trajectory over the four committed cycles (C1, C2, C3, C5; C4 aborted under the retention guard), reported in Section 5.1.4 alongside the retrieval-utility classifier ablation of Section 5.4.2 that decomposes the share migration into the self-improvement distillation contribution and the per-query routing refinement contribution.

G.7 When would CAEM not be worth deploying?

The architecture is not a free contribution. Each of the six differentiators above corresponds to an architectural component whose cost is non-zero: the verifier ensemble pays forward passes through MiniCheck, the cross-encoder, and the sentence encoder at every query; the labelled cycle-boundary batch requires ongoing labelling effort; the cycle policy requires operational discipline. The architecture is worth deploying exactly when at least one of the six differentiators is binding for the deployment, and is not worth deploying when all six relax simultaneously:

No abstain class. Deployments with no requirement for an abstain class.

No scale compounding. Deployments where memory-direct cost savings do not compound (very low traffic, or a product requirement that every query traverse the same path for uniformity).

Distillation traction absent. Deployments where the zero-shot tier's per-cycle improvement does not bind because the deployed workload distribution is too narrow for the SIL-driven distillation to gain traction, or too broad for a 3B-class generator to absorb at any cycle horizon.

Manual checkpoint sufficient. Deployments where a manual previous-checkpoint backup is sufficient and an external snapshot pipeline already solves the catastrophic-forgetting recovery problem.

Labelling abundant. Deployments with a labelling budget large enough to support frequent direct fine-tuning, where the verifier-as-sample-efficiency-multiplier value collapses because labelling cost is not the binding constraint.

Below parametric ceiling. Deployments below the parametric-capacity ceiling of the underlying generator, where direct fine-tuning still produces meaningful accuracy gains so the inference-time-error-catching value of the composite is dominated by what fine-tuning would also provide.

A deployment that satisfies all six conditions simultaneously is one where plain fine-tuning at the same labelling budget would deliver comparable user-facing quality with simpler implementation. A deployment that fails any one of these conditions is one where the corresponding architectural component is binding. The thesis's value proposition is the regime where at least one of the six conditions binds; the empirical receipts in Chapter 5 measure the contribution of each component on the registered five-benchmark workload, where every component is binding by construction of the failure-mode taxonomy in Section 2.1.2.

Part II: Design-justification: calibration discipline and cycle-boundary ordering

The architecture makes several non-obvious design choices in how the calibration fold is constructed, how the cycle boundary is sequenced, and how the deduplication discipline guards the disjointness of the data partition. Part II answers each as a direct "is this correct, and why not the simpler alternative" question.

G.8 Why use the same calibration fold across all cycles of the registered horizon rather than rotating folds?

The calibration fold is a fixed thousand-five-hundred-sample partition (five hundred samples drawn from each of the three training benchmarks) constructed once at pool-builder time and reused at every cycle boundary. The natural alternative would be to rotate the fold across cycles, drawing a fresh fifteen-hundred-sample slice each cycle from a larger reservoir. The same-fold protocol is preferred for four reasons.

First, the cycle-boundary recalibration is itself a measurement. The labelled fold is the reference distribution against which the verifier's calibration is read; if the reference distribution varies between cycles, then a cycle-to-cycle change in the fitted curves entangles two sources of variation (the verifier improving under self-improvement, and the reference set drifting) and the trajectory analysis cannot decompose them. A fixed reference fold is the variance-isolation lever that lets the per-cycle isotonic and composite-calibration readings be attributed cleanly to verifier change.

Second, the gold labels on the fold are constant across cycles. The model's verifier signals on a given fold question shift cycle-to-cycle (because the post-fine-tune model produces different token probabilities, dropout variances, semantic-entropy values, and so on), but the question's correct answer does not. Re-fitting the per-signal isotonic curves on the same (variable signals, constant labels) data keeps the calibrator matched to the current model, which is the regime in which the post-hoc calibration literature finds such calibration reliable; calibration fitted before a model or data shift degrades after it [25]. Rotating the fold would replace the constant-labels property with a moving-target labelling distribution that confounds the fit.

Third, the fixed-threshold empirical-precision anchor holds against the same-fold protocol. The anchor requires that the calibration-fold composite distribution at the registered cut-point projects forward as a one-sided lower bound on the cycle-$t+1$ candidate pool's storage precision under exchangeability between those two distributions. The protocol does *not* require exchangeability between cycle N's calibration fold and cycle $N+1$'s calibration fold; the same-fold choice across cycles is therefore not a contract violation, because the per-cycle composite refit absorbs any drift in the underlying signal distribution into the signal-to-probability mapping rather than into the threshold.

Fourth, the storage cut-points themselves are held constant across cycles at $\tau_{\text{store}} = 0.60$ and $\tau_{\text{defer}} = 0.45$. With the same fold every cycle and the same cut-points across cycles, any cycle-to-cycle variation in the realised storage precision at the cut-point is attributable to the per-cycle composite refit absorbing distribution drift, rather than to fold-resampling variance or to the cut-points themselves moving.

A defensible alternative for a substantially longer trajectory (say fifty or more cycles) would be a rotating-fold protocol drawn from a five-thousand-sample reservoir partitioned into ten overlapping subsets, where each cycle's fold differs but is still an honest sample from the labelled population. The variance-isolation argument weakens at long horizons because the same-fold protocol risks calibration over-fitting to fold-specific noise. The realised six-cycle horizon (the cycle-zero calibration plus cycles one through five, with cycle four aborted under the retention guard and cycle five closing the trajectory under the equilibrium-saturation gate) is short enough that variance isolation dominates, and the rotating-fold extension is registered as a future-work item should the trajectory horizon extend.

G.9 In production each calibration split would necessarily differ; how is this reconciled with the same-fold research protocol?

In production there is no pre-built calibration partition. The labelled batch at every production cycle boundary is drawn from accumulated serving traffic via stratified sampling (per the production runbook), so each cycle's fold is a fresh sample of three thousand or so labelled queries with content that varies cycle-to-cycle. This is a different operational mode from the research protocol but architecturally compatible with it.

Three points reconcile the two modes. First, the fitting procedure is fold-agnostic. The per-signal isotonic regression is a deterministic monotone fit on whatever (signal, label) pairs are supplied; sklearn's isotonic regression does not care whether the questions were the same as in a previous fit or different. The cycle-zero cut-point selection walks the labelled fold sorted by composite and picks the threshold at the registered precision target; this is run once at cycle zero and the resulting cut-points are held constant across cycles. Both procedures work identically on a fixed fold (research) or a rotating fold (production).

Second, production loses one variance-isolation property and gains a different fidelity property. The research protocol uses a fixed fold to isolate verifier-distribution change as the only source of cycle-to-cycle calibration variation. Production cannot use the fixed fold because the fold material does not exist. Instead, production gains a different property: the calibration fold matches the live deployment distribution exactly, because it is drawn from live deployment traffic. Research's fixed fold is a proxy for deployment (drawn from training-benchmark dev splits); production's rotating fold is the real thing. Each protocol is correct under its own labelling-availability constraint.

Third, the EMA smoothing on the thresholds carries forward the previous cycle's threshold value into the next cycle's fit at weight $\alpha_{\text{ema}} = 0.7$. In production this means a single cycle's traffic anomaly (an unusual spike of out-of-domain queries, for example) cannot drag the threshold to a degenerate value, because the next cycle's labelled batch will return the threshold trajectory toward the running average. The same EMA mechanism that prevents same-fold over-fitting in research prevents rotating-fold instability in production.

The practical consequence: an operator who wants production calibration to behave as close as possible to research-mode discipline can stratify the production sample to be representative of the five-benchmark workload composition and can hold the labelled batch size near the research-mode fifteen-hundred-sample point. An operator whose live distribution differs sharply from research can let the production fold reflect the live distribution and accept that the calibration trajectory will diverge from the research trajectory by exactly the distribution-shift magnitude. Both behaviours are correct; the choice is between distribution fidelity and protocol parity.

G.10 Why apply the cycle-boundary recalibration before retroactive re-verification rather than after?

The cycle-boundary order is: self-improvement fine-tune, then score the calibration fold under the now-updated generator, then re-fit the temperature scalar and per-signal isotonic curves and per-benchmark composite boost weights on the freshly-scored fold (the storage cut-points themselves are held constant across cycles), then reload the verifier from the new calibration files, then run retroactive re-verification against the now-recalibrated verifier. The natural alternative would be to run retroactive re-verification first under the previous cycle's calibration and recalibrate afterwards. The current order is the architecturally correct one for two reasons.

First, the post-fine-tune signal distribution falls outside the previous cycle's isotonic fit envelope on a non-trivial fraction of stored episodes. The fine-tune saturates token probability and collapses semantic entropy on chains the model trained on; the previous cycle's isotonic curves were fitted on the previous cycle's signal distribution and do not extrapolate cleanly past the upper edge of their training-input range. Running retroactive re-verification under the previous cycle's calibration produces calibrated probabilities on memorised chains that are artefactually low, and the resulting composite drops below the prune threshold for many EM-correct stored entries. The empirical evidence for this artefact is documented in the original cycle-one prune of a large share of the cold-seed memory under the stale-calibration ordering; the corrected order has the recalibration absorb the post-fine-tune signal-distribution shift before the prune rule fires.

Second, the corrected order preserves the architectural intent of retroactive re-verification. The re-verification pass is meant to upgrade the stored composite of episodes whose underlying correctness has improved under the new generator and to prune episodes whose correctness has genuinely degraded. The pass cannot do either if the calibration it reads through has not yet been updated to the new model. Under the corrected order, retroactive re-verification reads through fresh per-signal isotonic curves and fresh per-benchmark boost weights at the registered fixed cut-points; an entry whose correctness genuinely improves under the new model maps to a higher calibrated probability and stays in memory with an upgraded u_{stored}, an entry whose correctness genuinely degrades maps to a lower probability and gets pruned. The order is therefore not an implementation detail but a precondition on the theorem-licensing receipt for pool purity in Theorem 4.1.

G.11 Why disable the confabulation early-exit during retroactive re-verification when it remains active at query time?

The early-exit rule fires when the internal-confidence composite is high (above $\eta_u = 0.70$) and the top-passage entailment is low (below $\eta_g = 0.20$). At query time this conjunction is the architectural defence against confabulation: a model that is internally confident on a fresh question but has no supporting passage is generating a confident hallucination, and the rule routes the answer to retrieval-augmented regeneration rather than serving the hallucination. At retroactive re-verification time the same conjunction reads differently. The fine-tune just memorised the stored chains, so the internal-confidence composite is automatically near one for every entry the self-improvement loop touched; the first conjunct is no longer informative. Letting the rule fire under this regime collapses the conjunction into "low top-passage entailment, regardless of confidence," which is a much wider catchment than the registered confabulation rule and over-prunes EM-correct stored entries whose chain happened to have thin passage support.

The architectural fix is a query-time-only flag on the rule. The verifier's verify method takes an is_query_time parameter that defaults to true (preserving query-time inference and deferred-buffer reconsideration behaviour) and is set to false in the retroactive re-verification closure. Under the false setting the early-exit conjunction is short-circuited and the verifier falls through to the standard composite-and-threshold path, which still prunes any entry whose recalibrated u_{stored} falls below $\tau_{\text{retro}} = 0.50$. Confidently-wrong stored entries are still removed; EM-correct stored entries with thin passage support are no longer falsely pruned. The rule's intended semantics (catching query-time confabulation) is preserved; its inheritance into retroverify-time is corrected.

G.12 Why is the training threshold strictly higher than the storage threshold rather than equal?

The architecture registers two different thresholds on the calibrated composite. The storage threshold τ_{store} governs admission to the user-facing memory store and to the retrieval tier. The training threshold τ_{train} governs admission to the next cycle's self-improvement fine-tune pool; only entries whose composite clears τ_{train} enter the gradient signal. The convention is $\tau_{\text{train}} > \tau_{\text{store}}$, with the locked configuration setting $\tau_{\text{store}} = 0.60$ and $\tau_{\text{train}} = 0.70$ (held constant across cycles; the training-pool threshold was lowered from 0.75 to 0.70 on the trajectory-launch date to restore a non-empty verified-episode share while preserving the strictly-stricter invariant). The asymmetry is deliberate.

The reason is that the training pool contributes to all subsequent cycles through the generator's weight update, while the storage pool contributes only to the user-facing retrieval tier at the current cycle. A noisy entry in the storage pool serves at most a single Tier-1 retrieval and is re-scored at the next cycle's retroactive re-verification;

the architectural cost of a borderline storage admission is bounded. A noisy entry in the training pool propagates through the gradient signal into every subsequent cycle's generator and can compound the verifier's bias over the trajectory. The noise-tolerant-learning literature [9] bounds the corruption rate at which training under noisy labels still improves a model rather than degrading it; the architecture spends the strict-training-threshold budget to keep the training pool's empirical purity above that rate by construction.

A defensible alternative would be a single threshold $\tau = \tau_{\text{store}} = \tau_{\text{train}}$ that admits the same entries to both purposes. The simpler alternative loses the asymmetry between storage-class consequences (bounded, single-cycle) and training-class consequences (compounding, multi-cycle), and the noise-tolerant-learning bound on training-pool corruption is no longer enforced architecturally. The two-threshold protocol is the registered defence; the empirical purity of the resulting training pool is reported in Chapter 5.

G.13 Why deduplicate the calibration fold at three layers rather than relying on the dataset's native unique identifiers?

The deduplication discipline runs at three layers in the pool-builder. The first layer hashes every sample's question text into a sixteen-character content identifier and dedupes within each loaded dataset split, dropping ten thousand or more duplicate records on average per training-benchmark train split. The second layer dedupes within each individual partition slice (calibration, purity, training, evaluation) to catch any residual within-slice duplicates. The third layer cross-checks the calibration and purity slices against the evaluation slice and drops any sample whose content hash collides with an evaluation-slice sample. The natural alternative would be to rely on the dataset's native sample identifiers and assume disjointness by construction.

The dataset's native identifiers are unreliable for two reasons. The lucadiliello FEVER dump reuses integer claim identifiers across the train and dev splits, so a sample with id 13957 in the train split is content-distinct from a different sample with id 13957 in the dev split, and a join on native id produces spurious overlap warnings. The TriviaQA dump reuses sfq-prefixed and qw-prefixed and qz-prefixed identifiers within a single split, so two content-identical questions can both appear in the same split with the same id. Relying on native identifiers therefore both over-flags overlap (false positives on different-content same-id pairs) and under-flags overlap (false negatives on same-content different-id pairs). Content-hashing on the question text resolves both regimes by definition: identical questions hash to the same identifier regardless of native id; different questions hash to different identifiers regardless of native id collision.

The three-layer structure is the minimum sufficient cover. A single dataset-level dedup pass leaves intra-slice duplicates untouched. A single intra-slice dedup pass leaves cross-slice content collisions untouched (a calibration question that happens to also appear in the evaluation split). A single cross-slice pass relies on the per-slice slice having been itself dedup-clean; without the within-slice dedup, the cross-slice

check sees inflated counts and miscomputes the disjointness invariant. The empirical evidence for the three-layer requirement is reported in the loading-time logs of every cycle: each layer drops a non-trivial count, and the fold size after all three layers matches the registered fifteen-hundred-sample target rather than a stale-protocol nineteen-hundred-or-more sample inflated count.